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September 30, 2004

1. INTRODUCTION TO MATHEMATICS

As we've discussed human beings and their achievements and history, we've talked about them coming into an environment which had been prepared in such a way that they could meet their needs, to continue their existence.

Over the ages, as human beings have found their way and as they've made various lives for themselves, they have built up cultures and civilizations, different ways of life, according to what was found in the different environments. And we have to remember that the environment was built on law and order and pattern.

Human beings, when they came, had been given intelligence which allowed them to understand the law and order and pattern, which also allowed them to reason. They had been granted minds which could imagine, things that were not their in the concrete and an ability of making in reality those things which they had imagined.

As we begin to trace the history of the development of human beings, we see them using their environment to invent from, to create those things which would eventually make their lives better. That power of invention, that power of creation also rests on law and order and exactness. What is made in reality has to conform to the idea in the mind. As human beings invented tools, weapons, shelter, clothing, objects to use, artifacts we have found, and as they were working on making them, they were founding a science which was an exact science. They were using a power which was inherent in their minds – the power of mathematics.

John D. Barrow, in his book, “Pi in the Sky”, wrote the following: “If we look closer still at the technologically successful societies, we find them founded upon the language of mathematics. It lies at the heart of the scientific enterprise and at the root of the computer revolution. No practical progress can flourish without its sustaining presence.”

Mathematics has had to do with the lives of human beings since they first appeared on earth. It has to do with inventions, with discoveries and architecture, with machines and food, with clothing and shelter. The tendency of human beings to create and to invent is based on law and order and exactness. It is inherent in human beings. It belongs to them universally and therefore it also belongs to the child. That is why Maria Montessori always said that mathematics should not be hard for the child to grasp. It

shouldn't be any more difficult for the child to learn mathematics than it is to learn his language. Everyone expects the child to talk and to begin to express his ideas, to use language, but very often there isn't the same expectation there for mathematics. One of the problems is that often the environment does not provide the same opportunities for math as it does for language. This is an area that we need to be aware of and we need to make parents more aware of. The language of mathematics should be part of the young child's environment. It is easy to do if we are conscious of it. "Can you carry these three napkins to the table? One for dad, one for mom and one for you." Bring in counting often.

The child coming into the primary class should therefore have already developed an interest in numbers, just as he's developed an interest in language and should be happy in beginning to use it. Mathematics needs to be presented to the child in such a way that he can conduct a sensorial exploration of the facts about it, so that he can begin to use more and more of the language. The math materials, just like the other sensorial materials, allow the child to symbolize his explorations through the use of language. Without the possibility of a full sensorial exploration, without the possibility of assimilating and exploring the facts in the area of mathematics, without the language, the child enters the elementary level with gaps that need to be filled in. We need to help make sure that the children have access to the entire math program developed for the children's house when they are still in the first plane of development. The more the absorbent mind can be used to help the child take in those facts, that language, the easier it's going to be. By the time the children are in the elementary class, it becomes a memorization activity; they've got to be willing and to choose to memorize those facts. It becomes an act of the will and it becomes harder and harder the older they get. So, the younger they are, the easier the whole thing becomes – when they can take advantage of the absorbent mind.

We have to remember that the psychological characteristics change in the elementary level and this means the materials have to change. We still use some of the primary materials but in a different way. We have to make use of the appropriate characteristics of the child. The primary materials give a foundation for the exploration that belongs to the elementary level; that's why they are important at the first plane.

The materials that Maria Montessori prepared allow for sensorial exploration, a manipulative activity, a concrete experience, repetition of the experience, and then the possibility of putting to that experience a symbol that will allow the child to eventually make his own abstraction – to carry around with him, in his head, math facts. The primary math work follows a sequence that gives the child first of all, the possibility of coming into contact with quantity, after the experience with quantity, then being given the symbol for that quantity – and then putting the two together. This shows that quantity can be symbolized and the symbol can be quantified.

September 30, 2004

Math for the Primary Child

Remembering the math work from primary lessons, we recall it falls into certain groupings. The first thing done is to introduce the base of the decimal system. To do that we have the first group of materials; the numbers from 1-10. Within that group, under that heading, there are several materials, several presentations.

First, quantity is given with number rods, then symbol is given – sandpaper numbers. Then, we connect the symbol and quantity. The material used to do that is a set of cards (tiles) that are used together with the rods. Within this group of materials, there is also the spindle box which shows the idea of grouping and the idea of zero. Then we have cards and counters that introduce the odd and even numbers and to check that children have absorbed the idea of 1-10 and they can sequence those numbers. Then there is the zero-activity or the memory game of numbers, where each child pulls a number and the child will 'jump up and down x times.' This is the group of materials “From One to Ten.” We hope the child sees that this is relatively easy and is able to absorb these lessons readily.

A second group of primary materials comes under the heading of “Introduction to the Decimal System.” We first introduce quantity with the demonstration tray of the golden bead material. Then we introduce symbol using the large number cards. The third step is then putting them together, where the child matches quantity and symbol.

Once they have been introduced to the numbers in the decimal system, we can then introduce the operations using the numbers. All four operations are first introduced with the golden bead material and large and small number cards. We also have several other materials, including the stamp game. All four operations are done using the stamp game. This takes another step toward the symbolic since all the stamps are the same size – not a literal quantity-expressed representation. Also, the stamp game continues for individual work while the golden beads are used for small groups. There is one more material under the heading “Operations” used only for addition and the idea of carrying – the “Dot Game.”

After Operations, we have another heading in math materials called “Teens and Tens” that can be introduced in parallel to the Decimal System or Operations. The Sequin boards are used for this Teens and Tens work – using boards and beads together.

Skip and linear counting is also done, using bead chains – counting and labeling the chains. This work helps with conventions for transitions between 19 and 20, 29 and 30, for example.

Then there is the heading “Memorization.” This is where the materials include the “finger charts” and “strip boards.” There is an addition and subtraction strip board. There is also the addition and subtraction “Snake Game.” All these materials are meant to help the child become familiar with the language of mathematics.

They can't go through life carrying the materials. The children can work in groups with these materials in order to make it fun and exciting. Many children that are auditory should also be saying these facts in order to internalize the facts. Another major heading is “Passage to Abstraction.” This is where the bead frame is used. And finally, the last heading is “Parts of Numbers.” That group introduces fractions.

September 30, 2004

Math for the Elementary Child

In the area of math, the tendencies are continuing to operate. They are there to help the child's development but take a different direction in the elementary environment. The opportunities have to change.

The mathematical mind, the mind and its ability to work mathematically, the mind of curiosity, the mind of inventions, still continues. If the child has had the good fortune to come in contact with the full math program in the Casa dei Bambini, then exploring in a different way should be inviting to the elementary child – if he's realized that math is fun and not difficult to comprehend. He should then arrive at the elementary class enthusiastic about math, anticipating another exciting adventure in mathematics.

We need to remember the special characteristics of the child at this stage of development, particularly the reasoning mind and the desire for abstraction. The power of the reasoning mind is given to human beings to allow them to arrive at abstractions, which means to incarnate knowledge without the limitation of material.

We have to take care at the elementary level not to tie math down to curriculum, to the subject of mathematics, any more than we do any of the other areas. We have to keep math integrated into cosmic education. We have to show the children all of the variety that there is to explore. Math is part and parcel of the universe, of the world we know.

The base of the functioning universe is mathematics. The stars and their courses, the distances between stars, the shapes of their orbits, the formulas for chemical elements, the foundations of buildings, the angle of roof beams, the circumference of a wheel, the rainbow – all of these things have mathematical implications. To be able to explore the universe and to explore the achievements of human beings, we have to be able to make use of mathematics.

If we remember all of those facts the first plane child has been given, now at the second plane we need to help the child use those facts, use those tools, to lead him into abstraction in the world of mathematics.

We also need to remember that once again, the **materials need to be**

limited to the key materials – a key for each concept. We also need to remember that the work of the children required by us and the examples they use in their work provided by us must be limited. Beginning in the primary class, the children should be making up their own examples. In the elementary, they should still be making up their own examples. They will often go beyond what we've presented and often make it quite challenging. We have to have confidence that if they need help, they will come and ask us. One of the things we **need to learn to observe is children's struggles with solving challenges**. We must be careful not to step in too soon. It is our instinct to want to show them, but the children learn most in struggle – but before frustration sets in. If they become frustrated, they are prone to give up. We need to intervene just before this point.

Keep in mind the reasoning power of a child at this age. That power has to have the opportunity to work. This is where our providing the key is so important because then the **child can use the key to exercise the mind in generalization and problem solving**. If they've understood the key, they can take steps beyond this on their own. We can encourage this pursuit by offering little thoughts, "I wonder if...", at the end of presentations.

These are suggestions that intrigue the children. It is at this stage that we talked about one of the characteristics being the great and enormous potential of the intellect to understand and grasp. We've got to allow the intellect to exercise itself in its own stride. We have to keep challenging that intellect to move from step to step – not by holding it back in any way. We must provide opportunities for large projects that capture their imaginations. They need the challenges that engage the reasoning mind.

With that in mind, what else can we do with addition? With subtraction? With multiplication? Multiplication, in particular, gives us many avenues to explore. The child learns long multiplication which makes use of addition. He uses multiplication geometrically. He uses 'category multiplication.'

Another area of exploration related to multiplication, is the natural series of numbers. Suppose we are multiplying a number by itself, 6 times 6, for example. There is something special about what we get if we do that problem. Suppose we multiply by the same number again – we also have a special result in this. Where has the child had preparation for this work? In working with bead chains.

Suppose we multiply with the same number again? This is an idea that can capture the child's attention. The child will see some of the laws of multiplication, the commutative and distributive laws and what they mean. They are not only applied mathematically but also algebraically. We will show numbers that contain other numbers, i.e., multiples of numbers. We'll present what it means to find the least common multiple of numbers. We will break numbers apart and find their factors. These are numbers which

when multiplied together, give us our original number. We'll also discover prime numbers.

The division facts will be extended into long division. We'll show two ways of doing division – distributively and by grouping. In working out long division problems, we make use of subtraction. In the elementary class, we will not only work with whole numbers but also with fractions – two kinds. We'll also see that all four operations can be done with both kinds of numbers.

If we can do compound multiplication, we can do squaring and cubing. If we can square a number, we can take that resulting number apart and find its root, what it started from. If we can cube a number, we can take that resulting number apart and find its root. If we can have ten as the base of our number system, can 2 be used as a number base? Or can 5 be used as a base? These are just further explorations of numbers. We do all these things in the elementary level.

If we can use numbers for a formula, can we use letters and make a formula in a general form, so it will apply to all cases? Algebra, in its essence, is not harder than arithmetic but is rather a simplification of it. If we can have positive integers, can we also have negative integers? All of these are just different aspects of the law and order and formula of mathematics. The material is still concrete, to give the key, so that the mind can manipulate and bring out a pattern through the work of the hands and the manipulative activity. Now the concrete material is not so much to teach a process as it is to give just the next little dose of help that the child needs to understand the next step.

What we have to remember is that our help is needed in different doses by different children. We always have to look at the individual child and try to identify what help that child needs to get to the next level. It is the movement of the mind that we are concerned with now. Each child, when he is ready, will move into abstraction. We cannot dictate when that is going to happen. We cannot plan this progress on their behalf because the time taken varies greatly. The child must work with the concrete material as long as they need it, sometimes helping them to move away from it, step by step. Other children may just leap ahead; again, we cannot preconceive their progress in this case because we would hold them back in this case. Our help must always be customized and fluid. Arithmetic, algebra and geometry – all come from the same base, the needs of human beings. Wood-working and planes make use of mathematics.

If we can keep math related to those fundamental needs, we should be able to see that it is part of life. It is something that should not be difficult. It is often adults who have made it difficult, often because the adult already has developed a negative bias toward math. Many of us who have had a conventional education were presented math only in an abstract way. And many of us could not grasp it in this manner

and did not have a chance to ameliorate this problem. Hands on experience is a critical requirement for children as well as for teachers in training.

September 29, 2004

2. THE STORY OF NUMBERS

Prologue

This is another Great Lesson given about a week after the story of Communication in Signs, about the 6-7th week of school. This is not an introduction to a subject area like the earlier ones are. The children began work in math in the primary classroom.

This story will be an introduction into the history of numbers. New students shall attend and others may attend as they wish.

Begin

Human beings needed a language to talk to one another, to tell one another their ideas and their stories. So they had a language they could speak. Then last week we had a story about the language that they started to write down, the story about the alphabet.

In the same way, people needed a kind of language for their measurements and for their inventions and other things they made, to express how far they walked or how many children they had. Some time, we don't know exactly when, someone first thought of a language for numbers. Right from the very beginning of the time that human beings have been on the earth, it seems very likely that they needed to speak about numbers just as we do. How many children were around, how many people would fit into that cave, how many fish would it take to feed the whole group, how far is it to the river, when will the fruit come again on the trees – all these were things to do with numbers and survival.

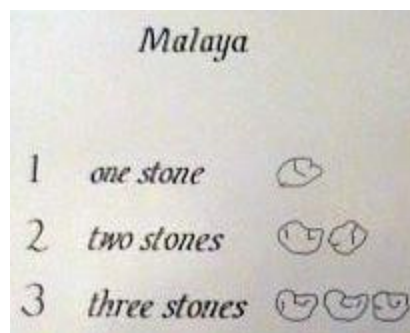
So no matter how far back we go to find out about people living long ago, we have never found a group of people anywhere who didn't have some words for numbers. Some people just had numbers for one and two. Other people, in their language, just had numbers for one, two and three. One group of people, who lived in what is now South America, didn't have words just for numbers, but to mean '1', they used their word for 'alone.' Everything else was their word for many.

So we can't go back to know more, because there aren't any written records for when the language for numbers first began. We think it is very likely that people

needed numbers to count just like we need numbers to count things.

There is one method of counting that's been used by many different people all over the world. That is using the fingers to count. Have you ever used your fingers to count? They are handy because they are always with you.

Other people used other things to count. Some people used sticks. Some used stones or pebbles. Some used fruit or shells. Some people made notches on a stick. Others tied knots on a rope.



Malaya people apparently kept count using stones

This chart shows how people in Malaya, kept track of the quantities one two and three, using stones. We can just imagine that people have needed to count all kinds of things that had to do with their daily lives. Sometimes they also needed to keep records about something or other. If they went on a hunt, how many deer were seen? The person who saw the deer would see the number or quantity, but supposing they wanted to remember that number to tell the other people in the group, how many had been seen. People who used stones, would collect that many stones and carry them around. Other people would make a pile of that many shells and take the shells with them. Others would take a stick, make one notch in the stick for each deer they saw and take it back to the group for later use.

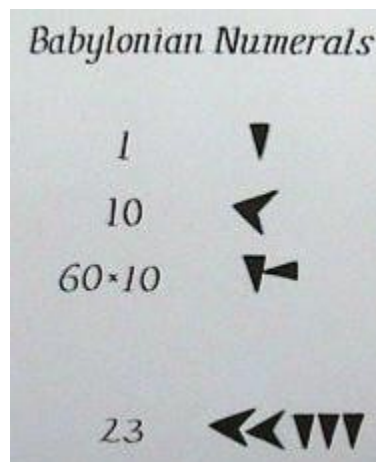
Just as it seems very likely that people were talking for a long time before they began writing down their words, in the same way, there were probably methods for counting long before they began to write down marks for numbers. When did people begin to make marks for their numbers. We do have some records of this.

About 5,000 years ago, the Sumerians, who were another group of people

like the Egyptians and the Phoenicians, could read and write, and they also had some marks for numbers. Their way of writing has been called cuneiform because of the shape of the marks they made. There was a lot of clay in the earth around where the Sumerians lived. They took this clay and made little tablets out of it. Then they used a stick to make their marks for numbers on these clay tablets. Then the clay would dry and a lot of these tablets have been found. So we've been able to see what these marks look like.

Remember we talked about the Egyptians in our last story? The Egyptians lived about the same time as the Sumerians. They grew lots of crops since they were farmers and were very successful at selling their crops. They had a need for very large numbers. So they developed a way to make marks for very big numbers.

The Babylonians came a little while after the Sumerians and the Egyptians. Here is a picture of how the Babylonians made their marks for numbers.



Babylonian numerals and today's equivalent numerals

Remember the story we had about the sandpaper letters and how in that story we said that the Greeks took the letters and changed them a bit. Do you remember? They must have liked doing that sort of thing because they also took the numbers and changed them a little bit as well. Then they decided to make their own marks for numbers. They began to write a mark for the number that was the first letter of their word-name for the number.

The Greek word for 5 was 'Penta', so they took their mark for the first letter from that name 'Penta' and that became their number 5. The Greek word for 10 is 'Deka'; so they took their mark for the first letter of that word-name and that became their number ten. Here is a picture of what some of the Greek numbers looked like.

<i>Greek Numerals</i>	
1	
5	⌒
10	Δ
100	Η
1000	Χ
2857	ΧΧ⌒ΗΗΗ⌒⌒

Greek numbers and translated values

This is how the Greek child would write 2857; a little complicated, right?

Then remember in the last story, I mentioned the Romans and how they also did a little changing of the letters. We also have Roman numerals. These are numbers you will still sometimes see today. The next time you go to the museum, look at the clocks and see how many of them have Roman numerals on their faces. Occasionally even today, you might look near the corner of a building and see Roman numerals telling us when the building was built.

Here is how the Roman child would write 1983.

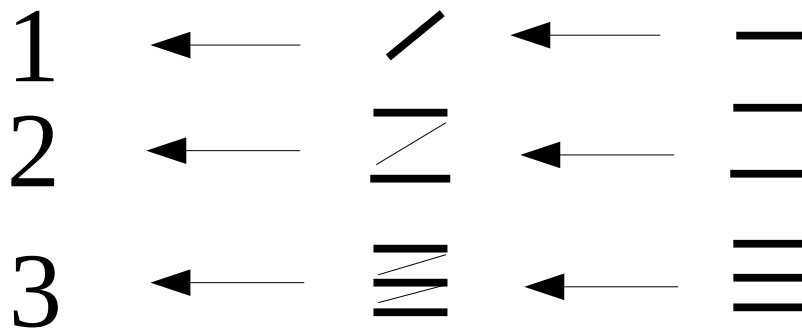
<i>Roman Numerals</i>	
1	I
5	V
10	X
50	L
100	C
500	D
1000	M
1983	MCMLXXXIII

Also, a bit complicated, right? Can you imagine doing multiplication in Roman numerals? That would be tough, right?

You may have noticed with each of these charts, that there is a number that looks more or less the same on each of them; it is the number 1. Isn't that funny. But if we look at the Chinese numerals? Yes, but it's on its side. It's laid down but it's still similar.

<i>Chinese Numerals</i>		
1	一	
2	二	
3	三	5625
4	四	
5	五	五
6	六	千
7	七	六
8	八	百
9	九	二
10	十	十
100	百	五
1000	千	

If we were to write down the Chinese numerals for 1,2 and 3, and imagine that we lightly connect strokes or hurriedly slant writing according to our wrist's inherent technique and anatomy, as we might do in practice, they kind of look like 1, 2 and 3 in our numeral system. Some people think that this is where our numerals came from.



Where do we find the numbers that we use? Where did they come from? Some of the first places where there have been found, signs like our numbers, are signs found in the side of caves in India, carved over two thousand years ago. The numbers carved there are very similar to the ones that we use. The numbers we write are called Arabic numbers but the Arabs didn't use them; that's confusing isn't it? The people in India traded with the Arabs. The Arab merchants took food and clothing and books home from India. Amongst the books was one on arithmetic that used signs almost identical to numbers we use.

Well, the people who were buying and selling thought this was quite a clever system. We know that the symbols were passed on to Europe because there is a page from a book that seems to have been written about 976 AD that shows these symbols and how very similar they are to the ones that we use. So the numbers came through the Arabs, even though the Arabs did not write them. This is how they got the name Arabic numerals.

It is not until books began to be printed that all numbers started to look more or less the same. They became what we think of as standardized. Before that, there were always differences – among different groups of people. There is another interesting thing. Do you remember the 'zero game' from primary. Jump zero times or walk to the door zero times? The Roman numerals don't have a zero but the Indian numbers had a zero. And that meant that you could write any number you wanted by just using the numbers from 1 to 9 and putting the zero in the right place. Well that's the story of our numbers that we use for our arithmetic work. 1,2,3,4,5,6,7,8,9 and zero. Very easy. So much easier – so much easier than some of the other numbers. Once again, we have someone, somewhere, whose name we don't know – to be grateful to – for giving us a way of recording our numbers.

End

Epilogue

We don't have to show all the charts we have. It's good to save two or three for follow-up exploration. In addition to the ones shown here, these are the ones from 976 AD – note the similarity between ours today and theirs. They wrote from right to left but the characters are nearly identical. There is a chart for the Egyptian numerals since children often get interested in Egyptian hieroglyphs. This chart shows changes in our numerals and how they developed.

Material making

These charts are in your charts. They only need to be cut and laminated.

September 30, 2004

3. VISUALIZING NUMERICAL HIERARCHY

Wooden Hierarchical Material - Quantity

Age

6

Purpose

To give the children a clear visual impression of the relationship of numbers and relationship of categories and hierarchies. Sub-consciously, create an awareness that a set of points generates a line and a set of lines generates a surface and a set of surfaces generates solid.

Materials

Floor rugs to hold the following:

Three small green cubes: $\frac{1}{2}$ centimeter cubed, 5 centimeters cubed and 50 centimeters cubed. Middle and large cubes are divided by red lines into 10 equal sections. Two blue prisms (bars): small one is $\frac{1}{2}$ centimeter squared by 5 centimeters, larger is 5 centimeters squared by 50 centimeters. Both are divided by green lines into 10 equal sections. Two red square prisms: smaller is $\frac{1}{2}$ centimeter by 5 centimeters squared, larger is 5 centimeters by 50 centimeters squared. Both are divided by blue lines into 10 equal sections. Set of tickets (labels) all the same size. Numbers are written in black and they go 1,10,100,1000 ... 1,000,000. The tickets and four smallest pieces are generally kept on a tray.



Beginning from a random layout, introduce each piece and arrange them according to the layout above. Children and teacher are on the far side of this illustration, viewing units to millions from right to left. Using a pregnant pause, ask - could there be tens of millions?

Prologue

We've already seen a direction that math work could go based on our story The History of Numbers. What symbol did the Egyptians use for the symbol 1000? They used a Lotus for 1000. The Lotus has many petals, like a thousand. For 100,000, they used a tadpole – since there are many many in a group. For 1,000,000, they used a man in astonishment. Some people used their words for the number two. There are many things of history about math that they will run across by investigating the past.

They will hopefully come to appreciate the universality of mathematics – how each human group had to develop some way of counting. Most people took things from their environment to represent numbers.

Another direction to take is to introduce the wooden hierarchical material. The wooden hierarchical material expands the children's knowledge of the decimal system. It is material that gives an impression of the quantities beyond 1000. It expands on their work with the golden bead materials. In a way, it summarizes their experience with the decimal system material in the primary class. It can be presented at the end of primary or the beginning of elementary, so some children may have already seen it. Even if they have, it should be presented again, a little differently, in the elementary level.

When you present this in the classroom, it would be presented to a small group of 3 or 4 children. The first thing to do is to have the children get out the floor mats and help carry materials to the workspace. Materials are placed into mixed order on the mats. The children would sit next to the teacher; children sit facing the same direction as the teacher. New students shall attend and others may attend as they wish.

Begin

Here's our simple unit (pick up small green). Being a unit it's all alone. **It walks by itself.**

When there are ten of them, they join together and form a line (blue prism of 10 held up).

When ten tens join together, they make a square (hold up red square). Here it is, one hundred.

(leave gap in between simple and thousands as you array the materials)

When ten hundreds are together, they again make a unit (hold up second larger green). Being a unit, it walks alone – 1 thousand.

When 10 thousands are together, they form a line (hold up larger blue prism). We call this 10,000. Here it is, 10,000.

When 10 10,000s come together, they again form a square (hold up larger red square). One hundred thousand.

And when 10 100,000s come together, they form this – one million. Once again it's a unit, and so it walks alone (point to largest green cube). This is one million units.

So we have a **unit, ten and hundreds** of the **simple numbers** (point to small versions).

We have **units, tens and hundreds** of the **thousands** (point to second larger),

and finally we have **units** of **millions** (point to largest cube).

Debbie, can you please show me 100,000? (points to large red square). That's right. How many of these (10,000s) are there in here? (Debbie counts ten of them). Yes, there are 10 of the 10,000s in 100,000.

Rob, can you please hand me 1000? (Rob hands the middle size green cube). That's right. **(This is the second part of a 3-period lessons. Do as much as you need to)**

(begin creating the layered layout – see caption below)

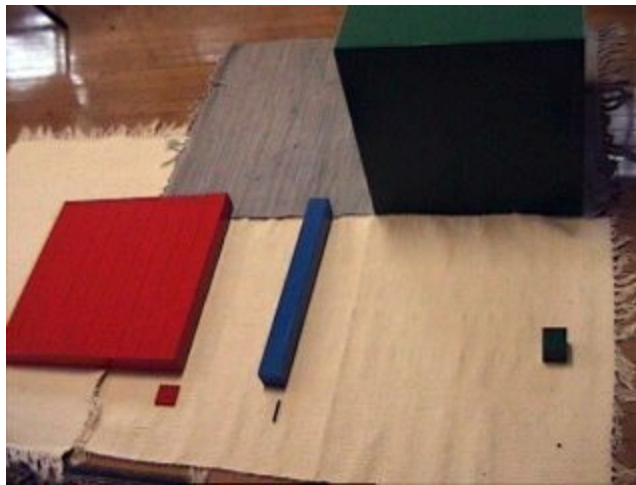
Melissa, can you please **take the 1 million and set it on the mat outside my office door?**

Carol, can you please take the 100 and set it on the shelf by the globe?

Melissa, Could you please take 10,000 and set it on the bell cabinet by the bells?
Carol, could you please take the unit and set it by the plant?

Melissa, can you please take the ten and set it by the classroom bell.
Carol, could you take the 100,000 and set it in the practice room.
Melissa, could you take the 1000 and set it by the flowers.

Danielle, can you please find the unit and bring the unit back?
(she retrieves it and places it into the same position to anchor a new layout).



Judy, could you bring the 1000 back? And we're going to put the 1000 right here.

We also need the 1,000,000, can you get it Judy? Now who's got the 10? Let's put that right here. Who had the 10,000? can you please get that one also?

Who's got the hundred? Can you place it right here? We also need the 100,000 – you have that one too.

So we have a **unit, ten and hundreds** of the **simple numbers** (point to small versions).

We have **units, tens and hundreds** of the **thousands** (point to second larger),

and finally we have **units** of **millions** (point to largest cube).

End first presentation of first exercise. (generally don't need third part of 3-period.)

Epilogue

Incorporating movement into the second period is a very important thing to do with 6 years olds. Get them moving and involved and they'll be much happier.

Do this slowly because this opens up the number system, introduces the pattern of unit, ten hundred – over simple and complex numbers – for the children. We can also view them as a family of units, family of tens, family of hundreds. Simple is a hierarchy, thousands is a hierarchy, millions is a hierarchy. Invite the children to work with the materials independently and show them how to put things away correctly.

September 30, 2004

Wooden Hierarchical Material - Compare Dimensions

Materials

Same as 1st wooden hierarchical material presentation

Prologue

This could be with the linear drawing 1 or the layered drawing 2. New students shall attend and others may attend as they wish.

Begin

Have you noticed that this piece will exactly fit here. How many of these (point) do we need to make one of these?

(hold up 10000 next to million to show ratio / proportion of dimensions)

(repeat other cases)

Epilogue

Show one or two and the children take off on this. Unit and million are proportional and can be offered as a picture of the earth and the sun!

September 30, 2004

Wooden Hierarchical Material - Symbols

Prologue

So far we've done the introduction of the quantity using the wooden hierarchical material. Many times we use the same pattern as in primary – quantity, symbol and then the two combined. Here, we introduce symbols using number cards. None of the material would be laid out yet. To this presentation, you would bring only the tickets (symbols) and a small group of children. The workspace would be on a table. They should know the first numbers from primary so we can just ask them what they are, holding them up. New students shall attend and others may attend as they wish.

Point a left finger (for righties) to area of card of interest as you go. The cards are then set on a table with units on right in order to million on the left, in a row. There is no need to space things in proportion because now we have commas to show this function.

Begin

What number is this? (hold up 1-card). And this is what? (hold up a 10-card, set number cards on the table in order). What's this? (hold up 100). What's this one? (hold up 1000) Let's look at this number for a moment. We have no units, no tens and no hundreds – the 3 for the simple units, then we have a comma – and what's in front of that comma – yes, three zeros, a comma and then and a one. That's how we know its *one* thousand.

Thousands come after the first comma.

(hold up 10,000) This number is 10,000. Let's look at it. We've got the 3 zeros for the simple numbers, then we have a comma – what's in front of the comma? Yes, a ten. That's how we know it's *ten* thousand.

(hold up 100,000) This is 100,000. Let's look at this one now. We've got the 3 zeros for the simple numbers, then a comma. What's in front of the comma? Yes, 100. That's how you can tell it's *one* hundred-thousand.

(hold up 1,000,000) This is 1 million. Three zeros for the simple numbers, a comma, three zeros for the thousands, another comma. What's in front of this comma? Yes, a 1. That's how you can always tell that this is *one* million.

(now we do second period of lesson.)

We assume there are only 3 new numbers, 10,000, 100,000 and 1 million. “Can you show me 10,000”?

(Extend the second period if child is shaky. Incorporate movement to keep high interest.)

Quantity and Symbol Combined

(We assume the second period was done well. we move into quantity with symbol.)

(hold up a number card)

Shilena, would you go and get this piece and bring it back and lay it out on the rug.



Becky, can you go and get that material. (point to a new card)

(hand tickets for quantities, for them to get it and to lay it into the layout above).

(repeat)

End

October 1, 2004

4. MULTIPLICATION

Large Bead Frame - Introduction

Material

The introduction and the first two exercises just use the large bead frame. The third and fourth will also use large bead-frame paper and a lead pencil. Beads start on the left and move to right of the frame.

Prologue

This is a material that can be introduced in the primary. At times, you will be introducing it to the child for the first time. It is a material that will enable the children to start to use those big numbers that we introduced to them with the wooden hierarchical materials.

We will see here an introduction to the large bead-frame and the first four exercises. How much you will present will depend on how much they have seen in the primary level. If they've seen the small bead frame, this will help. In the classroom you will do this with 2 to 3 children, not more. Gather them up, take them to the shelf, **show them where the materials are and ask them to help carry what we'll need to the table. This is a given for the rest of the math lessons.** The children would sit next to us. New students shall attend and others may attend as they wish.

Begin

We call this the large bead-frame. Here we have the units (point to the first green wire). Here we have the tens (point to blue tens, second wire), Here we have the hundreds. These are the thousands (point to the *wire* each time). These are the ten thousands.

Do you remember the 10,000 we saw with the hierarchical material? Now each one of these beads is going to be like one of those 10 thousands.

Here are the hundred thousands. Do you remember that 100,000 we saw with the wooden hierarchical material? Now each one of these beads is going to be like one of those hundred thousands.

Here we have the millions. (point to last wire). Do you remember the million we saw in the hierarchical materials? Now each one of these beads is going to be like one of those great big million cubes.

Focus on the three new categories that were just introduced with the wooden hierarchical material.)

End

October 1, 2004

Counting Numbers

Begin

Now, we're going to count using the large bead frame.

One unit (move one to right), two units, three units (move one each time) four units, six units, (continue to 10, moving each one to right – on the tenth bead, move all back to left and move one ten bead to right).

This is two tens, 3 tens, ... (at ten tens, move all left and move one hundred to the right). Two hundreds, three hundreds, four hundreds, ... ten hundreds make one thousands.

(Continue to end of the frame doing the entire frame)

End

Epilogue

Emphasize the point at which “ten units makes one ten” and all such transitions. This is the point of exchanging. Draw out the word teeeeennnn, in order to reinforce that ten of the category makes one of the next category. Do this slowly, deliberately and clearly. You are establishing the child's foundation. Take the time now to avoid problems later. Give each child a chance to count through the transition between 9 and the next category in the manner that you demonstrated it.

October 1, 2004

Composing Numbers

Prologue

This is to compose numbers on the bead frame. It is done another time and could be a slightly larger group. You may get 2-3 children working together on the large bead-frame.

Begin

Femina, can you make 560 on the bead-frame? (child does so).

Okay, 560. (point to the hundreds as you say 500 and point to the tens when you say 60)

Jung-en, can you please make 3700?

(child does so. Don't ask child to say it aloud.)

That's correct. 3700.

Pri Yanka, can you please make 86500?

(child does so)

That's right. 86 thousand 5 hundred.

Elisa, I would like you to make up a number on the bead-frame and then ask Danielle to read it.

(child does so. build up to the more difficult numbers. Give them as much practice and repetition as they need)

(while working on these numbers, there is an important thing about composing

numbers to include...)

Becky, can you please make “56 tens”. Right! Its the same as 5 hundred sixty!”

Emily, can you please make 72 ten thousands? Yes, that's the same as 7 hundred twenty thousand.

Judy, can you make 16 units? Yes, 16 units is the same as 1 ten and 6 units.

End

Epilogue

Developing facility to do this allows the child to master the next exercise. Do a lot of practice and repetition for this exercise beforehand. Let the child figure out their way of doing these numbers to avoid making a child go backward perhaps. 16 units may be counted as 16 separate units or the child may go directly to 1 ten and 6 units, for example. If they get it wrong, say “let me show you how I would do that.” We cannot at this point allow them to do it incorrectly and think that it is correct. All ways arriving at the correct answer are fine as long as you arrive at the correct answer. Different minds work different ways. This exercise probably gets the most practice among bead-frame presentations.

October 1, 2004

Counting and Writing Without Zeros - 3A

Purpose

The children see numbers from 1-9 , emphasizing the simplicity of using 1-9 in different categories.

Prologue

This one involves writing using the large bead-frame paper and the lead pencil. Don't make a big deal of the lines on the paper but note that they are colored to correspond to the beads on the bead-frame. They are an important guide in this way. We will write on top of the colored lines. This exercise involves writing without using any zeros. You might have one child move beads and one write it on paper, then switching roles next time, depending on how many bead-frames there are in the classroom.

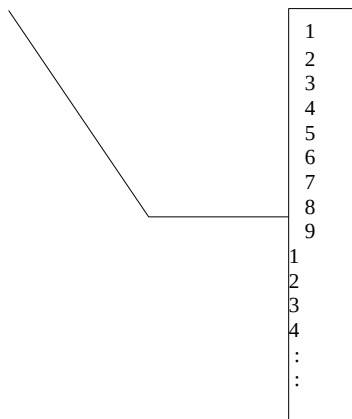
Begin

Today we're going to count and write down on paper the numbers we count. We're going to do that like this.

One unit (write it), two units (write it), ... continue entire frame or until the end of the page.(write 1-9 on unit line.)

(At 10 write '1' in the tens position on the paper. Do the same as you transition from one digit to the next, so that you write 1-9 over and over but in different digit places on the paper. Go all the way to the bottom of the paper on the left hand side (digits are progressing right to left). We are emphasizing the change of category.)

End



Write 1-9, shift left,
write 1-9, shift left,
repeat to bottom
of page.

Epilogue

Limitation in the elementary doesn't always mean just one as it did in the primary level. Use paper sparingly. Do part A and part B on one side. Use both sides. Put it back if the entire page is not used.

October 1, 2004

Counting and Writing With Zeros - 3B

Prologue

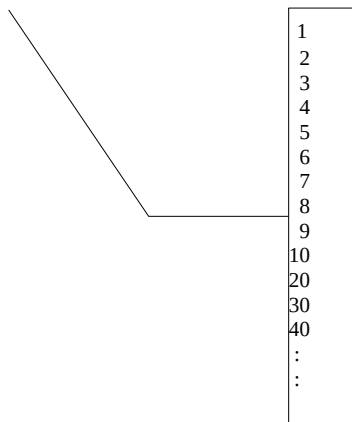
This exercise introduces zero.

Begin

We're going to count and write again but this time it's going to be a little different.

One (move bead and write), two (move bead and write), ...

(begins same as last time, this time we **write the zero** in the ones position and all other positions as we progress across the succeeding digit categories moving left)



Write 1-9, shift left,
write 10-90, shift left,
write 100-900,
repeat to bottom
of page.

October 1, 2004

Composing and Writing Numbers - 4

Purpose

Reading and writing of large numbers, consolidating the children's knowledge of the decimal system in areas such as place values, the function of zero and the use of numbers 1-9.

Prerequisites

Wooden hierarchical material (also presupposes all the primary work).

Prologue

This is making or composing numbers and writing them down on the bead-frame paper. Invite a child to make a number on the bead-frame “please make 4 hundred 76. Good, now we're going to write it on our paper.

Begin

Can you make 522? Good, let's write it down.

Let's do 5283. Fine write it down.

End

Epilogue

Again, you can have one child compose and handle beads and another writing it down. The goal is to have the child write a number down and simply read it without needing to make it on the frame. Don't keep insisting they make it on the bead frame if they want to skip this since this is our goal. Build up gradually from 2-digit, then to 3-digit and so on. Make sure to include empty categories to reinforce the use of zero in the number gradually. Again, we are writing on the lines, not in the spaces, in order to correlate the bead-frame and paper representations. We need to be enthusiastic about this work – make it fun. This concludes the introductory work with the bead-frame and leads to the next area of work with the large bead frame - multiplication

October 4, 2004

Large Bead Frame 1 - Multiplication

Purpose

To get a clear understanding of the processes involved in long multiplication. The process is much more important than getting the right answer. This is also preparation for the distributive law of multiplication.

Age

Between 6 and 7 years old.

Prologue

A logical follow-up to previous bead-frame exercises is to go on to arithmetical operations. We might guess this would be addition and subtraction; this could be the case. But we used the small bead frame to do this in the primary level. We assume the child has done this work so in that case, there is no need to repeat addition and subtraction at this time. We would do addition and subtraction as an exception only, using the large bead frame.

We will start with long multiplication, that is, multiplication using a 2-digit or greater multiplier. We should recall all of the preparation the children have had for this work, all the materials used for multiplication in the primary level. The child should have used golden beads, the stamp game, the multiplication boards (finger charts), and done memorization work with variations of factors using bead bars.

Sometime before doing multiplication on the large bead-frame, you want to briefly review the bead-bar work with multiplication. New students shall attend and others may attend as they wish.

Review of Primary Work

Material

Box of bead bars from 1-10, felt mat, pencil and paper.

Begin

Let's choose a number and write it on our paper. I would like you to set 5 out, 10 times. (child does so) How many beads does this make? (answers 50). Now let's look what I wrote down on my paper. What do I need to add to my paper to make our new number? (answers zero). Yes, that's right.

What would we get if I took 3 ten times? Yes 30. So again, we would just need to add a zero. What if I took 8 ten times? What do I need to do to my number?

Do you think this is true for any number when we multiply by ten? Yes, we just need to add a zero to get our answer.

What if we multiplied 4 by 100? What would that give us? (answer 400). So there we would need to add what to get our answer? Yes, 2 zeros!

What if we multiplied 7 by 1000? What would we get this time? Yes, 7000! What would we need to add to our 7 to get this answer? Yes, 3 zeros.

Do you think the same thing is true if we multiply by other numbers as well? Yes.

End

Long Multiplication with 2-Digit Multiplier

Material

Large bead frame, paper for large bead frame, a lead pencil and a red pencil.

Prologue

This is a small group presentation with children sitting next to me so they can see what I'm doing. Each category product is immediately added to the bead frame.

Begin

Today, we're going to do a multiplication problem on the large bead frame. Let's take the number 8 thousand 4 hundred 37, and let's multiply that by 34.

We write the problem here on the left side of our paper. (write the problem on paper)

This means we're going to take this number 8437, 34 times. First we'll take it 4 times. Then, we'll take it 30 times. The first thing is to see what we have here. (point to 4-digit)

I have 7 units (write on the right side of paper), 3 tens, 4 hundreds and 8 thousands. And I need to multiply all of that times 4.

(we write 8437 on left side of paper with 34 underneath - lining digits up on top of the colored lines of the paper in the usual green, blue, red pattern.)

Paper layout:

$$\begin{array}{r} 8437 \\ \times 34 \\ \hline \end{array}$$

$$\begin{array}{r} 7 \\ 30 \\ 400 \\ 8000 \end{array} \begin{array}{l} \text{---} \\ | \\ | \\ | \text{---} \\ \text{---} \end{array} \times 4$$

$$\begin{array}{r} 7 \\ 30 \\ 400 \\ 8000 \end{array} \begin{array}{l} \text{---} \\ | \\ | \\ | \text{---} \\ \text{---} \end{array} \times 30$$

Now we need to multiply the whole thing by 30. I have 7 units, 3 tens, 4 hundreds and 8 thousands. We need to multiply all of this by 30.

Can you imagine how long it's going to take us to multiply each of these numbers times thirty? We'll be here until tomorrow! But remember when we looked at multiplying a number times 10?

We looked at 5 taken ten times. How many did that give us? Yes, 50. What did we need to add to the 5 to get the 50? We need to add a zero, right?

I'm going to multiply each of these numbers by ten first. (write each number)

Then we'll only have to multiply them by 3. If we multiply 7 by 10, I need to add a zero. Seven units becomes 7 tens. **(add red zero)**

If I multiply 30 by 10, thirty becomes 300. (record answers on paper in a column to add *writing each of the zeros in red*). 400 multiplied by 10 gives us 4000. 8000 multiplied by 10 gives us 80,000.

So we'll multiply each of these numbers by 10. Then at the end we'll only have to multiply that answer by 3.

$$\begin{array}{r}
 8437 \\
 34
 \end{array}
 \qquad
 \begin{array}{r}
 7 \\
 30 \\
 400 \\
 8000
 \end{array}
 \times 4$$

$$\begin{array}{r}
 70 \\
 300 \\
 4000 \\
 80000
 \end{array}
 \times 3$$

Now we can do the multiplication. Looking at the multiplier on the paper and starting at the top (times 4).

Seven units taken 4 times: what does that give us? 28 units. Let's make that on the bead frame. Next we have 3 tens taken 4 times. 3 times 4 gives us how many? Yes, 12. This time it's 12 tens, since we're starting with 3 tens. 400 taken 4 times is sixteen hundreds or 1 thousand 6 hundred – let's move those over. Then we have 8000 taken 4 times. Well 8 times 4 give us 32 so we have 32 thousand (move bead from left to right to 32,000)

I'm done with the first part of my problem. So, let's now to do second part.

7 tens times three. Well 7 times 3 gives us 21 – so this is going to be 21 tens. (moves 2 hundreds and 1 ten bead). Next is 300 times 3. What's going to happen this time?

(we need to make change, so we use the 1 hundred remaining and move a 1 thousand, pushing all hundreds back to left, then continue counting the remaining hundreds needed).

Next we have 4000 taken 3 times. So we have 12 thousands.

80000 times 3. Well 8 times 3 gives us 24. So we have 24 'ten thousands' since that 8 is on the ten-thousands line. (move 2 hundred thousands and 4 ten thousands.)

We can read our answer now from the bead frame.

Let's write our answer on the paper now. (on paper on left side write answer under the original problem = 286,858).

We can read through our whole problem: 8 thousand 4 hundred thirty seven, taken 34 times gives us 286 thousand 8 hundred 58. This is how we do long multiplication on our large bead frame.

Before we start another problem, let's make a little decoration under our finished problem so we know we're starting another problem that's brand new.

Now, as you do bead frame problems, you can do three-digit multipliers!

End

Epilogue

Encourage children to do many examples.

Long Multiplication with 3-Digit Multiplier

For a three-digit multiplier, like 2836 times 324. Begin analysis for each digit just as was done in 2-digit multiplier problems. That is, on the right side, we have the multiplicand times 4, 20 and 300.

The process is identical with an extra analysis step for third digit. We must have **2 red zeros for our third hundreds-place** multiplier paper analysis. The last step is again to put the analysis on paper onto the bead frame.

October 4, 2004

Large Bead Frame 2 - Writing Partial Products

Prologue

This exercise begins to write partial products. We go back to a 2-digit multiplier. The first partial product is derived from multiplying just the multiplier's first digit and then stopping to record that answer. We'll clear our frame and multiply through the second digit to get our second partial product. Finally, we'll add the partial products, putting the child's focus on the paper and pencil work.

Begin

Let's take the problem 8734 and multiply it times 43.

(Analysis again is on right side of paper. This time we put a line under the multiplicand (in four rows as in exercise 1), to begin to have the look of an equal operator.)

Layout on paper:

$$\begin{array}{r} 8734 \\ 43 \\ \hline 26202 \end{array} \qquad \begin{array}{r} 4 \\ 30 \\ 700 \\ 8000 \end{array} \quad \times 3$$

we record our first partial product on paper

Layout on paper:

$$\begin{array}{r} 8734 \\ 43 \\ \hline 26202 \\ 349360 \\ \hline 375562 \end{array} \qquad \begin{array}{r} 4 \\ 30 \\ 700 \\ 8000 \\ \hline 26202 \\ 40 \\ 300 \\ 7000 \\ 80000 \\ \hline 349360 \end{array} \quad \begin{array}{l} \\ \\ \\ \times 3 \\ \\ \\ \times 4 \end{array}$$

we record our second partial and sum the final product on paper

(For the second digit (i.e., 40) begin by adding the red zero to each digit of the original multiplicand and skip the analysis step on paper that multiplies by 10.)

So, in this second step of analysis, the number is multiplied by 4 instead of 40.

Then we're ready to multiply and we go directly to the bead frame. 40×4 , 300×4 , 7000×4 , 80000×4 – each of these products are added directly to the bead frame. We now record the second partial product – taken from the bead frame (we earlier recorded the first partial product for the ones-place digit) – onto the paper.

The last step is to add the two partial products, each one under the original problem as is traditionally done, in rows lined up by digit. We add up the two partial products – for our final product.

End

Epilogue

In working with the children in the classroom, have a few problems ready to avoid having to do too much exchanging and to keep the problem simpler and with emphasis on the new procedures. Start with multipliers of 2,3 or 4. Build up with practice. Generally, there is one large bead-frame for every ten children.

Once the children are comfortable doing bead-frame work, if they haven't on their own, suggest they try something like 1346 times 206. This will introduce the handling of zero in the middle of the multiplier. There is no adjustment to the lessons required for this. A zero in the units place is another variation that will stretch the child's experience level. This gives them added confidence as well. Initially emphasize the process, not answers. Look for certain steps which make them arrive at the wrong answer, such as exchanging, if they often get a wrong answer but seem to understand otherwise.

These lessons should not be rushed through and must be prepared by working on the bead-frame for a long time. If they are struggling with this work, it is because they've not done enough introductory work making numbers.

October 5, 2004

The Bank Game

Purpose

Practice in long multiplication and reinforcement of multiplying by 10 and by 100. It provides another possibility of working with large numbers. It provides more practice with mental calculations and some fun moving around the classroom to make change.

Age

Somewhere between 6 and 8 years old.

Prerequisites

Students should have had the large bead frame by this time.

Materials

We need three tables for this activity: one for the workspace that contains the multiplicand and multiplier, a second for the bank and a third to keep the stores from which the multiplicand and the multiplier are drawn from.

There are also 3 different sets of cards: with white, colored and gray backgrounds. The first set with the white background are color-coded numerals ranging from 1 to 9,000,000. These white cards are used to report the product in a multiplication problem. The cards with colored background (green, blue and red) range from 1 to 9,000. They are used to report the multiplicand. Finally, there are a couple sets of the gray cards which range from 1 to 9 plus single and double zero labels; these are used to report the multiplier.

Prologue

There is no writing required here. It is a group activity, designed for a maximum of 3 children: the customer, banker and runner. This allows the children to work with really big numbers. At least one of the children in the group (the customer) has to know math facts and have some mental ability multiplying numbers. All 3 children don't have to be at the same level.

It's important not to do this activity too soon after doing the long multiplication work on the large bead-frame. It can run in parallel to the work with the flat bead-frame. It is good to have the tables a bit separated to allow some movement around the classroom.

Begin

We're going to play the bank game today. We need to set up our cards. On table 1, we'll set out the colored and gray cards. The white cards will be set up on table 2, remotely, as the bank.

Jasmine, could you set up these colored numbers in rows at table 3?

Now, there are three jobs in this game. There is a customer, a runner and a banker. The customer needs to know multiplication math facts well. This will be Jasmine. Barbie, the runner gets to move the most. Becky will be the banker then.

We're going to take the number 6327 to begin. We're going to multiply that by 23. Barbie, could you form 23 out of the gray cards. Table 3 will be our work table with materials we aren't using.

(both multiplicand and multiplier are placed on table 1.)

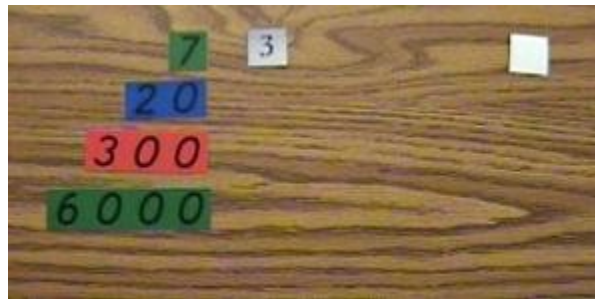
First we want to decompose our cards like this. (lay out in rows by digit)

7
20
300
6000

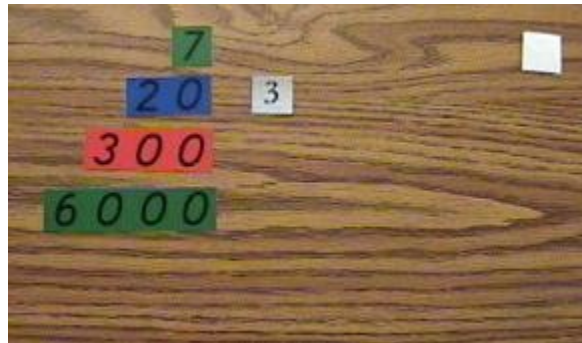
(set multiplier next to each row above as each row is multiplied)

(isolate the '3' digit of the multiplier)

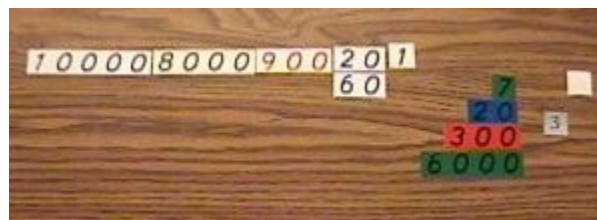
We need to begin multiplying with the units. Jasmine, you're the customer. Barbie, you need to go to the bank and get Jasmine 21 ($7 * 3$).



We'll set the white cards on table 1 separated by category. (put each category into its own column together on table 1. ($20 * 3$) in second row of white cards)



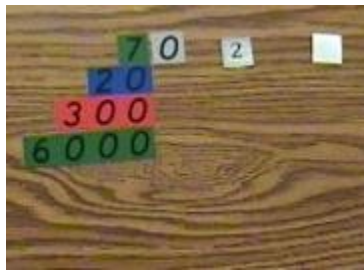
Then we need more white cards for ($300 * 3$) and ($6000 * 3$)



*after multiplicand has been multiplied by
multiplier units and product is recorded*

(turn over the 3 digit of the 23, and turn up the '2' to isolate the 20 in our next step.)

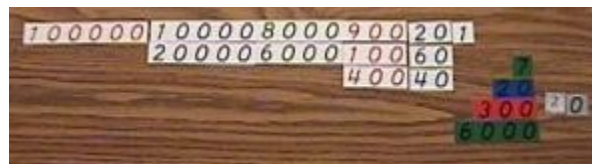
(now take the zero of the 20 – of the 23 and place it next to the multiplicand, row by row, as we multiply through the second digit of the multiplier, 2)



first digit



second digit



*after multiplying the multiplicand by the
tens of the multiplier - and the result at left*

Now we add up the columns of colored numbers with white backgrounds on table 1.

(addressing customer...)

You want to get your runner to get the sum of each category for you. First 1, then 120 (60 + 20 + 40) and so on.

(at this point, we exchange cards with colored backgrounds with white background cards)

(But we run into a problem along the way - there is not enough numbers cards at the bank. The amount that goes to the bank must be the same amount that is brought back to the workspace at table 1.)



We have 15,000 product in the thousands place and therefore a 10,000 value carryover. But the 10,000 card is already in use!

Well, let's see if there is another way to get our numbers. Can we combine anything?

(we need to combine the larger categories in order to free up the number card we need in order to do the summation. The runner and customer work together to figure this out. The banker will verify.)



Barbie and Jasmine agree to combine several categories – 15,000 in our thousands category sum with the existing 20,000 already in the answer cards – in order to retrieve a combined result, freeing up the white cards needed to form the new number we need



after summing all categories, we have a final answer in the white cards

Can you please put the card sets together - and we'll place our multiplicand, multiplier and product here in a vertical expression. (Slide digits of the three problem parts together into a magic slide number).

Now we can read through the problem. (Read through the problem as a group)

End

Epilogue

In the beginning be careful about the number you choose to avoid excessive exchanging. You can encounter problems while you're multiplying but one of the cards is already being used; so avoid this in the first examples in order to get the process going.

A 3-digit multiplier works the same. The difference is only that on the hundreds digit, two zeros are placed next to each of the multiplicand's digits.

October 5, 2004

The Bank Game Variations

Combine and Exchange by Digit

Each time you finish multiplying by a digit of the multiplier, exchange the white cards so that there is just one card in each category.

Combine and Exchange Always

Any time you have two or more cards in a category, exchange them before going on.

October 5, 2004

The Flat Bead Frame

Writing Final Answer Only

Purpose

To introduce the children to further work in abstract multiplication.

Age

Generally, between 6 and 7 years old; certainly no later than 8 years old.

Prerequisites

Large bead-frame. (This can run parallel to the Bank Game).

Material

The flat bead-frame and white number cards are kept in a clear plastic box – one for each of the digits. Strips of paper are cut to exactly fit the part of the flat bead-frame called 'the list'. These paper strips are $\frac{3}{4}$ inch by $10\frac{1}{4}$ inches. Paper (plain or graph paper as desired) and pencil.



Prologue

How is the flat bead-frame different from the large bead-frame? It lays flat on a table surface versus sitting up on an incline. The flat bead-frame's beads are all golden – no longer color-coded. Rows are vertical as opposed to the large bead-frame. Because the wires are vertical, the strip identifying the categories is horizontal. There are colored dots. There is no wider spacing between hierarchies as with the large bead frame. The flat bead-frame goes up to hundreds of millions as opposed to 1 million. Whereas, the large bead frame moves beads from left to right, the flat bead frame moves beads from top to bottom.

This material is a synthesis of the multiplication work done on the large bead-frame. **This material that often leads the children into abstract long multiplication.**

The work with this material can be **parallel to the Bank Game**. While the **Bank Game is a small group** activity, the **Flat Bead Frame is designed more for individual work, perhaps two at a time**. It provides for individual skill practice. It allows the children to **work with very large numbers**.

Be sure to **isolate each digit in the multiplier** as you progress through the problem laying each on its **respective position next to the green, blue or red dots at the right side** of the frame. As you multiply across the multiplicand, **use the left index finger to track which digit of the multiplicand is being operated against, by the multiplier digit**.

Begin

Today, we're going to use this. (hold up the flat bead-frame)

Here we have the simple numbers, units, tens and hundreds. Here we have the thousands, units, tens and hundreds (point to each hierarchy as you speak). And here we have the millions, unit, tens and hundreds – of millions.

We're going to do our multiplication problems like this. We're going to line our strip of paper like this next to the frame and we'll write our multiplicand on the strip of paper.

(place the paper strip across the bottom perimeter of the frame)

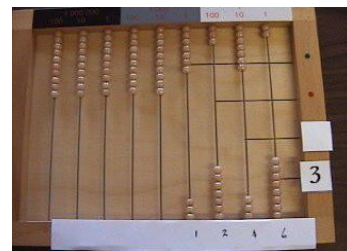
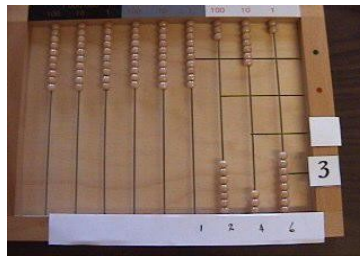
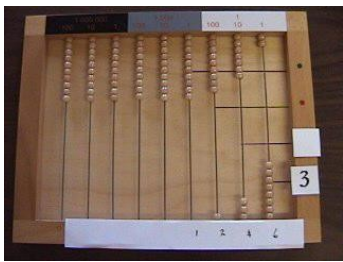
Let's begin with 1 thousand, 2 hundred and 46 (**1246**). Take great care to **write each digit under its respective wire on the frame**.

We're going to **multiply that number by 23. This is when we use our colored dots. 20 is two tens, so I'll put it on the blue dot. Three is 3 units so I put that on the green dot.**

We begin with units so I'll turn the 2 over. Six taken 3 times gives us 18.
 (move 1 ten and 8 unit beads downward from top – lining up the 8 with the wire corresponding to the multiplicand's digit we are calculating against.)



(repeat for other 3 digits)

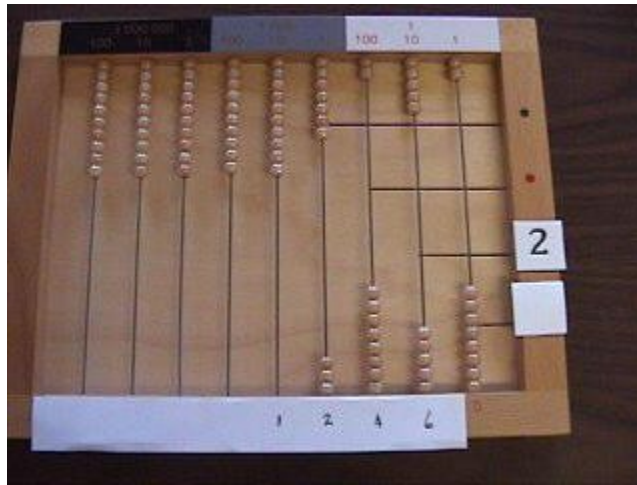


There, now I'm done multiplying by the units digit.
 (turn over the 3 and turn up the 2)

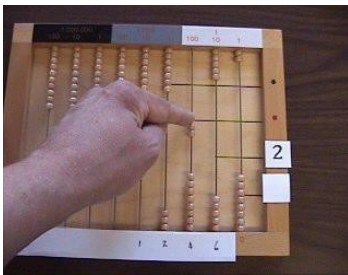
Remember when we multiplied by ten with the large bead-frame? What did we do? Yes, we added the red zero. Well, watch this.

(shift the paper strip to the left to simulate multiplying by ten. Continue as with the units-place digit in the multiplier as we did for '3' above, but use the 2 from 23.)

Six times two gives us 12. Remember the wire we use as units follows our left hand index finger. For 12, we'll take our 2 from just above our finger and the 1 from the next category over to the left. In this case, my finger is pointing into the board's tens category – even though I am using it for my units in the problem at this moment.



Next we have 4 times 2 – but we don't have 8 beads on the wire. We have to exchange. Count the 2 on the wire and then exchange – resuming the count with 3.



count 2, exchange thumb going up at right and index finger going down at left, count 6

(continue until all multiplicand digits are multiplied by each multiplier digit)

Now we're done. Let's read the whole problem together. (do so – adding answer)

End

Epilogue

After once or twice, have the child write the problem on paper before they do the calculations, filling in the answer at the end. With partial products, as we write them, don't put an addition sign – by convention.

October 5, 2004

Writing Final Answer with Partial Products

Prologue

Begin with a 2-digit multiplier and build up to 3 and 4 digits.

Begin

We're going to take the number (83256) and multiply this by 2 hundred 37 (237).

First, let's get our problem written down on the paper (write vertical expression).

On the bead-frame, first, we'll isolate my multiplier. (turn over the 2 and 3).

Now. We're ready to begin multiplying. ($6 * 7$), ($5 * 7$), ($2 * 7$), ($3 * 7$), and ($8 * 7$).
(move beads for each).

Today we'll do something different. We'll record a partial product on our paper. I have 2 units, 9 tens, 7 hundreds, 2 thousands, 8 ten-thousands and 5 hundred-thousands. So that's our first partial product. **(write the first partial product down on paper)**

$$\begin{array}{r} 83256 \\ 237 \\ \hline 582792 \end{array}$$

When we use partial products, I have to clear my bead frame. (clear all wires to zero)

Then we have to slide our paper strip to the left to multiply by ten.

(continue as above in the units digit, for this tens digits, placing beads as above).

Again, we need to stop because we have a second partial product. We need to record this number, lining up each digit with the digits of our first partial product. We have 0 units, 8 tens, 6 hundreds, 7 thousands, 9 ten-thousands, 4 hundred-thousands and 2 millions.

$$\begin{array}{r} 83256 \\ 237 \\ \hline 582792 \\ 2497680 \end{array}$$

So we can go on to our last digit of the multiplier. We turn over all digits of the multiplier except for the 2 in the hundreds place, and shift our paper strip to the left again.

We have $(6 * 2)$, $(5 * 2)$, $(2 * 2)$, $(3 * 2)$, and $(8 * 2)$. (move beads each time, lining up digits in the proper place accounting for carry-overs beyond value-9 into categories of next magnitude).

Now, let's record our third partial product and clear our frame again. What do we need to do to find our final answer? Yes, **we need to add them up on our paper**. Let's do that now.

$$\begin{array}{r} 83256 \\ 237 \\ \hline 582792 \\ 2497680 \\ 16651200 \\ \hline 19731672 \end{array}$$

Let's read our final product. "83 thousand 2 hundred 56 times 2 hundred 37 give us 19 million 7 hundred 31 thousand, 6 hundred 72."

End

Epilogue

We've already introduced commas, so they don't need special attention here. There is no need to include commas in the partial products. From the beginning, we want the children to move the beads and to write on the paper as much as is possible. The teacher is only facilitating if possible.

Be sure to get the children used to correcting their own work, corroborating by using another material such as the large bead-frame. (Calculators are banned at this point!) The children need to stimulate their own means for control of error.

Make sure the children realize when you slide the paper strip, that you are changing the category value of the numbers; they must realize this was just like writing our red zeros when using the large bead-frame.

For a variation, use a zero in the multiplier, in both units position and embedded in higher magnitude categories.

October 5, 2004

The Checkerboard

Carry at End Using Beads - 1

Materials

The checkerboard, a box of bead bars 1-9 and the number tiles. Gray tiles are for the multiplier and white tiles are for the multiplicand. (Sometimes the tiles and bead-bars come in a single wooden box set.) We can use other number cards if needs be.

Prologue

We'll do an introduction to the checkerboard and the first two exercises using this material. Be sure to isolate each digit of the multiplier as you move across the digits. This exercise is the most symbolic of the materials since the actual value is determined by a bead-bar but also by its placement in the grid. Note that diagonal magnitude values are determined by the label at the edge of the board – *not positionally on the board*.

Be sure to emphasize that the number for each digit of the product is determined by the number of beads in a given square as well as the category associated with that square. For example, a 4-bead bar in the tens place is equal value 40.

If a child has not mastered her math facts, this material is still usable since they can count the actual beads – as long as he also understands the value is ultimately determined using the square position also, for the magnitude of category involved.

Begin

This is called the checkerboard. If we look across the bottom and right side, we can see that we have numbers: 1, 10, 100 and so on. We also have category squares extending diagonally across the board.

I can take a 3-bar and the value of it will be determined by where I put it on the checkerboard. If I put the 3-bar on a units place square, it is value 3. If I place it on a tens place square, the 3-bar becomes 30 – and so on.

Suppose I put two 3-bars on the checkerboard like this? What is the value now? Yes, 3 thousand 3 hundred.

(now using tiles)

Let's take the number 1 thousand 2 hundred 46 (1246) and we're going to multiply this by 23. We set the multiplier here on the side. (both tiles look the same but multiplicand is in white tiles and the multiplier is in gray).

(for each square do the multiplication for the two digits to be multiplied at bottom and right side – beginning by isolating the digit of the multiplier to be used. Bottom right first square problem would be $(6 * 3)$, so the beads (representing the product for this square) would be 3 sets of six-bead bars. Continue right to left for each square in the row – each times the first digit of the multiplier. Then proceed with the tens digit of the multiplier - and so on, in the same manner – right to left, by completing the row, moving bottom to top by row for each multiplier digit.)

Combining Category squares

The second to Last step is this: we want to gather up all of our tens onto a single tens square. We slide the bead-bars on the tens square in the right column over and down into the tens square on the bottom row. And we do the same for each category of digit until all products are in a single units square, a single tens square, a single hundreds square and so on, in the bottom row.

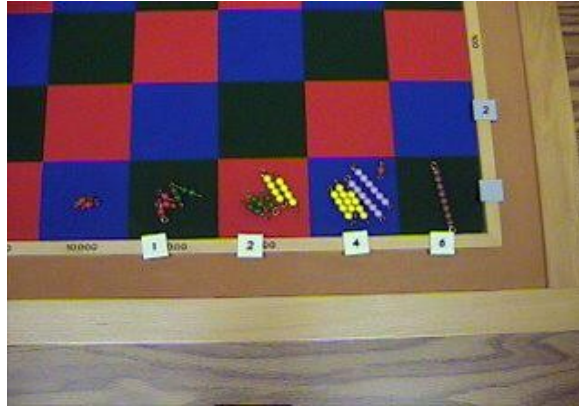


Here, we isolated the '20' and completed the second row of multiplication and then moved all beads from the second row down into the first row. For example, the two 6-bead bars in the tens square above originally came from the tens square above and to the right – the product of 6 times 20.

Combining Partial Products

So, we need to count up each square and combine our beads. Move right to left,

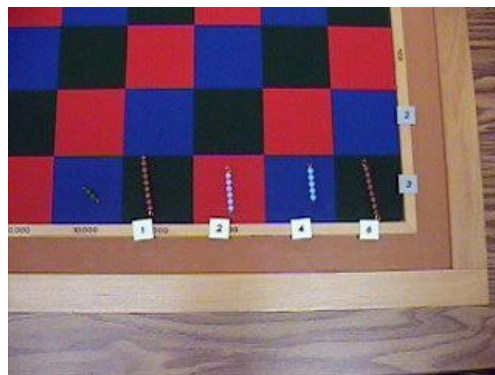
combining beads in each square and putting new beads in place of the ones in the square. So 3 sets of six-bead bars will become an 8-bead bar in the square and a 1-bead bar in the next greater magnitude category – that is, 18 spread across two squares versus the one original square.



Here, three 6-bead bars in the lower right units square have been replaced by a single 8-bead bar in that square and a single bead has been added to the tens square to its left.

Processing Addition Carryover

Finally, we go from right to left in the bottom row of squares, adding all the beads we have in each row and if, for example, we have 48 beads, the current square will receive a single 8-bead bar while the 4 will be represented in the next higher magnitude category to the left – where it will then undergo the same process.



Let's read our problem and final answer. 28 thousand 6 hundred 58.

End

October 5, 2004

Carry Using Multiplication Facts - 2

Materials

The checkerboard, a box of bead bars 1-9 and the number tiles. (Sometimes the tiles and bead-bars come in a single wooden box set.) We can use other number cards if needs be.

Prologue

Be sure to isolate each digit of the multiplier as you work through the problem.

Begin

(This is identical to the preceding example, “Delayed Carry”, with one change. There is no intermediate summing in each square. So a value of $(3 * 4)$ will not be represented as 4 3-bead bars but rather as 1 2-bead bar in the current square and 1 1-bead bar in the next magnitude category to the left of the current square.

End

Epilogue

These activities are not geared for writing.

October 25, 2004

The Geometrical Form of Multiplication

Purpose

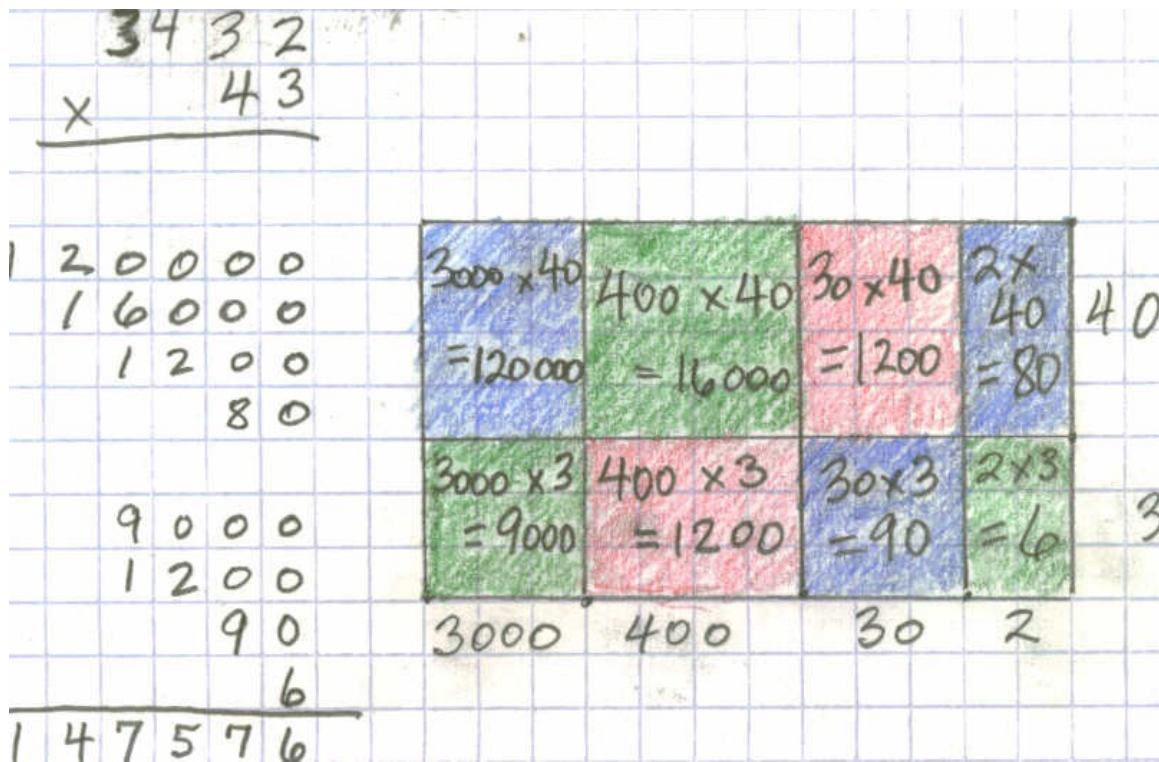
Further experience with long multiplication and preparation for category multiplication.

Age

Age 7 or 8

Material

Graph paper, ruler, lead pencil and 3 colored pencils – green, blue and red.



Final paper image

Prologue

After the children have become comfortable with the checkerboard, this lesson can happen. This does not use the checkerboard but relies on its understanding.

This is usually presented to a group of 2-3 children. Each child has his own supplies.

There is not a lot of talking – you are showing the children what to do and you are doing it together.

Begin

Today we're going to do a multiplication problem but in a different way. We'll do it on our graph paper.

We'll take 3, 432 and multiply that by 43. (write on left side)

$$\begin{array}{r} 3432 \\ \times 43 \\ \hline \end{array}$$

Notate Multiplicand

We'll start by putting our multiplicand on our paper and so we'll mark off squares, one for each value of the digit, and make a dot using our ruler.

(we set up the cells beginning bottom left corner of grid.)

I have 3 thousands. So I'll count off 3 squares over and make a dot.

Now we have 4 hundreds. So, I'm going to count over 4 squares and make a dot.

(continue from the left side of the paper following thousands, moving toward the right)

Now I have 3 tens. I count over 3 squares for the tens and make a dot.

Lastly, I have 2 units, so I count over 2 squares and make a dot.

Notate Multiplier

We make our squares upward on a horizontal axis now for the multiplier (same as multiplicand but beginning with lower right corner and making the multiplier on a vertical axis at the right side. (this should allow us to count squares within each square on the paper that, later, correspond to the product placed into each square)

(After grid is built, we want to color in green square into lower right, blue into tens of

multiplicand, red square of multiplicand. We could say and reuse the phrase, “These are tens, so what color are they”? Then, color in tens of multiplier blue -top row right, then color in red for the next square left - as the checkerboard does. Then color in green in third square from right on top row, finally color blue in top left square.)

Now we do the multiplication.

(Note: Language to use for each square is:

“Four taken three times is twelve, and how many zeros do we have? So that's 1200.”)

(multiply along the **bottom row first**, multiplier units (3), right to left. then proceed to top row, multiplier tens (40), right to left,

Write the products in a column underneath the original expression as a second problem.)

(after multiplying all, at top left square...)

Here we have 3000 times 40. **3 times 4 is how many? And How many zeros do we have?** Four zeros. So we have 120,000

Products from the first row:

120,000
16,000
1,200
80

Do the second row – units - in the multiplier (i.e., 3 units times each digit of the multiplicand) and add up the 8 partial products in an addition problem. **Add up all partial products and then write answer under the original expression.**

Products from the second row:

9,000
1,200
90
6

Variation

Coloring and gathering terms horizontally can also be done diagonally. Do all units together. When you sum at end, gather partial products from all thousands, then all hundreds, then all tens and finally units.

Epilogue

This give experience that when we multiply tens times ten we get one hundred. We are laying ground work for squaring and later on for square roots. It tells them what category each of the category combinations inherently produce.

October 25, 2004

Category Multiplication

Basic Form - 1

Material

Checkerboard, Box of bead bars 1-9, Number tiles, paper and pencil and some colored pencils, any variety, any colors.

Prologue

We do three exercises for category multiplication. **Every square that has a number tile at right and at bottom must receive a product. On a three digit by three digit problems, there must be nine partial products**, prior to sliding and combining.

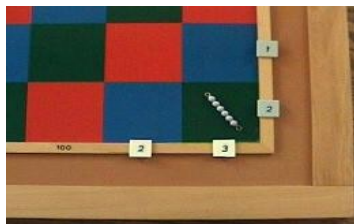
Begin

We'll do a multiplication problem today on the checkerboard. But we'll do it differently than we've been doing it.

So far we've set up our problem and then multiplied all the way through by our units and then all the way through by our tens. If we have a 3 digit multiplier, then we multiplied all the way through by our hundreds.

So we've been working horizontally. Today we work diagonally, by the kind of product we will receive. We'll do the units, then all of the multiplication problems that gives us tens, then all of the multiplication that give us hundreds and then if we have thousands, we'll do all of those.

Let's start with 23 times 12.



Where do we find our units? 3 times 2 is six (begin lower right square) Let's place our beads (1 6-bar).

(if product is 2 digit, bead span across 2 squares as an immediate carry.)



Now we'll do the tens. 2 taken 2 times is 4. (place beads) But we also have tens up here. 3 taken 1 time is 3. (We've done squares surrounding the lower right square).

Now we need to do hundreds. 2 times 1 is how many? Two.



Now we slide them all down so there is only one hundreds, one tens and units in the bottom row. Now we combine the beads for each category and place carry-over beads into following categories.



23 times 12 equals 276.

End

Epilogue

Start with a problem without changing and carrying so the children can focus on the process.

Go on to a three-digit multiplier once the children are comfortable with two digit problems.

October 25, 2004

Beads on Final Answer Per Category - 2

Prologue

We will do carrying in our head. We go back to a 2-digit by 2-digit problem without carrying, so the child can see what is different in the process.

Carry is always verbal and kept in memory now, recalled as we multiply the next category. We don't exchange or slide at this point.

Material

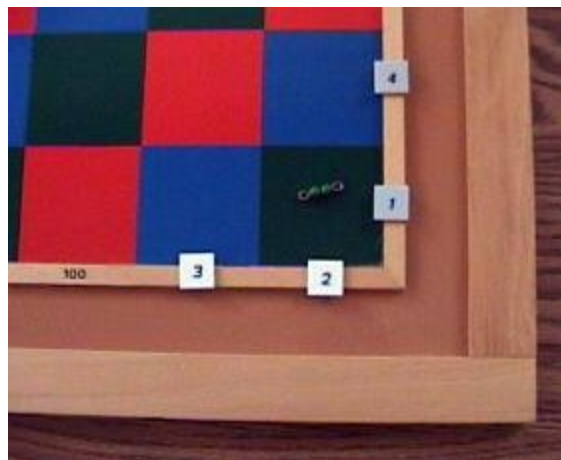
Checkerboard, Box of bead bars 1-9, Number tiles, paper and pencil and some colored pencils, any variety, any colors.

Begin

We take 32 times 41.

Today we're going to place just our final answer.

Where are our units? Yes, and 2×1 is two.



We have tens here and here. One product is 3 and the second is 8. 3 plus 8 gives us 11. We'll put down the 1 bead in the tens place and carry the other 1.



Where is our hundreds? Yes and 3 times 4 is twelve – plus the one I carried makes 13.



Beads laid out per square: 1B 3B 1B 2B

End

Epilogue

We're adding new wording. “**Plus the 2 I'm carrying...**” for example. Or, 7 x 4 gives us 28, **we put down the 8 and we carry the two**. We also have hundreds – **tens times tens gives us these hundreds**. These are all new language we want to introduce in this exercise. *We want to show what categories are used to derive which categories*, plus we account for carrying and adding products within each category **verbally**.

Always start on the bottom row and work upward, diagonally.

October 25, 2004

Final Answer Only - 3

Purpose

To speed up mental calculations and to focus attention on category multiplication. Indirect preparation for squaring of large numbers and eventually for square roots.

Age

Age 6 to upper elementary

Prologue

The children work for a long time with exercise two, including the verbalization of the category multiplication. This exercise has parts A, B and C.

Part 3A - Checkerboard

Be sure the children are becoming aware of category multiplication and the magnitude of the products for each combination of two categories.

Begin

Suppose we have a 2-digit by 2-digit problem. To find the units in this problem, what categories do you need to multiply? Units times units. So we say:

$$u = u \times u$$

What about tens? How many ways are there to find tens? Two ways:

$$t = t \times u$$

$$t = u \times t$$

How about hundreds? We have a fourth cell in our square. What categories do we use to get our hundreds? Yes, its the tens. So we can say.

$$h = t \times t$$

We will always do these four multiplication problems in a 2-digit by 2-digit

problem.



Now suppose we do a 3-digit by 3-digit problem. Let's look at what multiplication we must do.

We just added another number to the multiplicand and multiplier.

We start with $u \times u$ in the first cell.

Now let's find our tens. How many ways are there? Two.

$$t = t \times u$$

$$t = u \times t$$

In this problem how many ways are there to form hundreds. Yes. Three. Beginning at the bottom right, we have

$$h = h \times u$$

Then we have:

$$h = t \times t$$

Moving upward on the board:

$$h = u \times h$$

In this problem, how many ways are there to form thousands? Two.

$$th = h \times t$$

$$th = t \times h$$

Now we have one more cell to do – our ten thousands:

$$tth = h \times h$$

$$tth = t \times th$$

$$tth = th \times t$$

Any time you do a 3-digit problem by a 3-digit problem, these are the multiplication you do to get your answers – no matter what the numbers are.

You can use any size number you wants, there will always be a certain set of category problems that will occur, for a given multiplication problem.

Epilogue

Many times you must show the children this notation for the first time.

Part 3B - 2-Digit on Paper

Prologue

Again, begin with an easier problem – without carryover and borrowing.

Begin

Today, we're going to do a multiplication problem just on paper, just writing down the final answer. (No more checkerboard).

We'll take a problem 32 times 41. We want to do the units first. What categories do we multiply to get units?

(use colored pencils to draw a line between the numbers we multiply. Use a different color for each category, any color).

$$\begin{array}{r} 32 \\ \times 41 \\ \hline 1312 \end{array}$$

To find our units in this problem, what do we need to? Units times units.

What do we multiply to get tens?

Units times tens and tens times units.

To find our hundreds, we multiply tens taken ten times.

(verbalize the carrying always)

Epilogue

If you know the category multiplication, you can do all the multiplication and only write down the final answer. Use the colored pencils to show all multiplication for a given category.

Part 3C - Determining the Categories

Prologue

We're working on a 3-digit by 3-digit number. We may have trouble remembering what categories are needed. There is a convenient way to remind ourselves.

111	(3 categories)
x 111	(3 categories)

111
1110
11100

12321 => this is the number of ways, per category, to get each category in the product:

1 way to get 10 thousands

2 ways to get thousands

3 ways for hundreds

2 ways for tens

1 way for units.

This algorithm always works - telling us how many times a given category can be found.

October 7, 2004

Commutative and Distributive Law of Multiplication

Introduction to the Commutative Law - 1

Purpose

To introduce and make the children aware of the commutative and distributive properties of multiplication. Indirect purpose is preparation for algebra and preparation for squares of numbers and for square roots of numbers.

Age

Begin as early as possible (6+) and continue for a number of years; observe interest and capability to initiate.

Prologue

Multiplication has several different laws or properties which we will explore. This is very early work to do with the children. It can run parallel to the large bead-frame. The work is done with the bead-bars so it's great if children know their multiplication facts and they can focus on the laws themselves. If they do not know the multiplication facts, the bead bars are right there and they can count. This work can then reinforce learning the multiplication facts. We are going to present 5 exercises. The bead-bar work in the primary environment has hopefully created the foundation for these activities.

Materials

Box of bead-bars(B) 1-10, (graph) paper and pencil, felt mat. Number cards and operation signs. Note that multiplicand bead bars lie horizontally.

Intro & Tally

Let's take this - a 3B, 5 times. How much is that? Yes, 15.

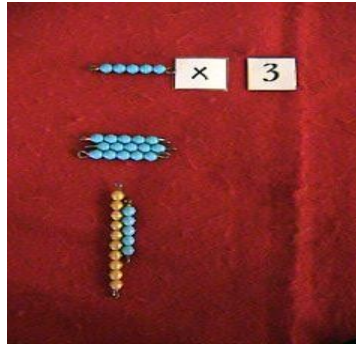


Write

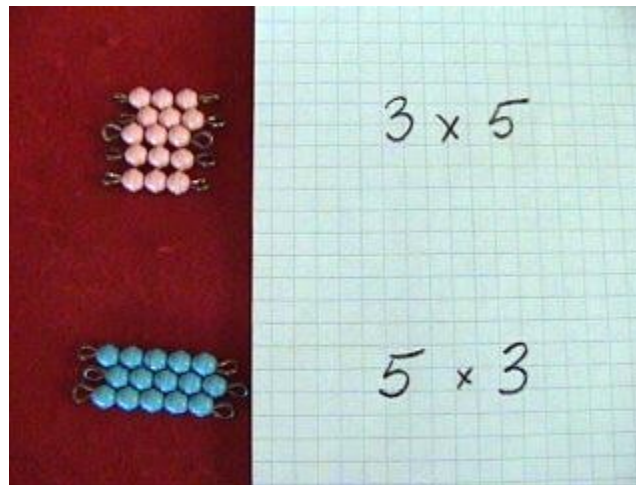
We can also write this: " $3 \times 5 = 15$ " (ensure they know the multiplication sign).

Commutate and Tally

We can also take a 5B, 3 times. How much is that? Yes, 15.

**Write**

We can write that down as well: " $5 \times 3 = 15$ ".

**Explore**

What do you notice about those answers? Yes, they're the same. Let's try it flipping numbers with some other numbers and see if the answers are still the same. (examples)

Summarize

So, when we are multiplying two numbers, it does not matter what order they are in. You always get the same answer. **We call that the commutative law of multiplication.**

End

Epilogue

Extended by writing down equations in color below or to the right - and coloring beads on graph paper to simulate the layout.

October 7, 2004

Introduction to the Distributive Law - 2

Material

A box of white number cards(C) 0-9, a box of bead-bars(B) 1-10, gray number cards (G) and two multiplication signs, two paper parentheses pairs, (graph) paper and pencil, colored pencils, two coin envelopes, and a large felt mat. Note that multiplicand bead bars lie horizontally; partial products lie horizontally in their first appearance only, always vertically when represented after that in addition, for example.

Prologue

Prepare two 3 inch x 5 inch coin envelopes ahead of time: a 2B and 4B in the first envelope and a white 2 card(C) and 4C in the second. Also set aside a 3C and a 3B. **Build the materials layout on the left half of the mat; the next lesson relies on it.**

Intro – 2B and 4B times 3C

Today, we're going to take what's in here 3 times. (point to envelope).
Don't peek!

(lay out “x 3” in cards)

Why don't you go ahead and look. What have you got?

A 2 bar and a 4 bar. That's 6 – let's put those aside. They came together in this envelope.

Let's show that the 4 and the 2 came together.

We're going to use those parentheses to show that.

(put parenthesis around the 2B and 4B).

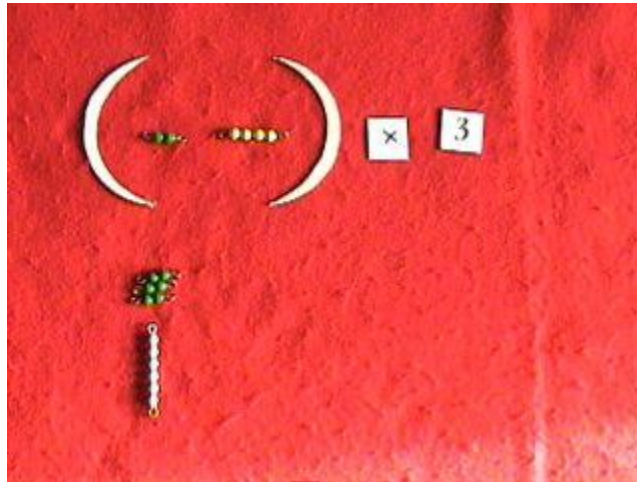
Interpret

(point to each value)

We'll multiply everything in the parentheses 3 times. We'll take 2 times 3 and then 4 times 3.

Execute & Simplify

How much is 2 taken 3 times? Yes, 6. Let's put 6 right here.



Place 3-2Bs and a subtotal 6B.

How much is 4 taken 3 times? Yes, 12. Let's put that out also.



Place 3-4Bs and a subtotal 10B and 2B.

Tally

Now we want to combine all these together to find our final answer for 6 plus 12.



Combine partial products to get 18 placed at bottom.

Intro – 3B times 2C and 4C

(present a 3-bar of beads)

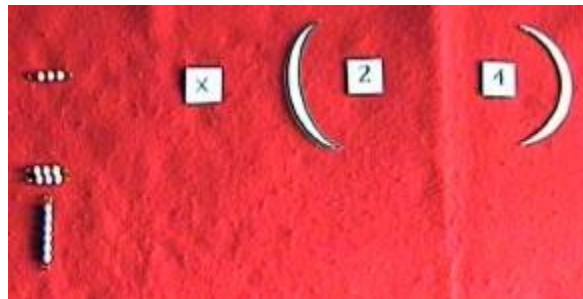
Now I want to take this 3-bar times whatever is in this envelope (a 2C and a 4C).

Interpret

So this says I want to take 3 2-times and 3 4-times.

Execute & Simplify

How much is 3 taken 2 times? Yes, 6. Let's put 6 right here.



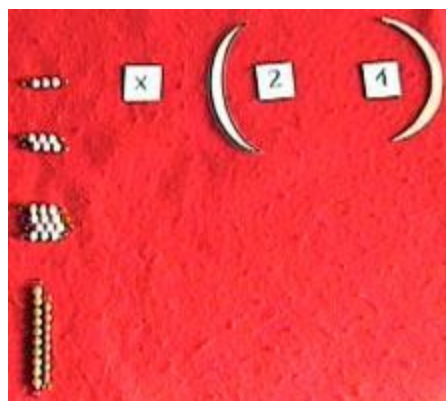
Place two 3Bs below original 3B and a subtotal 6-bar.

How much is 3 taken 4 times? Yes, 12. Let's put that out also.



Place 4 3Bs below and total of 10B and 2B.

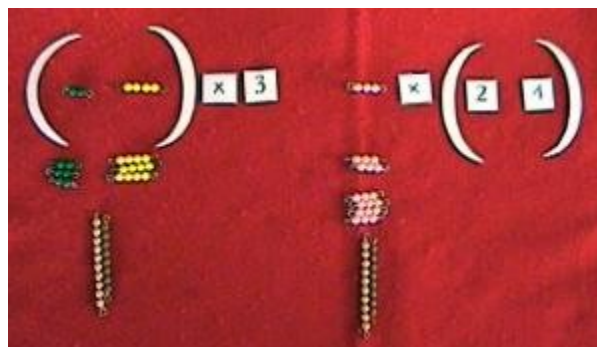
Now we want to combine all these together to find our final answer for 6 plus 12.



Combine to get 18 at bottom, 1-10B and 1-8B.

Summarize

Do you notice something about the two final answers we got? Yes, they are the same.



Commutative property of Multiplication concretely expressed

We multiplied 2 and 4, 3 times and we got 18. And we multiplied 3, 2 and 4 times and we again got 18.

Epilogue

If you have addend terms within the multiplicand, then you need to multiply each addend, both of them one after another, by the multiplier. The same is true for the multiplier – if it contains addends, each must be multiplied times the entire multiplicand. This is the distributive law of multiplication.

This can be explicitly stated after a matter of time, when the child has clearly seen how the law works in practice. Rarely, but sometimes, it is possible to state this earlier. Repeat with other examples to reinforce child's realization.

October 7, 2004

Multiplying a Sum by a Sum - 3

Material

A box of white number cards(C) 0-9, a box of bead-bars(B) 1-10, gray number cards (G) and two multiplication signs, two paper parenthesis pairs, (graph) paper and pencil, colored pencils, and a large felt mat. Note that multiplicand bead bars lie horizontally; partial products lie horizontally in their first appearance only, vertically in subsequent steps such as addition for the end product.

Prologue

This is a continuation of the commutative and distributive laws. Both operands of the multiplication operation have two addends. Put a 2B and a 4B into an envelope. Put a 3C and a 5C into a second envelope.

Intro

You've been working on the commutative and distributive laws.

I've got something in this envelope. There is a 2-bar and a 4-bar. Put them here and let's put parentheses around them – to show they came together.

I have a second envelope for you. Want to see what's there? What have you got?

A 3 card and a 5 card. Since they came together, let's put them to the right side and put parentheses around them again, showing they came together.

Interpret

This says we're to take the 2-bar and the 4-bar, 3 and 5 times. This means we take the 2 3 times and the 4 3 times and then the 2 5 times and the 4 5 times.

Layout:

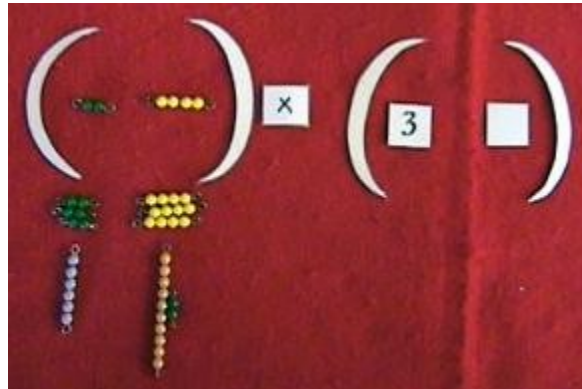
$(2B \ 4B) \times (3C \ 5C)$

(isolate the last 5)

Execute

So let's take the 2 3 times and place 3 2-bars below the first term.

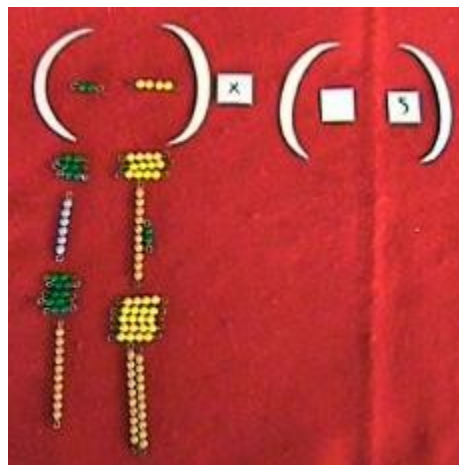
Now, take the 4-bar 3 times and place 3 4-bars also below the first term.



Products from the 1st addend of multiplier and both addends of multiplicand.

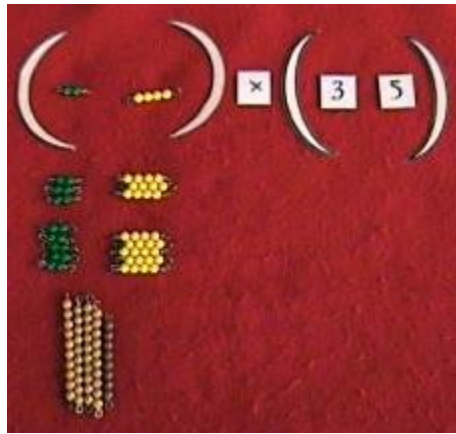
Now, let's take the 2 5 times and place 5 2-bars below.
(place under the first addend of multiplicand at bottom)

And let's take the 4-bar 5 times and place 5 4-bars also below on left side moving right.



How much do we have in all here?

(make a total sum in the 4th row)

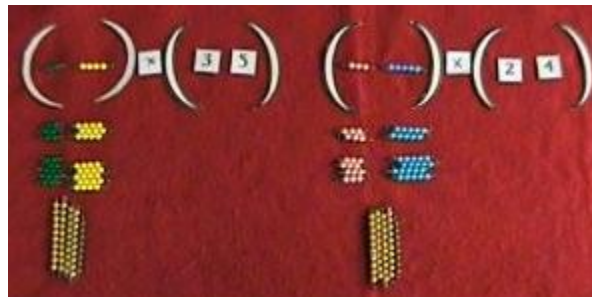


2 and 4 taken 3 times and 2 and 4 taken 5 times is 48.

Commutative Counterpart

(build a second problem – reversing the two parenthesized terms)
Here, we'll multiply 3 and 5, 2 and 4 times.

$$(3B \ 5B) \times (2C \ 4C)$$



Summary

We can see that 2 and 4 taken 3 and 5 times - equals 48.

We can see that 3 and 5 taken 2 and 4 times - equals 48.

So, our first problem gave a total of 48 and our second problem also gave a total of 48. Why? We used the distributive law to help show the commutative law is true.

End

Epilogue

At this point you've given everything, conceptually, the children need. What remains is to extend the concepts to more terms. For example, 3 digits times 3 digits.

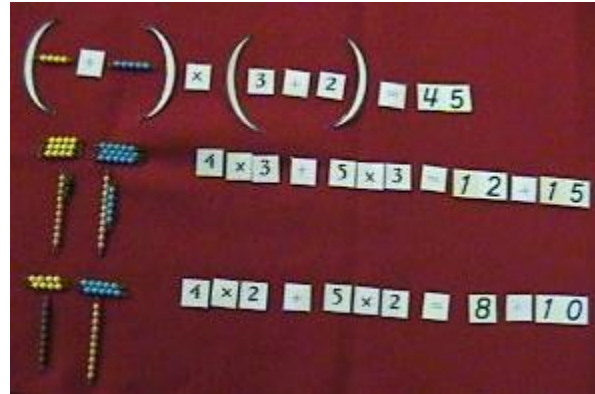
Initially, be around to help them as they begin extending the concepts into larger numbers. We are using the commutative and distributive laws in this exercise.

October 7, 2004

Introducing Cards and Signs - 4A

Material

We need 3 sets of decimal cards(D) 1 to 3000. A box of white number cards(C) 0-9, a box of bead-bars(B) 1-10, gray number cards (G) and two multiplication signs, two paper parenthesis pairs, (graph) paper and pencil, colored pencils, and a large felt mat. Note that multiplicand bead bars lie horizontally.



Final layout.

Prologue

Note that multiplicand bead bars lie horizontally; partial products lie horizontally in their first appearance only, vertically in subsequent steps.

Also note we are adding operation signs at this point. Our ultimate layout for this exercise is as follows:

$$(4B + 5B)$$

x

$$(3C + 2C)$$

=

$$45D$$

$$\begin{array}{l} 3-4B \quad 3-5B \\ 10B \quad 2B \quad 10B \quad 5B \end{array}$$

$$\begin{array}{l} 2-4B \quad 2-5B \\ 8B \quad 10B \end{array}$$

$$(4C \times 3C) + (5C \times 3C) = 12D + 15D$$

$$(4C \times 2C) + (5C \times 2C) = 8D + 10D$$

This is what is new.
Highlight
distributive aspect

Begin

Today, let's take 4 plus 5 (use beads and parenthesize using a sign also) times 3 plus 2 (use cards and parenthesize using a sign also).

We're going to find out what that equals.

Layout:

$$(4B + 5B) \times (3C + 2C) =$$

(fill in products and summations below as prior lesson)
(begin by isolating the 2 card)

Let's look at the first term times the first half of the second term. What did we do there?
We took 4 times 3. And then we took 5 times 3.

$$\begin{array}{rcl} (4B + 5B) & \times & (3C + 2C) = \\ 3-4B \quad 3-5B & & (4C \times 3C) + (5C \times 3C) = 12D + 15D \end{array}$$

Now how about the second addend of the second term? (isolate the 3 of the multiplier)
Let's multiply. Four times two (4 X 2) plus five times two (5 X 2) equals 8 plus 10.

Layout:

$$\begin{array}{rcl} (4B + 5B) & \times & (3C + 2C) = \\ 3-4B \quad 3-5B & & (4C \times 3C) + (5C \times 3C) = 12D + 15D \\ 2-4B \quad 2-5B & & (4C \times 2C) + (5C \times 2C) = 8D + 10D \end{array}$$

We add up our answers to get our final answer. We found that 12 plus 15 is 27 and 8 plus 10 is 18.

We can combine our beads and count them and we would find those numbers.
Our total was 45 (27 + 18). Should we count the beads?

We took (4+5) times (3+2) and that gave us 45.

You can practice with these materials whenever you like. Be certain to put them away carefully.

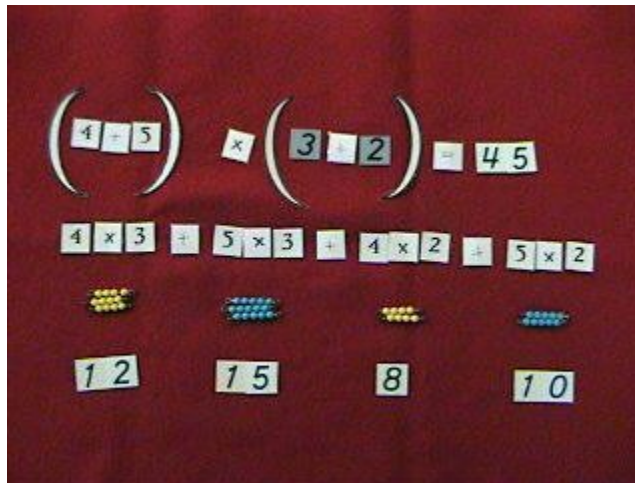
End

October 7, 2004

Laying out the Sums Using Cards- 4B

Material

We need 3 sets of decimal cards(D) 1 to 3000. A box of white number cards(C) 0-9, a box of bead-bars(B) 1-10, gray number cards (G) and four multiplication signs, four paper parenthesis pairs, (graph) paper and pencil, colored pencils, and a large felt mat.



Final layout.

Prologue

This is the next step into abstraction. It is the commutative counterpart to exercise 4A which has now had time to organize in the child's mind. This lesson happens days after 4A though later on it can be used in comparison to 4A since the answers show the commutative law. We extend exercise 4A by compressing two steps into one step, in one expression. We also extend 4A by using only number cards to begin; we use gray cards to show our multiplier.

Begin

Let's do our problem a little differently today.

Layout:

$$(4C + 5C) \times (3G + 2G) =$$

What's different? Yes, no beads! And we are using gray cards to show our multiplier.

So again we want to find out what this equals. (set out only cards this time)

(isolate 3G)

First is 4 times 3 plus 5 times 3

(construct next expression under the first expression)

Layout:

$$(4C \times 3C) + (5C \times 3C) + (4C \times 2C) + (5C \times 2C) =$$

(layout beads as the second expression. **Multiply each term out** and place beads for each term)

$$\begin{array}{ccccccc} (4C \times 3C) & + & (5C \times 3C) & + & (4C \times 2C) & + & (5C \times 2C) & = \\ 3-4B & + & 3-5B & + & 2-4B & + & 2-5B & = \end{array}$$

(finally, add the summations in decimal number cards as the last row)

$$\begin{array}{ccccccc} (4C \times 3C) & + & (5C \times 3C) & + & (4C \times 2C) & + & (5C \times 2C) & = \\ 3-4B & + & 3-5B & + & 2-4B & + & 2-5B & = \\ 12D & & 15D & & 8D & & 10D & = \end{array}$$

We see we get 45. So, we need to write our 45 as the original expression answer.

End

Epilogue

This is the first use of the gray number cards

October 7, 2004

Distributive Law and Hierarchy - 5

Part A

Materials

Golden bead materials units through hundreds including squares(S). Small decimal cards(D) – 3 sets. The paper parentheses and signs, felt mat (red is a good color with golden beads). It is possible to run out of cards with this exercise. Note that multiplier bead bars lie vertically and multiplicand bead bars lie horizontally; partial products lie horizontally in their first appearance, vertically as they carry forward to subsequent steps and are abstracted and operated on.

Prologue

There are four parts to this exercise

Begin

Let's take 40 plus 2 (gather beads to build expression) and let's take that 24 times (gather number cards).

Layout:

$$(4\text{-}10\text{B} + 2\text{-}1\text{B}) \times (20\text{C} + 4\text{C})$$

We need to multiply 40 times 20 and 2 times 20. And then we need to multiply 40 times 4 and 2 times 4. Let's isolate and begin with the multiplier 20. We start with 40 times 20. Forty is made from these 4 10-bars.

What happens when we multiply 10 times 10. 100. So what happens with 10 times 20?

It's 200, right? (key) So let's place 200 beads under this first ten bead-bar. And each of these ten-bars gets 2 100 squares.

(lay column of 2 100S under each 10B of the 40;

then proceed to multiply the two beads of the second addend in the first parentheses times the first addend in the second).

Layout:

(10B	10B	10B	10B	+	1B	1B)	x (20C + 4C)
100S	100S	100S	100S				
100S	100S	100S	100S				

Now let's take this bead (point to first one) bar times 20.

(lay 2-10B *vertically* under each single bead.)

(we are laying each product in beads under each part of the materials of the first term).

Layout 42 x 20:

(10B 10B 10B 10B + 1B 1B) x (20C + 4C)

100S 100S 100S 100S 10B 10B

100S 100S 100S 100S 10B 10B

Now we need to take all of this (point to first term) - times 4. (isolate the 4)

(place beads under each half of the first term, but in a 4th row now)

So everything in the first parentheses in the top expression gets four times as much.

(10 x 4 – four times - and then 1 x 4 – 2 times. Each product lines up under the first or second half of the first term. 2nd and 3rd rows are for left addend of the multiplier times the multiplicand and 4th row is for right addend of the multiplier times the multiplicand)

Layout – add 42 x 4:

(10B 10B 10B 10B + 1B 1B) x (20C + 4C)

100S	100S	100S	100S	10B	10B
100S	100S	100S	100S	10B	10B
4-10B	4-10B	4-10B	4-10B	4-1B	4-1B

Now what does all this equal?

How many hundreds are there? 8 hundreds.

(place 800D on the group top left)

How many tens are there? 4 tens.

(place 40D on the group to the right - 2 x 20)

How many tens are there? 16 tens.

(place 160D on the group bottom left - 40 x 4)

How many units are there? 8 units.

(place 8D on the group to the right - 2 x 4)

(lay number cards for the value of the 4 product groups on the beads in the center of each of the four groups)

800	40
160	8

So we need to add these up and see what we get.
(add up the number cards verbally as you sum them)

8 hundred plus 40 is 8 hundred 40. Plus 1 hundred sixty makes 1000. Plus 8 - makes 1008.

(add the “= 1008” at the end of the original expressions in decimal cards – replacing and summing the number cards from each of the 4 product groups.)

Layout – add 42 x 4 and final:

(10B 10B 10B 10B + 1B 1B) x (20C + 4C) =1008

100S	100S	100S	100S	10B	10B
100S	100S	800 S	100S	10B	40 B
4-10B	4-10B	160 0B	4-10B	4-1B	8 B

We took 40 times 20 plus we took 2 times 20 plus we took 40 times 4 plus we took 2 times 4. That's 8 hundred plus 40 plus 1 hundred 60 plus 8. All of this gave us 1008.

Let's read our complete expression now: “40 plus 2 times 20 plus 4 equals 1008.”

End

Part B

Talk through the steps of the two distributive steps, that created the bead product groups on the left. Build number-based representations of the two steps – to **show the whole process**. At end of each step add “=840” and “=168” in number cards – to correlate the bead layout at the left. Add them up to show that we still get 1008. Layout at right side:

$$40 \times 20 + 2 \times 20 = 800 + 40$$

$$40 \times 4 + 2 \times 4 = 160 + 8$$

(see exercise 4A)

Part C

This time work our problem with the cards first.

$$(40+2) \times (20+4) = \quad \quad \quad \text{(original problem)}$$

$$(40 \times 20) + (2 \times 20) + (40 \times 4) + (2 \times 4) = \quad \quad \quad \text{(see exercise 4B)}$$

(layout the 4 bead-based products under each of the 4 terms above)

Now let's show those values with number cards again.

(lastly, lay out number-card value for each of the 4 term groups)

(**verbally add up** the number cards and place a number-card representation of their sum after the equal-sign in the original expression at the top row.)

Part D

This is done on paper only. We're emphasizing the categories here. The children are familiar with this h,t,u notation already. (h=hundreds, t=tens and u=units)

$$(4t + 2u) \times (2t + 4u) =$$

$$(4t \times 2t) + (2u \times 2t) + (4t \times 4u) + (2u \times 4u) =$$

$$8h \quad + \quad 4t \quad + \quad 16t \quad + \quad 8u =$$

$$800 \quad + \quad 40 \quad + \quad 160 \quad + \quad 8 \quad = \quad 1008 \quad \text{(write this after the original expression)}$$

October 8, 2004

Multiples

Concept of Multiples

We're going back to multiples, an aspect of math that can begin very early. It builds on experience they've already had in the primary class. It will raise their awareness and give them the terminology related to multiples. Any time you take a number multiple times, we have a multiple. Skip counting was a preparation for this.

This would be a small group of 2 to 3 children. Choose one of the chains to begin with.

First we'll have a concrete representation for multiples followed by a few exercises that bring in additional details and also to give the children practice. We'll begin by presenting the concept. We make sure we choose two chains which are not going to show a common multiple – for example, 5 and 7 (4 and 8 would not be good examples).

Materials

Some of the short chains from the bead chains cabinet, some labels and a felt mat. There's a ticket for each bar – we won't use our first tickets for each bead on the first bar, just tickets for the last bead on each bar. So we'll set aside tickets 1-4.

Begin

Let's take the short chain of 5 (set it out on the mat).

Eric, can you ticket (label) the chain for us?

(she has laid out 5 5-bars horizontally and laid tickets underneath.)

Fine, these numbers, 5, 10, 15, 20 and 25 each contain 5 beads – an exact number of times. Therefore they are each what we call a multiple of 5.

5 is a multiple of 5 because it contains 5 one time.

10 is a multiple of 5 because it contains 5 two times.

15 is a multiple of 5 because it contains 5 three times.

20 is a multiple of 5 because it contains 5 four times.

25 is a multiple of 5 because it contains 5 five times.

That word multiple comes from two Latin words: **multus**, which means **many** – and **plus**, which means **fold** – so manifold or **many times**.

We take a number many times; that's why they're called multiples.

(now the second child lays out a series of 7-beads from the short-chain of 7s. Repeat skip-count dialog above)

These are all multiples of 7. Can you find a multiple of 5? (Vicki points to the top row of 5-bars). Can you point to a multiple of 7? (Vicki points to one of the 7-bars)

Can you find a multiple of 5? (Vicki points to 10 and is given a ticket. She lays a ticket below the 10th bead in the 5-bar chain)

(note teacher hides the color of the tickets since they are color-coded.)

(repeat calling multiples of 5 and 7; allow the child to identify answers and ticket each)

End

Epilogue

You could add in a third chain or do different chains. As a later test to make sure they've gotten it, ask them, "How about taking the number 6. Can you tell me a multiple of 6? Yes, 36." This will tell us if they have the idea.

October 8, 2004

Common Multiples

Material

Many bead-bars 1-10(B), a large felt mat, the box of white number cards(C), some bracket cutouts, paper and pencil.

Prologue

After the concept of multiples has sunk in, we'll go on to reinforce multiples and bring in a few nuances. This will hopefully be familiar to the children from primary activities. This lesson should be done on a table.

Begin

Remember that work we did the other day on multiples? Let's do some more today. We're going to begin by setting out a table of multiples of 2. We do that like this:

2 taken 1 time is 2 (each time set out beads, see below)

2 taken 2 times is 4

2 taken 3 times is 6

2 taken 4 times is 8... (continue to complete table of multiples of 2)

Do you remember doing this in primary? Good, who'd like to take over continuing?



So these numbers (point to each) - 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 - are all multiples of 2 because each of these numbers contain 2 an exact number of times.

Who would like to come up and set up a table of multiples of 3?

(continue with second child table of 3, third child table of 4)

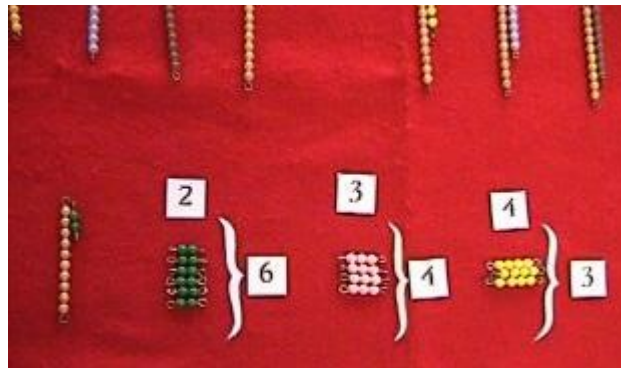
But there's something else we can see here. I see 12 as an answer here. Do you see 12 anywhere else. Yes, 12 is a multiple of 2 but is also a multiple of 3 and a multiple of 4.

Therefore we say that 12 is a common multiple of 2, 3 and 4. To emphasize that, I'm going to move one of these over here (form summary common multiple statement)

So, twelve is equal to 2 taken 6 times (place the beads for 12 in the 2-bar row to the right and use the number cards to label the factors 2 and 6)

It's also equal to 3 taken 4 times. (place beads and labels as for the 2-bar row)

And it's equal to 4 taken 3 times (place beads and labels as for the 2-bar row)



We can find common multiples in two numbers and sometime three of the numbers. Maybe we could build the tables for 5 and 6 and see what other common multiples we could find.

Now let's find multiples for larger numbers. For example:

12 taken 1 time – gives us 12.

12 taken two times equals 24

12 taken three times equals 36... (continue as above)

End

Epilogue

The beads are still here so they can count if needs be. We are giving them variety but also the repetition they need to learn. These are important activities that **can begin at the very beginning of the school year**. Teachers often hold the children back because they focus on the children having mastered abstractions before going on; in fact, these activities continue to reinforce what they need and also give rise to new discoveries within concepts already presented.

For added practice that moves more quickly for those ready to move faster, the children can also simply write the facts if they know them:

$$12 = 2 \times 6, 3 \times 4, 4 \times 3;$$

$$24 = 3 \times 8, 4 \times 6; \text{ etc.}$$

October 8, 2004

Common Multiples and Paper Patterns

Material

Multiples of numbers forms (small square – table E) and colored pencils.
Each child has his own form and his own pencils.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Multiples of 8

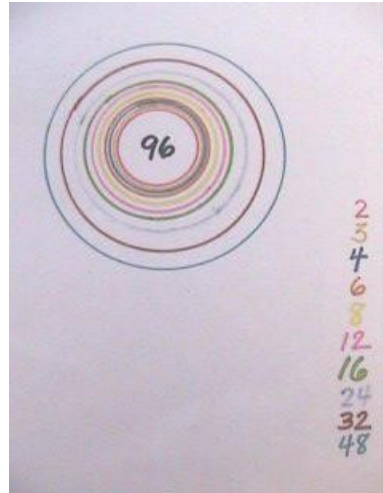


Chart for multiples of 96.

Prologue

This activity enables us to explore multiples more symbolically than we have up to this point. We start off using the forms to look up one multiple at a time; after they've had practice with that we can start looking for common multiples on the form as well.

Begin

Here we have all of the numbers from 1 to 100. We're going to use this table for our multiples work today.

Let's see if we can find all the multiples of 8 on this form. The first multiple is 8, right?

I'm going to make a triangle around the number 8 and I'll use brown since that's the color for 8.

The next multiple is 16. So let's make a triangle around that number. What's the next

multiple of 8? Yes, 24. So let's make a triangle around that one.
(continue with 32, 40, 48, 56, 64, 72, 80, 88 and 96)

Do you see a pattern on your forms now? Yes, it's diagonal. I'm going to connect the ones that have triangles around them. What pattern do we get finding multiples of 5?
(use this idea of patterns on the page to intrigue them)

We could even do two numbers on the same form, couldn't we? How about two numbers that have different patterns – like 2 and 3. Let's see what we get? We'll use different shapes for different multiples. Lines that intersect (after we've connected our triangles and circles) will show us our common multiples.

End

Epilogue

After ten times ____ the children might not know the multiples of a given number, so we should simply count to find succeeding multiples – e.g., count by one from 88 to 96.

If we do more than one number's multiples on the same form, write the numbers in the upper corner to track what we've already done on this form. A natural conclusion for this work is to choose a number like 96 and ask how many numbers is 96 a multiple of? Well, we answer, it's a multiple of 2, 3, 4, 6 but also a multiple like 32 and 48. Gradually, they can start to do more and more numbers on the same form.

Prime Number Variation

Today, we're going to find the multiples of all of the numbers from 2 to 10 on this one form. Since we're doing so many numbers on one form, we'd better just use circles. There's one other thing we'll do differently; the very first time a number appears, we will not circle it. In other words, we're going to start with the table of 2 but we'll not circle 2 but rather we'll start circling at 4 instead and circle all the multiples of 2 after 4. When we do multiples of 3, we'll start circling at 6 – and so on.

When we finish this, some numbers will have a lot of circles around them; some will have only one or two.

And some numbers won't have any circles around them. These numbers are called **prime numbers**. We can make a graphic to show this (see example chart).

October 8, 2004

Common Multiples and Math Facts

Prologue

We'll use tables A, B and C with this exercise, a lead pencil. Table C also uses a red pencil. Also, we want to have control charts for each of the tables available in the classroom. Now, forms come with the control charts. Table C used to come on two sheets back-to-back so we may encounter this.

We'll start off with tables A and B and a lead pencil. This is a small group presentation and each child has his own supplies.

	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Table A – Multiples 1-50

Begin

The other day, we looked at multiples on the multiples forms. You did all the numbers from 2 to 10 on that one single form.

Today, we're going to do the multiples of the numbers from 2 to 10 on these forms. They are called tables A and B. You will each have your very own collection of all these multiples.

	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100

Let's start with Table A. If we look at table A we can see here we have the table of 2; let's fill it in. **2 taken 1 time equals how many? Let's each fill that in on our form. 2 taken 2 times? 2 taken 3 times? Let's fill each one in. (continue) ... 2 taken 10 times? And so on.**

Make sure to take good care of these tables when you are finished, because we'll use them for something else.

When you are done, I need each of you to use these control charts to make sure you have all the correct answers. If they're wrong we'll make mistakes when we go to use them for a special investigation.

(children can work together if they wish)

End

Epilogue

Table A has all of the answers up to 50. Table B has the answers from 51 to 100. Our goal is for each child to fill in both tables. This is exciting and big work for them.

This is work that may take more than one day. It goes into their work folder or file as they work on them. They need to make sure they have the correct answers before the tables are complete.

October 8, 2004

Common Multiples and Prime Numbers

Purpose

Introduction to multiples and common multiples. With written work, we introduce the vocabulary and concept of prime numbers. This is also indirect preparation for fractions and factors.

Age

Age 6 years old.

Material

Completed tables A and B, a lead and red pencil and the blank table C.

List of factors		table c
1		
2		
3		
4	2×2	
5		
6	2×3	
7		
8	2×4	
9	3×3	
10	2×5	
11		
12	$2 \times 6, 3 \times 4$	
13		
14	2×7	
15	3×5	
16	$2 \times 8, 4 \times 4$	
17		
18	$2 \times 9, 3 \times 6$	
19		
20	$2 \times 10, 4 \times 5$	
21	3×7	
22	2×11	
23		
24	$2 \times 12, 3 \times 8, 4 \times 6$	
25	5×5	
26	2×13	
27	3×9	
28	$2 \times 14, 4 \times 7$	
29		
30	$2 \times 15, 3 \times 10, 5 \times 6$	
31		
32	$2 \times 16, 4 \times 8$	
33	3×11	
34	2×17	
35	5×7	
36	$2 \times 18, 3 \times 12, 4 \times 9, 6 \times 6$	
37		
38	2×19	
39	3×13	
40	$2 \times 20, 4 \times 10, 5 \times 8$	
41		
42	$2 \times 21, 3 \times 14, 6 \times 7$	
43		
44	$2 \times 22, 4 \times 11$	
45	$3 \times 15, 5 \times 9$	
46	2×23	
47		
48	$2 \times 24, 3 \times 16, 4 \times 12, 6 \times 8$	
49	7×7	
50	$2 \times 25, 5 \times 10$	

Multiples				
1 = 4	5 x 1 = 5	6 x 1 = 6	7 x 1 = 7	8 x 1 = 8
2 = 8	5 x 2 = 10	6 x 2 = 12	7 x 2 = 14	8 x 2 = 16
3 = 12	5 x 3 = 15	6 x 3 = 18	7 x 3 = 21	8 x 3 = 24
4 = 16	5 x 4 = 20	6 x 4 = 24	7 x 4 = 28	8 x 4 = 32
5 = 20	5 x 5 = 25	6 x 5 = 30	7 x 5 = 35	8 x 5 = 40
6 = 24	5 x 6 = 30	6 x 6 = 36	7 x 6 = 42	8 x 6 = 48
7 = 28	5 x 7 = 35	6 x 7 = 42	7 x 7 = 49	
8 = 32	5 x 8 = 40	6 x 8 = 48		
9 = 36	5 x 9 = 45			
10 = 40	5 x 10 = 50			

Prologue

If you look at table C, the first page has the numbers for 1 to 50 and the second page has numbers 51 to 100.

For table C we don't write down any expression like 24×1 . Rows of a number times 1, we omit. So we can see that 2 will not appear but once in table C so we underline this in red.

Begin

Let's start off by choosing a number like 24 - one that has a few combinations of multiples. Locate 24 on table C. Do you find it? Good.

Now, look at tables A and B and see if we find 24 as an answer anywhere.

(It's important as we do this, to do it sequentially and orderly. Start with table A and the first page.)

Do we find 24 as an answer here? Yes, 2×12 is 24. Let's write that down.

24 2×12

Now let's look at the table for 3. Do we find 24 here? Yes, 3×8 is 24. Let's write that.

24 $2 \times 12, 3 \times 8$

Now let's look at the table for 4. Do we find 24 here? Yes, 4×6 is 24. Let's write that.

24 $2 \times 12, 3 \times 8, 4 \times 6$

(continue table of 5, 6, 7, 8, 9 and 10. Add each expression that yields 24 to Table C)

After the table is complete, go back through it to find numbers that have only one combination of multiples that make the number. So, since we don't include a number times 1 on our list of combinations, we'll underline 2 since it is only made by 1 times 2 and no other combination works.

Finally, be sure to use the control charts to check your answers.

Epilogue

Prime numbers are underlined in red. Some numbers have many combinations and some with only one combination.

Numbers underlined in red don't have any other way of making them. These include prime numbers as well as those having only 1 among factors. So 2 and 13 are both underlined in red, for example. We are marking all common multiples.

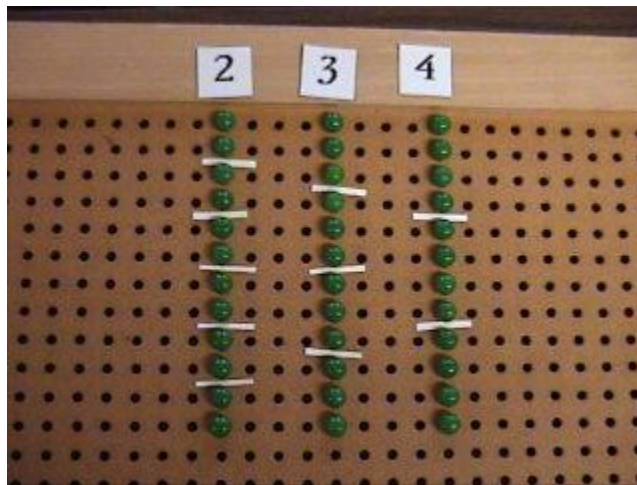
This work may go on over several days. Again, direct the children to use the control charts to check their answers.

October 8, 2004

Least (or Lowest) Common Multiples

Material

We use the peg board for this activity. We use hierarchically colored pegs. Little strips of stiff paper or thin cardboard $\frac{1}{2}$ inch to $\frac{1}{8}$ inch approximately. White number cards and paper and pencil.



Prologue

Up to now we've considered multiples and common multiples. This presentation will look at multiples further but also at the first multiple that numbers have in common. This is what the least or lowest common multiple is.

3, 6 and 9 or 4, 6 and 8 are two good groups to use. Be sure to use the paper strips to demonstrate the passage of each multiple.

Begin

Remember that work we've been doing with multiples. We've learned about common multiples.

We can also use the peg board to find common multiples. That's what we'll do today.

We're going to take the numbers 2, 3 and 4. (we've set the numbers out at the top of the

peg board with white number cards).

I'm going to form each of those numbers with pegs. Here's two, here's three and here's four (each time place green pegs of correct value in a column down the pegged column).



Are the columns even here? No. So, here's where I'll use my paper markers. Let's add to the shortest column. 2 is shortest, so we add that same number of pegs under the paper.



Under 3, I have to add three more. Are my columns even yet? No. Which one is shortest?



Yes, 2 and 4 are evenly shortest aren't they. Let's add to 4 first. (place paper strip between each group of the respective number – for 2, between each two on the pegboard).

(we are trying to get the pegged columns to become even among the 3 numbers 2, 3 and 4. Each time, we add a paper strip before adding another instance of the number. We

tweak each column until we get all three columns to an even length)

(continue adding to shortest column each time, placing paper strips between each group)

Wow, we've found three columns of equal length. It is on the 12th peg. Therefore we say that 12 is the least common multiple of 2, 3 and 4. We can write this on paper too.

We'll use LCM to abbreviate "Least Common Multiple." We would write "LCM 2, 3 4 = 12"

End

Extension

We're going to take the numbers 3, 4 and 5. Let's spread them out (place 3, 4 and 5 at top of peg board above distinguishable simple-hierarchical columns on the board).

Go ahead and begin filling in the pegs since you already know what to do here. We want our columns to come out evenly, right?

(student begins and continues for quite some time...)

Oh, we've run out of space on the peg board.

So, we're not going to use the markers anymore. Now, we're going to count and exchange.

(count ten green pegs and exchange it for one blue – so we get multiple blue pegs in one column with green pegs in the next column to the right.)

Now instead of looking for even columns, we're going to look for equal values.

We still add to the shortest column as long as it is also the lowest value, and when we get ten green pegs on any column, we have to exchange it for a single blue peg.

End

Epilogue

Always add to the shortest column or lowest value as you work down the peg board. We might use the peg board to find common multiples for 12 and 16. It is used for these larger numbers. At this point, they are using hierarchical values, no longer counting beads.

Conceptually, they have all they need related to multiples. We will return to least common multiples when factors are introduced.

October 12, 2004

Factors

Concept of Factors

Prerequisites

The concept of multiplication for the multiples and the concept of division for factors. Long division can run parallel to these lessons.

Age

6 years old. Entire set of lessons on multiples and factors continue until around 8.

Purpose

To introduce multiples, factors, lowest common multiples and greatest common factors. Indirectly, this prepares the children for doing fractions abstractly.

Material

The peg board and the box of pegs.

Prologue

In order to do least common multiples on paper, we need factors. The children have had experience with factors. Now we need to make them consciously aware of the concept.

Factors are taking numbers apart – looking at the numbers that go to make up other numbers. The children already had the presentation on multiples and have done some of that work before going into factors.

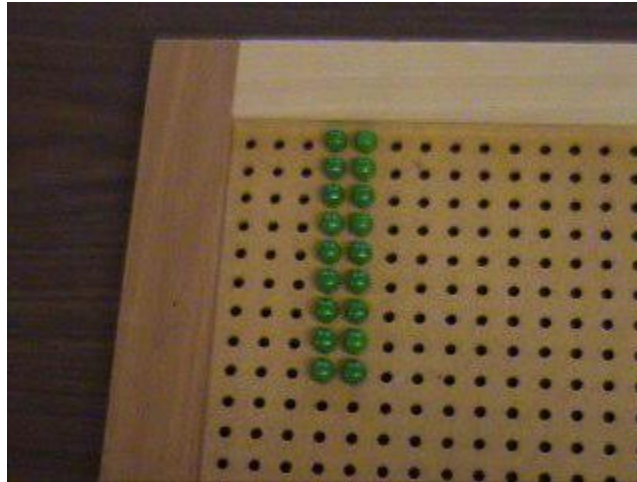
Begin

Today, we're going to look at something called factors.

We begin by choosing a number. Let's choose 18. We begin by putting 18 pegs into a cup.

I wonder if we can make exactly 18 out of groups of two. Let's see.

(begin putting pegs from the cup onto the board in groups of two that are visually striking centered around the formation of the groups)

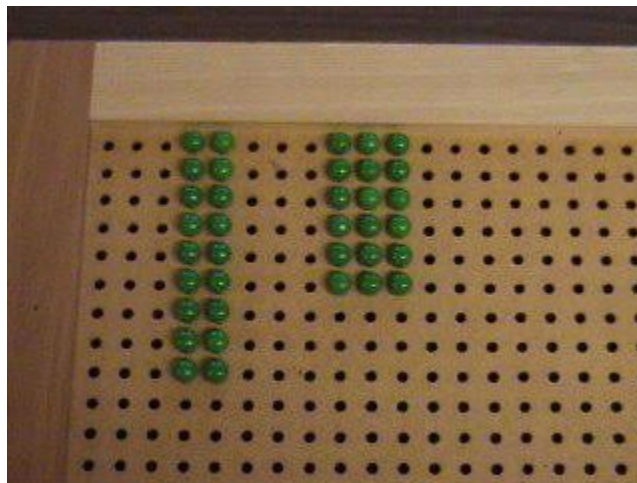


Yes, I can make exactly 18 with groups of two.

Since 2s worked, I'll leave them on the peg board and get 18 more pegs. (refill cup).

Now I wonder if I can make exactly 18 out of groups of three.

(begin putting pegs from the cup onto the board in groups of three)

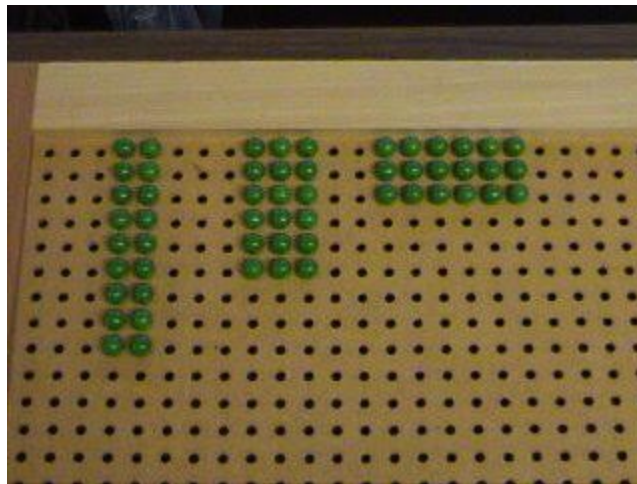


Threes work also don't they! We can make exactly 18 out of groups of three. Let's try fours.

(leave the threes on the board. take 18 from the peg-supply and put into cup. Begin making groups of fours on peg board)

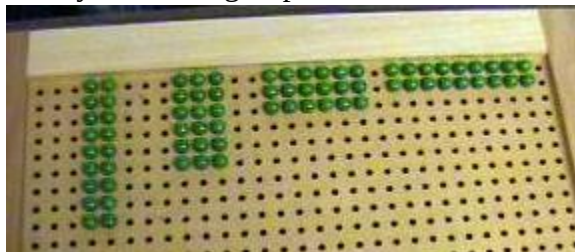
Huh? Fours don't work. We can't make exactly 18 out of groups of four. What about five? Can we make exactly 18 out of groups of five? Nope. Five doesn't work either.

What about six? Can we make exactly 18 out of groups of six? Let's see (as above)



Sixes work don't they? What about seven? (proceed as above) Nope, seven doesn't work. How about eight? Can I make exactly 18 from groups of eight? (as above) Nope. Groups of eight don't work.

What about nine? Can I make exactly 18 out of groups of nine? Nines do work!



What about ten. Can we make exactly 18 pegs using groups of 10? (let the child discover what does and does not work)

These numbers 2, 3, 6 and 9, with which we can make exactly 18 – are called the factors of 18. The word factor comes from the Latin word 'facere' which means 'to make.' Factors are the numbers, which when multiplied together, make other numbers.

Would you like to find factors of another number for yourself? You can choose the number and find the factors in the same way that we did here.

End

Epilogue

The child does not need to do any writing. This is only sensorial exploration and looking at the concept of factors.

October 12, 2004

Prime Factors with Paper

Material

A completed table C, paper and pencil.

Purpose and Preparation

In order to do multiplication on paper, we need factors. The children have had experience with factors. Now we need to make them consciously aware of the concept.

Factors are taking numbers apart – looking at the numbers that go to make up other numbers. The children already had the presentation on multiples and have done some of that work before going into factors.

Begin

Remember that work we did with factors. We saw how we could find the factors of numbers on the peg board. Today, we're going to find factors using table C.

We begin by choosing a number. Let's choose 24. Let's write 24 on our paper. Next, let's find 24 on table C and see how 24 is made.

The first combination we find is 2×12 . Let's write that down ($24=2 \times 12$)

The next combination we see is 3×8 . Let's write that just below our 2×12 .

The next combination we see is 4×6 . Let's write that just below our 3×8 .

The next combination we see is 6×4 . Do we need to write this one down?

No, because we have 4×6 and we saw that these are the same because of our commutative law.

Starting with the first one, 2×12 , if we check 2 on table C, we find that's one of the numbers underlined in red. Remember we called those numbers prime numbers because we couldn't break them down anymore.

Now we've got that number 12. Let's find 12 on table C. Go to that number and the first

combination after 12 is 2×6 . What do we know about 2? It's prime, right? We can just write that one down.

After 6 in the table we see 3×2 . We already know that 2 is prime but we see that 3 is also a prime number by looking at the table entry for 3 – since it's underlined in red. So, we can also write the 3 down since it cannot be broken down or made from any other number except 1.

Now we need to check 8. On table C the first combination we see is 2×4 . We know 2 is prime so we can write down 2. What about that number 4? Going to the table for 4, what do we see. 2×2 , right? We again arrived at all prime numbers.

Now, we'll do the same thing with 4×6 . Starting with the 4 we see 2×2 . Since they are prime, we can just write them down. Now we find 6 and see 2×3 . We know 2 is prime and we also saw 3 is prime. So we write them both and we have arrived at all prime numbers.

These are also known as prime factors. What we have done here is we have broken 24 down into its prime factors. When you break a number down into numbers that have no other combination or any other way of making that number, you have broken down the first number into its prime factors.

Paper layout:

24 =	$2 \times 12 =$	$2 \times 2 \times 2 \times 3$
	$3 \times 8 =$	$3 \times 2 \times 2 \times 2$
	4×6	$2 \times 2 \times 2 \times 3$

We write this P.F. $24 = 2 \times 2 \times 2 \times 3$ (convention is to write smallest first)

Let's find some factors of another number. What number would you like to pick?

End

Epilogue

After the children have worked with powers of numbers, you can show them that $2 \times 2 \times 2 \times 3$ is the same as $2^3 \times 3$.

October 12, 2004

Prime Factors with Pegs

Material

A peg board, box of pegs, a box of white number cards and signs for the operations. We also use long strips of cardboard, about 15 inches by $\frac{1}{4}$ inch wide – so they fit in-between the holes on the peg board.

Prologue

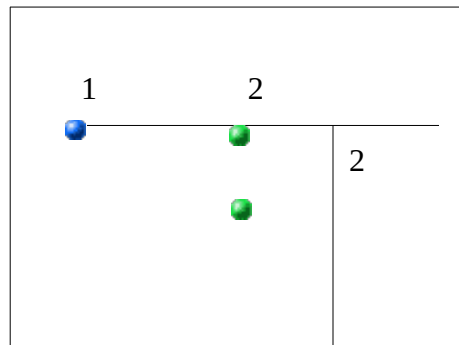
This is another way for children to see factors, again using the peg board.

Begin

We can find prime factors here on the peg board just like we found them using table C.

Let's take the number 12. We'll make 12 out of the white number cards and put them on the top of the peg board.

Next, we form 12, out of pegs, onto the board. We'll use 1-blue and 2-green pegs to show our twelve. Can 12 be evenly divided into 2 groups?

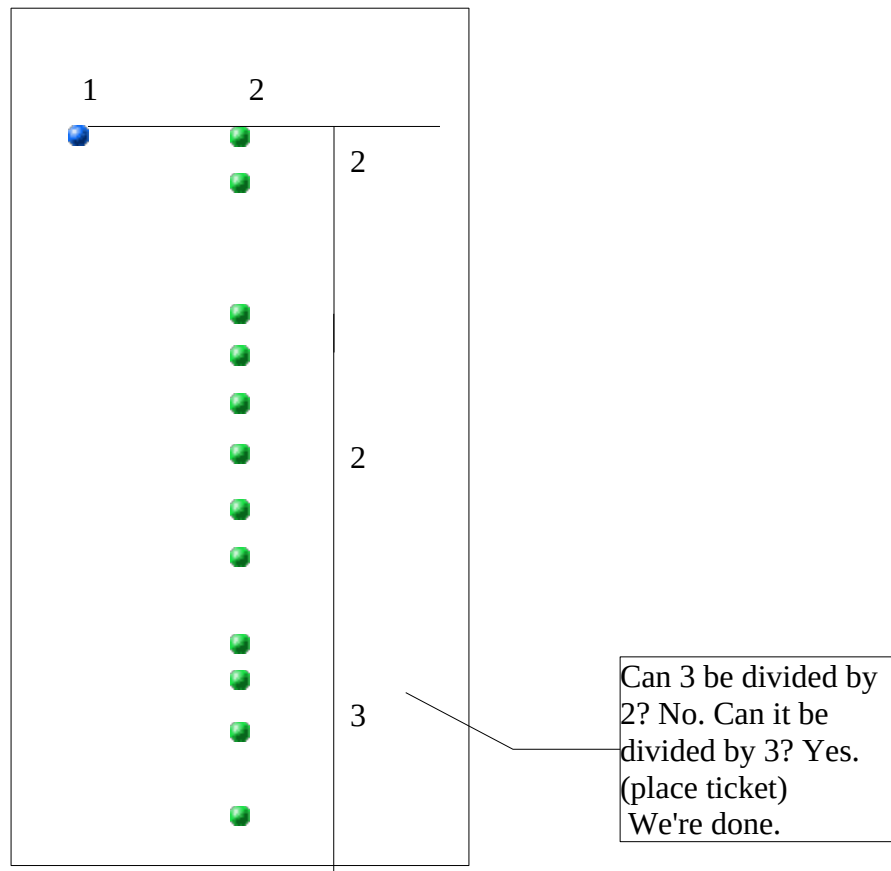


If we can do that, groups, how many would be in each group? Yes, 6.

separate 12 into 2 equal

(put one of the strips on the peg board to separate the pegs from the number cards. Put a 2C to the right of the pegs, continue below the 12 pegs to record the factor we've discovered – which is 6.)

Can 6 be evenly divided by 2? Yes, and if we do that how many do we get? Yes, 3 (place 3 pegs below the 6 pegs on the board)



(pull down tickets to form expression: $12 = 2 \times 2 \times 3$)

So, the prime factors of 12 are 2, 2 and 3

End

October 12, 2004

Greatest Common Factor with Pegs

Material

A peg board, box of pegs, a box of white number cards, paper and pencil and signs for the operations. We also use long strips of cardboard, about 15 inches by $\frac{1}{4}$ inch wide – so they fit in-between the holes on the peg board.

Prologue

After sufficient practice with earlier factor lessons.

Begin

Remember we worked with multiples and we found that numbers can have multiples in common.

We also found that we could find the lowest common multiple. The first multiple that two or three numbers have in common.

Working with factors, we can also find that numbers can have factors in common. With these factors, we can find the highest or **greatest common factor**.

Today we're going to see how to find the greatest common factor of two numbers.

Let's start with the number 20. The first thing we want to do is break the number down into its prime factors.

(do **Prime Factors with Pegs lesson** to find prime factor for 20.)

("Can 20 be divided by 2? Yes, just below the 20 pegs on the board, we put a 10 peg in tens category on the board and left of the vertical cardboard strip partition; plus a white 2-card to the right of the partition. Can 10 be divided by 2"? Yes, - etc., until we reach 2, 2, and 5 in white number cards to right of partition and respective pegs.)

Now, we want to do the same thing for our second number – 36.

(do **Prime Factors with Pegs lesson** to find prime factor for 36. *Can 36 be divided by 2? Yes, we put 18 pegs and 2C at right of partition – just below 36 on the board. Can 18 be divided by 2? Yes, we put 9 pegs and a 2C to the right of the strip. Can 9 be divided by 2? No. Can it be divided by 3? Yes, place 3 pegs and a 3C to right of the strip -etc., until we reach 2, 2, 3, 3 in white number cards to right of partition and respective pegs)*

(at end, using white number cards at right of cardboard strip partition, we construct $20=2 \times 2 \times 5$ and $36=2 \times 2 \times 3 \times 3$ at the bottom of the table under the board)

When we look at these cards, do we see that these two numbers have something in common? What do they have in common? Yes, 2×2 .

That's the part they have in common - 2×2 . What do we get if we multiply 2×2 ? Yes,

4. Therefore we can say that 4 is the greatest common factor of 20 and 36.

We can write this $\text{GCF } 20, 36 = 4$ (write in pencil)

Would you like to find the greatest common factor of two other numbers?

End

Epilogue

Always begin by dividing by 2, then 3 and up – when trying to find factors.

October 12, 2004

Greatest Common Factor with Table C

Material

A completed table C, paper and pencil.

Prologue

After sufficient practice with earlier factor lessons.

Begin

We can do the same kind of work we did on the peg board, using table C and writing on paper. If we want to find the greatest common factor, we first have to break numbers down into what?

Yes, we break down numbers into their prime factors to begin with. What numbers should we pick? Okay, 36 and 28. $(36 = 2 \times 2 \times 3 \times 3, 28 = 2 \times 2 \times 7)$

Now, we'll use our completed table C like we did before to find 28. The first combination I find is 2×14 . We know 2 is prime so we can write it down already. How about 14?

14 is 2×7 . Well we know 2 is prime and we can look here on the table to see that 7 is underlined in red and is therefore also a prime number. So let's write it down.

How about 36? What do we see on the table?

2×18 . We know 2 is prime so write it down. How about 18? We see 2×9 . 2 is prime so write it down. What about 9?

9 is 3×3 according to our table. So we've broken 36 into its prime factors.

What factors do we have in common between 28 and 36? It's 2×2 , right?

So the greatest common factor of 28 and 36 is 4. (write $\text{GCF } 28, 36 = 4$)

End

Epilogue

Always begin by dividing by 2, then 3 and up – when trying to find factors.

October 12, 2004

Last Steps to LCM Abstraction with Table C

Material

Completed table C, paper and pencil.

List of factors			
51	3×17	76	$2 \times 38, 4 \times$
52	$2 \times 26, 4 \times 13$	77	7×11
53		78	$2 \times 39, 3 \times$
54	$2 \times 27, 3 \times 18, 6 \times 9$	79	
55	5×11	80	$2 \times 40, 4 \times 20$
56	$2 \times 28, 4 \times 14, 7 \times 8$	81	$3 \times 27, 9 \times 9$
57	3×19	82	2×41
58	2×29	83	
59		84	$2 \times 42, 3 \times 28, 6 \times 14$
60	$2 \times 30, 3 \times 20, 4 \times 15, 5 \times 12, 6 \times 10$	85	5×17
61		86	2×43
62	2×31	87	3×29
63	$3 \times 21, 7 \times 9$	88	$2 \times 44, 4 \times 22$
64	$2 \times 32, 4 \times 16, 8 \times 8$	89	
		90	$2 \times 45, 3 \times 30, 5 \times 18, 6 \times 15, 9 \times 10$

Prologue

Before we can do abstract work with LCM, the children need factors. Now we can go on to lowest common multiples.

Give the children enough time and practice so they are comfortable finding greatest common factors – before introducing this lesson.

Begin

Remember we were working on finding the lowest common multiple of numbers. I wonder if we can now use our knowledge about finding factors to help us find the lowest common multiple of numbers.

Suppose we want to find the LCM of 12, 24 and 36.

(write L. C. M. 12, 24, 36 = on the paper)

Remember how we used table C to break numbers down into their prime factors? Let's do that for each of these numbers.

(write '12=')

Looking at 12 on the table we see 2×6 . 2 we know is prime. Let's look at 6. We see in the table 6 is 2×3 . We already know 2 and 3 are prime. So the prime factors of 12 are:

$$12 = 2 \times 2 \times 3$$

(continue as above for 24 and 36 to get $2 \times 2 \times 2 \times 3$ and $2 \times 2 \times 3 \times 3$. Write the expressions)

$$12 = 2 \times 2 \times 3$$

$$24 = 2 \times 2 \times 2 \times 3$$

$$36 = 2 \times 2 \times 3 \times 3$$

So looking at these factors, we can see that each one has 2 twos and a three. What does 24 have that 12 does not have? Yes, an extra 2. What does 36 have that 24 doesn't have? Yes, an extra 3.

We need to find each factor as many times as it appears in any of these numbers. That is, we need to take each factor as many times as it appears in any of these numbers.

So, we have 2 and what is the maximum number of times it appears? Yes, three times so we have $2 \times 2 \times 2$. For 3, we see it appears only two times, that is, 3×3 . We want to build a string of these factors to find the lowest common multiple.

So $2 \times 2 \times 2 \times 3 \times 3$ represents our lowest common multiple. What do we get when we multiply $2 \times 2 \times 2 \times 3 \times 3$? We get 72.

So we write L. C. M. 12, 24, 36 = 72

How can we check this? We can use the peg board or simply multiply the numbers together.

Would you like to choose some other numbers to find the LCM of?

End

Epilogue

Invite the children to use the peg board to find that 72 is the first value that all 3 numbers will arrive at equal rows on the peg board.

October 12, 2004

Last Steps to LCM Abstraction with Paper

Material

Besides using table C, we've used the peg board to find prime factors. After lots of practice with this, at some point we help the children realize they will not always have the peg board or a table C with them. So we move to working on paper.

Prologue

For this, we substitute the paper for the peg board. To find the LCM of three numbers, take the **maximum** number of **each** factor it appears in **any** number.

Begin

LCM 12, 24, 36 =

12

12	2
6	2
3	3
1	

24

24	2	(can 24 be divided by 2?)
12	2	
6	2	
3	3	
1		

36

36	2
18	2
9	3
3	3
1	

So we've broken 12, 24 and 36 into their prime factors. To find the LCM of these three numbers we take the **maximum** number of **each** factor it appears in **any** number.

If we multiply them ($2 \times 2 \times 2 \times 3 \times 3$), we again arrive at 72.

End

Epilogue

In the classroom, every time you introduce a new concept in math, and after the children have had experience with the materials, give the children some word problems that use that concept.

In training, we cannot stop to do word problems though we deal with them as a certain topic along the way. But in the classroom, we should use word problems for each and every concept – so that word problems permeate the classroom. We can find samples in text books or develop our own set of word problems for the different concepts.

October 6, 2004

5. DIVISION

Distributive Division with Racks and Tubes

Age

Depending on the primary preparation, generally this starts somewhere around 6 or 6 ½ years old and goes on until around the age of 8 years old. They should be done with this by around 8 years old or it becomes too cumbersome for those older.

Purpose

Lead the children to abstract long division.

Writing Final Answer Only - 1

Material

The Racks and Tubes set, come with groups of seven racks (used as a store from which our dividend is created). Racks are color-coded: 3 white racks for the simple hierarchy, 3 gray racks for the thousands hierarchy and 1 black rack for units of millions. Within each rack – the beads themselves have the familiar green, blue and red for units, tens and hundreds, respectively. Each rack holds 10 tubes and each tube contains 10 beads. We have seven cups colored on the outside to correspond to hierarchy and colored on the inside to correspond to category.

The material also includes 4 boxes of skittles, 2 of green, 1 of blue and 1 of red. Each box contains 9 skittles (used to create a single digit's value in the divisor). We also have 4 boards: 2 have a green stripe across the top, one has a blue stripe and the last has a red stripe. The boards are used by the divisor, one board per digit, meaning with four boards, we can have divisors up to 4 digits. (Some sets have 3 skittles and boards.)

Finally, we will use paper and pencil.



Set up to begin: 8 thousand 9 hundred 42 divided by 34

Prologue

Remember, in primary, we began with the golden bead material. There, we emphasized that each one gets the same amount. If we had a divisor greater than 9 (or a zero in the units position of the divisor), we had to use category bows with golden beads division. And then, when sharing out was done, that child would have to get 9 friends and the one with a ten-skittle had to again divide his lot into ten portions. There were also the exercises for memorization for division.

What is most important here is the process; the children must understand the process we use for division. Hopefully, the children have been introduced to racks and tubes in the primary environment – with a 1-digit divisor.

At this level, we'll start with a 2-digit divisor and focus on the process. We'll extend this to using a 3-digit divisor. Then we introduce the steps used in writing a division problem that the children need to understand to arrive at an abstraction capacity.

Because there are several steps, we delineate them along the way. The process extends across those procedural descriptions (they are for the use of the teacher only).

It also helps if we use correct terminology: dividend, divisor and quotient. The dividend is the number to be broken apart. The divisor is the size of the chunks broken away from the dividend and the quotient is the number of parts broken off. A remainder is an incomplete leftover chunk. Initially, we want the child focused on the procedure and not the terminology; gradually, work in the terms.

If the children have been introduced to this material in the primary, they will be familiar with the basic process of sharing out, recording the answer found in the value at the unit-level, clearing the board and repeating that process. We" use the same process with additional steps to accommodate multi-digit divisors.

Overview of the process is:

1. use skittles to represent the divisor (one board per divisor digit, 1 skittle per divisor digit-value)
2. use beads to represent the dividend – with colored beads (quantity) placed loosely in the colored cups (one cup per digit of the dividend) – bead contents and cup colors do not match here since we exchange along the way of sharing these beads out.
3. share out the quantity beads beginning at greatest magnitude category, 1 person and 1 board for each digit of the divisor, taking turns once per row of distributed beads – until some category in the divisor can no longer be evenly distributed among skittles in that category-owner's row on his board.
4. when any person cannot share out an entire row, record the per-skittle value found in the units board of the divisor - above the written digit of the dividend - corresponding to the smallest category of the dividend being currently shared out. [We look to the units board of the divisor. The category being shared there is the digit, over which, we record the value found in the unit board's per-skittle value].
5. The person with the highest magnitude digit in the current round of sharing gives his cup's contents to the board to his right – the next lower magnitude digit.
6. clear the boards (place shared beads back into the store for each rack and board)
7. shift in the next smaller categories in the dividend, by shifting the racks and cups to the next person to the left, in effect, shifting-in the next lower category of the dividend into the process of being shared and shifting-out the completed category.

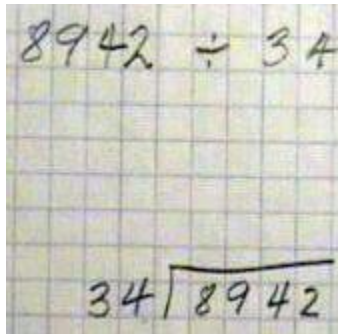
Note we can only share out beads that are the same color as what is in our cup; this forces us to exchange. Also note that in a round of sharing, only the highest magnitude board actually exchanges (from the unused store); other divisor boards are in fact always borrowing from the next higher category.

Begin

Today we're going to do a division problem. We're going to use this material to work out our problem. Let me show you our problem.

We're going to take 8 thousand 9 hundred and 42 – and we're going to divide this by 34. Remember seeing problems like this with this division sign? (write the problem in a horizontal expression on paper) We can also write a division sign like this. (write it

horizontally)


$$8942 \div 34$$
$$34 \overline{) 8942}$$

Here (point to the racks) we have units, tens and hundreds for the simple numbers, units, tens and hundreds for the thousands, and units only for the millions. Let's put these racks on the table since we'll make our dividend from these.

(give each child, each digit of the divisor, a store, a cup and the board. then push tray to top of workspace at right).

Along with each rack we also have a cup that is placed right in front of your rack – each rack-cup pair corresponds in color by category in the dividend – and we begin with the two largest magnitude categories in our round of sharing out the dividend.

We want to form this quantity (8942) in our cups. (children form dividend quantity in the cups) Now, that we have our number set up, we're going to push our categories-in-waiting over here to the right topmost area of the table. The ones we'll share out now remain in front of us.

For this dividend, we're going to divide by a divisor of 34. This means we're going to use 2 boards – one for each digit. Elisa, you're going to get the blue board to share out the 30. Emily, you're going to get the green board to share out the 4. One for the tens and the other for the units.

Emily can you get the box of green skittles and blue skittles? Now lets place the skittles on our boards, blue skittles on our tens board (3) and green skittles on our units board (4) - one skittle per digit value. Great, now we're ready to start sharing.

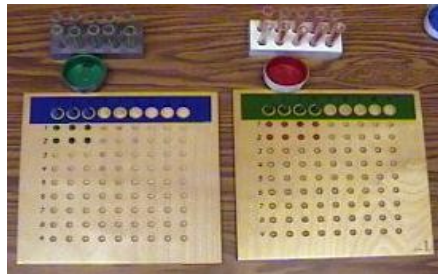
We have two boards, so we want to share out two categories per sharing round. Elisa, you've got our thousands rack (8) and Emily, you're going to get the hundreds rack (9);

these are the first two categories in our dividend we are going to share since they are the largest.

Now we're going to share out; one bead to each skittle evenly and fairly for all. (beads from the cups are distributed into each of the divisor's digits, under each skittle-row on all the boards, until there is not enough beads to go around for all skittles among the boards.) Take turns for each row between each board (digit of the divisor), always completing every row among all boards. **Each time ask, "Do you have enough beads to go on?"**.

(until...)

Ah! Since Elisa can't share again equally, we have to do something different.



The thousands place, after two rounds of sharing, has only two beads

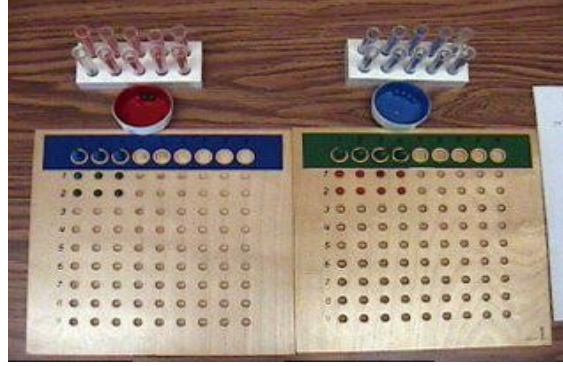
Did all of the skittles get the same number of beads on all the boards?

Emily, on your board, how many did one unit skittle receive? (two hundreds). Yes, since we had 2 hundreds, we're going to write it right there above the hundreds place in the dividend on our paper. (hundred -red- beads are currently on the units board)

Since the units board is currently sharing the hundreds category, we write the '2' on our expression – just above the hundreds place of our dividend – in our quotient.

Eliza, do you have any thousands beads in your cup? Yes, okay – please place those in Emily's cup. (pass incomplete row from thousands cup to the hundreds cup)

Now let's shift the cups – Emily's to Elisa and Elisa gets the next lower category from our dividend categories-in-waiting - up here to the right on our table.



*When sharing is not possible,
leftover thousands are dumped into the hundreds cup,
boards are cleared and shifted left.
(thousands shift out, tens shift in)*

Now you can **turn your cup upside down**, since we are done for thousands.

(Repeat process until quotient is filled in and no more beads remain in any cup - no remainder.)

Let's read our problem. 8 thousand 9 hundred and 42 divided by 34 is 2 hundred 63.

Epilogue

As we record digit answers, we always ask – “did all of the skittles get the same number of beads on all the boards”? This assures we have shared the current categories correctly; every skittle must have the same value across all the boards – the same number of rows.

For three digit variation, note that only the highest board ever dumps into lower category cups - once sharing is no longer possible in a lower category. For the highest category, as with other categories, it is the highest category itself that determines whether that board can go on sharing.

Begin with several problems with no remainder and then let the children come up with their own problems until a remainder is encountered and handled. We have a **remainder** when we cannot share equally among the cups across all the boards, there are no more categories in the dividend's categories-in-waiting – and there are beads remaining in one or more divisor cups.

October 6, 2004

Writing Final Answer and Intermediate Remainders - 2

Prologue

We're going back to Racks and Tubes using a 2-digit divisor.

The difference between this example and the previous one is this: when recording, we check to see what beads remain in the other dividend cups plus the cup for the next category of categories-in-waiting; this forms our ***intermediate remainders that are written in successive rows under the dividend.***

Begin

We're going to take 7 thousand 8 hundred 46 – and we're going to divide this by 32. We're dividing by 32 so we get two boards. Form your divisor by adding your skittles to your board.

We have two boards so we start with the highest two categories of the dividend. Categories-in-waiting remain there to our right at top of the table.

(When we get to the point where sharing is not possible for all categories, record the digit of the quotient on the units board and **record the amount left in the cups plus the amount brought down from only the next category of the categories-in-waiting, as an intermediate remainder – to be written below the original dividend in a second row.**

(Repeat rounds of sharing and each time we record a digit, write down a new intermediate remainder in a new row under the dividend)

$$\begin{array}{r}
 245 \text{ r}6 \\
 32 \overline{) 7846} \\
 \underline{64} \\
 144 \\
 \underline{128} \\
 162 \\
 \underline{160} \\
 6
 \end{array}$$

End

Epilogue

We have a final **remainder** when we cannot share equally from the dividend cups to each skittle across all the boards, and/or there are no more categories in the dividend's categories-in-waiting – and there are beads remaining in one or more divisor cups.

October 6, 2004

Calculating / Recording What is Shared Out - 3

Prologue

This exercise introduces the new aspect of calculating and recording what has already been used. Instead of writing intermediate remainders, we'll **record what is shared out through each of the products formed by multiplying one digit of the quotient with the multiplicand** – in the manner of long division.

Begin

Problem is 7846 divided by 32. Set up as in previous examples: dividend in cups, divisor split by digit among children – each with boards, skittles and a store.

End of first round, write 2 over the hundreds place in dividend.

I wonder if we can figure out how many we've used up? Katarina, how many all together have you used up? 400. And Judy, how many have you used? 6000, yes.

(write 64 under the dividend)

Control of Error

Now I should be able to subtract and know how many beads I have left. Kat, you should have 1 bead and Judy, you should have 4 beads. (derived from the difference)

(shift-in new category)

Kat, how many beads do you have in the new cup you just took from the categories-in-waiting? That's right 4.

(write the 4 after the '14' difference that was recorded below the first amount that was used and the 'equals' operator – in effect “bringing down” the next digit – '144')

(repeat the process until end. At each end of round, look at all beads on the board, to determine what amount will be subtracted from the last round's difference (or dividend the first time).

$$\begin{array}{r}
 245 \\
 \hline
 32 \overline{) 7846} \\
 \underline{64} \\
 144 \\
 \underline{128} \\
 166 \\
 \underline{160} \\
 6
 \end{array}$$

(Each round's difference (after the first intermediate already-used value was subtracted during the first dividend digit) is determined by the number of beads remaining in the cups plus the next lower category from the categories-in-waiting). Already-used beads are taken from the boards (the successive subtrahends in the problem) and the (successive) differences are taken from the beads in the cups plus the next categories-in-waiting category (next intermediate remainder).

Epilogue

Is there any easier way to find what we used? Now, instead of counting each bead on the boards and multiplying each by its category value, we **count the rows per skittle and multiply that by our divisor** to find the already-used value to be subtracted from our (or dividend) intermediate remainder. **This is a new and second variation for this exercise.**

We can use the old method of counting all the beads on the boards in their respective category values and adding them up – as a control of error – a corroboration for our answer found by multiplying skittle rows times the divisor.

October 6, 2004

Moving to Paper - Non-negotiated Quotient Digits

Prologue

This activity provides the last bit of help the child needs in order to make the complete transition from the Racks and Tubes to doing division on paper. The new part of this activity is focused on using paper and pencil.

Set up the materials as usual but draw attention to the paper and have them **verify it using the materials. Give them the rationale of what you do step by step**, allowing them to prove it with materials.

Begin

Let's do the problem 7846 divided by 32.

And let's see if we can determine how many times we can share out. How many times will we be able to share 7 out to 3? 2 times, yes.

Do we have enough beads to share out 8 2 times also? Yes. And the units board is using hundreds so we will put a 2 into the hundreds position above the dividend.

Now we want to ask how many we've used up. We learned we can find this by multiplying our quotient digit times the divisor. Doing this we get 32 times 2 and we get 64 as our product – so this number 64 is the beads we've used up.

So now we can subtract and we have 14 plus we brought down 4 tens from the categories-in-waiting. So we have 144 beginning in the thousands place remaining to share out.

$$\begin{array}{r} 2 \\ 32 \overline{) 7846} \\ \underline{64} \\ 144 \end{array}$$

How many times will we be able to share out 14 to these 3 skittles? That's right 4 times. How many will be left over if we do that – yes, 2. They would be left over for the units person to borrow.

Is 24 (144-120) enough for us to divide by 2. Yes. So I think 4 will be the next digit of our answer in the quotient. (write 4 in tens position in quotient).

We think we'll be able to share out again 4 times. Now we can multiply 4 times our divisor and we have 128 as our next already-used amount. If we subtract, we should have 160 left for our next intermediate remainder. (inspect the cups to corroborate)

$$\begin{array}{r} 24 \\ 32 \overline{) 7846} \\ \underline{64} \\ 144 \\ \underline{128} \\ 16 \end{array}$$

(clear the boards, shift and bring in next category)

When I subtract 128 (4 x 32) from 144 I have 16. And how many beads did I just bring down? (look at next category cup) 6 units. So I write 166 as our next intermediate remainder.

How many times can I get 3 from 16? Yes, 5. Would there be any left over? Yes, so that 1 left over will go to the next lower category which again makes 16, but 16 tens this time. Can I also multiply 2 5 times and get 16 or less? Yes, so we think 5 will be the next digit of our answer.

So multiplying 5 times our divisor of 32, we get 160 and this is our new already-used amount. We again subtract and this will be our next intermediate remainder – or in this case a final remainder.

$$\begin{array}{r}
 245 \\
 \hline
 32 \overline{) 7846} \\
 \underline{64} \\
 144 \\
 \underline{128} \\
 166 \\
 \underline{160} \\
 6
 \end{array}$$

End

Epilogue

What is **new here is the reasoning we use in our head**. We have not yet seen a case here where - a given intermediate remainder is smaller than a given quotient-digit times both divisor digits.

October 6, 2004

Moving to Paper – Negotiated Quotient Digits

Prologue

This activity helps the child move to working on paper only and ***introduces the fact that both divisor digits must be multiplied by the same quotient digit – and the sum of the two products must be smaller than the whole current intermediate remainder.*** The current minuend is our intermediate remainder and the summation of the two products (one for each digit of the divisor times candidate quotient digit) become our current subtrahend.

We wait until the children are quite ready for this – until they have been working well with previous Racks and Tubes problems.

Begin

Suppose we have 17,564 divided by 38.

1 bead to share out 3 times. No can do. So we dump our one ten-thousand bead to the thousands-cup, shift and bring in the new category from categories-in-waiting.

Negotiation

Now, how many beads do we have above the tens board. We have 17. How many times will we be able to share out 17 to these 3 skittles. 5 times, yes. How many is left over?

Yes 2 is left over from the tens board. So the units board will use these two – giving the units board 25 to use – not just the 5 hundreds.

Is 25 enough to share out to 8, 5 times? No. I think we should try 4. 8 times 4 gives us how many? 32. So 2 is written under the 5 of our dividend. How many is 4 times 3? 12, yes ... plus the 3 we carried over – makes 15. We write 15 below the 17 of our quotient.

So let's write a 4 – not a 5 – over the hundreds place above our quotient.

(Repeat process)

End

Epilogue

Try this 7548 divided by 37 for practice. Does a zero in the dividend pose a problem? Not really. How about the problem 51,252 divided by 207 – a zero in the divisor. Try this one as well. How about 78,143 divided by 350? (this is the case where answers must be taken from the tens place instead of the units place. What about a divisor of 500?

Do all these special problems in practicals. Illustrate these as necessary and mention their special handling in albums.

Remember that these problems with Racks and Tubes are done over many months. This also assumes many months of practice with beads and other prior materials.

What about checking their answers? We don't want to bring answers home to check them – that's not the job of Montessori teachers. Let them know to check a division problem, one must multiply the quotient by the divisor and add a remainder if there is one. This provides control of error on their own. They can also use bead-frames, checkerboard or the Bank Game to check their multiplication.

November 19, 2004

Group Division with Stamp Game Material

Age

This begins around age 8.

Prerequisites

Racks and Tubes work.

Materials

Stamp game materials (not using the skittles), one small disk, long thin strip of cardboard for a layout separator, strips of paper to write problems on, blank tickets, pencil and later on, regular paper to write on.

Prologue

Here we go back to division as a process. We will do some exercises to show group division. When we do division with racks and tubes, we share out a quantity; that is distributive division. Here we will make groups.

Group division does not share out. We ask how many groups of 13 can I make out of 39 – for example. We form groups of 13 – as opposed to sharing one by one into 13 groups. These are older children.

Begin

(lay out thin cardboard strip on table)

Let's do $12 \div 3$

(write $12 \div 3$ on the paper strip)

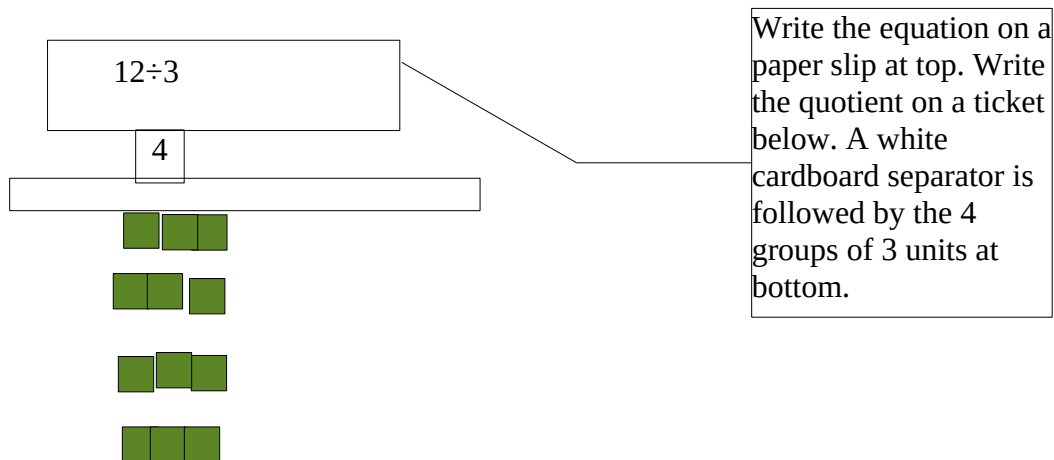
(place problem on paper above cardboard strip)

I want to see how many groups of 3 I can make out of 12.

(count out the 12 green stamps to begin. Now begin making groups of 3)

Let's see, here's a group of 3. (... continue making groups of 3)

How many groups of 3 did I make (4). I write that on a ticket and place it above my cardboard strip below my problem)



I'll write in my answer on my paper strip

($12 \div 3 = 4$)

Now

Contrast Distributive and Group Division

This is another thing that division can tell us. We can take a quantity and make groups of a certain amount with it. Here, we started with 12 stamps and we make it into groups of 3 – 4 of them. Before we would take 12 and share each one to a skittle; in my problem, how many skittles would I have had? (3) And what would my answer have been? (4) Yes, 4 – so we get the same answer.

We can check our answer too, right? How much is 4 times 3? Yes, 12. That was our dividend we began with.

===

How about this problem? Let's take $1842 \div 7$ (write problem on paper strip)

So, I need to get out 1842 in stamps. (build the dividend with the stamps at left)

This time, I'm going to **use this disk** to help me keep track of which category we are considering. I put it between my problem paper strip and the cardboard strip.

I have 1000. Can I make a group of 7 from that? No. So, I'll move my disk over and we'll consider hundreds. (we only have paper strip, disk and cardboard strip, and

dividend now)

(for each category we consider, record the number of groups of 'divisor' and move disk)

Let's see **how many groups I can make with my hundreds?** And I can exchange my thousand for hundreds. (do so and make groups of 7 from the hundreds)

How many groups of 7 did I make from my hundreds? (2) I'll write 2 on my blank ticket.

Now **how many groups of 7 can I make out of my tens?** (move disk right)

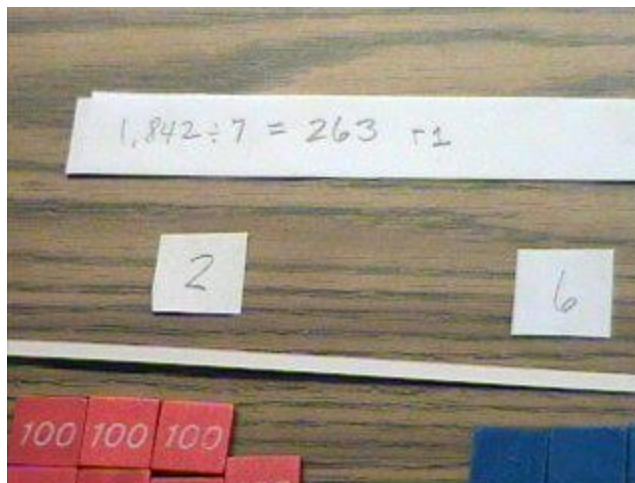
I can exchange my leftover hundreds for tens also, right? (do so and make groups of 7s using all tens possible)

Now how about my units? I can still exchange tens for units, right? (do so)

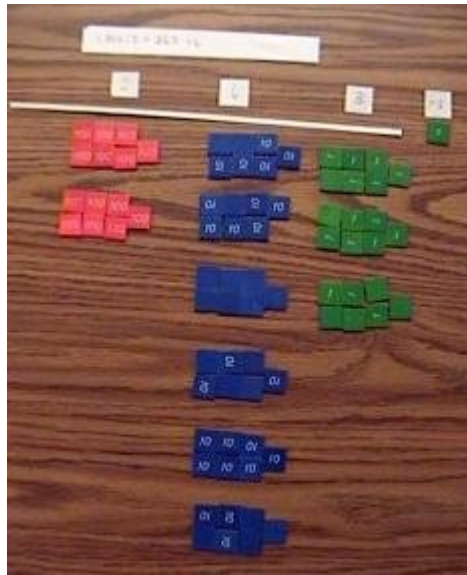
Now, **how many groups of 7 can I make with my units?** (build next column to right of groups of 7 in units)

How many groups of 7 did I make using my units? (3) Yes, 3. Let me write that on my ticket. (take disk away to far right – out of picture)

I have 1 left over as a remainder. So I'm going to write 'r1' on a ticket to show that.



I have 2 groups of 7 out of the hundreds. I have 6 groups of 7 out of my tens. I have 3 groups of 7 out of my units and I have 1 remaining – a remainder of 1.



So 263 remainder 1 is our answer. (write answer on the problem strip)

Epilogue

Have children work at introductory level for a time, using a 1 digit divisor. Note that in group division, we don't clear anything like we do in distributive division (using skittles). Everything remains on the table to see and this aids in control of error.



Two or More Digits in the Divisor - 1

Let's do the problem $3936 \div 32$

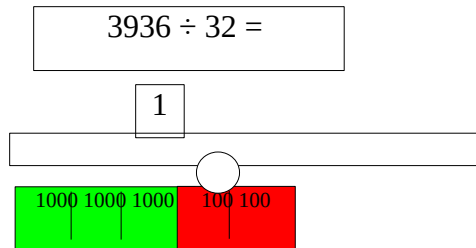
(count out dividend quantity as before and lay out on the table)

How many groups of 32 can I make from 3 thousands? (place disk at left)

None. I need to consider two categories – so let's move the disk to the right and consider hundreds.

How many groups of 32 can I make with my 39 hundreds? Let's see.

(form 3 thousands and 2 hundreds with disk over the hundreds)



I can make 1 group in my hundreds category. Let me **write 1 on my ticket**. (do so)

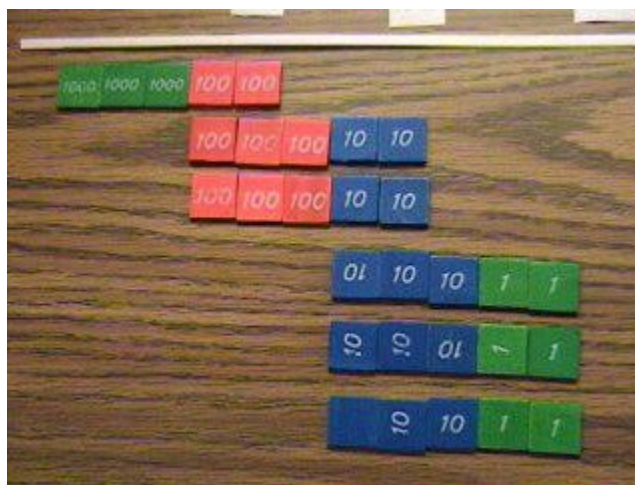
Now I'll consider tens, so I'll move my disk to the right after the hundreds place below.

How many tens can I make from what I have left? I have 7 hundreds and 3 tens. I want to see how many groups of 32 I can make from my tens.

I'll exchange 1 hundred and form groups of 32 tens. How many groups did I form? (2) I need to write that on a new ticket. (do so)

How many groups of 32 can I make with the units I have remaining? I can make 3.

(write on a new ticket)



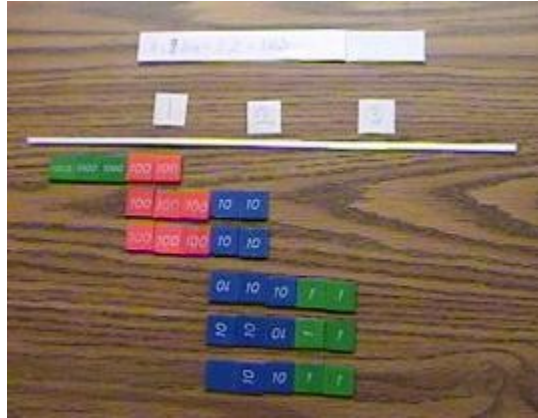
32 hundreds, 2 times 32 tens, and 3 times 32 units

How many groups of 32 did I make from hundreds? (1)

How many groups of 32 did I make from tens? (2)

How many groups of 32 did I make from units? (3)

We have no remainder. So my answer is 123. (add answer to the problem strip)

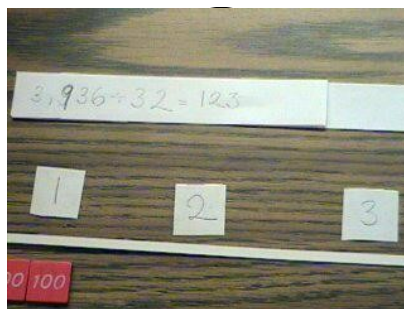


Let's check our work. My first group is 1 group of 32 hundreds. (write 3200 for new expression)

We made 2 groups of 32 tens. 32 times 2 is 64 tens. So we have 640 (write 640)

We made 3 groups of 32 units and that is 96. (write to right again in the expression)

When we add them up what do we get? Yes, 3936. Is that our dividend? Yes. So this is a correct answer.



Extension: Zero in the quotient - 2

Let's do $4848 \div 48$ (write problem on problem strip)

So, I'll build my dividend quantity from my stamps. (do so)

How many groups of 48 can I make out of thousands? (disk is at 1000s)

I have to consider 2 categories so let's move our disk to the hundreds place.

Now, How many groups of 48 can I make of hundreds? (4 thousands and 8 hundreds with disk above hundreds)

I made one group of 48 out of my hundreds so I'll write that on a ticket (do so)

Now how many groups of 48 can I make out of my tens? None! So I'll write zero.

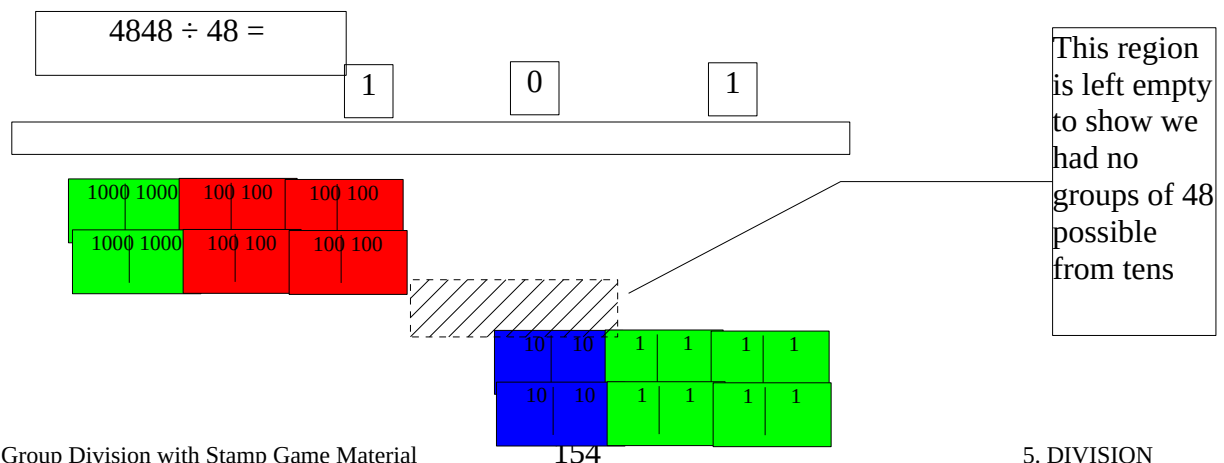
(write a ticket with zero for the tens)

Now how many groups of 48 units can I make? Yes, 1, so I'll write that

(write ticket with 1 on it and move disk aside)

So I have 1 group of 48 hundreds, no tens and 1 group of 48 units, with no remainder.

So my answer is 1 hundred 1.



Epilogue

Help the children create their own word problems with group division. Do a few with them to give them the idea. Encourage them to make up their own and to do them with their friends. “I have 64 pieces of fruit. I want to make baskets with 8 pieces of fruit in each basket. How many fruits do I need in each basket”?



Steps to Abstraction – 3

Prologue

Typically this exercise in group division is not presented until you are sure the children will be able to do it on paper relatively soon.

(No tickets used here – paper and pencil using long division replaces the tickets)

Today, let's do a problem like this.

$$3936 \div 32$$

(write the problem on a separate paper to right as well as on the problem strip)

Let's build out dividend as usual. (take 3936 in stamps)

Let's look. I only have 3 thousands. Can I make a group of 32? No. We need to consider two categories. (move disk to right to consider hundreds)

How many groups of 32 can I make out of 39 hundreds? (1) (do so)

(write 1 above hundreds on the big piece of paper to right – not on the problem strip)

Let's look to see how many I will have used up now. I've used up 32 hundreds. So I'll write that down (write on the large paper as a partial product in long division).

Now, I'll subtract to see what I have left. (perform subtraction as in long division)

(corroborate that calculated difference by comparing with the stamps waiting to be grouped from the dividend)

And I'll bring down my next number from the dividend, which is 3.

$$\begin{array}{r} 123 \\ 32 \overline{) 3936} \\ \underline{32} \\ 73 \\ \underline{64} \\ 96 \\ \underline{96} \\ 0 \end{array}$$

Now I'll consider my tens. I move my disk over. How many tens do I have here? (3)

How many groups of 32 can I make from 73 tens? (2) Okay, 2 – let's see if that works.

Yes, 2. So I'll write this above my tens place on my paper. How many tens did I use up?

Yes, 64 is what I used. (write 64 as a partial product on the long division problem)

And now I'll subtract (on the large paper) and I get 9 tens left. Does that match our remaining stamps? We should have 9 tens left.

Now I'm finished with my tens and I must consider my units. So, I'll bring my units down on my paper (do so on expression - bring down the 6)

How many groups of 32 can I make from my 96 units? (3). Three – okay let's try that out.

How many units did I use? Yes, 96. So I'll write that in my problem as a partial product.

$$\begin{array}{r}
 123 \\
 32 \overline{) 3936} \\
 \underline{32} \\
 73 \\
 \underline{64} \\
 96 \\
 \underline{96} \\
 0
 \end{array}$$

So my answer is 1 hundred 23 and there is no remainder.

Epilogue

First time, teacher does the writing and student manipulates the materials. After a time, you can switch roles. Many children come to the abstract level from the stamp game rather than from racks and tubes but they reinforce each other.

Group Division with Fractions - 4

Materials

Circle fraction insets additional pieces. See illustration at end.

Begin

Let's do this problem.

$$2/3 \div 2/9$$

Today, let's think of it this way. How many groups of $2/9$ can we make from $2/3$?

(lay $2/3$ s unit on the table)

What can I do? Yes, let's exchange. How do I know it's ninths? Yes, from my denominator. (form $6/9$ after exchange)

How many groups of $2/9$ can I make out of $2/3$? How many did I make? (3)
(split the 6 pieces forming $6/9$... into 3 groups of $2/9$ each)
(fill in answer on paper)

We know the rule for dividing fractions. Let's see if we would get the same answer?

What do I do to divide by a fraction? Yes, I invert my divisor and then multiply.

$$2/3 \times 9/2 = 18/6 \quad \text{(write this on paper in front of us on table)}$$

And that's an improper fraction so let's form a proper fraction and we get 3.

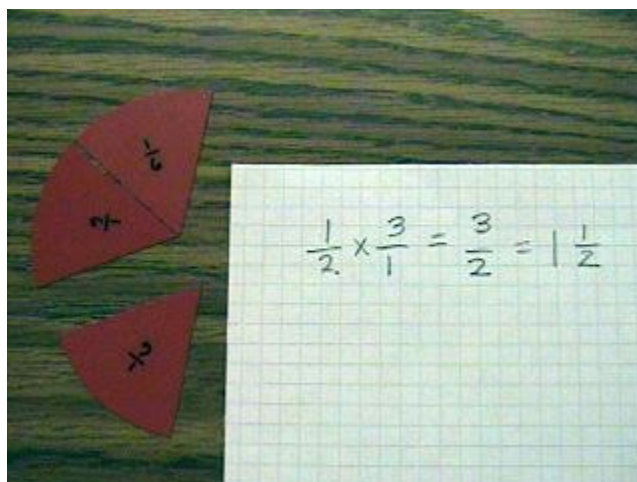
Did we get the same answer? Yes.

===

Let's do another. How about $1/2 \div 1/3$? Let's do it as a group division problem.

Our question is then, how many groups of $1/3$ are contained in $1/2$? Well we need to exchange for what? Yes, $3/6$.

$2/6$ make $1/3$. (move it away as a group) $1/6$ is only half of a third – so I made 1 and a half groups since $1/6$ is only half of $1/3$.



One group of $\frac{1}{3}$ (actually $\frac{2}{6}$) and $\frac{1}{2}$ group of $\frac{1}{3}$ (actually $\frac{1}{6}$) - made up of our original $\frac{3}{6}$ (originally $\frac{1}{2}$ before exchange)

(write)

$$\frac{1}{2} \div \frac{1}{3} = 1 \frac{1}{2}$$

Let's use our rule for dividing a fraction to check it.

$\frac{1}{2} \times \frac{3}{1} = \frac{3}{2}$ which is improper. Making it proper we get $1 \frac{1}{2}$.

November 19, 2004

Word Problems

Begin

Here, we'll do some word problems using group division with fractions. For example, suppose you have $\frac{3}{4}$ cup of salad oil and your cookie recipe calls for $\frac{1}{8}$ cup of oil. How many batches of your recipe can you make? (do problem with materials and abstractly for each)

What are we asking here? How many groups of $\frac{1}{8}$ are in $\frac{3}{4}$?

$$\frac{3}{4} \div \frac{1}{8}$$

We'll need to exchange fourths for eighths. So how many batches can I make (count them) 1,2,3,4,5 and 6. Since we have a word problem my answer must include the unit of measure so it is 6 batches.

Let's check it with our rule for dividing fractions.

$$\frac{3}{4} \times \frac{8}{1} = \frac{24}{4} - \text{which equals 6 batches.}$$

===

If it takes 1 hour to fill a tub $\frac{3}{8}$ full, how long does it take to fill the whole bathtub?

How many groups of $\frac{3}{8}$ are in 1? So our problem is $1 \div \frac{3}{8}$

October 26, 2004

Divisibility

Material

Golden beads, blank tickets, pencil and paper, a large (red) felt mat.
(Applies to all exercises for divisibility.)

Divisibility by Two - 1

Prologue

General procedure is:

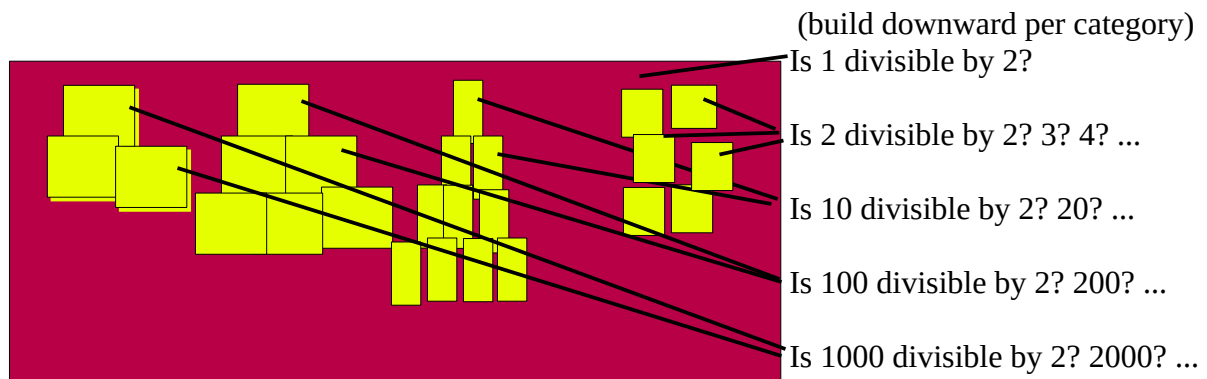
1. Pose question
2. Category exploration, recording what was divisible.
3. Build original number
4. Divide each category into 'divisor' equals parts
5. Answer question

We do the **first five exercises here – groups I and II**. The groups are:

1. Divisibility by two, five and twenty-five
2. Divisibility by four and eight
3. Divisibility by three, nine, a number's prime factors and eleven

Exercises are **grouped by reasoning ability and simplicity**. Leave out two **groups on the mat at the very end, created from the initial value (3584 in this case)**.

As we investigate each category, one by one, we will build bead values on the mat downward.



Begin

(pose (and write) a question...)

Is 3 thousand 5 hundred 84 divisible by 2? This means, can we separate it into 2 equal groups. That's what that word divisible means – to separate into groups.

Let's look at this number a minute. We chose 3584. It has thousands, hundreds, tens and units in it. We're going to investigate each of these categories here.

We'll start our investigation with the units. Is 1 divisible by 2? No. I can't make two equal groups from one can I.

Is 2 divisible by 2? Yes, we can see that here. (place a bead for each question, see layout)

Is 3 divisible by 2. No

Is 4 divisible by 2? Yes, it is.

Is 5 divisible by two.

No Is 6 divisible by 2. Yes – we can see it here.

Is 7 divisible by 2. No.

Is 8 divisible by 2. Yes. Again we see it here.

How about 9? No.

So in our units we find 2, 4, 6 and 8 are divisible by two. That's interesting.



Leave 2, 4, 6 and 8 as units divisible by 2.

Now, let's look at tens.

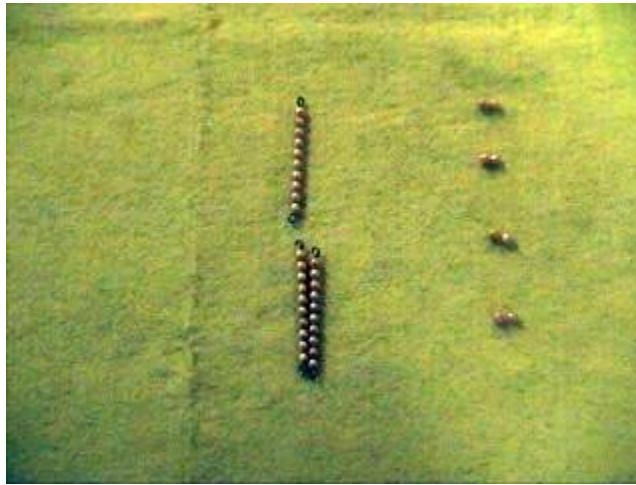


Is 10 divisible by 2. (place a 10B) Yes. **How do you know? How could we prove it?**

Yes, **I could exchange** this 10-bar for ten single beads and now I can make two equal piles. I can see from our beads this is possible.

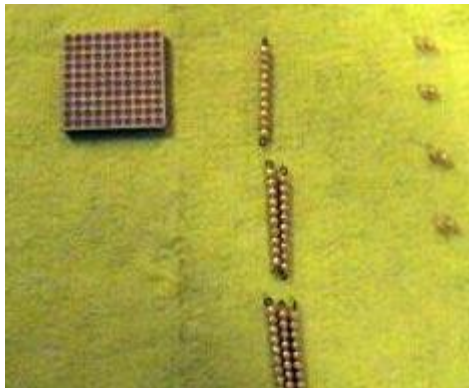


Is 20 divisible by two? Yes, because we can always pull apart our two bead-bars and if we needed to we could even split up a bar of ten, right?



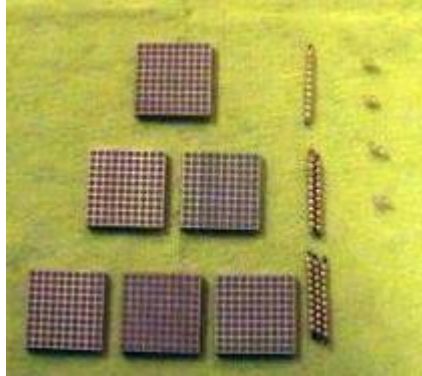
What about 30? 40? 50? 60? 70? 80? 90? Can we conclude that all of the tens are divisible by 2?

Yes?

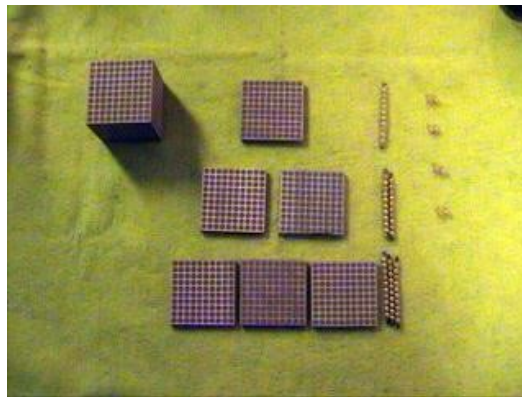


What about 100? Is 100 divisible by 2? How do we know? (we can exchange this for 10Bs and make two equal groups of 50 each). That's right, we can prove it that way.

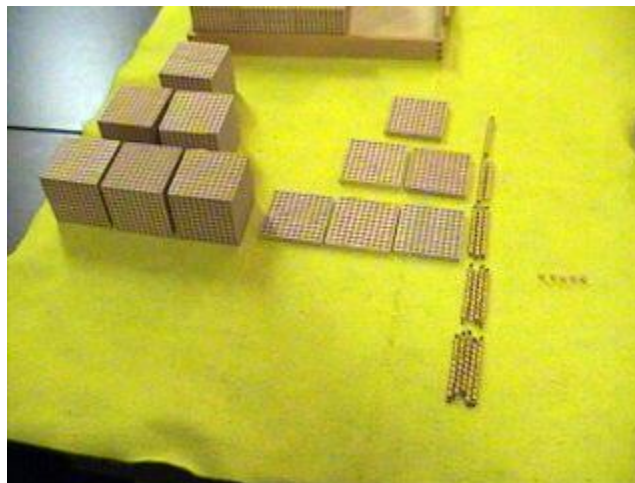
What about 200? How do we know? What about 300? 400? 500? 600? 700? 800? 900? What can we say about all of the hundreds? They are all divisible by 2!



What about 1000? Is 1000 divisible by 2? How do we know? We can exchange it for what? How many 10 hundreds, yes. Then we could make equal piles.



What about 2000? 3000? We can again conclude that all thousands are divisible by 2.

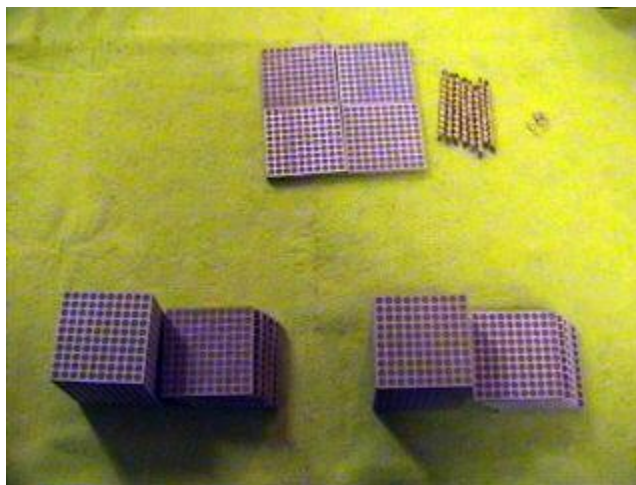


What did we find out? Thousands, hundreds and tens can always be divided by 2 but with units, only 2, 4, 6 and 8 are.

Let's go back to our first problem. Let's form our number from our golden beads.

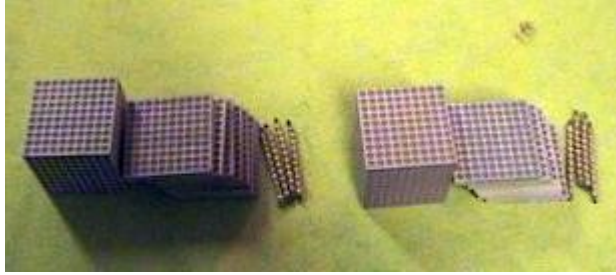


What did we discover about thousands? (exchange and make two groups)



How about our hundreds (exchange in order to make two groups)

What did we discover about tens? (exchange in order to make two groups)



How about units? Is 4 divisible by two? Yes, so 3584 is divisible by 2!

We know 3584 is divisible by 2 from our investigation. For example, we exchanged our 3 thousand cubes for 2 groups of 1 thousand, 7 hundreds and 5 tens, each – to prove it.

Maybe you'd like to pick a new number to see if it is divisible by 2. (more examples)

End

Epilogue

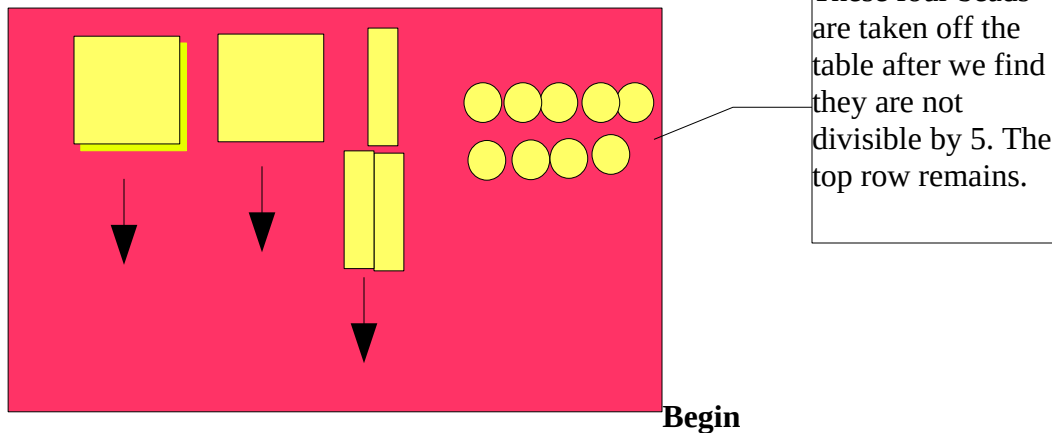
Through the child's experience, he will discover the rule for when numbers are divisible by 2. They've had some of this in the primary. **Let them discover the rule.** Have them keep doing numbers as above until they can express some form of our rule that is correct. A number is divisible by 2 if the last number ends in 2, 4, 6, 8 or zero.

October 26, 2004

Divisibility by Five - 2

Prologue

Start each exercise with our question. Pick numbers only up to thousands since the beads only go up that high. Only leave on the mat what is divisible by 5 at the end. Alternatively, if the children are a bit slower, choose a number and split it into 5 groups as in exercise I.



Is 8,995 divisible by 5?

(write it on paper)

We want to know if a number is divisible by 5. Remember when we asked if a number was divisible by 2, we looked at each category. Let's do the same thing.

Is 1 divisible by 5? No. Is 2 divisible by 5? No. Is 3 divisible by 5? No. Is 4 divisible by 5? No. Is 5 divisible by 5? Yes!.

(begin second row)

Is 6 divisible by 5? No. Is 7 divisible by 5? No. Is 8 divisible by 5? No. Is 9 divisible by 9? No.

So in our units, 5 is divisible by 5.

Now we need to look at tens. **Is 10 divisible by 5? How do you know?** Okay, we can exchange a 10-bar for five groups of 2. Ten is divisible by 5.

What about 20? Yes? How can we prove it? Yes, we can make 5 groups of 4 if we would

exchange our 2 10Bs for 5 4-bars or 20 units.

What about 30? **How can we prove it?** Yes, we can make 5 groups of 6 if we would exchange our 3 10Bs for units.

What about 40? How can we prove it? Yes, we can make 5 groups of 8 if we would exchange our 4 10Bs for units.

(continue to 90 or until the preception that all multiple of tens will be divisible by 5 arises)

So what can we say about all of the tens? They are all divisible by 5 aren't they!

What about 100? (yes) How do we know? Yes, we could exchange our 100 square for 5 groups of 20 beads each.

What about 200? (yes) How do we know? Yes, we could exchange our 2 100 squares for 5 groups of 40 beads each.

(repeat until the idea comes out that all hundreds are all divisible by 5)

So we can say all the hundreds are divisible by 5!

What about 1000? Is 1000 divisible by 5? How can we prove it?

(repeat as above)

Now, let's see if our original number is divisible by 5. (do previous exercise)

End

Epilogue

A number is divisible by 5 if a units digit ends in a 5 or in 0. This is what we want them to discover on their own.

Extension

Create a table of divisibility. The row category is the divisor. The column category is the dividend. The table cells are filled in with Y for yes (the number is divisible by the given divisor) or N for no (the number is not divisible by the given divisor).

Divisibility
by:

	2	5	25	Et cetera =>
2648	Y	N	N	:
9875	N	Y	Y	:
3015	N	Y	N	:
:	:	:	:	:

October 26, 2004

Divisibility by Twenty-Five - 3

Prologue

Start each exercise with our question. Pick numbers only up to thousands since the beads only go up that high.

Begin

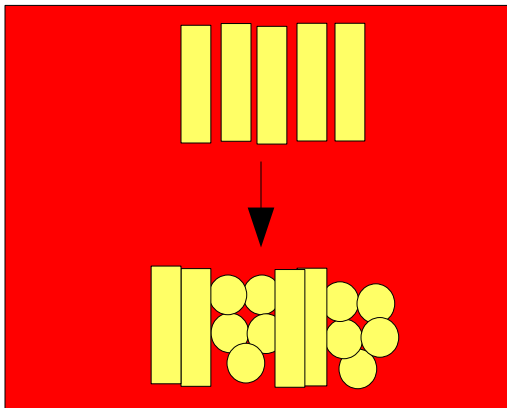
(invite children to come with a number for the dividend. i.e., Is X divisible by 25?)

We want to know what numbers are divisible by 25. Let's again look at each category.

Is 1 divisible by 25? (continue to 9)

Is 10 divisible by 25? (continue to 40)

Is 50 divisible by 25? Yes. **How do you know?** We can see this using our beads again with 2 groups of 25 beads each, right?



Exchange for 2 groups of 25. Leave this out, explore remaining tens and then go on to hundreds

Is 60 divisible by 25? (continue to 90)

What about 100? Yes, How do you know?

(repeat until we find all the hundreds are divisible by 25)

So, all the hundreds are divisible by 25, aren't they.

What about 1000? Yes, How do we know? Because all the hundreds are all divisible by 25.

**So we discovered all the thousands and all the hundreds are divisible by 25.
in the tens, we found that only 50 is divisible by 25.**

Examine Two Categories

**What about 25? Why didn't we find that numbers? (we only looked at even tens)
We had to have looked at 2 categories.**

I only looked at 1 category at a time. So, is 25 divisible by 25. Yes. Can you think of another number that is divisible by 25? Yes, **75 also.**

End

Epilogue

So in order to tell if a number is divisible by 25, we have to look at 2 categories: the units and the tens. A number is divisible by 25 if a number ends in 25, 50, 75 or 00. This is the final exercise of group I. Again, let the children discover these rules.

October 26, 2004

Divisibility by Four - 4

Prologue

Pick numbers only up to thousands since the beads only go up that high. This is the beginning of group II. Now, we must begin to **look at 2 categories at a time**.

Lay units category exploration in **horizontal rows** of 4. Stack hundreds for each multiple.

Prerequisites

Multiples must have been covered by this time.

Begin

Is 9748 divisible by 4?

I would like to know if a number is divisible by 4. (9748).

We know now we want to **investigate each category**.

(Lay individual beads out in two rows of 4, left to right - 1-4, second row 5-8)

For units, is 1 divisible by 4? No. Is 2 divisible by 4? No. etc.

Is 4 divisible by 4. Yes, (continue)

So, in our units, we discover that 4 and 8 are divisible by 4.

How about tens? Is 10 divisible by 4? No. Is 20 divisible by 4? **How do you know?** Yes, we could exchange and make 4 groups of 5 beads.

Is 30 divisible by 4. Why not? Right, I can't make four even groups. How about 40? Yes, (continue)

(each time, place the proposed multiple on the mat, then remove the ones that don't work)

What about hundreds? Is 100 divisible by 4? How do we know? Yes, I can make 4 groups of 25 beads.

How about 200? Well, if 1 hundred is divisible by 4, aren't you going to have 4 groups of 25 in each and every hundred. So, each and every hundred is going to be divisible by 4.

What about 1000? Yes, How do we know. If all the hundreds are – well this is ten hundreds. So all of the thousands are divisible by 4 also.

We've discovered all thousands and hundreds are divisible by 4. In the tens, we discovered 20, 40, 60 and 80 are divisible by 4. And in units, 4 and 8 are divisible by 4.

What about a number like 16? Yes, so we missed some numbers.

(Write additional numbers divisible by 4 up to 100 on paper)

We have 12 also. What about 24? (write each while speaking – 28, 32, 36, 44, 48, 52, 56, 64, 68, 72, 76, 84, 88, 92, 96)

End

Epilogue

A number is divisible by 4 if the last two digits of the number are divisible by 4 or are 00.

October 26, 2004

Divisibility by Eight - 5

Prologue

Start each exercise with our question. Pick numbers only up to thousands since the beads only go up that high. Must have had multiples by this time. See prior divisibility lessons for more detailed wording.

Lay units category exploration in **horizontal rows** of 8. Stack hundreds for each multiple.

Begin

Is (choose a number) divisible by 8?

Let's look at our units. Is 1 divisible by 8? (continue)

So in units we see that 8 is divisible by 8.

How about tens? Is 10 divisible by 8? Why not? Etc.

Is 40 divisible by 8? Yes, Why?

So in our tens, we discover that 40 and 80 are divisible by 8.

What about hundreds? Is 100 divisible by 8? Why not. We can't make 8 equal groups can we. Is 200 divisible by 8. Yes, because you make 8 groups of 25 out of this. (continue)

Is 400 divisible? Yes, we can make 8 groups of 50. (continue)

What about 600? Yes, why. Yes, It's a multiple of 200 so it must also work for 600.

How about 1000? Yes, Why? Yes, It's a multiple of 200. So 1000 is divisible by 8. If 1000, what can we say about all the other thousands? Yes, they are all divisible by 8.

Examine Three Categories

But what about 16? Are there others we've missed? Yes. What do we call these numbers (multiples). How about 408?

End

Epilogue

To arrive at the rule for 8 we'd have to look at three categories because after 400, certainly 408 is divisible by 8. So how could we word a rule for 8? A number is divisible by 8 if the value of the last three digits is divisible by 8.

October 27, 2004

Divisibility by Three - 6

Prologue

This begins **group III** and requires more knowledge and sophisticated understanding of division. Note **new pattern**: “**if I took one away...**” We begin with our question:

Begin

Is 9,738 divisible by 3?

Start with Tens and Take One Away

What have we done before. We looked at each category, right? So let's do that but **this time let's start with tens.**

Is ten divisible by 3? (hold up a 10B) No.

But let me ask you this: if we took one bead away from this 10B, would what is left be divisible by 3. Yes! Nine is divisible by 3.

Is 20 divisible by 3? (hold up 2-10Bs). No, But if I took 1 bead away from each of these tens, would what is left be divisible by 3? Yes, 18 is divisible by 3.

How about 30? (hold up 3-10Bs) Yes, it is. But I also have to know this: if I take 1 bead away from each bar, would what is left still be divisible by 3? Yes, 27 is also divisible by 3. Yes, so this seems to work when we take away 1 bead from each 10B.

How about 40? No, but if we take 1 bead away from each 10B, is what is left divisible by 3? Yes, 36 is divisible by 3.

What about 50? No, but if we take away one bead from each 10B, how many would be left? Yes, 45. Is 45 divisible by 3? Yes, it is.

So it seems that no matter how many tens we have – if we take one bead away from

each one, what is left is divisible by 3. Let's remember that.

What about 100? Is 100 divisible by 3? (hold up bead square) No. But let me ask you this.

If I take one bead away from this hundred, would what is left be divisible by 3? Yes. We'd have 99 left and **99 is divisible by 3.**

So with hundreds, it seems to work also. If I take 1 bead away from the hundred, it is divisible by 3.

Let's look at 1000. Is 1000 divisible 3? No. But if I take 1 away, what do we have left? **Yes, 999. Is that divisible by 3? Yes!**

So what could we say about thousands? As long as we take 1 bead away from each thousand, what is left is divisible by 3. So let's try to remember that also.

Now let's look at our number. 9,738. Let's set this out in golden beads.

I want to know if this number is divisible by 3?

What do we remember about thousands? If we take away 1 from each, they are divisible by 3. So **I'll place these 9 beads here below the thousands to represent the 9 I will take away from this 9 thousands.**

(place 9 individual beads under the thousands category of the dividend)

What do we remember about hundreds? If we take away 1 from each, they are divisible by 3. So **I'll place these 7 beads here below the hundreds to represent the 7 I will take away from these 7 hundreds.** This helps us remember. So what's left up here is now divisible by 3.

(place 7 individual beads under the hundreds category of the dividend)

What do we remember about 10s? If we take away 1 from each, they are divisible by 3. So **I'll place these 3 beads here below the tens to represent the 3 I will take away from this 30.** This helps us remember. So what's left up here is now divisible by 3.

(place 3 individual beads under the tens category of the dividend)

So we're no longer counting our original number – we're only concerned with units now.

(combine ALL units on the mat $(9+7+3+8)$ and see if you can make 3 equal groups)

Now, all we have to consider are these units beads. If I put them into 3 equal groups I know they are divisible by 3.

Now you can choose a number and do this in the same way.

End

Epilogue

The beads in the units at the end represent the digits of the number 9, 7, 3, 8. So we can word the rule for divisibility by 3 by saying, **“A number is divisible by 3 if the sum of its digits are divisible by 3.”**

October 27, 2004

Divisibility by Nine - 7

Begin

(exactly the same process and wording as divisibility by 3, substituting 9 for 3)

We've investigated divisibility by 3 and now by 9. Can I divide 9 by 3? Yes! So, if I know some number is divisible by 9, what can I say about that number being divisible by 3?

If a number is divisible by 9, can I always say that number is also divisible by 3? Yes.

Can I also say any number divisible by 3 is also divisible by 9? No...

For example, 30 is divisible by 3 – but not by 9.

But any number I pick that is divisible by 9 is always divisible by 3! Why is that?

End

Epilogue

A number is divisible by 9 if the sum of all the digits is divisible by 9.

October 27, 2004

Divisibility by a Number's Prime Factors - 8

Prerequisites

Prime factors must have been covered already.

Prologue

After doing divisibility by 3 and 9, we find that any number divisible by 9 is also divisible by 3. The reverse is not true. So at some point, we rely on this fact too. Present this idea when possible. This ties in with multiples and can be highlighted.

Begin

We're going to pick a number – I've chosen 30.

The first thing we'll do today is to **find the prime factors of 30**.

30		
30	2	Is 30 divisible by 2
15	3	Is 15 divisible 2. By 3?
5	5	
1		

So we have the prime factors of 30. Now we use tickets.

The fact that these are the prime factors means $2 * 3 * 5$ equals 30. (write each number on a ticket as you speak and set it out on the mat)

$$2 \times 3 \times 5 = 30$$

And here's my 30 in golden beads. (lay below 30 in the expression)

(to slow it down, we'll refer back to the 30 each time)

Let's start with 2 (pull down the number 2 a bit to isolate it)

Is 30 divisible by 2? Yes, let's verify it (take one 10B, exchange for 10 beads and make two equal groups from the 3 ten-bars).

(change back to 3 10Bs)

Is 30 divisible by 3? (pull down the 3 from the expression)

Sure, we can easily see that. (separate the 3 10Bs on the mat)

(change back to 3 10Bs)

Is 30 divisible by 5? Let's prove it. (exchange the 3 10Bs and make 5 equal groups – of all individual beads) So, we know the answer is yes, 30 is definitely divisible by 5.

(change back to 3 10Bs)

We've seen 30 is divisible by 2, 3 and 5. **Now we'll do the combinations of them.**

For example, 2×3 is 6. Is 30 divisible by 6? (exchange the 3 10Bs and make 6 equal groups – of all individual beads) So, we know the answer is yes, 30 is definitely divisible by 6.

(change back to 3 10Bs)

How about 2×5 is 10. Is 30 divisible by 10? (separate the 10Bs to show ten groups) So, we know the answer is yes, 30 is definitely divisible by 10.

Finally we have 3×5 is 15. Is 30 divisible by 15? We can exchange 1 10B and then show that we can make groups of 15 (make two sets of 10B and 5-1Bs)

So we've seen 30 is divisible by its prime factors but also by any combination of its factors. So if you know a number is divisible by 2 and 3 – then you also know that number is divisible by 6.

If a number is divisible by 2 and 4, then you also know that number is divisible by 8.

So now we can add divisibility by 6 to our divisibility table.

End

Epilogue

So now we can add divisibility by 6 to our divisibility table. This is why it's useful to look at divisibility by prime factors.

Note: Divisibility by 7. The rule is, “A number is divisible by 7 if the difference between the sum of the odd number periods and the sum of the even number periods is divisible by 7. Take the number 654,076,790,241 for example. (here, periods are same as hierarchies)

$$654+790 = 1444 \text{ and}$$

$$076+241 = 317$$

then, take the smaller from the larger – does not matter which is which.

$$1444 - 317 = 1127$$

Is 1127 divisible by 7? If so, 654,076,790,241 is also divisible by 7.

$$1127 / 7 = 161$$

So, both are divisible by 7

October 27, 2004

Divisibility by Eleven - 9

Prerequisites

Must have done divisibility by 3 and 9 – and understanding them well. Don't try this one otherwise.

Prologue

We'll eventually combine units category into a new store with more units, one for each hundred; from this store, one unit is taken for each thousand and ten. The difference, if 0 or divisible by 11, tells us the original dividend is also divisible by 11.

Begin

Is 2849 divisible by 11?

We know we have been investigating the categories and we'll do that again, starting with units again.

Is 1 divisible by 11? 2, 3, 4, 5, 6, 7, 8, 9 – No we find nothing in units divisible by 11.

So units seem to be extras.

I'm going to write a plus sign above this unit category to help us remember units are just extras.

(there is a ticket and beads below)

What do we know about tens. If I added 1 to this 10B, would it be divisible by 11?

How about 20? Is it divisible by 11? No. but if I added 1 bead to each 10B, would it be divisible then? Yes. So it seems we're short some.

I'm going to write a minus sign above this ten category to help us remember tens are short somehow.

(there are two tickets and units and tens beads below)

Is 100 divisible by 11? No, but if I took 1 bead away we'd have 99. Is 99 divisible by 11? Yes, it is. So if I take 1 bead away from each 100, what is left will be divisible by 11.

So in hundreds, again it seems like we have extras. **I'm going to write a plus sign above this hundreds category to help us remember hundreds give us extras.**

(there are three tickets and units, tens and hundreds beads below)

Is 1000 divisible by 11? Well let's think.

For tens I had to add a bead. With hundreds, I had to take one away.

So let's first **try adding 1. Is 1001 divisible by 11?**

Yes, it is 91. So **if we just add 1 bead to each thousand, it is then divisible by 11.**

If we did that for each thousand, however many we had, that number would be divisible by 11. So it seems we are short again.

I'm going to write a minus sign above this thousands category to help us remember thousands are short somehow.

(there are four tickets and units, tens, hundreds and thousands beads below)

So let's return to our first problem. Let set out 2 thousand 8 hundred 49. I've kept our plus and minus signs up here to help us remember what to do to make each divisible by 11.

Short, Extras, Short, Extras

Let's review what we know so far. What we discovered is that in units and hundreds, we have extras.

And in thousands and tens, we are short. So in this number 2849, in the units we have 9 that are extras.

(pull unit beads down to form a new store at bottom of the table)

I have 8 hundreds so I have 8 extras from that 8 hundred.

(take 8 from the original store and place at bottom under the hundreds.)

We still have the thousands and the tens and they each have a shortage.

The tens each need - how many more – a total of 4. I'll use some of these extra beads at bottom to give the tens. It does not matter which pile we pick from.

(place the 4 under the tens but far above the two extra stores at bottom)

We'll again use 2 extra beads for our thousands, taken from our extras.

(place the 2 under the thousands but far above the two extra stores at bottom)

So how many extra units do I have left? 11 – so our number is divisible by 11.

End

Epilogue

A number is divisible by 11, if the difference between the sums of every-other category is divisible by 11.

Units and hundreds are extras. Tens and thousands are shortages. So we added the digits in the units and hundreds, added the digits in the tens and thousands and then subtract the smaller sum from the larger sum. If that difference is 0 or is divisible by 11, then we know the number is divisible by 11.

October 27, 2004

General Pattern for Divisibility

For divisibility by 2, introduce the concept since children are in the first year lower elementary; they need the security of knowing what to do.

Divisibility by 2 is structured:

1. Is a number divisible by ____
2. Examine categories of the decimal system – one at a time – asking the same question.
3. Examine some numbers, going over results from the category exploration. This restates what it means to be divisible
4. After some experience, the children will come to the rule themselves. Try to get their words, their understanding – not the way we've stated it in our rules.
5. Add the rule to their divisibility chart that we've been accumulating. Also add some new numbers to this chart to help them review the previous rules experienced.

6. FRACTIONS

Introduction to Fractions Using Circle Fraction Material

Prologue

When we look at numbers we've looked at whole numbers. But there are also fractions. We'll start sensorially, moving from the concrete materials into the abstraction of fractions. We'll see the same four operations we've already seen, using the fractions.

The children may have used these materials in the Casa dei Bambini. They may have been introduced to the names of the fractions.

Material

Ten green metal plates, each of which has a circular inset, ten centimeters in diameter. One is a whole circle. The others are divided respectively into 2, 3, 4 and so on – equal pieces. Each piece has a knob. A box of extra fraction pieces, identical to the original ones in the insets. Black strips made of construction paper. Blank tickets and a pencil. Labels for all of the segments (e.g. $\frac{1}{7}$ piece) of the insets.



Prerequisites

The children need an understanding of whole numbers and operations using whole numbers. So children can be pretty young when they begin.

Begin

(lay materials out from left to right, beginning with unit (whole), $\frac{1}{2}$, $\frac{1}{3}$, ... up to $\frac{1}{10}$ arrayed in a straight line)

This is a unit. (hold up the whole)

(pick up 2 - $\frac{1}{2}$ pieces, holding them together as a whole)

This is also a unit. But if I break them into 2 equal pieces, these pieces are now called fractions.

Fractions comes from the Latin word frangere, meaning to break. The past participle is fractus. Fractions are broken pieces of equal size.

We can show fractional pieces are equal by superimposing one fraction onto another. We can show they are equal by doing this.

These equal pieces are called **halves** (hold up $\frac{1}{2}$ pieces).

How many equal pieces has my unit been broken into here? (point to $\frac{1}{3}$)
Yes, three equal pieces. We call these **thirds**.

Here, we have 4 equal pieces. We call these **fourths**.

(continue on in order to $\frac{1}{10}$, introducing each one. Ensure they pronounce / hear the 'th' at the end. Ensure they know there are equal parts. Ensure they know the terms half, third and so on. Do a 3-period lesson on these as necessary)



Let's talk about just one of these pieces. This is one fourth (hold up $\frac{1}{4}$ piece)

Here, with my halves, if I wanted just one, I would ask for 'one half.'

November 16

Writing Fractions

Purpose

To show that if you break a unit unto equal pieces, the pieces are called fractions.

Material

Fraction insets. Charts C1 and C2.

Begin

I can see a line between my two halves. We're going to use that line to help us write fractions.

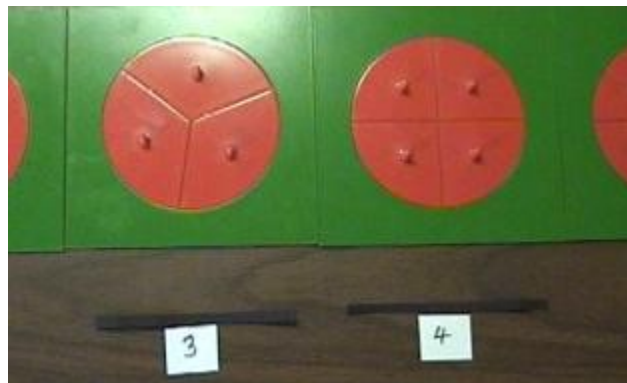
I have this black strip that is like that line. If we are talking about halves, how many pieces are in the whole? Yes, 2. I'm going to write two and put it below the line.

(write 2 on a ticket and place below the black strip, all below the inset.)

How many pieces is my 5th fraction going to have in a whole? Yes, 5. So I'll write my 5 below the line.

(write 5 on a ticket and place below the black strip, all below the inset.)

(repeat so there are several denominators set out preparing for the numerator introduction following)



How many pieces in the whole would you like to learn to write?
(accept an answer and write it out)

Sometimes, people call these families. This is a family of thirds. (point to thirds inset). This is a family of fifths (point to fifths inset).

This bottom part has a special name. It's called **denominator**. It comes from a Latin word, nomen, meaning **name**.

So the denominator tells us its name – whether it is fifths, sevenths or tenths.



Here's my family of fifths. Suppose I wanted just one of them. I'm going to write 1 on my piece of paper and I'm going to put it here above my black strip.

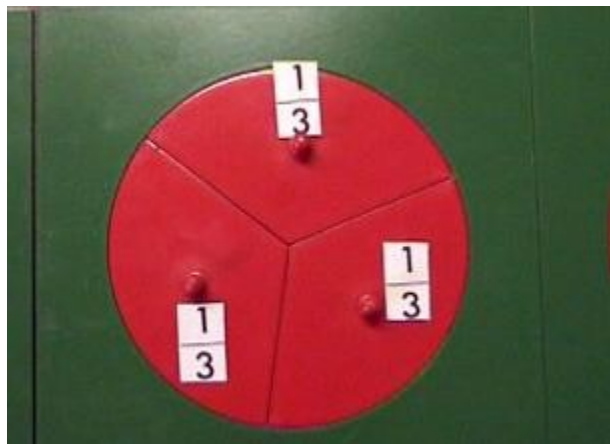
(write 1 on a ticket and place above the black strip already present, all below the inset.)

How about tenths? Suppose I wanted just one of them? What would I write. Yes, I'd write 1 and I'd place it above my line.

(write 1 on a ticket and place above the black strip already present, all below the inset.)

(do several more to get the idea of numerator across)

Here, I have some tickets. (take out and spread them on the table) This says '1/7.' Who'd like to put the tickets in the correct place on the table, underneath the fraction inset?



Here in the unit envelope, the tickets say '1/1.' Who'd like to label all the parts we have

here?

(allow child to come up and put the correct tickets right on the pieces. Next, remix them all up and let the children relabel all of them, classifying the fractions as they go.)



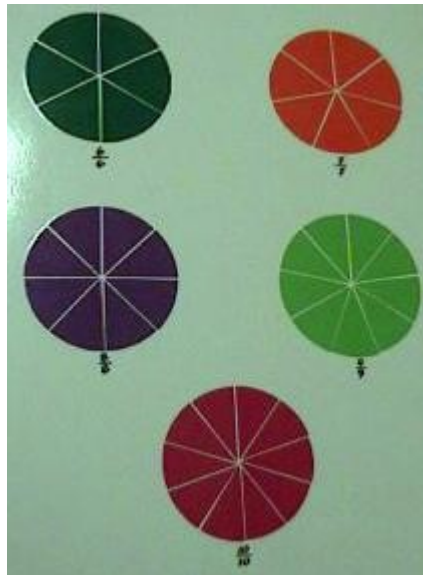
Let's look at these fourths. Instead of $\frac{1}{4}$, I want two pieces. So what do I write? I'll write ' $\frac{2}{4}$ ' instead.

(write the fraction and place the ticket below the inset with the number of pieces in the numerator)

This number we write above the line has a special name. It's called the **numerator**. It comes from the Latin word numero, meaning **number**. It tells us the number of pieces taken from the whole.

I have charts to show you. These are the fraction charts.

(we can make any colors we want but if $\frac{1}{2}$ is colored orange, all references to halves must be orange)



Maybe you'd like to make some fractions like these and color these like you want.

Epilogue

Be sure to pick up pieces one at a time; they can scratch otherwise. Children can trace and color in the fraction charts; halves should look like halves so let them color if they cannot cut out halves that look like halves. Have them label the fractional parts afterward.

So far, we presented the concept of fraction, we named them and we learned to write them.

November 16

Equivalence of Fractions

Purpose

Direct purpose is to show the principle of equivalence and this is secondarily preparation for finding common denominators and reducing fractions.

Material

Fraction insets. Charts C3, C4 and C5 (optional).

Prologue

Do this work very slowly. Have children repeat it as many times as possible. Make charts, booklets and so on until the ideas are very solid.

Begin

Suppose I take this ($\frac{1}{2}$ piece). What **I want to know is if anything else will fit exactly in this space** I have left in the inset.

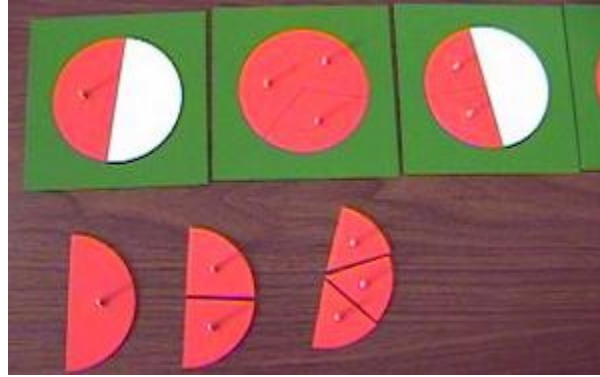
Let's start here with the frame next door. Let's see if thirds will fit. (try it) No. Doesn't work.

How about the next one – the fourths. Yes, they fit exactly when I put two of them. (place $\frac{2}{4}$ pieces below the frame)



The next frame we'll try is fifths. No, this does not work.

How about sixths? What do you think? Yes, we can see the sixths fit exactly.
(place $\frac{3}{6}$ pieces below the frame)



(continue with 7th, 8th, 9th, and 10th pieces, **placing the ones that fit down below the frame**, arraying the number of pieces that fit into $\frac{1}{2}$)

(Note: when you try out if it fits, actually place the pieces from the frame we're trying into the $\frac{1}{2}$ inset area we have taken the first $\frac{1}{2}$ piece from.)

So what did we find out? In this space, we could fit the $\frac{1}{2}$, but also $\frac{2}{4}$ and $\frac{3}{6}$ and $\frac{4}{8}$ and $\frac{5}{10}$ pieces. So we can say those fractions are equivalent.

We found out that $\frac{1}{2}$ is equivalent to $\frac{2}{4}$ which is equivalent to $\frac{3}{6}$ which is equivalent to $\frac{4}{8}$ and which is equivalent to $\frac{5}{10}$ pieces. They all fill the same space.



Well that was fun. Let's see if this works with any other fraction besides $\frac{1}{2}$. Let's try to see what is equivalent to $\frac{1}{3}$.

(invite student up to test each frame one at a time to make something equivalent to $\frac{1}{3}$)

Ah, so how many did we find that were equivalent to $\frac{1}{3}$? $\frac{1}{3}$ is equivalent to $\frac{2}{6}$ and is also equivalent to $\frac{3}{9}$ pieces.

Epilogue

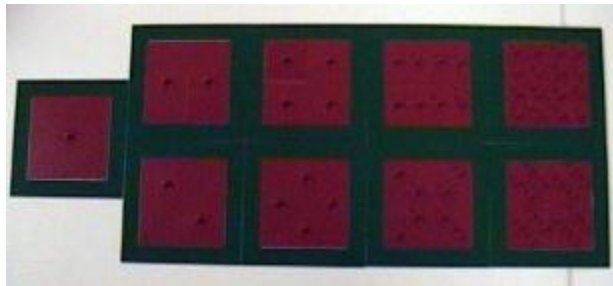
Be sure to place pieces in the same positional orientation to make the maximum visual impression. This is the time for handwork and materials as much as possible. This is followed by charts, tracing, booklets, scrolls and so on.

Another idea children often come up with is to take $\frac{2}{6}$ or $\frac{3}{9}$ pieces and to search backward – toward the $\frac{1}{3}$ – as opposed to always searching toward smaller fractions. Show the charts C3-C5 as follow up which show equivalence also.



Chart C3 - Equivalence

You can also array the square fractional material in the same manner as above, doing roughly the same kind of presentation. This reinforces the idea irrespective of the shapes involved. Then progress to the triangle fractional insets to concretize the ideas. This tells them the idea can be generalized; units are not always circles.



November 16

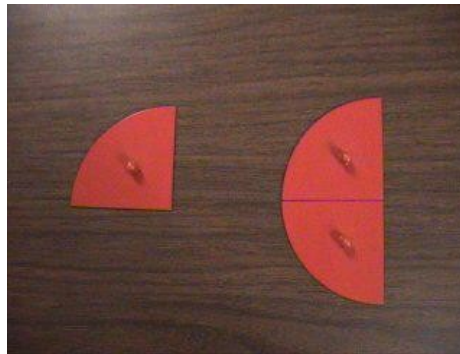
Addition & Subtraction of Fractions with Same Denominator

Material

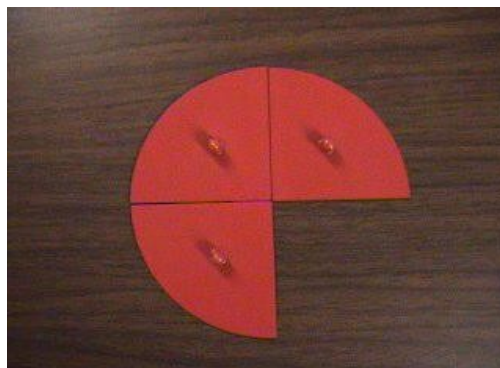
Circle Fraction insets. Paper and pencil. Chart C7 showing addition and C12 showing subtraction are optional follow-up.

Begin

Suppose I have $\frac{1}{4}$ and I have $\frac{2}{4}$ here. What happens if I add them together?
(place the pieces in one figure)

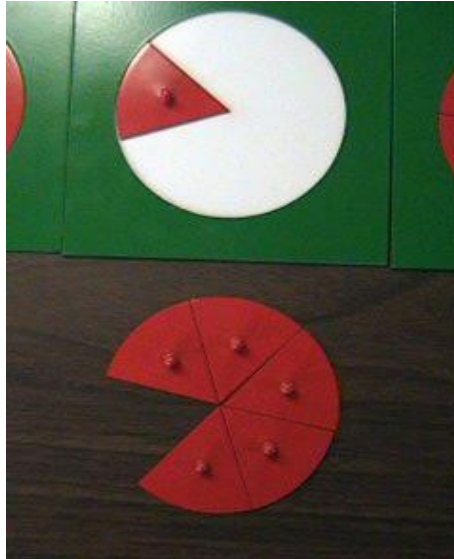


I wonder how many fourths that will leave me? Yes, $\frac{3}{4}$. Who'd like to come up and do another one like this?



How about taking out $\frac{2}{6}$. Now let's add to that $\frac{3}{6}$. (lay out pieces as above each time)

How much does that give us all together? Yes, we have $\frac{5}{6}$ as a result.



(give each child a turn to do one like this. Repeat until the idea is grasped)

Now let's take $\frac{1}{3}$. And we want to add another $\frac{1}{3}$. Let me show you how we would write this:

$$\frac{1}{3} + \frac{1}{3}$$

What do we get when we put them together? Yes, we get $\frac{2}{3}$.
(add answer to expression)

$$\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

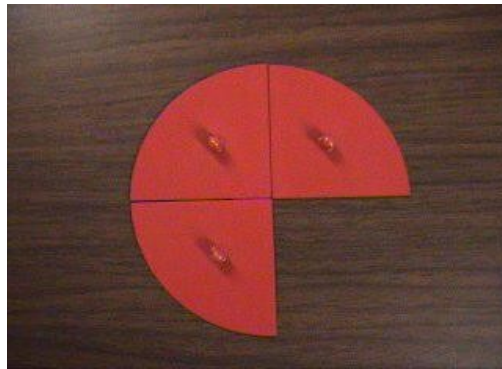
Who'd like to try one? (allow each child to create a problem and write the expression.)

$$\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$$

(repeat and invite them to work independently)

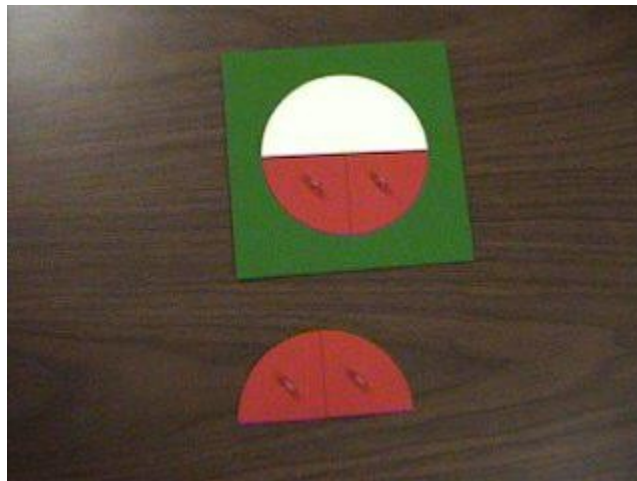


Today, I'm going to take away from $\frac{3}{4}$ (place the pieces on the table)



And I'm going to take away $\frac{1}{4}$ (put one of the pieces back into the frame)

What do I have left? Yes, $\frac{2}{4}$.
(put 1 piece back into frame)



I'm going to take out $\frac{5}{7}$. How many should I take? (three)

I'm going to take away $\frac{1}{7}$. What will I have left? (take $\frac{1}{7}$ piece and put into frame)

(repeat and give each child a chance to do it themselves)

Let's write this too. I'm going to take out $\frac{6}{10}$ and then I'll take away $\frac{5}{10}$ from that.

We would write this:

$$6/10 - 5/10$$

What do I have left (fill in answer above)

$$6/10 - 5/10 = 1/10$$

(repeat many problems of both addition and subtraction before going on)

Epilogue

If you have a child who moves to writing very quickly, let that child work ahead of the group. Don't hold that child back. We write the fraction bar horizontally. Later they will write the line diagonally but not at this point.

November 16

Multiplication & Division of Fractions by a Whole Number

Prologue

This lesson comes only after much experience with addition and subtraction of fractions. Note that when you place the pieces on the table, before the multiplication occurs, the pieces are in separate groups – to visually show they are numerically separate to begin with.

Multiplication

Material

Circle fraction insets. Extra box of plastic fraction pieces. Chart C17 is optional.

Begin

Today, let's take $\frac{1}{3}$ 2 times. ("one third one times, one third two times" put each piece down separately)

What do we do to get the answer in multiplication? We put them together, right?

What do I end up with? Yes, $\frac{2}{3}$. So $\frac{1}{3}$ taken 2 times equals $\frac{2}{3}$.

(ask for a volunteer)

I'd like you to take $\frac{1}{6}$ 3 times. (she does so)
How many do you have now?

So $\frac{1}{6}$ times 3 equals $\frac{3}{6}$. (ask for next volunteer and do many examples)

What about $\frac{1}{2}$ times 10? (now we introduce the extra pieces)

(put them separately initially as above. Then combine in the multiplication process.)
(some see that the answer is 5 right away. Others will want to write $\frac{10}{2}$.)

$$\frac{1}{2} \times 10 = \frac{10}{2}$$

In fractions, we want to express our answer in the simplest terms – that means the least number of pieces. So I could trade these 2 halves in for a unit, and these 2 for a unit ...

(continue to 5 units)

These are the least number of pieces we can use to show this answer.

Epilogue

Don't jump into a problem like $\frac{1}{2}$ times 10. But if they bring it up, explain it. This introduces an **improper fraction** – a fraction in which the numerator is larger than the denominator. Show them using the materials how to reduce to the lowest term. It also provides an intro to **mixed numbers** – a value with a whole number and a fraction. Show them how to write it: $4\frac{1}{2}$ instead of $\frac{9}{2}$, for example.

Division

Material

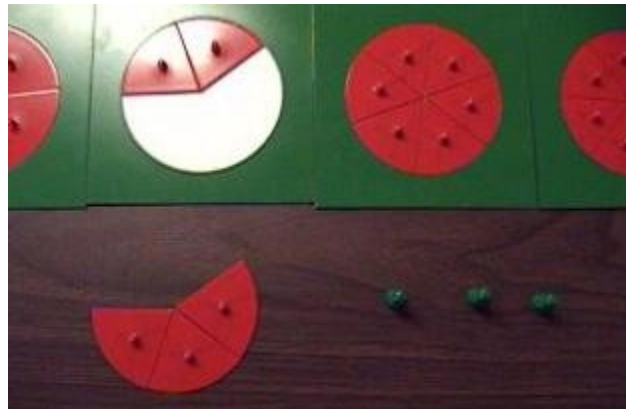
Unit skittles, circle fraction insets. Extra box of plastic fraction pieces.
Chart C18 is optional.

Begin

Today, let's take $\frac{3}{5}$. And I want to divide it by 3.

(place three fifths pieces on table and 3 skittles to the right)

How do I find my answer? Right, we share out. (share out pieces to each skittle)
So per skittle we have $\frac{1}{5}$.



(Ask for volunteer)

Let's take $\frac{6}{8}$ and divide that by 2. (she takes her pieces placed together on the table)

and her 2 skittles to the right)

What did we get? (3)

So $6/8$ divided by 2 equals $3/8$.

(another volunteer)

Let's write this one down. How about $9/10$ divided by 3?

(forms one figure of $9/10$ on the table and arrays 3 skittles to the right)

What did we get for each skittle? ($3/10$)

So $9/10$ divided by 3 is $3/10$. Can you write it on the board (she does so)

Epilogue

If the children have not begun reducing fractions by this time, this is another opportunity to introduce and use it. Again, tell the children it's nice to use the least number of pieces and show how $4/10$, for example, is equivalent to $2/5$ - using the materials: the area in the beginning frame filled with other types of pieces ($4/10$ area filled by $2/5$ pieces). Trade one piece at a time to drive home the equivalency. Other problems are $1/3$ divided by 3 - which forces one to exchange the $1/3$ for $3/9$. Or, $2/5$ divided by 4.

An important thing in the fraction lessons is to help the children realize that all four operations can be done with fractions. This work spreads over a long period of time. The children should continue to do design work with both fraction and metal insets.

Because we are working within the limits of the material, it's a good idea to have teacher-made problem cards. These are laminated cards with 6-8 problems to guide the child along the path of the lessons, without extra difficulties.

November 17, 2004

Addition & Subtraction with Different Denominators

Introduction

Addition

Prerequisites

Firm foundation with equivalence in fractions. Lots of experience with addition and subtraction of fractions with like denominators. They may have experience with multiplication and division with whole numbers.

Prologue

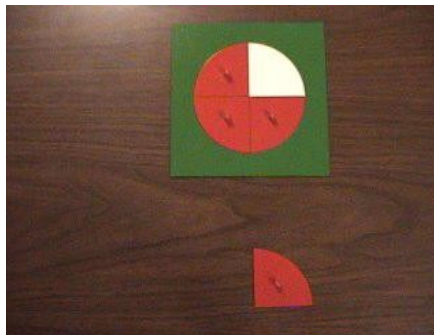
Up to this point, we adhered to problem within the confines of our materials. We as teachers are making up problems. Eventually, we need complexities like $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$. This problem cannot be worked out using our materials. We'll do a series of exercises here that lead the children to solve problems outside the limits of the materials.

Material

Circle fraction insets, paper and pencil.

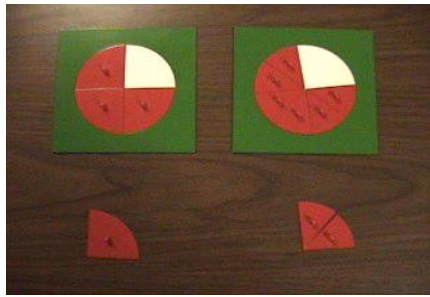
Begin

Today I have $\frac{1}{4}$ and I want to add to that $\frac{3}{8}$. Addition means putting them together but if I put them together I don't know what I have since my denominators are different; they are from different families of fractions.

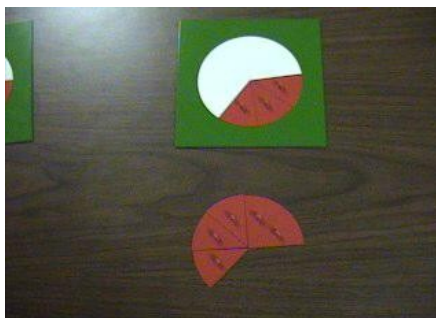


When we are adding fraction – **in order to put them together they must have the same**

denominator.



And what do we get? Yes, $5/8$.



So $1/4$ plus $3/8$ equals $5/8$.

If we are not sure what to change, we can put other pieces into the frame to see what will fit. Sometimes we know about equivalent fractions that will let us skip trying pieces in the frame.

Let's try the problem: $3/10$ plus $2/5$. What do we do first? We need the same denominator, right? In this problem, we'd like to change the tenths to fifths or to change fifths to tenths. How can we do it?

Yes, we can change $2/5$ into an equivalent fraction – $4/10$. And then we simply add the 4 and the 3 from both numerators. So $3/10$ plus $2/5$ equals $7/10$. Or we could also say $3/10$ plus $4/10$ equals $7/10$ – since $4/10$ and $2/5$ are equivalent fractions.

When you add fractions you must have the same denominator. That denominator is called the common denominator.

Epilogue

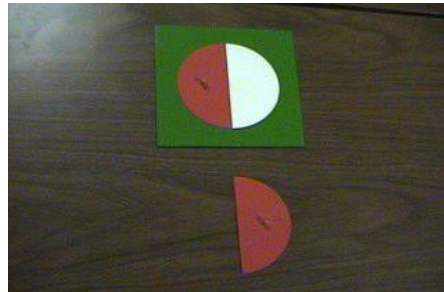
When you ask, “Can you put them together”, this is at a point where the concrete materials have taken the common denominator; get the child to this point of choosing the materials that establish the same denominator when the pieces are together.

Note that problems like $\frac{2}{6} + \frac{1}{3} + \frac{1}{9}$ require several steps to reach the common denominator. Also, $\frac{1}{4} + \frac{1}{4} + \frac{3}{9}$ requires several changes to equivalence: $= \frac{1}{2} + \frac{2}{6} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

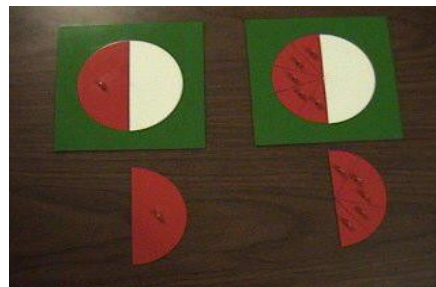
Subtraction

Begin

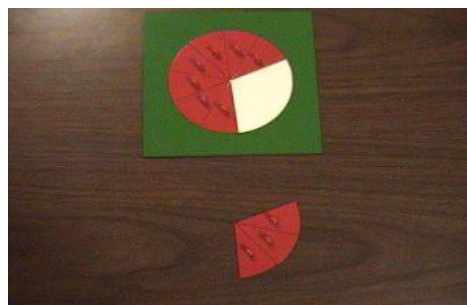
I have $\frac{1}{2}$ and I want to take away $\frac{2}{10}$. What can I do?



Let's exchange $\frac{1}{2}$ for $\frac{5}{10}$.



So now I can just subtract 2 from my 5. So, $\frac{1}{2}$ minus $\frac{2}{10}$ is $\frac{3}{10}$.



(note that with the materials in subtraction, we do not put out the subtrahend)

Let's do another: $\frac{2}{3}$ take away $\frac{3}{6}$. What would we do. Okay, yes, we can exchange our $\frac{2}{3}$ and exchanged it for $\frac{4}{6}$. So what do we get subtracting our numerators? $\frac{2}{3}$ take away $\frac{3}{6}$ equals $\frac{1}{6}$.

How about $\frac{1}{2} - \frac{1}{5}$? What can we do? Yes, we can exchange the $\frac{1}{2}$ for $\frac{5}{10}$. Then what's next? We still don't have a common denominator do we? So we can change the $\frac{1}{5}$ for $\frac{2}{10}$, right? What now?

Yes, we subtract the numerators – since we have a common denominator.
So, $\frac{5}{10}$ minus $\frac{2}{10}$ is $\frac{3}{10}$.

November 17, 2004

Abstraction: Multiply by Fraction Equal to One - 1

Material

Circle fraction insets, paper and pencil.

Prologue

This work is based on all the equivalence work we've done so far. It is a method to exchange one fraction for another equivalent fraction.

Begin

We've done a lot with fractions and equivalence. We've seen for example, that $\frac{4}{8}$ is equal to $\frac{1}{2}$, right? And that $\frac{2}{6}$ is equivalent to $\frac{1}{3}$.

I want to show you today how to do this problem just on paper and we'll use materials only if we need to.

Let's start with $\frac{1}{4}$. Do we know any fraction that's equivalent to? ($\frac{2}{8}$). Yes, this is equal to $\frac{2}{8}$. (lay out pieces to reinforce they are equivalent)

$$\frac{1}{4} = \frac{2}{8} \quad (\text{note the space -- is important})$$

We want to look at the denominator. I have 4 and 8. I could multiply 4 by 2 to get 8 in my denominator. (write it as below)

$$\begin{array}{r} 1 \\ \hline 4 \end{array} \times 2 = \frac{2}{8}$$

If I do this, I only get $\frac{1}{8}$ and we said it was equal to $\frac{2}{8}$. So let's look at the numerator. What do I need to do? What if I multiplied that by 2?

$$\begin{array}{r} 1 \\ \hline 4 \end{array} \times \boxed{\frac{2}{2}} = \frac{2}{8} \quad (\text{draw square around the } \frac{2}{2})$$

Let's look at this number $\frac{2}{2}$. That number is equal 1 isn't it? So when I multiplied $\frac{1}{4}$ by this number, I did not change the value – only how it looks.

Let's try this one:

$$\frac{2}{5} = \frac{4}{10}$$

Let's start with our denominator.

What do I do to the 5 to make it 10? Yes, multiply it by 2. Let's do that.

(lay out pieces to reinforce they are equivalent)

$$\frac{2 \times 2}{5 \times 2} = \frac{4}{10}$$

But that will only give me 2/10. So what do I need to multiply my numerator by to get 4/10? Yes, multiply by 2. And since I multiplied the value 2/5 by 1, I did not change the value – only the way it looks.

(repeat for problems $1/3 = 3/9$, $4/8 = 1/2$ (division by 4))

(Always point out we are multiplying and dividing by 1 in the form of an equivalent fraction – 4/4, 2/2, 3/3 and so on all being 1)

How about $1/2$ plus $2/3$?

$$\frac{1}{2} + \frac{2}{3} =$$

What would be a common denominator? (6). Okay so what do I need to do? I multiply my first fraction by $3/3$ to get an equivalent fraction with a denominator of 6.

$$\frac{1 \times 3}{2 \times 3} + \frac{2}{3} =$$

What do I need to do with second fraction to get a denominator of 6? I multiply by $2/2$, right?

$$\frac{1 \times 3}{2 \times 3} + \frac{2 \times 2}{3 \times 2} =$$

So after multiplying out these, I get a problem with a common denominator that I can now do my addition on.

$$\frac{3}{6} + \frac{4}{6} = \frac{7}{6} = 1 \frac{1}{6}$$

November 17, 2004

Abstraction: Product as Common Denominator - 2A

Material

Pieces of clear plastic (2 rectangles of same size upon which lines can be drawn), an additional piece of plastic for making more of the same, circle fraction insets, paper and pencil.

Prologue

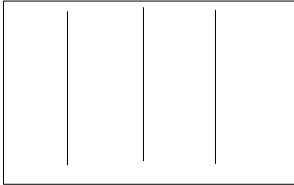
Everything in exercise 1 is based on the children already knowing equivalent fractions. It is limited to what is in our materials. For this lessons, we will look at a method to find a common denominator – beyond using the materials.

Begin

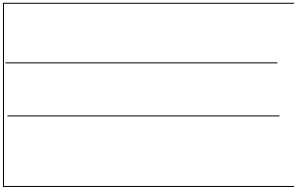
Suppose I want to add $\frac{1}{3} + \frac{1}{4}$.

(children may know a common denominator but have them follow you)

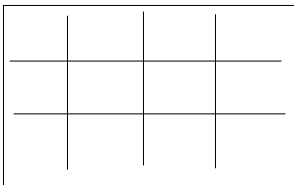
Look I have this piece of plastic and I have put 3 lines on it to make 4 equal spaces.
(lines are vertical) So here I have 4 fourths



I have another piece of plastic that is the same size – but this time I have 3 equal parts.
(lines are horizontal) So here I have 3 thirds.



What if I do this? (**overlay the two pieces together**) What do I have? Yes, **I have 12 twelfs now**. Where were my thirds? (show how first plastic is still underneath)



How many twelfs are in that third? (count on the paper) I have 4/12 that are equivalent to my 1/3 .

How many twelfs s are in one fourth? (count in the paper) I have 3/12, so 3/12 is equivalent to 1/4 .

So do I have a common denominator now? Yes, 12. Now I can add them.

$$\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

===

How about 1/5 plus 2/3.

From experience we can see we can get a common denominator by multiplying our denominators together. So 5 x 3 is 15 – our common denominator then is 15. So our problem becomes different:

$$\frac{1 \times 3}{5 \times 3} + \frac{2 \times 5}{3 \times 5} = \frac{13}{15}$$

So we can see again, we multiply each fraction by 1 (3/3 and 5/5) so the values do not change. But this allows us to add the numerators to solve our problem.

Epilogue

Children quickly realize they can multiply the denominators to get a common denominator, instead of using the graph paper. Don't forget the verbage: **“What do I need to do to my [numerator | denominator] to get my common denominator?”**

This method gets us a common denominator but not necessarily the lowest common denominator; their work with multiples and LCM prepares them for that. Before going on to 2B, make sure they are comfortable finding the LCM.

Abstraction: LCM as Common Denominator - 2B

Begin

Let's do 1/4 + 1/6 + 1/8 . If we multiply these denominators to get one that is common, we'll get 192; that's a big number to work with!

We get 192 since that is 4 x 6 x 8. This means 192 is a multiple of 4 and 6 and 8, right?

If we can find a multiple that is common, can't we find a common multiple that is the lowest among all of them? What would we need to do?

$$\frac{1}{4} + \frac{1}{6} + \frac{1}{8}$$

$$\begin{array}{r|l} 4 & 2 \\ 4 & 2 \\ 2 & 2 \\ 1 & \end{array}$$

(find factors for 6 (2x3) and 8 (2x2x2) also)

What do I need to find the LCM? Yes, **take each factor the most number of times it appears among the 3 numbers.**

LCM is $2 \times 2 \times 2 \times 3$. So our lowest common multiple then is 24.

So, 24 is a lot easier to work with than 192, right? And it is our lowest common denominator. So what must we do for each fraction to make each denominator 24?

(work through each fraction asking, “**What do I need to do to this denominator? What do I need to do to my numerator?**”?)

$$\frac{1 \times 6}{4 \times 6} + \frac{1 \times 4}{6 \times 4} + \frac{1 \times 3}{8 \times 3} =$$

I could have done this problem with 192 as my common denominator. I would have gotten the same value, though it would have looked different until I reduced the fraction I got for my answer.

(repeat problems: $2/3 + 2/6 + 5/9$. LCM = $3 \times 3 \times 2 = 18$.)

(Show the children upon receiving the answer $28/18$, we'll reduce the terms before looking for the LCM and adding. i.e., $2/3 + 2/6 + 5/9 \Rightarrow 6/9 + 3/9 + 5/9 = 14/9$)

Epilogue

Emphasize the opportunity to reduce beforehand to avoid extra work and big numbers. “In order to find our lowest common denominator, **first check the original problem and see if any of the terms can be reduced.**”

Add subtraction problems to this exercise, the second and subsequent lessons.

Summary of Steps for Working on Paper

1. Reduce any term that can be

2. Find the factors to get the LCM – which is also the lowest common denominators
3. Calculate the LCM
4. Use the lowest common denominator to calculate new numerators and denominators for each term
5. Do addition or subtraction calculations
6. Reduce answer to lowest terms in non-mixed number format

November 18, 2004

Multiplication & Division

Whole Number Times a Fraction

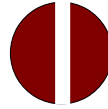
Material

Circle fraction insets, box of extra fraction pieces, paper and pencil.

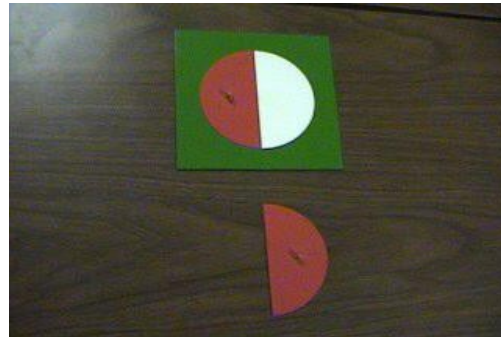
Begin

Suppose I have a unit (take whole and place on table) and I want to take it $\frac{1}{2}$ times.

I can't do that with this whole unit. Let's exchange it for 2 halves. (do so – both halves are together on the table now)



Now I can take it $\frac{1}{2}$ times. I don't need this one – I'll put it away (place the half back into the frame and leave the other on the table)



Now I've taken a unit $\frac{1}{2}$ times.

===

Let's take $3 \frac{1}{4}$ times. (Lay out the whole and 2 extra pieces for 2 more whole units)
So let me exchange. (replace each whole with 4 fourths)

My problem is to take $3 \frac{1}{4}$ times. (take $\frac{1}{4}$ from each whole and put them together on the table separately)

So I have $\frac{3}{4}$ here in my answer. (refer to the pieces removed from the 3 whole units) I don't need these. (put the remaining pieces of the whole units away to isolate the result)

Let's see how we **write this**. $3 \times \frac{1}{4} = \frac{3}{4}$

===

Let's take 4 times $\frac{2}{3}$ s. **(write it on paper)**

$$4 \times \frac{2}{3} =$$

Using my multiplier as a guide, I'm going to change each of my units to the thirds family. (replace them) Now I have 4 instances of $\frac{3}{3}$ s.

My problem says to take 4 $\frac{2}{3}$ times. So I'll take this one, this one, this one and this one. (take a $\frac{2}{3}$ portion of each and place them together)

I don't need these so I'll put them away (put remaining pieces away to isolate answer)
I'll put my thirds together and what do I have? Yes, 2 and $\frac{2}{3}$ s.

$$4 \times \frac{2}{3} = 2 \frac{2}{3}.$$

(note line between numerator and denominator is
horizontal when written)

===

How about 5 taken $\frac{1}{4}$ times?

(repeat as above: Replace 4 whole units with units of the 4th family – all laid out on table. Take $\frac{1}{4}$ from each unit and separate out together. Put back the ones not needed. Restate problem with resulting pieces in the answer)

Fraction Times a Fraction

Let's do this one: $\frac{1}{3} \times \frac{1}{3} =$

What do we do? My **multiplier tells me I need 3 pieces**. Let's see, I'll **trade my $\frac{1}{3}$ in for $\frac{3}{9}$** .



Now, I can take $\frac{1}{3}$ of the inset). So, $\frac{1}{3}$ times $\frac{1}{3}$ equals

that. I don't need these (put the other $\frac{2}{3}$ back into



What does the denominator of the multiplier tell us? Yes, how many pieces we need in each whole. What about the numerator of the multiplier? What does it tell us? Yes, how many pieces to take.

Do you notice **our answer is smaller than the multiplicand? Does that make sense?**
 Yes, we are taking a part of it – not even 1 times it – less than 1 times it, right?

Do problems $1/3 \times 1/3$, $2/3 \times 2/3$ and $2/3 \times 1/3$ – and compare the answers for logical sense. Which should be smaller, which should be greater – and why. (leave the pieces out so the children can see the concrete fact of what is larger and smaller)

Epilogue

Make sure that this appears as taking a number of times - versus taking away – since we don't want to confuse this with subtraction.

Involve the children - each gets a turn. Ask them if answers are logical.

Sharing Out Multiplicand Symbolically

After lots of practice, give them a problem such as $2/3 \times 3/4$. This can't be done with the materials.

What does my denominator tell me? I have to divide my $2/3$ into 4 equal pieces. We have been seeing that the fraction line means to divide, right?

So that means I can simply multiply the denominator by 4.

$$\begin{array}{r} 2 \\ \hline 3 \end{array} \times \begin{array}{r} 1 \\ \hline 4 \end{array} = \begin{array}{r} 2 \\ \hline 3 \times 4 \end{array}$$

We know that what I do with my denominator, I need to do the same thing with my numerator in order to keep the same value.

$$\begin{array}{r} 2 \\ \hline 3 \end{array} \times \begin{array}{r} 3 \\ \hline 4 \end{array} = \begin{array}{r} 2 \times 3 \\ \hline 3 \times 4 \end{array} = \begin{array}{r} 6 \\ \hline 12 \end{array}$$

This gives me $6/12$.

(note this is what we've been doing concretely when we exchange)

I can rewrite this $6/12$ this way:

$$\boxed{\begin{array}{r} 2 \quad 2 \quad 2 \\ \hline 12 \quad 12 \quad 12 \end{array}} + \frac{2}{12}$$

My numerator says I need to take 3 of these. So, how many are there in 3 of them?

Yes, 6.

So $\frac{2}{3} \times \frac{3}{4}$ is $\frac{6}{12}$ and when we reduce this fraction our answer is $\frac{1}{2}$.

Applied General Procedure ($\frac{1}{7} \times \frac{4}{9}$)

1. My multiplier's denominator tells me I need to break my multiplicand into 9 pieces.
2. Multiply the denominator of multiplicand times 9.
3. What I do to my denominator, I have to do to my numerator
4. Add "x 9" to multiplicand numerator as well. Multiply to get result for numerator and denominator

$$\frac{1}{7} = \frac{1 \times 9}{7 \times 9} = \frac{9}{63}$$

5. Now separate into 9 $\frac{1}{63}$ rd pieces and take 4 of them. We are left with $\frac{4}{63}$ rd s.

Summary of Steps for Working On Paper

The outcome of this lessons is the abstract comprehension of the rule for multiplying fractions. If the children don't state this in so many words by the end of a lesson, do more repetitions of this lessons. At some point, the children will offer up this rule to us as a discovery. If not, at some point, we'll want to move on and we can state the rule. **The rule in multiplying a fraction by a fraction is: multiply the numerators, multiply the denominators and reduce to find the final answer.**

November 18, 2004

Whole Number Divided by a Fraction

Material

Four large skittles (one whole, halves, thirds, and fourths), circle fraction insets, box of extra fraction pieces, paper and pencil.

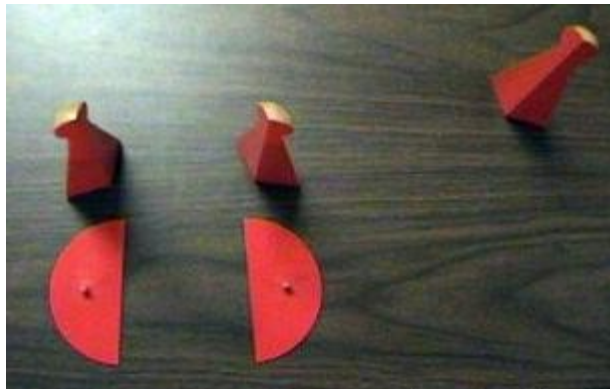
Prologue

The skittle represents the multiplier; the multiplicand is the inset. We need to make an entire unit skittle to get our final answer. General procedure is:

1. Break skittle into number of pieces indicated in the divisor's denominator
2. Use a number of skittle pieces for sharing out, equal to divisor's numerator.
3. Break inset dividend equally, into the number of pieces in the divisor's numerator.
4. Share out each equal part of the inset dividend to each part of the skittle.
5. Add portions of skittle needed to make 1 whole (include new inset portion with each)
6. Add up the inset pieces for all skittle pieces that make a whole skittle as the answer

Begin

I would like to divide 1 by $\frac{2}{3}$ s. (**write expression on paper** $1 \div \frac{2}{3}$)



My numerator tells me how many pieces I want and my denominator tells me I want 3 equal pieces. (choose a skittle divisible by the denominator)

I'm going to set the 2 thirds (skittle) side by side. I need to share this unit out to 2 pieces. How can I do that? Yes, I can exchange it for 2 halves and then I can share it. (place $\frac{1}{2}$ unit under each part of the skittle in use (2 of the 3 – the 3rd third is set aside)

When doing division, how do I find my answer? I look under one unit. So in order to make this a unit, I need to bring this over (bring the third from the skittle back into the

picture)

Let's imagine this skittle also gets the same as my others. (add a 3rd $\frac{1}{2}$ piece inset to associate with the 3rd third that was set aside – since we need an entire skittle for the answer)



Epilogue

Be sure to reform a whole skittle at the end to get your answer. Explain away how we added new fractional pieces to associate with the skittle parts we had set aside - in any way the children will accept. Fairness is a good ploy to justify adding in inset pieces for skittle parts.

Fraction Divided by a Fraction

Begin

Let's do $\frac{1}{4} \div \frac{2}{3}$

(choose the thirds skittle and take 2 pieces of it, setting the other one aside)

(set out the $\frac{1}{4}$ inset underneath to the left)

I need to share this $\frac{1}{4}$ out to 2 pieces. What can I do? Yes, exchange for $\frac{2}{8}$.

Now I can share it out. (do so – $\frac{1}{8}$ for each part of the skittle)

But our answer is in one unit - so I'll bring back my third piece of the skittle and also add my 1/8 inset piece to be fair.

Now I put them together and my answer is altogether 3/8. Let's write this:

$$\frac{1}{4} \div \frac{2}{3} = \frac{3}{8}$$

===

repeat for practice the problem: $\frac{1}{3} \div \frac{3}{4}$

Our answer is bigger than our dividend. Does that make sense? Is that logical? You divide the dividend into an equal number of parts and that number is given by the numerator of the divisor. And then you take the number of times given by the denominator of the divisor to reform a whole skittle; this makes it bigger than the original dividend.

Abstraction: Concrete Process Multiplies by Reciprocal

Take original problem:

$$1 \div \frac{2}{3}$$

The first thing I did was to take the 1 and divide it by 2.

$$1 \times \frac{\text{-----}}{2}$$

This is saying I'm going to divide this 1 by 2. But then I needed to take it 3 times.

$$1 \times \frac{3}{2}$$

What do you notice about the 3/2 and my divisor of 2/3. They are flipped.

If we multiply this we get 3/2

$$1 \times \frac{3}{2} = \frac{3}{2}$$

So we can write our answer:

$$1 \div 2/3 = 3/2$$

===

New problem: $\frac{1}{4} \div \frac{2}{3}$

$$\begin{array}{r} 1 \\ \text{----} \times \text{----} \\ 4 \quad 2 \end{array} \quad \text{(divide my dividend by 2 pieces)}$$

$$\begin{array}{r} 1 \quad 3 \\ \text{----} \times \text{----} \\ 4 \quad 2 \end{array} \quad \text{(take the product above, multiply by the number of skittle parts)}$$

Epilogue

To divide by a fraction, invert the divisor and multiply. This can be stated at some point but let them discover this on their own if possible.

November 18, 2004

Fraction Charts

We've talked about different types of charts. These are factual charts. They are designed to use after a lessons – brought out to show a visual impression of what you have done with the materials or as an idea of what the children can do. They might be used to summarize a lessons or used to help children recall a lesson.

You can bring out several and leave them on display for a few days to spur the children on to do their own handwork (charts, booklets, etc.). You don't need to put out or use all of the charts. They should be stored where they are available to the children for their exploration even if you do not present them.

The chart pieces are the same size as the inset pieces. **Masters will be given that should go into your final album.** We'll get copies of these charts that fit into the album to get an idea of the color scheme. We'll find some assistant or parent and let them use the inset pieces, give them fade-less art paper and they can make these charts. Though the insets are all red, we choose a color for each fraction type and then final albums – fraction charts use these colors consistently. If $\frac{1}{3}$ is blue all $\frac{1}{3}$ pieces must be blue.

December 6, 2004

7. DECIMAL FRACTIONS

Prerequisites

Children need to have knowledge of multiplication facts, especially multiplying by 10. Plenty of practice in activities like the bank game. They need a concept of parts of numbers, such as common fractions. They need awareness of decimal system categories – gained through wooden hierarchical material, racks and tubes and so forth.

Quantity

Material

Two of the circle insets: the whole circle and the tenth round frame of fractions. Beads in the hierarchical colors (in a silver box here) with green, blue and red beads inside. Box of decimal fraction material – color coded – used to represent the quantities. (One entire shelf has decimal fraction materials in the math area here).

Large felt mat (yellow is good for color contrast)

Prologue

This is a small group presentation. We may have to do the same presentation several times in the classroom and this slows the presentation down – as compared to the pace we see in learning the presentation. There is repetition required in the classroom but also it is built into the lesson.

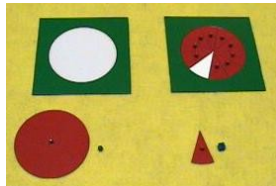
Begin

What is this (hold up the whole circle inset)? It's a unit.

What is this (hold up a green bead)? It's also a unit.

What is this (hold up a $1/10$ piece from inset)? It's a tenth.

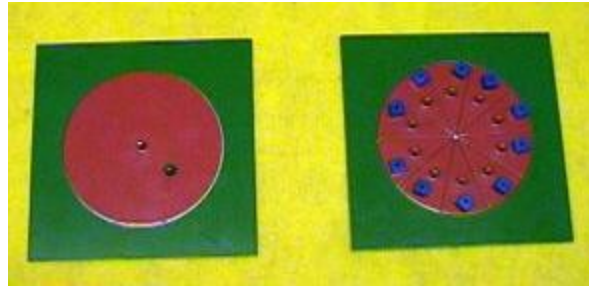
This is also a tenth (hold up blue piece of fraction material).



So, here we have a unit and this is also a unit (refer to the inset whole and the green bead)

This is a tenth and this is also a tenth (hold up blue bead and the inset piece again)

Would one of you like to put one of these tenths on one of these tenths?
(place bead on whole and tenth blue cubes on each piece of the inset)



Let's put these away. (note pattern, 3 instances in italics to follow)

If Divided by 10

Here we have a unit (hold up green bead). If we divide the unit into 10 equal pieces, we get tenths.

Here is one tenth (hold up a blue cube).

If we were to divide one tenth into ten equal pieces, our unit would be divided into 100 equal pieces. And each one would be a hundredth. (place red cube)

Recapitulate

Ten hundredths make one tenth. Ten tenths make one unit.

*If I were to divide each hundredth into ten equal pieces, then the whole unit would be divided into 1000 equal pieces. (hold up a green cube)
Each one would be a thousandth.*

Ten thousandths make one hundredth, ten hundredths make one tenth and ten tenths make one unit.

(continue pattern)

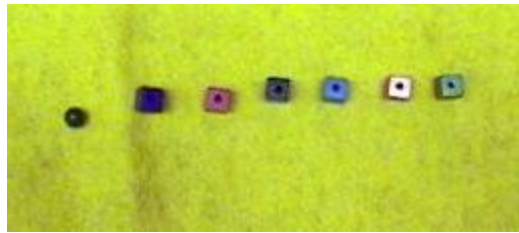
If we divide each thousandth into ten equal pieces, the entire unit would be divided into ten-thousand equal pieces – each one would be a ten-thousandth. (place light blue cube)

Ten ten-thousandths make one-thousandth. Ten one-thousandths make one hundredth. Ten hundredths make a tenth. Ten tenths make a unit.

But we're not done. If we divide the ten-thousandths into ten equal pieces, the entire unit will be divided into 100,000 equal pieces. One of them is a hundred-thousandth.

Ten hundred-thousandths make one ten-thousandth. Ten ten-thousandths make one-thousandth. Ten one-thousandths make one hundredth. Ten hundredths make a tenth. Ten tenths make a unit.

If we divided each of these hundred-thousandths into ten equal pieces, the entire unit will be divided into 1 million equal pieces. One of them (hold up lightest green cube) would be a millionth – and here it is (place it next to the other cubes on table at right).



Ten millionths equals 1 hundred-thousandths. Ten hundred-thousandths make one ten-thousandth. Ten ten-thousandths make one-thousandth. Ten one-thousandths make one hundredth. Ten hundredths make a tenth. Ten tenths make a unit.

Epilogue

The colors of the cubes are still green, blue and red as we get into the thousandths and millionths levels – but lighter shades of colors for 1 tenth, 1 hundredth and 1 thousandths.

We don't ask in the lessons – we tell them since this is all new for them. A logical follow up is to repeat the lessons. Some children may want to label these new fraction categories while others may want to go back to work on other things. We have to return to this material a number of times on other days.



Multiply and Divide Unit by Powers of Ten

Here we have a unit. If I multiply this unit by ten, what do I get? Ten, yes. Here it is (place blue bead to left of green bead)



If I divide that unit by ten, what do I get? Yes, $\frac{1}{10}$. And here it is. (place blue decimal material cube tenth to right of green bead)



If I multiply the unit by 100 what do I get? We get 100. And here it is (place red bead at left of blue ten bead)



If I divide the unit by 100 what do I get? Yes, 1 hundredth. And here it is (place red decimal material cube to right of blue tenth cube)



If I multiply the unit by 1,000 what do I get? We get 1,000. And here it is (place green bead at far left)



If I divide the unit by 1,000 what do I get? Yes, 1 thousandth. And here it is (place green cube at far right)



If I multiply the unit by 10,000 what do I get? We get 10,000. And here it is (place blue bead far left)



If I divide the unit by 10,000 what do I get? Yes, 1 ten-thousandth. And here it is (place light blue cube far right)



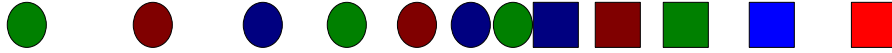
If I multiply the unit by 100,000 what do I get? We get 100,000. And here it is (place red bead far left)



If I divide the unit by 100,000 what do I get? Yes, 1 hundred-thousandth. And here it is (place light red cube at far right)



If I multiply the unit by 1,000,000 what do I get? We get 1,000,000. And here it is (place green bead far left)



If I divide the unit by 1,000,000 what do I get? Yes, 1 millionth. And here it is (place light green cube to far right)



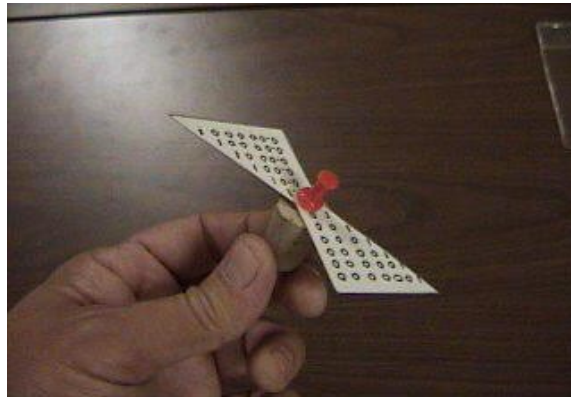
If we multiply by the powers of ten, we get these categories that go to the left of the unit – tens, hundreds, thousands, hundred-thousands and millions.

If we divide by the powers of ten, we get these categories that go to the right of the unit: tenth, hundredth, thousandths, ten-thousandths, hundred-thousandths and millionths.

This is our decimal system. If you multiply the unit by powers of ten, numbers get larger and larger. If you divide by the powers of ten, the numbers get smaller and smaller.

Epilogue

There is something which can be used here called a “help to understanding.” It is a plastic strip with the categories written on it horizontally; it can be placed under the beads and cubes on the table layout - to help identify them.



*Twirl the propeller around on the thumb-tack axis saying,
“The decimal system revolves around the unit.”*

December 6, 2004

Symbol Linked to Quantity

Prologue

Remember when we talked about the pattern of introducing quantity, then symbol, and then symbol linked to quantity. Here we skip presenting symbols by themselves.

As they are introduced, label all decimal fractions using the whole number cards, flipped over and adding a decimal point to each. Then announce you were teasing (see Ticket and Tease) and label all decimal pieces using the decimal fraction cards from the box of decimal fraction materials – reallocating the whole number cards to their original locations also.

Material

Hierarchically colored beads, decimal fraction material, large felt mat, product cards from the bank game (taking the one-card of each category – 10, 100, 1000, etc.). We need a black decimal point (taken from a box of them – made by a hole punch)

Begin

Describe System, Middle Alternating Outward

Here we have a unit (place green bead on felt mat).

If we multiply the unit by ten, what do we get? 10 – and here it is.

(place blue bead representing 10 to left side of unit - leaving room for tickets later)



If we divide the unit by 10 what do we get? 1 tenth.

(place dark blue cube representing tenth to right of unit)



If we multiply the unit by 100 what do we get? 100. (place red bead left of ten)



If we divide the unit by 100 what do we get? $1/100^{\text{th}}$ (place red cube right of tenth)



If we multiply the unit by 1000 what do we get? 1,000. (place green bead)

If we divide the unit by 1000 what do we get? $1/1,000^{\text{th}}$ (place green cube) Here it is.

If we multiply the unit by 10000 what do we get? 10,000. (place blue bead)

If we divide the unit by 10000 what do we get? $1/10,000^{\text{th}}$ (again, light blue cube) Here it is.

If we multiply the unit by 100000 what do we get? 100,000. (place red bead)

If we divide the unit by 100000 what do we get? $1/100,000^{\text{th}}$ (place light red cube) Here it is.

If we multiply the unit by 1000000 what do we get? 1,000,000. (place green bead)

If we divide the unit by 1000000 what do we get? $1/1,000,000^{\text{th}}$ (place light green cube) Here it is.



Ticket and Tease

I've got some cards here so we can label the quantities of these beads. We'll put the labels above the beads. (place labels, spreading beads out, under each category to left of unit)

1,000,000 100,000 10,000 1,000 100 10 1



I have tickets for categories greater than 1, but I need tickets for our fraction categories!

(turn over the 10 and place over $1/10$ cube – placing decimal point after the first zero - “0.1” - then working way to “0.000001” using the same technique).

1,000,000 100,000 10,000 1,000 100 1 0.1



(continue using whole number cards and decimal points to label remaining fractions)

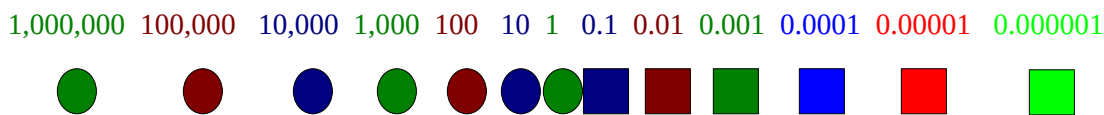


(after all decimal fractions are labeled...)

I was only teasing that I didn't have enough cards to label all our categories!

Here I have categories for decimal fractions also – so we'll be able to lay out all of the categories.

(take decimal fraction cards from box of fraction material, laying each over the cubes, for successive categories and put whole number cards back into their correct locations).



Epilogue

For repetition, we turn over all cards, **mix them and re-ticket** the pieces. We want the children to manipulate and practice ticketing. We **need repetition**.



Help To Understanding

We might notice a child having trouble labeling the categories and recognizing the tickets. If we notice this, we can do this:

(using previous layout from Symbols linked to quantity)

Begin

Let's look at what we have here. Here in the middle we have 1 – that's just 1.

Now let's look at this ticket – 10. We have zero in the units category and 1 in the tens category. That's how we know this is ten.

Now let's look here at $1/10$. We have zero in the units category and 1 in the tenths category. That's how we know this is $1/10$.

Now let's look at this ticket – 100. We have zero in the units and tens categories and 1 in the hundreds category. That's how we know this is one hundred.

Now let's look here at $1/100^{\text{th}}$. We have zero in the units and tens categories and 1 in the hundredths category. That's how we know this is $1/100^{\text{th}}$.

(And so on ... continue through one million)

Epilogue

We could also show a second “help to understanding” called a propeller. Numbers go from 1 million down to 1 millionth on the small propeller. In the middle is 1. We use a push-pin to balance the propeller.

Begin

“In the decimal system all the numbers are balanced around 1” (twirl propeller). That's the decimal system for you. Categories keep going in both directions – either getting bigger or smaller from the 1 in the middle. (on shelf, the pin from the propeller is pushed into a cork - the propeller is in a plastic bag – carry all on a tray.)

December 6, 2004

Formation and Reading of Quantities

Material

Decimal board, (yellow columns), hierarchically colored beads, box of decimal fraction material, a miniature crown and a paper candelabrum.



Decimal Board

Prologue

This is our first use of the Decimal Board. We gather 2-3 children.

Begin

Tour Board Top to Bottom

This is called the decimal board. If we look across the top, we can see numbers – the same numbers we say when working with our decimal fraction material.

Would someone like to read these off? (1 million, 1 millionth – left to right)

Under each of the numbers, we have colored circles on our board. Do you notice how the colors are shaded?

(color shades are darker at left end (i.e., $10^{(x)}$); lighter at right end (i.e., $10^{(-x)}$))

What about the unit – do you notice something? Yes, there is a black circle around both the number and the green circle corresponding to one. That's because the unit is king.

(place green bead on units and put crown on the unit insignia in the center of the board)

Introduce Kingdom, Tens-Tenths, Hundreds-Hundredths

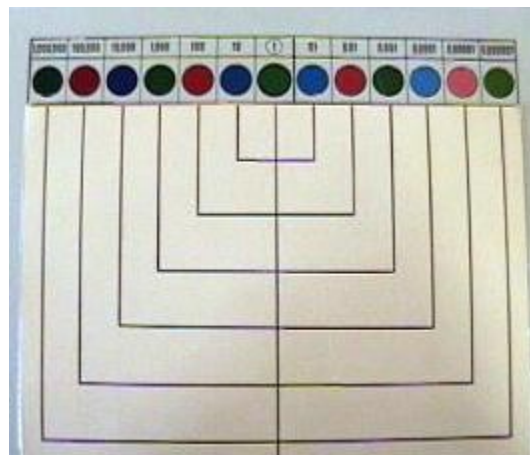
As you know, in this kingdom the unit has tens and tenths, hundreds and hundredths (place beads and cubes as you speak)

(thousands and thousands ... continue to million and millionths)

Candelabrum

If in this kingdom, these circles were flames burning at the top of this candelabra, **the higher the category, the bigger the number, the darker the flame is** . The lower the category, the smaller the number, the lighter the flame is.

(place candelabrum on the decimal board lining up the flames)



The millions are the darkest and the millionths are the lightest in color.

Epilogue

The candelabra is another help to understanding. It provides a visual clue for visual learners about the relative values of the categories on the board. It provides one more opportunity to say the names of the categories.



Using the Board

Prologue

Begin with the previous setup on the table. Set up cards from highest (unit) to lowest (millionths) categories top to bottom, lined up with the decimal board on its left side.

Begin

We're going to talk about units and categories that are less than a unit. Remember that with whole numbers, in a category, tens, for example – 6 tens would be 60 – if you have 3 tens, you have 30, right?.

We aren't limited to just one. It's the same with these decimal categories.

I'm going to put some tenths on our decimal board. I put 4 tenths on our board.

I can also place the card for that 4 tenths next to our board. (place on right side of board 0.4 – left side still has 0.1. Line them up horizontally. left side has all decimal categories top to bottom, largest to smallest)

(repeat and let the child practice – giving the 9 or less of whatever category they choose)
(one child can place the quantity and the other can find the corresponding cards)

Epilogue

If children have trouble here, we know we need more repetition on the previous 2-3 lessons. Let them work on this lessons for a time – choosing the categories, getting the matching cards, etc.



Extend

Now it's my turn. I'm going to put a decimal quantity on the decimal board. I'll put some

hundredths. First, how many hundredths did I put down? (placed on the board under hundredths column)

I have 16 hundredths. **Do we have a card that says that? No. We need to do something new. (count out 10 hundredths). We have to count ten hundredths and exchange.**

What are ten hundredths equal to? Ten hundredths would make 1 tenth. So I'm going to take all ten of these off the board and replace them with a single 1 tenth cube.

Now how many hundredths are left? (6) Can you find a card for that? (places card). But that doesn't express our whole number. What else do we need? 1 tenth. Yes.

We'll align our units to combine our cards. (place 0.1 over 0.06 left justified) Let's lay this on our right of the board. (place the shorter card on top)

Our quantity is 16 hundredths. (when we assign number-names to a quantity, always begin with the smallest decimal fraction. We use the name of the last category on the right – 16 hundredths in this example)

Epilogue

Put away the decimal fraction cards in sequence.

With the children, build up to more and more categories. After they understand the concept and usage, show them how we say decimal numbers “point 41771” as opposed to 41 thousand 7 hundred 71 hundred thousandths. Tell them this is how we speak of these numbers normally.

Also, in addition to building up to big numbers, use a quantity like 2.43 and tell them it is called “two and 43 hundredths”. This is the only place in verbalizing numbers where we should say “and” in our numbers. Give them whole numbers to emphasize when and how “and” is used in verbalizing numbers.

Combine cards from the bank game if you want to use whole numbers greater than 1. Decimal cards from bank game and from the decimal fraction box.

Children can dictate to each other. Challenge a friend to build a number from a quantity or vice versa.



Game

There is also a game to play for children having trouble with the size of the categories. Give cubes to several children. “Jung-En, what did I give you? ($\frac{1}{10}$) Judy, what did I give you? ($\frac{1}{100^{\text{th}}}$). Who has more”? Point out that it is not always the person with the most cubes since we have to consider what categories are being used.

Note: There are three places for decimal cards. Decimal cards of 1-3000, bank game decimal cards, and decimal fraction cards

December 7, 2004

Operations

Material

Decimal board, box of decimal fraction material, hierarchically colored beads.

Prologue

Depending on the children's experience, we may start with simpler problems than what is here. Don't do a problem involving carrying for example.

Addition

Begin

First let's form the two addends with the cards and place them to the right of the boards, one under the other. Can you read what they say?

(child reads the numbers)

We'll take $54/100^{\text{th}}$ and add 95 hundredths.

Can you form the first addend? Can you form $54/100^{\text{th}}$ on our board? Who else can form our $95/100^{\text{th}}$ on our boards? (space the addends cubes quantities apart horizontally)



(the two addends appear in correct categories but on two horizontal levels corresponding to the addends at the right side of the board)

It's like any addition problem so we want to put our numbers together. We do that like this.

(push all the cubes together into one level on the board, still under the right categories)

(begin adding with the smallest categories as we always do with addition)

Now let's count tens. 1, 2, 3 ... what happens at ten? Oh, we have to exchange. What do we exchange these for? Yes, 1 unit.



$54/100^{\text{th}}$ plus $95/100$ gives us 1 and $49/100$.

Let's do more problems!

Epilogue

Initially we choose addends with the same categories.

If you run out of cards at left to form your answers as you combine, you can create a ticket, use a different card from the left but place it under the addends (in addition) so the number you need is lined up with the addend categories (even though the number card itself is the wrong category); the third option is to use a number from one of the addends and then remember that that number was used in both places (answer and addend) – the least elegant solution. Often, we'll borrow cards from the left side – but just make sure the colors match what you need; if you need 10 thousandths – use tenths, for example – but line up the categories so that the tenth will be interpreted correctly as a ten thousandth, as we add.



Moving Addition to Paper

Material

Decimal board, box of decimal fraction material, hierarchically colored beads, paper and pencil.

Begin

We'll setup our board and addends as usual. Today, we're also going to use paper.

Suppose we have this problem:

" $2.21 + 3.32 =$ " (write on paper horizontally)

(setup board for this problem as a support, build addends, combine and count)

We have 3 hundredths, 5 tenths and 5 units – so we'll place our decimal point after the units on its right.

Let's write out problem on paper.

When we form this problem on the decimal board, we put all units in the units column. We put all of the tenths in the tenths column. We put all of the hundredths in the hundredths column. Now we'll do the same on paper.

Units have to be written right under units. Tenths have to be under tenths. All categories are the same – they have to align. We put down our equal sign and our addition sign.

```
2.21
+3.32
- - - -
5.53
```

And when we write our answer, our 3 hundredths needs to be written under the other hundredths categories in the two addends.

How do I know where to put my decimal point? Right under the decimal points! That's how we know it's all lined up correctly. Now as you work, you'll want to write down your

problems like this.

Epilogue

This is the key to knowing how to add decimal fractions. If they've abstracted regular addition, keeping categories aligned and decimal points aligned is not much of a problem after a try or two.



Subtraction

Begin

Let's do this problem:

578 thousandths take away 187 thousandths. We'll do subtraction today.

Can you form the minuend on the board? .578

We want to subtract 187 thousandths. .187

Where will we take this from. We take it from the minuend on the board.

(place the cubes, etc. that we take away at bottom of the board)

Starting with 7 thousandths: 1, 2, 3, 4, 5, 6, 7.

Now subtracting our hundredths: 1, 2, 3, 4, 5, 6, 7 - I don't have enough

I need to take one tenth and exchange it for 10 hundredths. Now I need one more hundredth to take away our 8 hundredths.

And I'll take 1 tenths away for the tenths category. (collect subtrahend at bottom board)

Now what do I do to find my answer. Yes, count what is left on my board.

We have 1 thousands, 9 hundreds and 1, 2, 3 ... tenths. (forms cards on right)

.578 plus .187 give us .391

Now we can check our answer by adding it to our subtrahend that we've gathered at the bottom of our board.

Epilogue

Working on paper here is the same as for addition.



Multiplication

Prologue

Here, we'll do the first stage – multiplying a decimal fraction by a unit multiplier. We're doing stage I here.

Material

Decimal board, box of decimal fraction material, hierarchically colored beads, paper and pencil. Add in **gray number cards, used for the multiplier**.

Begin

Today, we'll take the number 63 hundredths. (form multiplicand at right)

And we'll multiply this by 3.

(place gray 3-card to right of multiplicand – 2 inches to right at same horizontal level)

Secondly, we want to decompose our card – making sure to keep lowest categories at the top. It's the longest card – but still has to go to the top of the cards.

0.03

0.6

Now we have to multiply each category by our multiplier.

3 hundredths times 3 – what does that give us? 9 hundredths.

(place 9 hundreds on board)

(slide multiplier digit (gray card) down to the next category)

6 tenths taken 3 times give us 18 tenths. How will we form that on our board? 1 unit and 8 tenths, yes. (if child chooses 18 tenths, don't correct him)

(form 1.8 on the board)

Now all we have to do is count and find our product cards.

1, 2, 3, 4, 5, 6, 7, 8 tenths and - - 1 unit.

(Reading through the whole problem at end) ... 63 hundredths times 3 gives us 1 and 89 hundredths



Division

Prologue

This is stage I: **Division of a whole number or a decimal fraction by a unit divisor**. We could work out any number of problems such as $3 \div 2$

If we want to record the problem, we record on paper.

Materials

Decimal board, box of decimal fraction material, hierarchically colored beads, paper and pencil. Add in **gray number cards, used for the divisor**. Green skittles and a felt mat. *Start out with only skittles, felt mat and fraction materials though.*

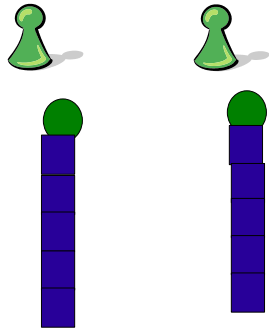
Begin

Let's do this problem: $3 \div 2$

My dividend is 3 and my divisor is 2

(begin to share out each unit under each skittle – the first two units go immediately)

For my third unit, I have to exchange for ten tens. And now I can keep sharing out.



Where do we find our answer? Under one unit skittle. How many did each one receive?

1 and 5 tenths.

$$3 \div 2 = 1.5$$

Who would like to come up and do another one?

Epilogue

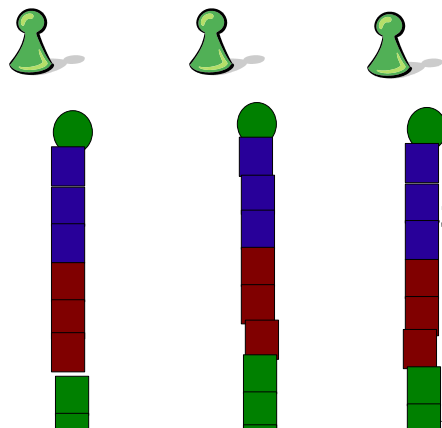
If we have a problem in which we run out of materials, such as a quotient with a repeating decimal, we just stop and write the answer. See extension below.



Extend

Let's try the problem: $4 \div 3 =$

(distribute 4 among 3 skittles)



Division

3

7. DECIMAL FRACTIONS

3

What do you notice about that answer? It's the same number again and again.

Rather than writing 1.33333 ... and on and on, we write 1.3 with a line over the repeating part of the number – or:

$\overline{1.3}$

If there are more than one number in the repeating pattern, the line goes over them all. For example, 1.331331331 and so on becomes:

$\overline{1.331}$

December 7, 2004

Conversion of Common Fractions to Decimal Fractions

Material

Circle insets, the “instrument for conversion of common fractions to decimal fractions”, green skittles, felt mat, bead and cube fraction materials, paper and pencil.

Prologue

This lesson consolidates the relationship of common fractions to decimal fractions.

Begin

Here, we have a black frame (hold up instrument) that has some numbers on it. Let's look at it carefully. (this looks like the tool for measuring angles but is not)



It starts at the top with zero and as we go around the numbers are marked in groups of tens. 0 – 90, 90 is that last printed number. In between all the numbers there are 9 white lines. So our **circle has been divided into 100 equal pieces. Each one is called a hundredth.**

If I take our unit (hold up whole inset) and place our unit circle into this frame (it fits), we could say that the unit is divided into 100 equal parts called hundredths.

I can write that down. $1 = 100/100$ (“one equals 1 hundred hundredths”)

Let's take 1 half. Just as with our other instrument, there's a white line coming down vertically under the zero. So, our half circle will be lined up with the straight edge on our white line within the frame. We look at the bottom of our instrument for our answer.

We see it is 50. So we say that 1 half is 50 hundredths since it has the same value. We can write this $\frac{1}{2} = 50/100$.

We've done a lot of work with decimal fractions. There's something special about decimal fractions. "Deci" means ten. When we have fractions that have 10 or 100 or 1 thousand in the denominator, we have decimal fractions. I can write this number as a decimal fraction.

If we say 50 hundredths – well, let's think of our decimal fractions. We could write 50 hundredths like this:

(write 0.50)

So we know three ways to write the same amount (write on paper)

$$\frac{1}{2} = 50/100 = 0.50$$



Begin

Let's do another problem. We'll put $\frac{1}{2}$ away and work with $\frac{1}{4}$ now. Let's fit it into our frame. How many 1 hundredths does $\frac{1}{4}$ equal? We can see on our instrument it is 25 hundredths.



Do we know how to write 25 hundredths as a decimal fraction? Yeah – 0.25

So again we can write the same amount in three ways: $\frac{1}{4} = 25/100 = 0.25$

Would you like to try one now?

Epilogue

This frame only works with hundreds. This lesson is preparation for work with percentages. Do example 1/7 to illustrate that inexactness in measuring of okay here.



Material

Felt mat, bead and cube fraction material, green skittles.

Begin

A common fraction is simply a decimal fraction with a denominator of a power of ten.

Suppose we wanted to take an common fraction and convert it to a decimal fraction.

How could we do that?

Let's start with a fraction of 1 half. But I want to express 1 half as a decimal fraction to the tenths. My frame only works for hundredths. What can I do.

A common fraction like $\frac{1}{2}$ means 1 divided by 2. This little line (between numerator and denominator) is really a division sign. So this really means $1 \div 2$.

So using our beads and cubes and skittles, we can work a problem like this out.

I'll put out 2 skittles. What do I need to do with my unit. I need to exchange for 10 tenths, right?

Now I can share then out. And what do I get? Yes, 5 tenths.

$\frac{1}{2}$ expressed as a decimal fraction to the tenths equals 5 tenths.

(write $\frac{1}{2} = 0.5$ - and include the radix form for the problem so there are 3 forms total)

Who'd like to do the problem $\frac{1}{8}$? (put out 8 skittles, exchange and share out)

$\frac{1}{8}$ expressed as a decimal fraction to the thousandths is 0.125

We'll do one more; let's take $\frac{1}{7}$ and I want to express it as a decimal fraction to the hundredths.

(we have to share out, exchanging and so on up to thousands. When we arrive at thousands (ignoring remainder), we round up or down depending on the value of the thousands – greater than 5 we round up (0.15) but if it is less than 5 we round down (0.14). if the digit is a 5 many people round up.)

To get decimal fractions to the hundredths, we have to share out to the thousandths (the category after the precision we want) if sharing does not come out evenly by the time we reach hundreds sharing. Then we round. But there is something to know about rounding.

Some people, when the rounding digit (thousandths) is 5, they round up when hundredths place is an odd number and leave the hundredths digit the same for even numbers.

So 0.145 would be 0.14 but 0.135 would also be 0.14 – after rounding.

(If we have a common fraction $\frac{60}{100}$, we can make decimal fraction as well. It first becomes $\frac{6}{10}$ s and then it is reduced to $\frac{3}{5}$, which provides a quotient of 0.6. Likewise, we can convert decimal fractions to common fractions; 0.36 becomes $\frac{36}{100}$, reduces to $\frac{9}{25}$. How quick this is introduced depends on the child. We can begin to introduce this as the children are ready.)

December 8, 2004

Effects of Multiplying or Dividing Numbers by Powers of Ten

Material

Decimal board, fraction material and hierarchically colored beads. Paper and pencil. Skittles for division.

Prologue

We don't have to set up the number cards – just the fraction material and the decimal board.

Multiplication

Begin

Suppose we have this problem : 23×10 (write it on paper)
3 times 10 give us 30 (place beads on the board under correct category)
20 times ten give us 200 (place beads on the board under correct category)
Let's write down the answer (“write = 230”)

How about this problem: 2.3×10 (invite a child to set out answer on the decimal board)
What is $3/10$ taken 10 times? It's 3. (place beads on the board under correct category)
If we take 2 ten times we get – 20. (child sets beads on the board)
So what's our answer? 23 (write it on the paper “=23”)

Suppose we have this problem: 0.23×10 (write it on paper)
First we need to take 3 hundredths 10 times. What do we get? (0.3)
(child sets answer on the decimal board)
We need to take 2 tenths 10 times. What do we get? (2)
(child sets answer on the decimal board)
What's our answer on the decimal board? Two and three tenths (02.3)

Let's do one more. Suppose we have this: " $0.023 \times 10 =$ " (write it)

3 thousandths times ten gives us $3/100$. (set cubes on board)

And 2 hundreds times ten is 2 tenths. (place cubes on board)

So our answer is 23 hundredths **(write on paper "0.23")**

Paper should say:

$$23 \times 10 = 230$$

$$2.3 \times 10 = 23$$

$$0.23 \times 10 = 2.3$$

$$0.023 \times 10 = 0.23$$

Remember, we said anything times ten means we add a zero. Is that what we did here each time? No – not at all.

We did this the first time. But here, it has more to do with the decimal point than with the zero. Our decimal point moved one place each time.

Did it happen in our first problem? Yes, even though we didn't see it. Each time, the decimal point moved one place to the right as we multiplied each number by 10.

Epilogue

The work in primary of multiplying by ten comes of help here. What rule could we use to work for both whole numbers and decimals?

"When multiplying a number by 10, whether whole or decimal, move decimal point one place to the right, adding a zero if necessary."

Once the children have been able to express this rule and the idea of this lesson, we might throw out "what about multiplying by 100? Is there a rule for that?"

"When multiplying a number by 100, whether whole or decimal, move decimal point two places to the right, adding a zeroes if necessary."

Lastly, when multiplying a number by any power of ten, move the decimal point to the right as many places as there are zeros in the multiplier, adding zeros if necessary.

Division

$100 \div 10$ - what's the answer to this problem (they will know this one)

How about this one – $10 / 10$ (1)

$1 / 10 =$ (0.1)

$0.1 / 10 = 0.01$

What's happening here?

(decimal point moves left one place)

That's right, when we divide by ten, we move the decimal point to the left one place – adding a zero if necessary.

What do you think would happen if we divided by 100?

(it would move two places to the left and add zeros if necessary)

Good. When dividing a number by 100, move the decimal point to the left 2 places – adding zeros if necessary.

So what about dividing a number by any power of ten? Is there a general rule applying to any power of ten? (let them formulate the rule if possible)

(when dividing a number by a power of ten, move the decimal point left the number of places as there are zeros in the divisor, adding zeros if necessary)

December 8, 2004

Multiplying a Decimal Fraction by a Decimal Fraction

Exercise I

Prologue

In this case our multiplier will be a decimal fraction. When we did multiplication before, our multiplier was always a whole number – a unit.

Cards are set out to the left of the decimal board and working numbers will be set to the right side (per convention)

Material

Decimal board, box of decimal fraction material, hierarchically colored beads and paper and pencil.

Begin

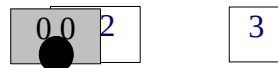
Let's start with 2 tenths x 3 tenths (0.2 x 0.3)

When you form your multiplier, we'll use gray cards (use gray double-zero card with a decimal point on it, with a 3 placed over the top to get 3 tenths)



I cannot do this. How can I take 2 tenths of 3 tenths? Before we were multiplying by a whole number and we had no problem doing those problems! So maybe if I make my multiplier into a whole number – I'll be able to work the whole problem out.

(take the double zeros from under multiplier and place it beneath the multiplicand to change the problem to 0.02 x 3.



2 hundredths times 3 is 6 hundredths.

There's my answer. (place cubes on board)

Do you remember multiplying $\frac{2}{2}$ or by $\frac{3}{3}$ where it was really like multiplying by 1. We didn't change the value – we just changed how it looks. Right? That's what I did here to my problem. I took the whole problem and I multiplied it by $\frac{10}{10}$ – which has the value of 1.



I divided the multiplicand by ten and multiplied the multiplier by 10 – to arrive at 0.02×3 .

(write this expression on paper)

$$(0.2 \times 0.3) \times (10/10) = (0.2/10) \times (0.3 \times 10) = 0.02 \times 3 = 0.06$$

This is a problem we can work out with our material.

===

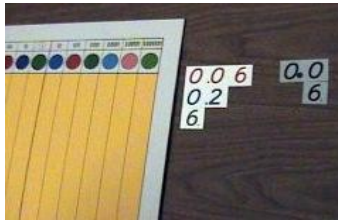
Let's do another one. How about 6 and 26 hundredths times 6 tenths.

$$(6.26 \times .6)$$

(build up multiplier with the double-zero gray card)



First thing I need to do is to decompose our multiplicand – which we always do. The lowest category is always at the top.



With this material I can't multiply by 6 tenths. We need to make the multiplier a whole number. I can do that by multiplying the multiplier by 10.

(slide the gray cards up away from the multiplier)

But I don't want to change the value of my multiplier. So what do I need to do?

Yes, I need to divide my multiplicand by ten. So I slide my gray cards over my first multiplicand part.



So I get .6 thousands – times 6 – gives my 36 thousandths

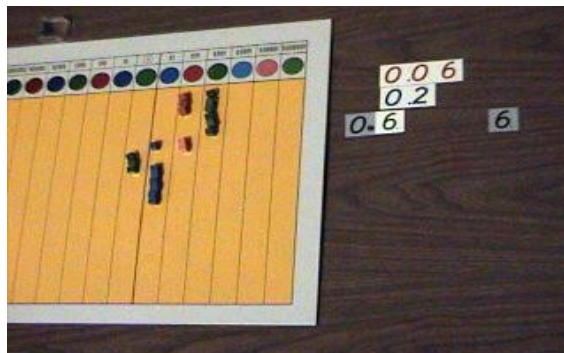
(put 3 pink cubes in hundredths column and 6 light green cubes in thousandths column)

Now to go on further, we have to slide the multiplier down, but also the zeros on our multiplicand – because **again we have to divide the 2 tenths of our multiplicand by 10, since we multiplied our multiplier by 10.**

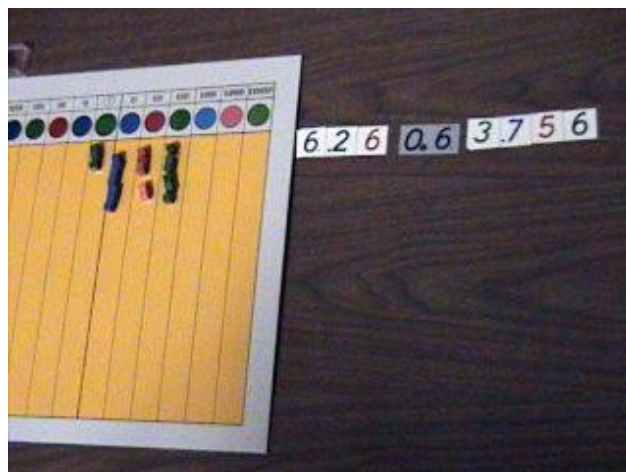
2 hundredths times 6 gives us 12 hundredths
 (put 1 cube in tenths column and 2 cubes in hundredths column)



(move gray cards but they now go on top of the multiplicand card to form .6)
 6 tenths times 6 gives us 36 tenths
 (put 3 beads in units column and 6 blue cubes into tenths column)



I'm done multiplying. Now we must find our final answer. (count and combine)
 (form original equation and form final answer in magic slide fashion)



6.26 times 0.6 gives us 3 and 756 thousandths.

Is this a reasonable answer? (use rough estimating – a good self-check to practice)

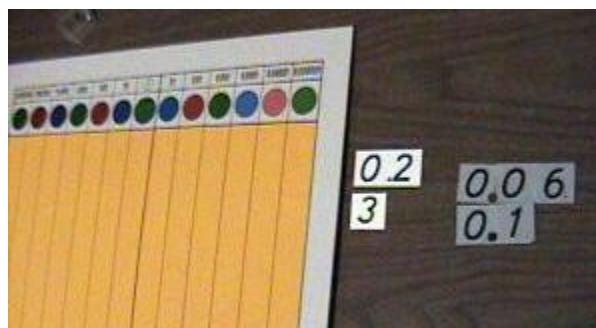
We start out with about 6. We're taking it a bit more than half. So is it logical to get 3 and a little bit more? Yes!

===

What about 3.2×0.16 ?



Format operands



Decompose operands



Isolate and multiply each multiplier digit to whole number, divide each multiplicand digit by the same value before multiplying each partial product



Multiply each multiplier digit times each digit of the multiplicand. Here we've already multiplied 6 times 2 thousandths, slid the gray cards down and then multiplied 6 times 3 hundredths – placing cubes into appropriate columns for each partial product.



Isolate, then multiply second (and subsequent when applicable) multiplier digit to whole number, divide each multiplicand digit by same value before multiplying each partial product – placing cubes into appropriate columns on the board each time.



Multiply second multiplier digit times second digit of the multiplicand



Combine fraction material and complete product at top, far right

Let's read our problem. 3 and 2 tenths times sixteen hundredths equals 5 hundred 12 thousandths.

Is this a reasonable answer? We took about 3 and multiplied it by about 1 tenths. That would give us about 3 tenths and we got about 5 tenths – so yes this is close.

Epilogue

Note addition and subtraction answers go below the operand cards in a vertical arrangement (all three of them to the right of the board). Multiplication answers go to the right of the operand cards in horizontal arrangement (all three to right of board).

Gray cards are included in the decimal fraction materials and are used for multipliers – as usual. In the beginning give them the problems to avoid materials-related problems – at least until they get the process down.

December 8, 2004

Exercise II

Prologue

This lesson now moves the child to paper and records partial products; the cards need to be written vertically for this.

Begin

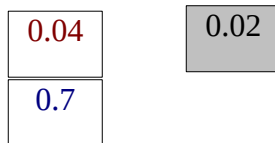
We do 0.74×0.32

(work problem same way **using the fraction material and board** at each step - but **use paper and pencil to record** each partial product along the way too)

Cards

Paper Image

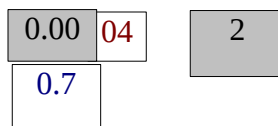
Partial Product 1



(this step is omitted and assumed in subsequent partial products illustrations – using the appropriate power of ten each time to make a whole number of the multiplier digit)

After multiplying multiplier $\times 100$ and dividing multiplicand digit (0.04) by 100.

Becomes ...



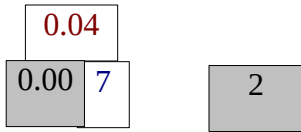
$$\begin{array}{r} 0.74 \\ \times 0.32 \\ \hline 8 \end{array}$$

Push the cubes together to form the concrete partial product. Leave a space and then begin the second partial product of cubes.

Product is 8 ten-thousandths. Align 8 into far right column under first column. Keep columns aligned but don't attempt to align products with the operands in any way.

Cards

Partial Product 2

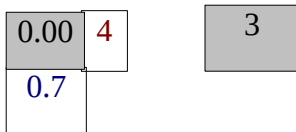


Paper Image

$$\begin{array}{r} 0.74 \\ \times 0.32 \\ \hline 148 \end{array}$$

Product is 14 thousandths. Align second partial product with the categories in the first partial product (i.e., 4 thousandths comes just to left of 8 ten-thousandths)

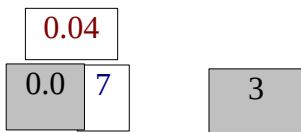
Partial Product 3



$$\begin{array}{r} 0.74 \\ \times 0.32 \\ \hline 148 \\ 2 \end{array}$$

Product is 12 thousandths. Align third partial product with the categories in sum of first and second partial products. Carry the 1 hundredth into the calculation of 4th partial product.

Partial Product 4

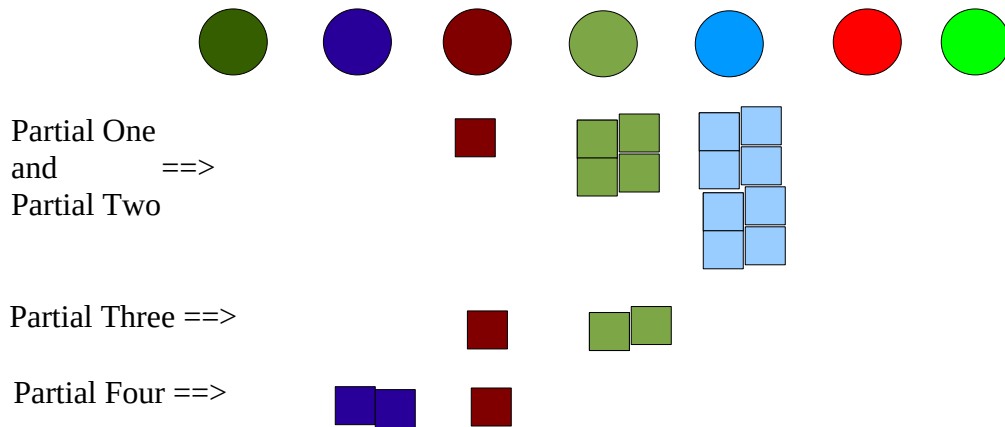


$$\begin{array}{r} 0.74 \\ \times 0.32 \\ \hline 148 \\ 222 \end{array}$$

Product is 21 hundredths. Add the carryover from partial product 3 to make it 22 hundredths. Align fourth partial product with the categories in sum of first and second partial products

Let's **verify** our answer using our fraction materials now.

Board Image



Let's slide up all cubes and find our sum.

Final Paper Image

$$\begin{array}{r}
 0.74 \\
 \times 0.32 \\
 \hline
 148 \\
 222 \\
 \hline
 0.2368
 \end{array}$$

Decimal point is placed based on the position of the fraction material on the Decimal Board.

December 8, 2004

Decimal Checkerboard

Introduction to the Decimal Checkerboard

Prologue

Start in the middle using a unit – the king. Build up horizontal axis – kingdom's expand. Build up vertical axis – kingdom's expand.

Then fill in the 4 quadrants verbalizing the category multiplication rules – upper left, upper right, lower left and lower right – in that order.

Within each quadrant, begin at vertical axis and work outward horizontally. Think of the horizontal axis as the multiplicand and the vertical axis as the multiplier.

Material

Felt squares: each 7.5 centimeter square, 12 dark green, 9 dark blue, 7 red, 9 light blue, 7 pink and 5 light green.

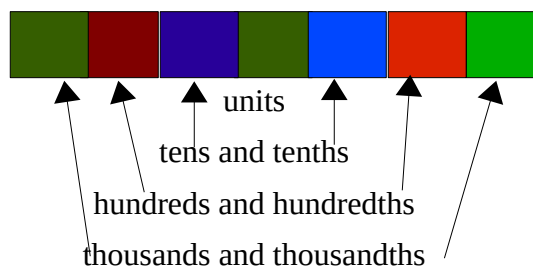
Begin

In our kingdom, the unit is king. But kingdom's expand. (**build horizontal axis**)

We have tens but we also have tenths. We have hundreds but we also have hundredths.

We have thousands but we also have thousandths.

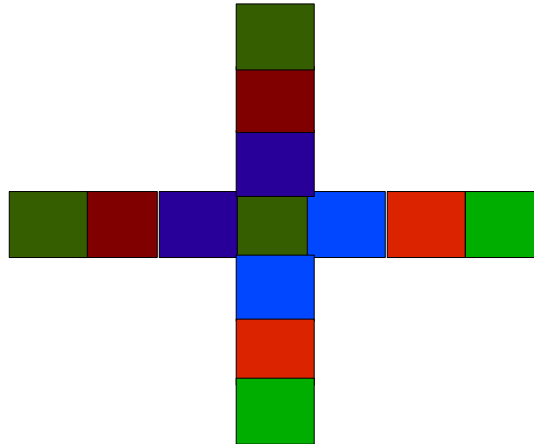
(place felt pieces down as the category names are given)



But kingdoms can expand this way too. (add on new axis - **build vertical axis**)

We have tens but we also have tenths. We have hundreds but we also have hundredths.

We have thousands but we also have thousandths.



(after set up of initial “unit borders” – horizontal and vertical, verbalize “we can multiply tens by tens and that gives us hundreds”)

tens times tens gives us hundreds

(add red felt square, upper left quad)

tenths times tenths gives us hundredths

(add light red square, lower right)

:

:

tens times tenths gives us units

(add dark green square lower left)

tens times tenths gives us units

(add dark green square upper right)

:

:

(continue to fill in entire grid with appropriate colored felt squares)

(at end, **bring out the actual decimal checkerboard** to show the children)

Epilogue

The pattern always guides you to see what the next answer is. You don't have to memorize each category combination abstractly. Follow the pattern.

December 9, 2004

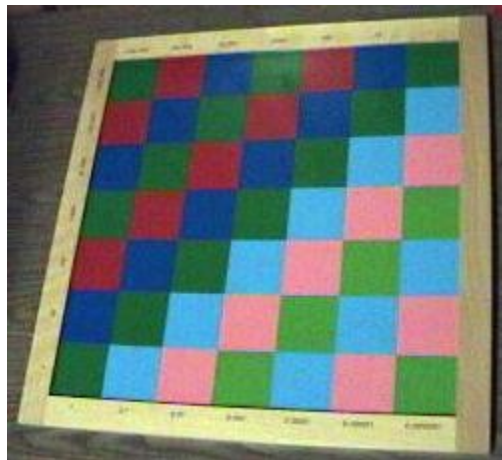
Exercise I

Prologue

On the decimal checkerboard, we can multiply any kind of numbers, whole or fraction in either multiplicand or multiplier. In order to do these, we need to know how to set them up. (illustrate each example – the board and the cards)

Material

Decimal points, decimal checkerboard, box of bead bars 1-9, number or tile cards



Prerequisites

Regular checkerboard

Begin

We'll work with the decimal checkerboard today.

We begin by setting up our board **(place 1 red bead on each square)**

I would like to know what the value is of all these beads.

Like the regular checkerboard – here are all of our units (point to green diagonal – begin

at unit and choose diagonals up and leftward up to corner), here are all of our tens, here are ... continuing ... all of our millions.

Here are tenths, here are hundredths ... and millionths (point to each row beginning with the tenths diagonal and proceeding down and right to the corner)

To find out what we have, we have to do this now (move all beads down into lowest column – combining categories). I gather them not only across the bottom but along the left side also.

Now I'll **exchange the beads so I have just one bead bar on each square** (at bottom and left side)

Now we **read our answer down the left side and across the bottom.** So I can read the value of all of my beads now.

I have 1 million 2 hundred 34 thousand 5 hundred 67 and 654 thousand 3 hundred 21 millionths.

We can do multiplication on this. Does anyone have any questions before we continue.

===

We start off with 427 thousands (gray cards top-right side for multiplier, white cards at top for multiplicand).



initially lay multiplicand at top toward right and multiplier at right side toward the top

Now we need to make sure units times units equals units.

Is our multiplicand lined up correctly with its units on a unit square? (no)

When we multiply multiplicand units and multiplier units together, do we get units? (no)

Then we'll need to move the multiplier to that units times units equals units – it must fall on a green square.



Pull multiplicand and multiplier down to units times units equals units

Now we can multiply. Start with lowest category.

(multiply all beginning units times units)



decimal point should be omitted as child comes to understand board categories

Now we slide beads to lowest row and exchange so we have one bead on each category.



Our answer is 13 and 6 hundred 64 thousandths (13.664)

===

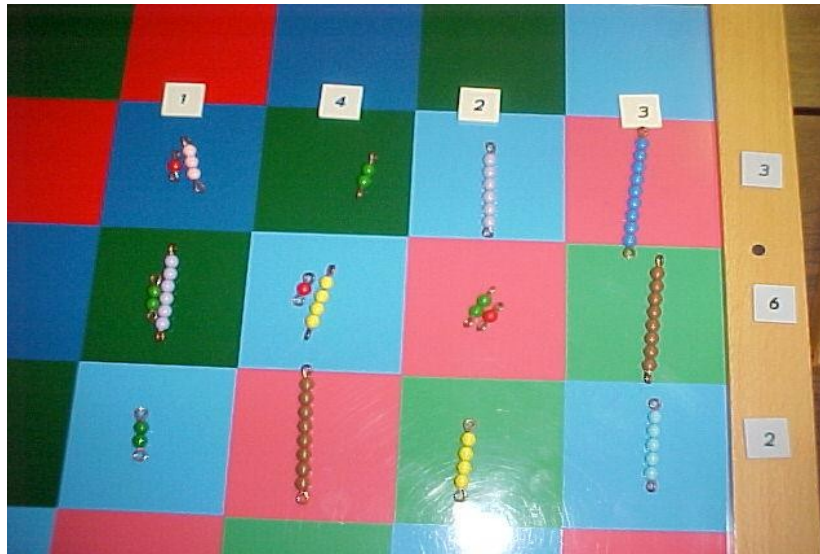
5,467 times 21 hundredths (0.21) = 1148.07



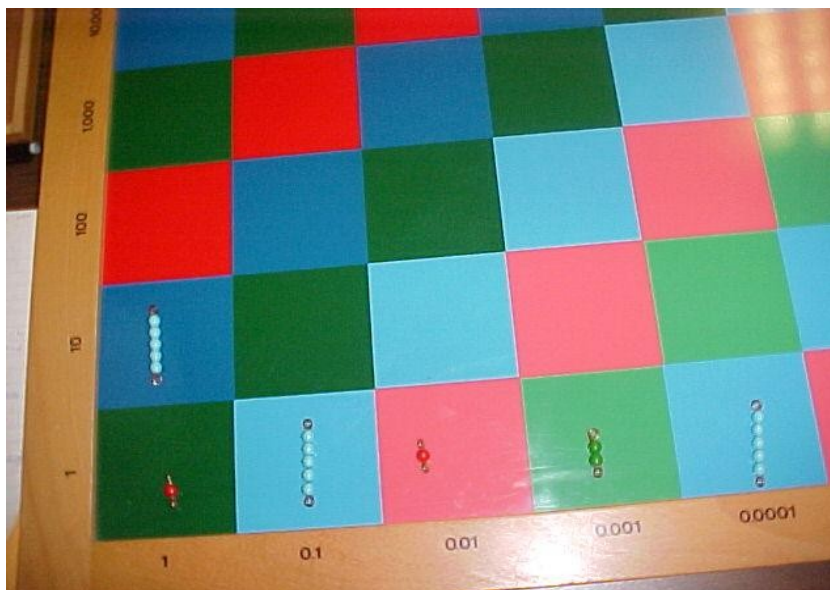
In this case neither multiplicand nor multiplier need to move

===

$$14.23 \times 3.62 = 51.5126$$



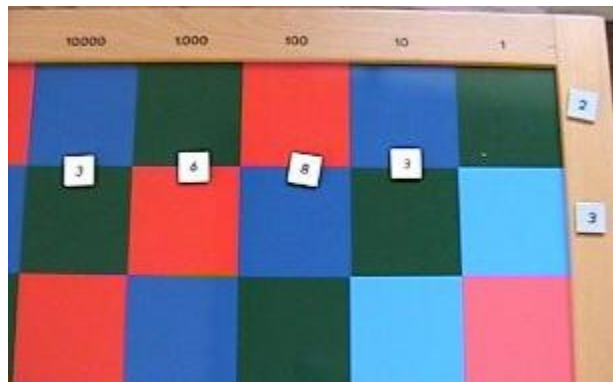
For units times units equals units, we move multiplicand down, then multiplier if needed, then multiply out all multiplicand digits by each multiplier digit.



At end, gather beads to lower left corner and process carry-overs.

===

$$3683 \times 23 = 84,709$$



In this case we move the multiplier down – in order to get units times units equal units.

===

$$37.64 \times 0.63 = 23.7132$$



In order to establish units times units as equal to units, bring multiplicand down.

December 9, 2004

Exercise II

Prologue

Here we move to paper and record partial products. The real difference between this material usage is the setup. We first put multiplicand on the top border of the board, right justified. Then, multiplier goes on right border, top justified, laid vertically. Then, look at the multiplicand to see if units are on a units cell; move down if necessary – to establish that units times units yields units – multiplicand, multiplier and product units categories must all fall on green cells.

Begin

Let's do 37.64 times 0.63

$$\begin{array}{r} 37.64 \\ \times 0.63 \\ \hline 11292 \\ 22584 \\ \hline 23.7132 \end{array}$$

Record first
partial
products –
times 0.03.

Record second
partial product –
times 0.6

I wonder if we can look at our problem on paper and figure out where the decimal point goes. We start with how many – almost 40 right? We're taking it about $\frac{1}{2}$ times. So a bit more than half of 40 means the decimal will come right after the 3 (23.7132)

Epilogue

Note that the first time we write a partial product, we simply align the first (lowest category) digit of the partial product on the far right column of the vertical expression. Thereafter, we take the lowest category of the new partial product – and align it with the same category in the old (first) partial product. This is how we explain it to the children.

December 9

Relative Size of Terms in a Multiplication Problem

Prologue

This uses paper and pencil only. It is a theory-type of lesson. Not every child needs this lesson; some get this understanding on their own.

Begin

A multiplication problem has 3 parts: a **multiplicand**, **multiplier** and a **product**. We're going to look at these terms – the size of these in relation to each other.

There are 3 cases. We'll look at the value of each term in relation to the others.

Less Than One Times Greater Than One

The **first case is multiplying a fraction that is less than 1 times a number greater than 1.**

0.25 x 4 for example

What's the answer to this problem? (1)

Let's take 0.5 times 6.

What's the answer to this problem? (3)

How about take 0.21 times 2.

What's the answer to this problem? (0.42)

$$0.25 \times 4 = 1$$

$$0.5 \times 6 = 3$$

$$0.21 \times 2 = 0.42$$

Write each new problem and answer.

In each of these problems, we started with a decimal fraction that was less than 1. Also, we had a multiplier that's greater than 1. My question to you is – **what can we say about the product in each of these cases with respect to the multiplier and multiplicand?**

(it is more than the multiplicand and less than the multiplier)

Does that make sense? If we take a number less than 1 but take it so many times we can see it does make sense. The product will be in between the multiplicand (greater) and multiplier (lesser)

Let's look at this: 0.32×7

If I multiply I get 224. Where are we going to put the decimal point? (after the first '2')

Why does that make sense? If we want the product value between the multiplier and multiplicand, the decimal must follow the first '2.' Does our generalization hold true for this problem also? **“With multiplicand less than 1 and multiplier greater than 1 or vice versa, the product is in between those values.”**

How about 0.47×8 ? After multiplying I get 376. **Where does the decimal go? After the first '3' because this gives us a value that is in between the multiplicand and multiplier.**

$$0.25 \times 4 = 1$$

$$0.5 \times 6 = 3$$

$$0.21 \times 2 = 0.42$$

$$\mathbf{0.32 \times 7 = 2.24}$$

$$\mathbf{0.47 \times 8 = 3.76}$$

Now for our second case.

Less Than One Times Less Than One

Suppose we have a fraction less than 1 times a fraction less than 1. An example like this:

$$0.25 \text{ times } 0.2$$

So our multiplicand is less than 1 and our multiplier is less than 1, right? What will happen? If we multiply across we get 050.

Where will we put our decimal point? Would it make sense to put it after the 5? No since we are multiplying a fraction times a fraction – how could we end up with 5?

How about before the 5? Does that make sense? We're taking a part of a part – could we end up with more than we started with? No.

So let's put the decimal point before the first '0' (0.050)

$$0.25 \times 4 = 1$$

$$0.5 \times 6 = 3$$

$$0.21 \times 2 = 0.42$$

$$0.32 \times 7 = 2.24$$

$$0.47 \times 8 = 3.76$$

$$\mathbf{0.25 \times 0.20 = 0.050}$$

What about 0.50×0.3 . If I multiply I get 15 and where can we place the decimal point?

Tenths times tenths give us hundreds. So it has to go before the '1.'

$$\begin{aligned} 0.25 \times 4 &= 1 \\ 0.5 \times 6 &= 3 \\ 0.21 \times 2 &= 0.42 \\ 0.32 \times 7 &= 2.24 \\ 0.47 \times 8 &= 3.76 \\ 0.25 \times 0.20 &= 0.050 \\ \mathbf{0.50 \times 0.3} &= \mathbf{0.15} \end{aligned}$$

So in these examples, the **multiplicand is less than 1 and multiplier is less than 1**. How can we describe the product related to those first 2 terms?

It has to be less than the multiplicand) That's right, we're taking a part of a part. It's also less than the multiplier. **Because we're taking a part of a part the product will always be less than both of them.**

Now for our third case.

Greater Than One Times Greater Than One

We have a **decimal fraction greater than 1 times a decimal fraction greater than 1**. For example:

$$\begin{aligned} 3.5 \times 1.5 &= 5.25 \\ 0.25 \times 4 &= 1 \\ 0.5 \times 6 &= 3 \\ 0.21 \times 2 &= 0.42 \\ 0.32 \times 7 &= 2.24 \\ 0.47 \times 8 &= 3.76 \\ 0.25 \times 0.20 &= 0.050 \\ 0.50 \times 0.3 &= 0.15 \\ \mathbf{3.5 \times 1.5} &= \mathbf{5.25} \end{aligned}$$

Where will we put the decimal point? If we put it anywhere but after the first 5, **our answer would be way too much or way too little.**

What about $3.68 \times 4.1 = 15.088$?

We can **look at whole number here! So we can multiply 4 times 4 and the answer has to be around 16**. So we know to put the decimal point between the 5 and 0.

$$\begin{aligned} 0.25 \times 4 &= 1 \\ 0.5 \times 6 &= 3 \\ 0.21 \times 2 &= 0.42 \\ 0.32 \times 7 &= 2.24 \\ 0.47 \times 8 &= 3.76 \\ 0.25 \times 0.20 &= 0.050 \\ 0.50 \times 0.3 &= 0.15 \\ 3.5 \times 1.5 &= 5.25 \\ \mathbf{3.68 \times 4.1} &= \mathbf{15.088} \end{aligned}$$

Greater Than One Times Greater Than One

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7. DECIMAL FRACTIONS

So in this third case, a **decimal fraction greater than 1 times a decimal fraction greater than 1**, we can say **the product is greater than both of them**.

We want to keep these rules in mind as we do decimal fraction work to see if our answers make sense.

December 10, 2004

Relative Size of Numbers When Dividing

Age

Upper elementary

Purpose

To help children see some aspects of the size of numbers when we divide.

Prologue

This is the last section of Decimal Fractions.

Material

Decimal board, blank tickets, lead pencil, some decimal points.

Begin

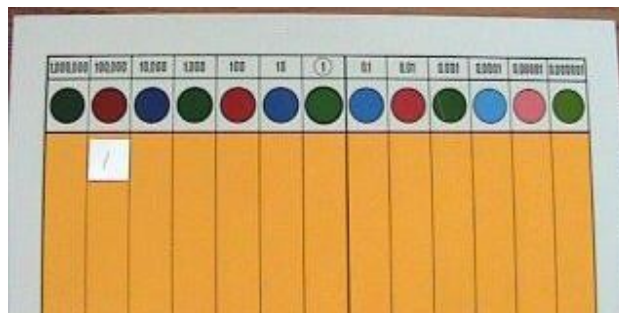
You've been hearing for years now that if we have 10 in one category, it's equal to 1 of the category to the left of it. For example, 10 units equals 1 ten. Or, 10 tens equals 100.

I can also say 1 unit is equal to 10 tenths. Right?

The question I have is this: I want to know **how many tens there are in 1 hundred thousand.**

In order to know this, I'm going to ask a series of questions.

How many hundred thousands are there in 100,000? (1)



(write 1 on ticket, place on the hundred-thousands category of the decimal board at top)

How many ten thousands are there in 100,000? (10) We just add a zero.



(write and place zero as before in correct category (ten-thousands category at top)

How many thousands are there in 100,000? (100) We just add another zero.



(write and place as before in correct category)

How many hundreds are there in 100,000? (1000) We just add a zero.

(write and place as before in correct category)

Now finally, the question I started with. How many tens are there in 100,000? (10000)

There are 10,000 tens in 100,000. Again, we just add a zero.



(write and place as before in correct category)

(now **choose any of larger category and ask how many of a smaller categories are in it**)

– --- -- now do fraction -- --

Now let's find out about decimal fractions.

How many tenths are there in ten-thousand?

Well, how many ten-thousands are there in ten-thousand? (1)

(write and place on board in 10,000 category – moving right)

How many thousands are there in ten-thousand? (10) (write and place on board)

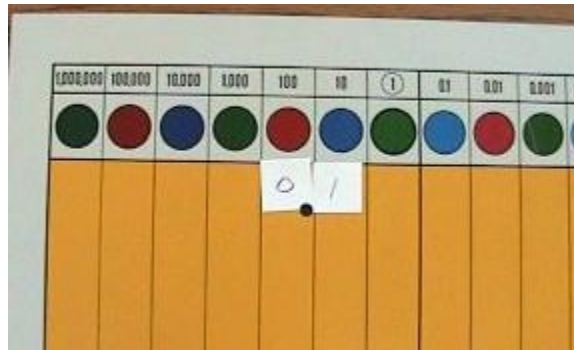
How many hundreds are there in ten-thousand? (100) (write and place on board)

How many tens are there in ten-thousand? (1000) (write and place on board)

How many units are there in ten-thousand? (10,000) (write and place on board)

So, how many tenths are there in ten-thousand? (100,000) (write and place on board)

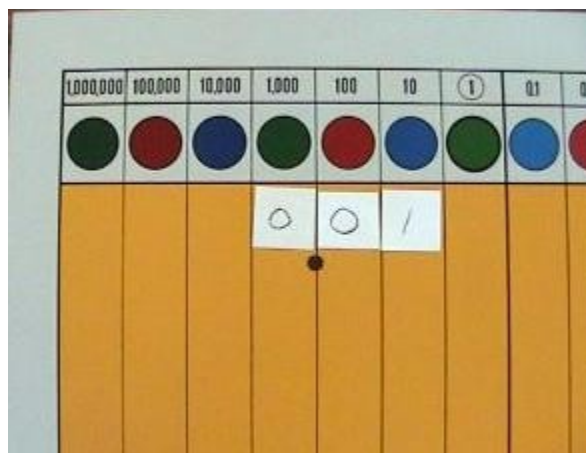
How many hundreds are there in ten? (0.1) One tenth.



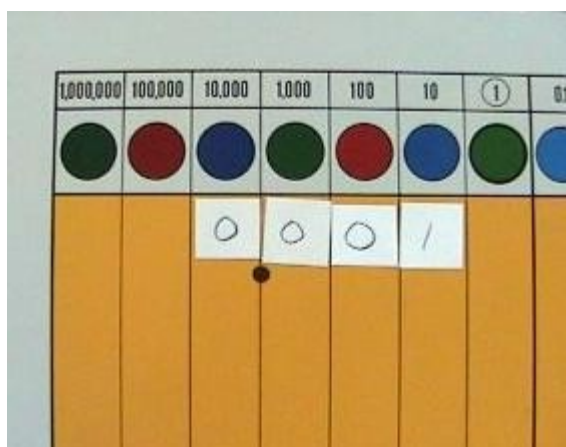
- (place **decimal to right of 1**,
- then **add a zero** under 100s category,
- then **move the decimal point to left of the 1** – to get 0.1, beginning in the hundreds category)

How many thousands are there in ten? (0.01)

How many 1000s in 10?

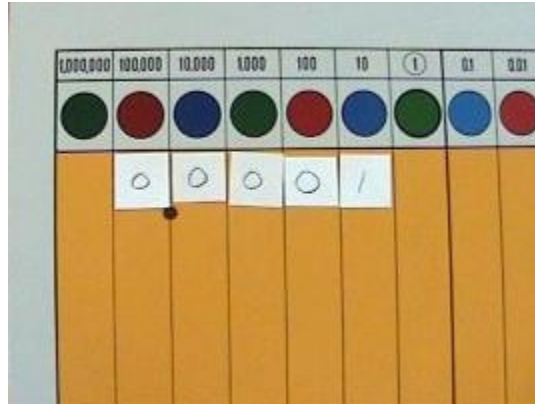


How many ten-thousands are there in ten? (0.001) (lined up beginning left underneath the 10,000s category)



How many 10,000s in 10?

How many hundred-thousands are there in 10? (0.0001) One ten-thousandth.
(lined up '0.0001' - beginning leftmost zero underneath the 100,000s category)



When you divide a smaller number by a larger number the answer will be smaller than the dividend.

Epilogue

There are two ways of looking at division problems: distributive and group. Here, we're asking a group division problem each time. This is not the sort of lesson that is given to every child. It should be clearly illustrated so it is done correctly when we do get around to it.

December 10, 2004

Division of a Decimal Fraction by a Decimal Fraction

Prologue

Up to now, we've only done division by a whole number divisor.

Material

Decimal fraction material, hierarchically colored beads, felt mat, skittles (red, blue and green), some colored disks from the stamp game, pencil and paper.

Begin

Let's take this problem – 8 and 6 tenths divided by 4 and 3 tenths:

$$8.6 \div 4.3$$

(set up 4 green skittles and 3 blue skittles at top of felt mat)

(form the divisor – 8 green beads and 6 tenths cubes – at bottom of mat)



Now we have to share out from our dividend.

When we share out, units get 10 times more than tenths, right?

(share out the green beads to the green skittles, the blue tenths cubes to the blue skittles)

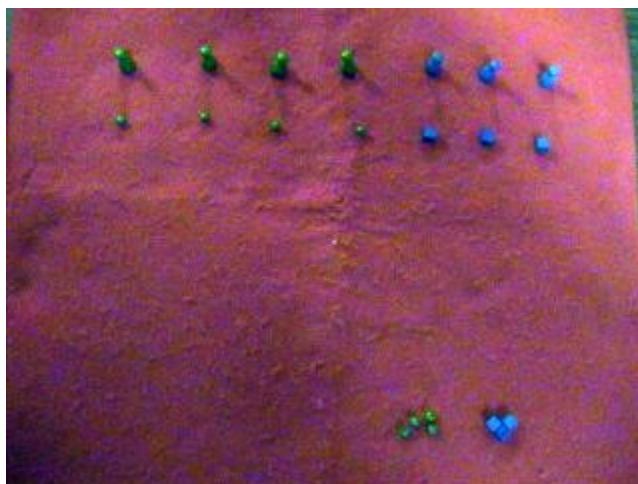


image after first round of sharing – share out to all skittles – each turn

Where do we find our answer? We find our answer in division under the unit.



Under our green skittles we have shared out 2 green beads so our answer is 2.

===

What about four tenths divided by twenty-five hundredths?

(write this down on paper for the child too – or allow him to write if he's ready)

$$0.4 \div 0.25$$

(use fraction material to form dividend at bottom of felt mat)

We find our answer under each unit but we don't have any units! This is why I brought these colored disks. (place the green disk in the units position)



(during sharing out – to 1 green disk, 2 blue and 5 red skittles)

My tenths go under each tenths-skittle - but I have to exchange for hundredths to share out to the hundredths-skittles since they get 10 times less.



(exchange along the way, several times)

(at end – each blue = 1 tenth and 6 hundredths, each red = 1 hundredth and 6 thousandths)



This problem doesn't have units so we'll look at how much 1 tenth received and we know the unit would get 10 times more. How many does a tenth have? (0.16)

So what does a unit receive? (1.6)

- - - -

Note: this is a repeater example.

$$0.36 \div 0.027 = 13.3$$

(during sharing out – repeating pattern has begun)

Well we've run out of materials. We can't go any further. So let's see what we have.

How many did hundredths receive? We can't count it in the usual way since it goes on and on.

(at end)

We don't have units or tenths. But we can look at one-hundredths and a unit would get 100 times more.

If hundredths got 0.13, to find our answer, what would a unit receive?

Well, 100 times as much would be 13, right? And we see we get 3 of everything after this. **So we have a repeating number – 13.3.**

December 10, 2004

Abstraction: Multiplication of Decimal Fraction on Paper

Prologue

Some children would come to this understanding in this lesson on their own and won't need this lesson. Others will need help getting to this understanding.

Material

Pencil and paper.

Begin

How about this problem today: $0.3 \times 0.15 =$ (write on paper)

We could get out the decimal checkerboard or the decimal board to work out this problem. But we'll look at it differently today.

Suppose I write each of these numbers as a common fraction? Do we know how to do that? Yes.

How do we write 0.3 as a common fraction? (3/10)

How do we write 0.15 as a common fraction? (15/100)

$$0.3 \times 0.15 = \frac{3}{10} \times \frac{15}{100} =$$

Now looking at this as common fractions what do we do?

We multiply the numerators and we multiply the denominators, right?

(we get 45/1000)

Do we know how to write that as a decimal fraction? Yeah. We write 0.045.

$$0.3 \times 0.15 = \frac{3}{10} \times \frac{15}{100} = \frac{45}{1000} = 0.045$$

===

How about $0.4 \times 0.6 = 4/10 \times 6/10 = 24/100 = 0.24$

===

What about this one? 0.106×3

Can we work this out the same way? Yeah.

$0.106 \times 3 = 106/1000 \times 3/1 = 318/1000 = 0.318$

(repeat as many example as needed to get the idea)

I wanted to bring to your attention to the following:

In our first example, we multiplied tenths times hundredths and we got thousandths.

In our second example, we multiplied tenths times tenths and got hundredths.

In our third example, we multiplied thousandths times units and got thousandths.

If I were to write one of these problems vertically like this, we would do our normal multiplication. **But then we would have to find where to put our decimal point.**

$$\begin{array}{r} 0.4 \\ \times 0.6 \\ \hline 24 \end{array}$$

Where would we place this? (before the 2) That's right. You've noticed that we have 1 place to the right of the decimal in the multiplicand and 1 place to the right of the decimal in the multiplier. So **there are two places total to the right of our decimal place.**

I wonder if this will work in all cases?

(do 0.3×0.15 vertically)

$$\begin{array}{r} 0.3 \\ \times 0.15 \\ \hline 45 \end{array}$$

Now how many decimal places do we have to the right of our decimal point?

There is 1 in the multiplicand and 2 in the multiplier – so our answer has to have a total of 3 places to the right of the decimal point – so our answer is 0.045.

$$\begin{array}{r} 0.3 \\ \times 0.15 \\ \hline 0.045 \end{array}$$

(repeat examples to verify the rule and drive the idea home)

Now as you go on doing your multiplication problems, **this is the rule you can use for placing your decimal point.**

Epilogue

Continue to encourage the children to ask the question if the answer is reasonable.

December 10, 2004

Abstraction: Division of a Decimal Fraction on Paper

Prerequisites

Fraction inset materials and prior division lessons (decimal fraction, racks and tubes, group division, stamp game, etc.). As always, interest and readiness are the key factors.

Prologue

Since it may have been a while since we used the material, We'll do one simple problem with the material to start off. The point of this lesson is to know where the decimal point is placed after dividing using a decimal fraction divisor.

Start with easy problems; the children should be able to look at the problems and see the answers without struggle.

Material

Fraction materials for review section. Pencil and paper for the main section.

Begin

Let's review how we divided using our materials.

$$3.3 \div 3$$

(form dividend on felt mat, place beads and cubes on felt mat as the dividend, share out as usual, put materials away)

Now **what if I take this problem to work out on paper:**

Well if I divide 3 by 3 I get 1. And if I divide 3 by 3 I get 1 again.

$$3.3 \div 3 = 11 \qquad \text{(leave off the decimal point at first)}$$

Where is a reasonable place to put our decimal point? (after the first 1)

Why is it reasonable? Because you are starting with a bit more than 3 and are sharing it out to 3 – so a reasonable answer is a bit more than 1.

(fill in the decimal point to make 1.1)

===

Suppose we have $4.6 \div 2$

If we divide 4 by 2, we get 2. If I divide 6 by 2 I get 3.

And where is a reasonable place to put the decimal point? (after the 2)

Why is that reasonable? We are dividing a bit more than 4 by 2 – so a reasonable answer is a bit more than 2.

(All of these examples have a unit divisor. Now we get into fractional divisors)

What if we had $4 \div 0.25$. Can we just look at this and know the answer?

Not as easy as when we had a divisor of a whole number, right?

Maybe if we make our divisor a whole number, it will be easier – so what would we have to do? (multiply by 100)

$$4 \div (0.25 \times 100) = 4 \div 25$$

But, that would change my problem if I did only that.

Well what about 10 divided by 2? If we multiply the divisor here by 100 what do we get? (200) In order to get the same answer to the problem, what would we need to do to the 10? Yes, we have to also multiply that by 100.

If we multiplied 10 by 100, we get 1000. And 100 times our 2 gives us 200.

1000 divided by 200 gives us 5. and 10 divided by 2 gives us 5 also.

$$10 \div 2 = 1000 \div 200 = 5$$

Let's use this idea to solve our first problem.

So, 4 divided by 25 hundredths. We need to multiply by 100 in order to make the divisor a whole number. What do I need to do to my dividend? (multiply by 100)

So that means I need to take 4 times 100, giving us 400 – and 25 hundredths times 100 giving us 25 (do decimal point shifting to show new values)

$$400 \div 25 = 16$$

We see we get 16 using the same technique but that was a lot of work. So here's a trick

If we write that number $0.25\sqrt{4}$, to make our divisor a whole number, we need to multiply it by 100. Which means we need to move the decimal point over 2 places to the right.

(shift decimal point in expression)

But - if I only do this to the divisor, I change my problem. So I also have to do it to my dividend.

And since we've found where our decimal point goes, let's write it in the answer right away.

$$\begin{array}{r} 16 \\ 25\sqrt{400} \end{array}$$

Now I've multiplied my dividend and my divisor by 100. Now if we divide ...

(do the division problem)

So we see that our answer is 16.

===

Let's try another problem like this: 72 divided by 0.36

What do we need to do to our divisor to make it a whole number?

(multiply by 100)

What do we have to do then to our decimal point? (move it right two places)

$$72 \div 0.36 \text{ is not equal } 72 \div 36$$

But if we do that to the divisor, so as not to change our problem, we have to multiply multiplicand also by 100.

$$72 \div 36 = 2$$

So we also have to move the decimal point 2 places to the right and add our decimal point in the answer.

(do the long division)

$$36\sqrt{72}^2 \quad \times 100 = 200$$

So we can see that 72 divided by 0.36 is 200.

So now you can do your division like this. This is how people do their division.

Epilogue

We move to using the radical expression since this is the more advanced technique. Go slowly with obvious problems since they may have not abstracted all previous lessons.

October 28, 2004

8. INTRODUCTION TO SQUARING AND CUBING

Prerequisites for Squaring and Cubing

Materials

Short bead chains and large felt mat. We set them out horizontally, even at the left end to highlight the length difference.



Lay out short bead chains horizontally, aligned at left

Prologue

Work in primary with skip counting provides foundation for elementary work. It laid a foundation for multiples. It also lays another foundation for squaring and cubing of numbers. It's important that the children have enough experience with the chains.

If for any reason the children have not done **linear and skip counting** in the primary, they can do it in the elementary classroom.

Activity 1

I can't do much with the 1 chain or the 2 chain. But I wonder what I can make with the 3 chain. I can make a triangle.

I wonder what I can do with the 4-chain. I can make a square.

I wonder what I can do with the 5-chain? Yes, I can make a pentagon. What shape can we make with the 6-chain? Hexagon.

How about the 7 chain? A heptagon. With the 8-chain an octagon. And with the 9-chain a nonagon. What can I build with the 10-chain? (decagon)

(this is one activity they can do with the short bead-chains)

End

Activity 2

Something else we can do is to take the 10-chain, we'll make a nice decagon. Now let's take the 9-chain and build it inside. "What did we say we could make with this – a nonagon. Let's build that right inside."

"And then let's take the 8-chain – what shape can we make? Yes, an octagon; again we'll build that right inside."

End

Variation - Activity 2

(circumscribe (not inscribe here) figures around others. start with the 3-chain and build outward.)

End

Epilogue

These can also be done using the long chains; they allow a larger group to work together since it's more tricky to make the angles on larger chain figures. Especially with the long chains, we can also count and label multiples.

Begin

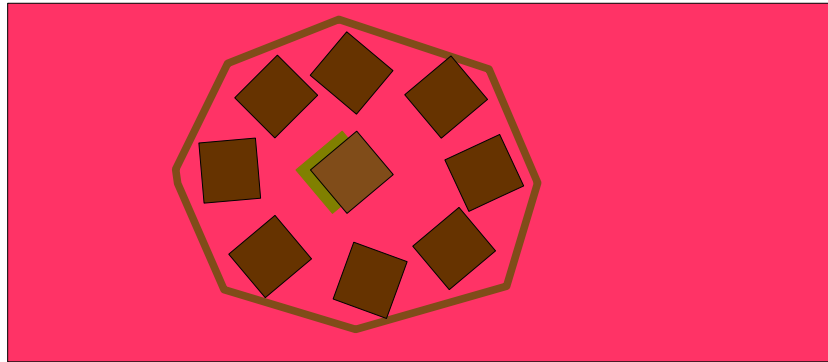
The first thing I want to show you is how to carry the bead-chains. I think of my hand as a shelf and lay it across my hand. I take the next one off and hand it right there over my hand. If you have one of the odd number chains, you can carry the end part of the chain

with the right hand.

Now on our mat, let's see if we can make a great big octagon.

Each side of our octagon will continue up to the big loops at the end and if we run off the end of the mat, that's okay.

Now let's get the squares and put one at each angle. Then we put a cube (see illustration 2 for this date)



End

Epilogue

Sometimes the children do one chain at a time. They can work with any length chains they want. These are indirect preparation activities to make sure the children have had enough experience with the chains.

October 28, 2004

Notation of Squares

First Presentation

Materials

The short chains from the bead chain cabinet, squares of the numbers 1-10 (one square each), blank tickets and a pencil, a felt mat and 3 sets of prepared tickets. They are in a box labeled “squares of numbers.”

The first set of tickets starts at 1×1 and goes up to 10×10 . The second set starts at 1^2 to 10^2 . The third set of tickets has the value of each square – starting at 1 and the last ticket is 100.

Prologue

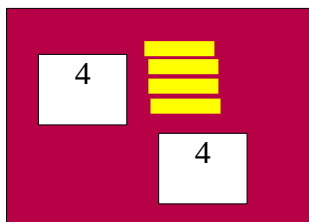
There are two exercises. For this first presentation, choose one of the chains that is in the middle – say the 4, 5 or 6-chain.

Begin

(lay out chain across the felt mat)

I have here the short chain of 4. If I do this with my chain (make a square 4), what do I get? Yes, a square – it's a square of 4.

I have 4 taken 4 times. Let's write that down. (make a ticket for 2 of the adjacent sides saying '4'. What did we say this gives us? Yes, a square of 4.

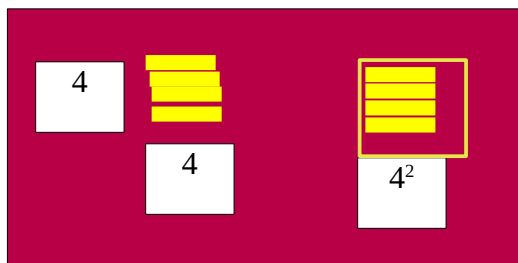


Who would like to go to the cabinet and get a real square of 4?

(direct child to the cabinet and have him retrieve the 4-square)

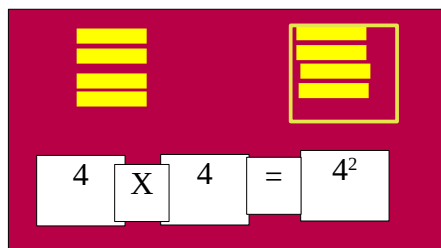
Do you know how we write a square of 4 down? This is how we write 4 squared.

(write on the ticket – 4^2)



We write that little two because we have 4 here and 4 here (point to the sides)

We can say that 4 times 4 equals 4 squared. Let's **pull our tickets down** here to the bottom and form an expression that says how we've formed our square of 4.



(write a ticket '9²')

Femina, can you bring that from the bead-chain cabinet? (do this to see if they have caught on)

How did you know to bring chains of 9? Yes, because I wrote down **square of 9**.

Carrie, can you go and get this one please? (write 6^2)

How did you know to bring chains of 6? Yes, because I wrote down **square of 6**

(If you know they have had lots of math experience ask)

... and how much is this worth? (36 beads) (otherwise, tell them its worth 36)

End

October 28, 2004

Second Presentation

Materials

In addition to the materials listed for the first presentation, we need all of the short chains (ch) from the cabinet and 1 square of each number.

Prologue

We form squares from each short bead chain to show the idea of x , x times as a square of x . As usual, there is the material, then symbol.

Begin

1 taken 1 time gives us the square of 1.

(place separate bead bars at left and its square at right for each number 1-10)

2 taken 2 times gives us the square of 2.

3 taken 3 times gives us the square of 3.

4 taken 4 times gives us the square of 4.

5 taken 5 times gives us the square of 5.

6 taken 6 times gives us the square of 6.

7 taken 7 times gives us the square of 7.

8 taken 8 times gives us the square of 8.

9 taken 9 times gives us the square of 9.

10 taken 10 times gives us the square of 10

Eliza, you're first – I'll give you the first set of tickets. Danielle, you're second, so I'll give you the second set of tickets, (continue for each child, 5 of them sharing the tickets).

We want to lay them down like this. **(get the children lined up and point to where they should set the tickets)**

layout (down left side of mat – 3 columns of tickets at right):

1ch	1S		1 x 1	1^2	1
2ch	2S		2 x 2	2^2	4
3ch	3S		3 x 3	3^2	9
:	:	:	:	:	

(continue to ten;

- in far left column we have folded the chain to make a square; in next column right we have placed the square)

End

Extension

After the children have had this experience, instead of doing this exercise with the bead material, they could do it on graph paper. (on left, draw circles for the folded chains, with the figure outlined at right to show they are then squares – then at right they can write what the tickets would say.)

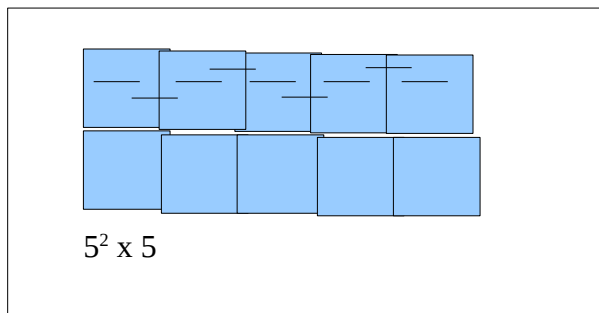
October 28, 2004

Notation of Cubes

First Presentation

Material

Large felt mat, **one of the long chains**, we chose 5, **all of the squares of that number** and the **cube of that number**. Blank tickets and a pencil.



Final layout: Chain of 5 built into 5 squares by folding the chain into 5 sections streaming left-right 5 turns. Then, 5 true squares below that, and tickets at bottom.

Prologue

This is after Notation of Square work. Note that we fold the chain differently now (folding 25 beads to the right, 25 beads to the left, etc.) We make a series of 5 squares. This makes 5 squares of 5 from the single chain.

Begin

(begin by spreading out the long chain on the mat)



I'm going to fold my chain into 5 parts to form 5 squares.



Then I'll place squares under each one. (place actual squares under folded chain squares)

I wonder what I get if I stack the squares up? (stack them 5 high) Yes, I get a cube.

(stack up actual squares into a cube)

Would you like to go to the cabinet and get the cube of 5?

We started with a 5 squared (point to squares in long chain).

How many did I pile up here? Yes, 5 also. That's 5 squared times 5.

(place cube next to the 5 stacked squares. write $5^2 \times 5$ on a ticket and place it below)

Do we know another way to write 5 squared?

Let's write 5×5 on a ticket and substitute 5 times 5 for 5 squared – it's the same thing.

(“ $5^2 \times 5$ ” becomes “ $5 \times 5 \times 5$.” move 5^2 ticket to the side)

Do you know how to write a cube of 5? We write it like this (ticket says 5^3)

It means that right here, you have 5 (point to bottom of one square) times 5 (point to side) taken 1 times, 2 times 3 times, 4 times, 5 times. 5 times 5 times 5. (each time place another square on top)

That's why we write “ 5^3 ” - because $5 \times 5 \times 5 = 5^3$.

Tickets now says “ $5 \times 5 \times 5 = 5^3$.” (other tickets have been put away. Tickets appear below the final 5 squares of beads)

End

Epilogue

This can be followed up with a **3rd period activity** to assure they know this. Carrie, can you please go and get this one? (hand exponential notations on tickets and they retrieve the cube for that number). Hand one of the sharper children a squared exponential notation on a ticket to keep him on his toes.

October 28, 2004

Second Presentation

Material

- Large felt mat and four sets of labeled tickets (for the cubes of numbers).
- First set starts at $1^2 \times 1$ and goes to $10^2 \times 10$
- Second set starts at $1 \times 1 \times 1$ and goes to $10 \times 10 \times 10$
- Third set starts at 1^3 and goes up to 10^3
- Fourth set is the values for these – 1 to 1000.
- All the squares and all the cubes from the bead cabinet



Layout continues from 5, in this pattern up to 10.

Prologue

Just as with the presentation of squares, we set out two columns. Squares are piled on the left and real cubes are placed on the right.

Begin

The square of 1 taken 1 times gives us the cube of 1. (place square and then cube)

The square of 2 taken 2 times gives us the cube of 2. (place square and then cube)

The square of 3 taken 3 times gives us the cube of 3. (place square and then cube)

The square of 4 taken 4 times gives us the cube of 4. (place square and then cube)

The square of 5 taken 5 times gives us the cube of 5. (place square and then cube)

(continue up to ten)

I have tickets to show what we've done. Let's take the number 1. We took a square of 1

and we multiplied it by 1; that's what our first ticket shows.

On this second card, we show that this is the same as $1 \times 1 \times 1$.

As we already saw, we can write that using a small 3 to show we've multiplied 1 by itself 3 times. This is what our third tickets tells us.

And the last ticket gives us the value of $1 \times 1 \times 1$. So all of these tickets say the same thing in different ways.

Now, you may finish filling in all the tickets for all of our cubes.

(Give each of four children one each, of four sets of tickets. Lay them out to right of the two columns of beads. This is the same setup as notation with squares exercise)

(tickets laid out to right of cubes)

$1^2 \times 1$	$1 \times 1 \times 1$	1^3	1
$2^2 \times 2$	$2 \times 2 \times 2$	2^3	8
:	:	:	:
$10^2 \times 10$	$10 \times 10 \times 10$	10^3	1000

End

October 28, 2004

Games with Squaring and Cubing

Game I

Prologue

We present each of these on different days. We call them games because they are not key presentations, they are not mandatory. They are in the vein of extensions.

Material

One square for each of the numbers 1-10, one cube for each of the numbers 1-10, a felt mat, paper and pencil.

Begin

Let's build a pyramid. (stack each square on top of each other concentrically, beginning with the 10-square.)

I wonder what it's worth. What is the value of all the beads in this pyramid?



What do we have at the bottom? 10 square. And that's worth how many? Yes, 100.

What do we have next? 9 squared – and that's worth 81. Let's write it. ($9^2=81$)

What do we have next? 8 squared – and that's worth 64. Let's write it. ($8^2=64$)

What do we have next? 7 squared – and that's worth 49. Let's write it. ($7^2=49$)

What do we have next? 6 squared – and that's worth 36. Let's write it. ($6^2=36$)

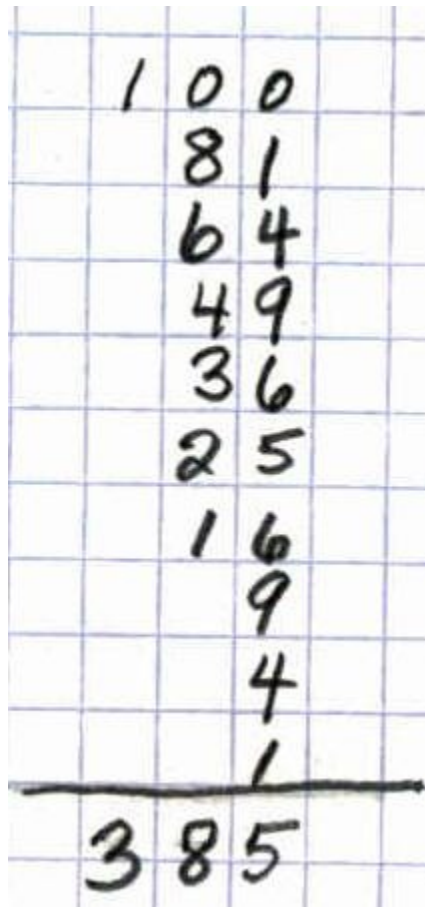
What do we have next? 5 squared – and that's worth 25. Let's write it. ($5^2=25$)

What do we have next? 4 squared – and that's worth 16. Let's write it. ($4^2=16$)

What do we have next? 3 squared – and that's worth 9. Let's write it. ($3^2=9$)

What do we have next? 2 squared – and that's worth 4. Let's write it. ($2^2=4$)

What do we have next? 1 squared – and that's worth 1. Let's write it. ($1^2=1$)



Now all we have to do is add these up. (385)

Now let's do this. (make a tower from the cubes in similar fashion)

What did I build? Yes, a pyramid of cubes?



What do we have on the bottom and how much is it worth? Yes, 10 cubed is 1000.

Next we have 9 cubed? Yes, 9 cubed is 729.

Next we have 8 cubed? Yes, 8 cubed is 512.

Next we have 7 cubed? Yes, 7 cubed is 343.

Next we have 6 cubed? Yes, 6 cubed is 216.

Next we have 5 cubed? Yes, 5 cubed is 125.

Next we have 4 cubed? Yes, 4 cubed is 64.

Next we have 3 cubed? Yes, 3 cubed is 27.

Next we have 2 cubed? Yes, 2 cubed is 8.

Next we have 1 cubed? Yes, 1 cubed is 1.

1	0	0	0
	7	2	9
	5	1	2
	3	4	3
	2	1	6
	1	2	5
		6	4
		2	7
			8
			1
3	0	2	5

So what did we get? (3025)

We've compared the value of squares 1-10 with cubes of the values 1-10.

End

October 28, 2004

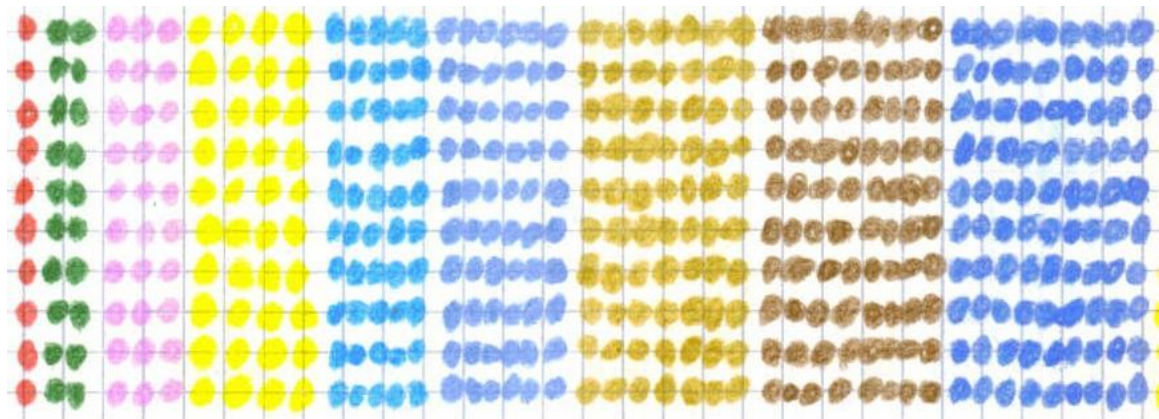
Game II

Prologue

There is no hidden fact to discover here; it is just for the fun of seeing the patterns in the numbers. We note at the end, all the differences came out to two.

Material

Bead bars 1-9, 1 square for each of number 1-9. Felt mat and paper and pencil.



Begin

(start by setting out multiplication tables.

One take 1 time is 1 (begin far left column of 1-bars, proceed top to bottom)

1 taken 2 times is 2

:

2 taken 1 time is 2 (begin second column from far left, top to bottom)

2 taken 2 times is 4

:

3 taken 1 time is 3 (begin third column from far left, top to bottom)

3 taken 2 times is 6

3 taken 3 times is 9

:

(continue with 4, 5, 6, 7, 8, 9 building a grid of bead bars)

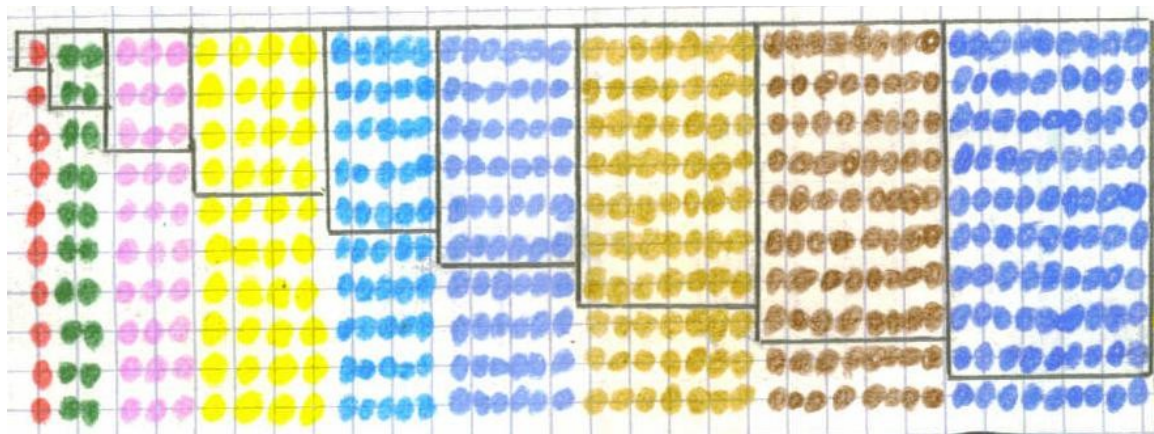
(replace all squares on the top of the grid, beginning top left)

Here I see the square of 1. Let's exchange it for a real square of 1.

Here I see the square of 2. Let's exchange it for a real square of 2.

: : : : :

(continue to square of 9)



Here we have the square of 1 and 1 taken 9 times. (point to 1st column)

Here we have the square of 9 and 9 taken 1 times.(point to 9th column)

Here, we have the square of 2 and - 2 taken 8 times. (point to 2nd column)

Here we have the square of 8 and - 8 taken 2 times. (point to 8th column)

Here, we have the square of 3 and - 3 taken 7 times. (point to 3rd column)

Here we have the square of 7 and - 7 taken 3 times. (point to 7th column)

(**continue** pattern for 4 and 6, **NOT** 5)

Let's write down on paper what we have here. (see layout below)

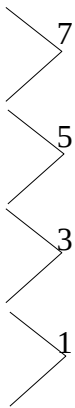
We have:

$1^2 + (1 \text{ taken } 9 \text{ times, gives us } 9)$
 $2^2 + (2 \text{ taken } 8 \text{ times, gives us } 16)$
 $3^2 + (3 \text{ taken } 7 \text{ times, gives us } 21)$
 $4^2 + (4 \text{ taken } 6 \text{ times, gives us } 24)$
 $5^2 + (5 \text{ taken } 5 \text{ times, gives us } 25)$
 $6^2 + (6 \text{ taken } 4 \text{ times, gives us } 24)$
 $7^2 + (7 \text{ taken } 3 \text{ times, gives us } 21)$
 $8^2 + (8 \text{ taken } 2 \text{ times, gives us } 16)$
 $9^2 + (9 \text{ taken } 1 \text{ time, gives us } 9)$

Starting here at 5^2 and working our way up. What the difference between 24 and 25? (1)

What's the difference between 24 and 21? (write it and continue upward)

$1^2 + 9$
 $2^2 + 16$
 $3^2 + 21$
 $4^2 + 24$
 $5^2 + 25$



$6^2 + 24$

$7^2 + 21$

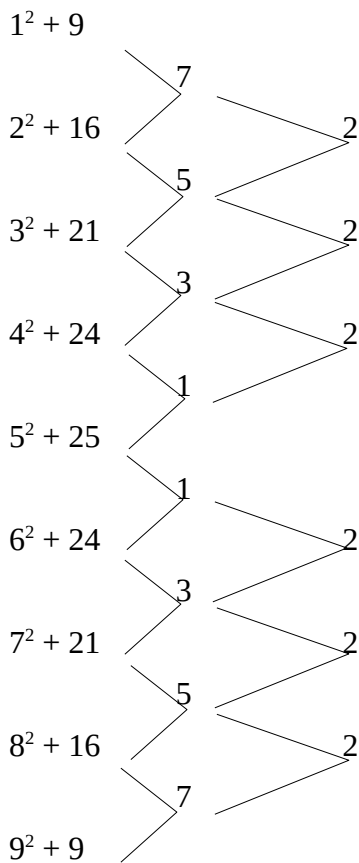
$8^2 + 16$

$9^2 + 9$

Now, what's the difference between 1 and 3? (write bracket and 2)

What's the difference between 3 and 5? (write another bracket and 2)
 (continue for all of top half of the squared numbers)

(we began at 5th row and worked up; now begin with 6th row and work down. Do this pattern for last two subtraction steps)



End

Epilogue

It is these kinds of patterns that children are fascinated with.

October 28, 2004

Game III

Material

Paper and pencil.

Prologue

This is game three using the squares of numbers. Other exponents and numbers can be used for exploration.

Begin

We start across the top of our paper writing $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2, 10^2$

Let's find the value for each now. (write values underneath)

What the difference between 1 and 4? (write difference between values for all underneath)

: : :

What the difference between 81 and 100?

Now we'll find the difference again, between the numbers in the differences we just wrote.

What's the difference between 3 and 5? (first two)

: : :

What the difference between 17 and 19? What do we get. Yes! A difference of 2 for each.

1^2	2^2	3^2	4^2	5^2	6^2	7^2	8^2	9^2	10^2
Find values of expressions above									
1	4	9	16	25	36	49	64	81	100
Find difference between adjacent numbers above									
3	5	7	9	11	13	15	17	19	
Find difference between adjacent numbers above									
2	2	2	2	2	2	2	2	2	

Now, let's see what we get with cubes. What do you think we will get this time?

Let's find the value for each cube first (write values underneath)

Now let's find the different between the value like we did with squares.

What the difference between 1 and 8? (write difference between values for all underneath)

: : :

What the difference between 729 and 1000?

Now we'll find the difference again, between the numbers in the differences we just wrote.

What's the difference between 7 and 19? (first two)

: : :

What the difference between 217 and 271? What do we get. Yes! 54.

Well, we did not get the same difference among all number like we did before. Let's try this one more time. We'll find the difference again, between these new numbers.

What's the difference between 12 and 18? (first two)

: : :

What the difference between 48 and 54? What do we get. Yes! A difference of 6 for each

1^3	2^3	3^3	4^3	5^3	6^3	7^3	8^3	9^3	10^3
Find values of expressions above									
1	8	27	64	125	216	343	512	729	1000
Find difference between adjacent numbers above									
	7	19	37	61	91	127	169	217	271
Find difference between adjacent numbers above									
	12	18	24	30	36	42	48	54	
Find difference between adjacent numbers above									
	6	6	6	6	6	6	6	6	

.

Maybe you'd like to experiment with numbers like this! See what new things you can find.

End

Epilogue

If we were to do this for all the numbers (exponents to values to difference to difference ...) what other patterns would we see. What are the difference with other numbers and how many layers do we need to go down – finding the differences between successive rows for a given number – to find an even result each time?

The pattern is:

Power (or # layers) times previous constant different equals new constant difference.

<u>Power</u>	<u>Number of Layers Required to Arrive at Constant Different</u>	<u>New Constant Difference</u>
2 nd	2	2
3 rd	3	6
4 th	4	24
5 th	5	120
6 th	6	720
7 th	7	5040
:	:	:

October 29, 2004

Decanomial Square

Age

6

Material

Many bead bars 1-10(B), felt mat, all squares and cubes from the bead chain cabinet, paper and pencil for later extensions.



First stage layout – multiplication facts table

Prologue

This is a game called The Decanomial Game. Use about 3 children to build the first stage layout which is a grid of beads – 1B to 10B left to right and 1B to 10B top to bottom. Layout carefully to line up so as to appear visually as the Decanomial square.

The procedure for the three stages is as follows:

1. Layout all beads in a multiplication facts table
2. Replace all squares along the diagonal from 1 to 100 with an actual bead square.
3. Verbalize commutative equalities in the table while replacing horizontal beads along left edge – turning the new beads vertically
4. Work along each row from the left edge to each diagonal square (e.g., 5x5), then work downward from top to that same diagonal square; in each case, form new squares from bead bars – then replace squared-bars with true squares, including the diagonal square itself.
5. Stack squares above and left of each diagonal square onto the diagonal square
6. Replace stacked squares with real cubes and stack cubes into a tower

Begin

1 taken 1 time is 1, 1 taken 2 times is 2, continue...

(lay beads horizontally but each value laid in a column – moving down the leftmost column on the table)

1B (1 taken 1 times is 1)

1B (1 taken 2 times is 2)

1B

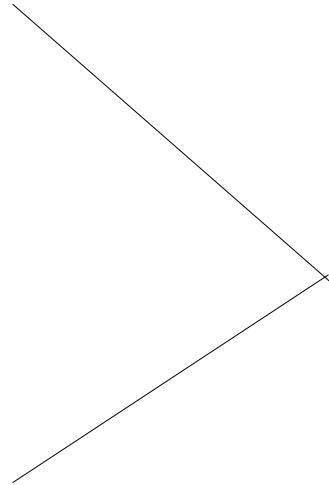
1B (1 taken 3 times is 3)

1B

1B

:

1 taken 10 times is 10



First column

Who would like to do the table for 2? (repeat as above for 2)

2B (2 taken 1 time is 2)

2B (2 taken 2 times is 4)

2B

2B (2 taken 3 times is 6)

2B

2B

:

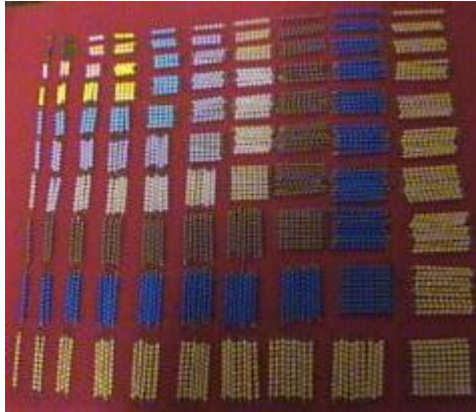
(continue as above to 10)

(continue building table for 3 and so on moving left to right **saying math fact out loud**)

Now let's replace the squares we can see on the diagonal with a true square.

(end of stage 1)

(**stage 2** – proceed from 2-1B in left column – moving left to right up to the diagonal square in each row – row by row, top to bottom. As you go, verbalize the commutative equality for each corresponding expression (e.g., 1x2 in second row left column and 2x1 in top row second column) and replace 2-1B with 1-2B vertically)



1 taken 2 times is the same as 2 taken 1 times (replace beads at left, vertically)
(stop at square)

1 taken 3 times is the same as 3 taken 1 times (replace beads at left, vertically)

2 taken 3 times is the same as 3 taken 2 times (replace beads at left, vertically)
(stop at square)

1 taken 4 times is the same as 4 taken 1 times (replace beads at left, vertically)

2 taken 4 times is the same as 2 taken 4 times (replace beads at left, vertically)

3 taken 4 times is the same as 4 taken 3 times (replace beads at left, vertically)
(stop at square)

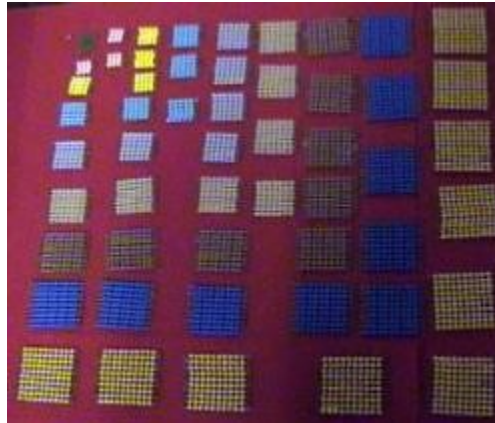
1 taken 5 times is the same as 5 taken 1 times (replace beads at left, vertically)

: : : : : :

(continue up to 10)

(end of stage two)

(**stage 3** - original squares stay on the diagonal. Form new squares to top and left.)



(**point to second row, second column from top left corner of grid**)

Here I see a square of 2. Let's exchange these for a real square.

In the 3s, I can see three squares of 3 altogether.

If I push these together (above square) I get a square of 3, let's exchange for a square of 3

And if I push these together (left of square) I get a square of 3, let's exchange that too.

In the 4s, I see two squares of 4 if I combine starting from the top and left edges.

If I push these together (leftovers from left and top) I get a third square of 4.

Let's exchange those for a square of 4 and place it above our fourth 4-square on the diagonal.

Who would like to continue forming squares for 5s?

(Finally – to finish stage 3)

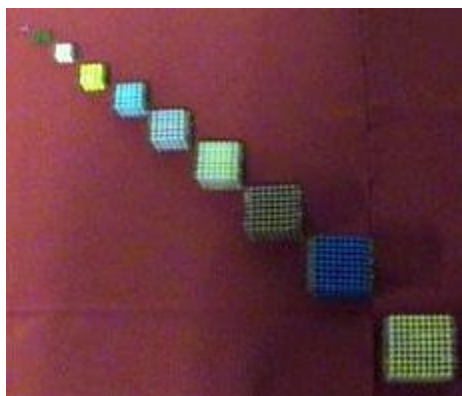
The square of 2 taken 2 times gives the cube of 2 (place 2 2-squares stacked)

The square of 3 taken 3 times gives the cube of 3 (place 3 3-squares stacked)

The square of 4 taken 4 times gives the cube of 4 (place 4 4-squares stacked)

The square of 5 taken 5 times gives the cube of 5 (place 5 5-squares stacked)

(up to ten)



Now watch. (replace each stacked squares with cubes and stack them into a tower)



What law was I using here? (commutative law)

Maybe you'd like to start at this point and work backwards!

End

Extension – angles of color

Another repetition of multiplication facts. Draw stage 2 layout on graph paper. Remember layout after exchanging using commutative. We have angles of color. Draw that on graph paper and express the numeric multiplication right on the paper underlying each replacement of beads. “2 times 1 is same as 1 times 2.” etc.

Extension – 1+2+3

Recalling the initial layout: $(1+2+3+4+5+6+7+8+9+10)^2$. This is what we had and this looks like $(1+2+3+4+5+6+7+8+9+10) \times (1+2+3+4+5+6+7+8+9+10)$

We could start by multiplying through by 1. $1^2 + (2 \times 1) + (3 \times 1) + (4 \times 1)$, etc. We do not use beads at this point.

Extension - a+b+c

Instead of using numbers, let's use letters. So I would have

$$(a+b+c+d+e+f+g+h+i+j)^2$$

This means we'd have

$$(a+b+c+d+e+f+g+h+i+j) \times (a+b+c+d+e+f+g+h+i+j) =$$

(multiply each in the multiplicand times each multiplier, combining like terms)

$$a^2 + ab + ac + ad + ae + af + ag + ah + ai + aj + \quad (a \text{ times all})$$

$$ab + \quad b^2 + bc + bd + be + bf + bg + bh + bi + bj + \quad (b \text{ times all})$$

$$\quad ac + \quad bc + \quad c^2 + cd + ce \dots (c \text{ times all})$$

October 29, 2004

The Paper Decanomial

Material

Ten envelopes: first labeled Decanomial squares are squares of each number – like the original squares that appeared on the diagonal in the Decanomial game. These squares are yellow and the value is printed in the middle of each.

The other 9 envelopes are labeled Decanomial-1, Decanomial-2 ... up to 9. In each of these are any of the combinations that would appear vertically and horizontally from each particular square 1-9. All of these are rectangles, are white and list their values.

Prologue

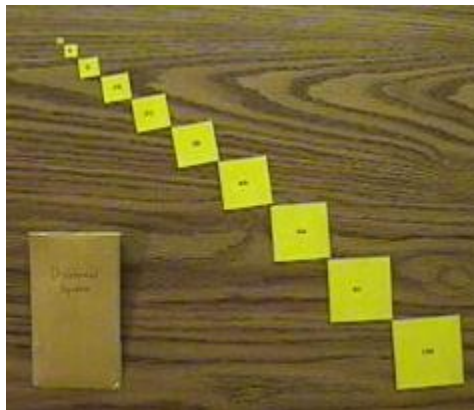
Begin with envelope labeled 'Squares.'

Begin

Let's see what we have in this envelope. I have the square of 1 which equals 1. I have the square of 2 which equals 4. The square of 3 equals 9.

(place each square on a diagonal as was seen on the Decanomial square.)

... The square of 9 is 81 and the square of 10 is 100.



First completed envelope – Decanomial Squares

(we set squares on the diagonal and now choose whichever envelope we want to do next)

(now begin filling in rows and columns surrounding the squares – **to right and bottom**)

Let's find the envelope for 5. (dump out all the contents)

6 taken 5 times is 30. 5 taken 6 times is 30

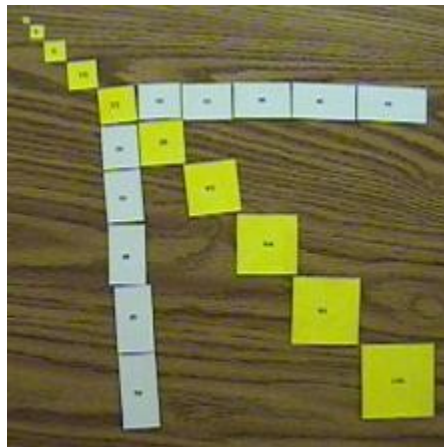
(place rectangles to right and bottom of 5 square)

7 taken 5 times is 35. 5 taken 7 times is 35.

8 taken 5 times is 40 and 5 taken 8 times is 40.

9 taken 5 times is 45 and 5 taken 9 times is 45.

10 taken 5 times is 50 and 5 taken 10 times is 50.



Second completed envelope – “Decanomial-5”

Finish by doing envelopes 2-4 and 6-10 in the same manner – rows and columns for each – **to right and to bottom**)



Completed Paper Decanomial

End

Epilogue

For a challenging follow-up, mix up all the white rectangles and the yellow squares and then reconstruct the Decanomial square from scratch.

Be sure the children know how to pick up and repack the squares and rectangles. Choose any square (e.g., 6) and take all rectangles to its right and bottom, placing them into its envelope (i.e., 6). Repeat for all squares. Lastly, pack up all squares and place them into the envelope labeled “Decanomial Squares.”

October 29, 2004

Sums Using Squares and Cubes

Purpose

Gives the idea that all four operations can be done with squares and cubes of numbers. Indirectly, this provides the children further experience handling squares and cubes and prepares them for doing this work abstractly and algebraically.

Age

Decanomial work can begin around 6 and this lesson occurs around 7-8.

Material

Felt mat, some bead squares and cubes of your choice, paper and pencil – at a certain moment we need an index card or a piece of paper like that, green skittles from the racks and tubes set.

Prologue

Present addition and subtraction on one day and do multiplication and division on a different day.

Begin

Addition

Suppose we have $3^2 + 5^2 =$

(child writes on paper what we write on the board)

Well here is 3 squared and here is 5 squared (set each square on the mat)

We want to add them together. I can't put these together since they are squares.

But do we know the value of each? How many are here? Yes, 9. How many in this? Yes, 25. Now can we add?

Yes, 9 plus 25 equals 34. Is that how many beads I have on my mat? Yes, you could count them if you like. 34. (write $9 + 25 = 34$)

$$3^2 + 5^2 =$$

$$9 + 25 = 34$$



Suppose we have $4^2 + 8^2 = ?$ Will this present a difficulty? I can set it on my mat again. I can't just put them together can I? But we know the values. What's the value of this? (16)

And what's the value of this one? Yes, 64. (write $16 + 64 = 80$)

So if we add them I have 80. Is that how many beads I have on my mat? Yes, so we just find the value of each and then add them together – simple as that.

What about a problem like this?

$$2^2 + 5^3$$

Again I can set them on my mat but I can't just put them together. But we know their values. 2^2 is 4 and 5^3 is 125. Altogether we have 129. Is that what I have on my mat? Yes. (write $4 + 125 = 129$)

You can make whatever problem you want to make and work it out in the same way.

Subtraction

What about this problem: $9^2 - 2^2$

Can we do subtraction? What do I need on my mat? Just the square of 9 since we only take out what we start with for subtraction. This is why I brought my piece of paper.

We want to take away a square of 2. So I'll just cover it up; I've taken it away.

(the paper covers the corner of the square of 9 – covering a square of 2) Now what is our answer? We can just count them.

We can also work out this problem using what we know about the values. We can subtract 4 from 81 to get 77. Is that how many remains uncovered there? Yes.

(write “ $81 - 4 = 77$ ”)

What about $8^2 - 5^2$? What do we need on our mat? Yes, the square of 8. Using my piece of paper we can again show this – covering a square of 5 from one corner of the 8-square.

Can we use the values too? Yes, $64 - 25$ is 39. And that's what remains uncovered on my mat.

(write “ $64 - 25 = 39$ ”)

(let them work repeat new problems for practice)

End



Begin

Multiplication

Suppose we had $5^2 \times 3$?

Well we can set out 5 squared 3 times. We could count all those beads if we wanted to but we already know what 5 squared is worth, right. 5^2 is 25 and we're multiplying that times 3 – so we get 75.

Is that how many beads are on the mat? Yes. We can count it.

We could also do this: $8^2 \times 2^2$

What's the next thing to do? I know what kind of squares but do I know how many to set out?

I first have to find the value of my multiplier. This happens to be what 2^2 is 4.

So we can restate the problem as $8^2 \times 4$. Now we could count up the value of the beads or find the value of the multiplicand – which is 64 – and multiply by 4 – and we get 256.

Is that how many beads I have on my mat.? Yes, again we could count them.

(within limitations of the materials, let the children make examples for multiplication)

Division

Let's look at a division problem.

(be sure to choose a problem you can actually work out with the material.)

So let's take $4^3 / 4$. Well I know I begin with my cube of 4. What else do I need? Yes, 4 skittles. So let's set those out.

(set a row at top of mat of 4 skittles - in this case I can exchange my cube of 4 for 4 squares of 4 – be sure you can work out your problem before beginning)

We can share these out by exchanging - and see that the answer is 4^2 since each skittle got a square of 4 or 16.

So we can say $4^3 \div 4 = 4^2$



Let's try this problem.

$3^3 + 2 \times 3^2 - 6^2$ (write equation on paper)

We need to find values to work this out, right?

$27 + 2 \times 9 - 36$.

There's something you need to know about the order of working out a problem like this.
When there are different operations in one problem there is an important order that must be followed. We do the multiplication and division first and then do any addition or subtraction that remains.

So we must do our 2×9 first. And we are left with $27 + 18 - 36$. And we get 9.
(write the equation and answer - here we introduce precedence – PEMDAS)

End

Epilogue

As time goes on, use parentheses to indicate the precedence when the problem is first introduced so they begin to automatically see it.

November 1, 2004

9. SQUARING

Transformation of a Square

Part I

Age

Begins around 7 or $7\frac{1}{2}$. If a child has had a solid primary foundation, it could start earlier.

Prologue

This work often runs parallel to multiples and factors. We begin with a square of ten transformed into a binomial. With a small group of children.

Material

A few squares of ten – plastic or glass if possible. Some rubber bands of various colors. Felt mat (red), paper and pencil.

Purpose

Preparation for true squaring and for square roots.

Begin

Here we have square of ten (place on mat).

I wonder if we can get any other squares out of this square of ten? Let's find out.

(begin in the upper left corner when counting, left to right)

I'm going to count and place a colored rubber band (count to 4, place rubber band around the square at location of 4th bead.)

Beginning in the same place, let's count down 4. (do so) And again, I'll mark this place with a rubber band. (rubber bands now form a cross when looking at the square.)

Let's see what we have. What do I have in this corner. (a square of 4). Yes, a square of 4. Do you see any other squares here? Where?

Yes, on the diagonal of the big square, we have another one. It's the square of 6, right? Let's count it and make sure.

What else do I have besides these two squares? I have 2 rectangles, don't I. I have one in this corner and one in the other one. (the two corners that are not square)

Shall we do another one? Let's count across to 7 this time. (count and mark with rubber band)

Now in the same way, I'll count down 7. Hold your finger there and let's put a rubber band on it. (again rubber bands form a cross atop the original square).

What squares have I made here? Square of 7, right. And also a square of 3 on the other corner. Besides the 2 squares, again I have 2 rectangles.

I started off with a square of 10. And I changed it into a square of 7 and a square of 3. Let's check that. Here we have 7 + 3. Here we have 7 + 3. Here we have 7 + 3 and here we have 7 + 3. So we have a square of (7 + 3).

Let's see how we'd write that on paper.

$$10^2 = (7+3)^2$$

Now let's see what all makes up our square of 7 + 3. What do we have here? Yes, 7 squared. And here we have what? 3 squared. And what else did we say we had?

Two rectangles, yes. What's the size of our two rectangles. We have 3 x 7 and this is also 3 x 7). We have 2 rectangles each, 3 x 7.

$$\begin{aligned} 10^2 &= (7+3)^2 \quad (\text{write this on paper}) \\ &= 7^2 + 3^2 + 2(3 \times 7) \\ &= 49 + 9 + 42 \\ &= 100 \end{aligned}$$

End

Epilogue

We present this as a square of 10 because this presents the most possibilities. As the children begin working on their own, you don't have use 10. You can use a square of 8 or 9. Encourage them to try the other squares as well.

November 1, 2004

Part II – From Square of Ten into a Trinomial

Begin

Let's see if we can get any more squares out of our square of ten.

Let's start here and count over 2 – and put a rubber band.

Let's count over 3 more and put another rubber band (two going the same direction)

Now let's count down in the same way. One two, and we place a rubber band.

And again we count down 3 – and place another rubber band.

(we now have a grid of 9 spaces on the original square)

This time we started with a square of ten and we transformed it into a square of 2 plus 3 plus 5. Shall we check each side. $2 + 3 + 5$. $2+3+5$. Again, $2+3+5$. $2+3+5$.

So we have a square of $2+3+5$. (we go right into writing since we did it already)

Let's see what we have.

$$\begin{aligned} 10^2 &= (2 + 3 + 5)^2 \\ &= 2^2 + 3^2 + 5^2 \text{ (point to each square on the grid)} \end{aligned}$$

And how many rectangles do we have this time? Can we identify them?

The first one is 3×2 (3 by 2) and there is a second the same

So we have two rectangles 3×2 . If we look at the other rectangles we see we have two more that are the same and finally a third set of rectangles the same in size.

$$\begin{aligned} 10^2 &= (2 + 3 + 5)^2 \\ &= 2^2 + 3^2 + 5^2 \\ &= 2^2 + 3^2 + 5^2 + 2(3 \times 2) + 2(5 \times 2) + 2(3 \times 5) \end{aligned}$$

What is the value of 2 squared? The value of 3 square? (verbalize solving the equation as you write it on paper)

$$\begin{aligned}
10^2 &= (2 + 3 + 5)^2 \\
&= 2^2 + 3^2 + 5^2 \\
&= 2^2 + 3^2 + 5^2 + 2(3 \times 2) + 2(5 \times 2) + 2(3 \times 5) \\
&= 4 + 9 + 25 + 12 + 20 + 30 \quad (\text{"If we add them all up what do we get"}) \\
&= 100
\end{aligned}$$

End

Epilogue

Leave $3 + 3 + 4$ for the children to discover. Or $4 + 4 + 2$? These are special because there will be 4 rectangles that are identical. If you do a binomial, $5 + 5$ would also be interesting since it would be 2 squares and 2 rectangles. This leads to a little discussion. We express it the same – two squares and two rectangles – even though there is 4 squares; we are establishing a pattern we need to emphasize.

November 1, 2004

Part III

Prologue

This extends the work with the pattern we established in the first two exercises. Construct a binomial square abstractly by drawing and coloring.

Materials

Printed circles (100) on a small square paper and colored pencils.

Begin

Here we have a paper square of ten. Let's see if we can make some squares out of this too.

(begin with a binomial)

Let's see if we can make this into $3 + 7$ squared. (choose first colored pencil)

(again start upper left)

We'll color as we go. 1, 2, 3. And I'll count down and color in 3 also. (all one color)

I have made my first square now. (color in the whole square)

I said I wanted to make $3+7$ squared, so what do I need now? Yes a square of 7. **Where do I need to start coloring?** (begin on the diagonal – at the corner of first square)

How do we know this is where we start? In all the work we've done in the previous exercises, we always began counting on the diagonal. (we counted differently this time, not along the sides as before - but we begin from the next point diagonal (down and right) . Now we have to know – the rubber bands don't form it – we do)

Now let's color in the circles for our second square. (do so)

Let's choose a third color and color in our two rectangles. (do so)

End

Epilogue

Because the children have been writing the previous two exercises, they may want to write here as well. After coloring, paste your square to one side of a half piece of paper and then, to the right of it, do the writing as we did in prior exercises; this leads to making a booklet of these.

November 1, 2004

Part IV

Material

Graph paper, a ruler, lead pencil and colored pencils.

Prologue

This provides even more experience on polynomials.

Begin

We've been doing all kinds of work transforming a square of 10 into other squares. Today, we'll draw the same idea into graph paper.

(choose whatever square we want)

Let's start with a square of $4+6$.

On graph paper, let's use our ruler to count across 4 – and make a dot.

(count as we did in the first two exercises – along the top edge to right)

And now I'll count from there over another 6 and make another dot.

Now from my first starting position, I need to count down 4 – and make a dot.

And then I'll count down 6 more – and make another dot.

Now, I'll complete my square. (fill in outline of main square). And then I need to connect my dots to complete my two squares. (... connecting each point at positions 4 – with lines to opposite sides)

Let's choose our first color and color-in our square of 4. I'll choose a second color for my square of 6 and a third color for my two rectangles. (color quadrants uniquely among the three colors – the two rectangles are the same color)

Now right on top of each square, we can add notation.

(write on right side of the paper)

4^2	4×6
6×4	6^2

$$\begin{aligned}(4+6)^2 \\&= 4^2 + 6^2 + 2(4 \times 6) \\&= 16 + 36 + 48 \\&= 100\end{aligned}$$

(the value 100 is our control of error since we began with a square of 10)

End

Epilogue

What's fun about introducing this on graph paper is that we can go beyond squares of 10. We can do a square of 12 and so on. We can do trinomials and quadrinomial – whatever we want at this point, since they have the concepts involved.

November 1, 2004

Part V – Algebraic Formula for a Binomial

Material

Bead squares of ten, colored rubber bands, felt mat, blank tickets, pencil and parentheses.

Prologue

This is a lessons that introduces assigning variables to the binomial work we have done – generalizing what we have learned and moving into algebra.

Begin

Can you please make a square of $3+7$? (we expect the child to be able to do this now)
(count over 3, place rubber band, count down 3 and rubber band.)

We've changed our square of 10 into a square of $3+7$ now.

Now, instead of 3, suppose we call this a (point to the edge of the square of 3). And instead of 7, we're going to call this b (point to the edge of the square of 7).

If we do that, what does this first square become?

It becomes a^2 . Let's write that on a ticket and place it on the square itself.
(place ticket ' a^2 ' on our square of 3)

What is this second square called then? Yes, it's called b^2 . (place ticket ' b^2 ')

What would this rectangle be called then? Yes, $a \times b$ – let's write a ticket for that.

I've got one more rectangle. Again, it's also a times b , so let's write another ticket for that.
(place two tickets ' $a \times b$ ' on each rectangle)

(now we've labeled each quadrant – a^2 , b^2 , $a \times b$ and $a \times b$)

Now let's look at what we have all together. (pull tickets down onto mat)

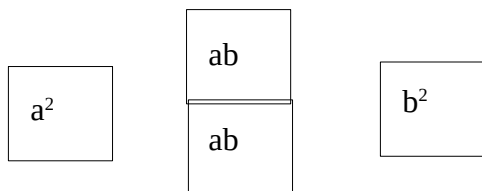
When we use algebraic notation, we don't need a multiplication sign. All we have to write is ab . So, let's get rid of these multiplication signs.

(turn over the tickets “a x b” and write “ab” instead)

So let's see what our square of $3+7$ has become.

(write ticket “ $(a + b)^2 =$ ” and place it on mat below the square)

(pull tickets from the squares and arrange as follows)



How many times do we have ab ? Yes, two times. So we can write this $2ab$.

And this gives us (verbalize the expressions as you write “ $2ab$ ” and the two plus signs)

$$a^2 + 2ab + b^2$$

End

Epilogue

Now we would expect the children do other examples; the algebraic notation, ticketing the pieces, pulling them down and deriving the same algebraic formula each time.

November 1, 2004

Part VI – Algebraic Formula for a Trinomial

Begin

Here's our square of ten. Can you make it into $3+5+2$ squared?

I'm here to help you if you need help.

(Count beads over and down the sides and place rubber bands at positions 3, 5 and 2)

We now have $3+5+2$ squared. (Just like we did in Part II)

Now instead of 3, we'll call this a (point to the edge of the square of 3)

Instead of 5 we'll call this length b . (point to the edge of the square of 5)

Instead of 2, we'll call this c . (point to the edge of the square of 2)

Let's write tickets for each quadrant now.

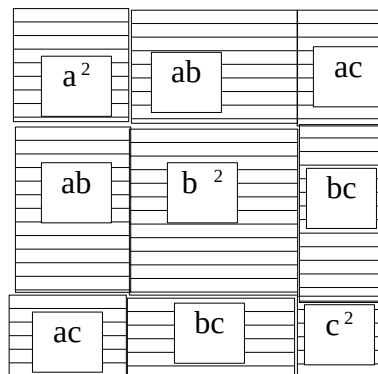
The first square is a^2 .

The second is b^2 .

And we have a third square. Let's write a ticket – c^2 .

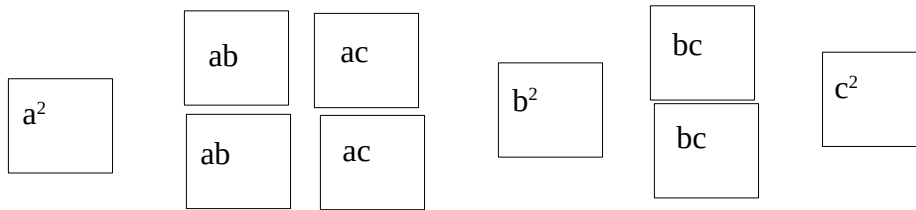
Now we have some rectangles to ticket also. Let's start with this.

(ticket two ' ab ' rectangles, two ' ac ' rectangles and two ' bc ' rectangles)



Let's see what we have here altogether.

(pull tickets down) a^2 , ab , ab , ac , ac , b^2 , bc , bc and c^2 (place like tickets in columns)



So let's write out our expression. (write ticket " $(a + b + c)^2 =$ " and plus signs)

(verbalize our expression as you construct it with existing tickets)

$$(a + b + c)^2 = a^2 + 2ab + 2ac + b^2 + 2bc + c^2$$

Epilogue

For follow-up work, they can use other bead squares. They can draw it on graph paper, but now using algebraic notation.

Every time the child divides a square into a binomial, he ends up with 4 quads – two squares and two rectangles. And two times two gives us four. Every time the child does a trinomial, he ends up with 9. Out of those 9 pieces, he gets 3 squares and 6 rectangles. And 3×3 gives us 9. If we try a quadrinomial, we'll end up with 16 – four of which are squares. And 4 times 4 equals sixteen.

Some children will begin to notice these properties. They will have other opportunities to see these things after this, if not – there are other chances to see these things. More than likely they will notice that the squares always appear on the diagonal. Other spaces are always filled in by rectangles – their sizes determined by the size of the squares.

November 2, 2004

Passing From One Square to Another

Exercise I

Purpose

Further experience with squares. Preparation for square roots.

Age

Exercise I begins around 7-7 ½.

Material

One square for each of the numbers 1-10. Bead-bars 1-9. Felt mat, paper and pencil.

Prologue

This is another exercise to prepare the child for true squaring and square roots. Begin with a square sort of in the middle. We'll start with a square of four and we want to build this square of four into a square of five. Begin with the two squares on the mat. **Always begin in top left corner when aligning and counting squares.**

Begin

We want to compare these two squares.

(lay 4-square on top of 5 square aligned at upper left corner.)

To build this four-square into a square of five, I need to add a four-bar here (counts along top of square). I also need to add a four-bar here (count along side). We are still missing a single bead in the lower right corner. So, I'll add that too.

(add 2 4Bs and a single bead to create a square of 5)

Now we have a square of five. Let's do another one.



A square of 4 turned into a square of 5 – adding 2 4-bars and a single bead

Let's take a square of six and make it a square of seven. First thing we do is superimpose. It's very clear what we need when we superimpose.

In order to build a square of 6 into a square of 7, we need to add a 6-bar here (bottom) and a 6-bar here (at right edge). We also need to add a bead in the lower right corner. Now let's count the sides.

We have 1,2,3,4,5,6,7 on this side. (repeat for other three sides). Seven on all sides. We do in fact have a 7-square now.

Let's write this down. I started with a 6-square (Speak to each variable in expression)...

(write expressions on paper as you say them)

$$6^2 + (6 \times 1) + (6 \times 1) + 1 \quad \text{(if children say we added a 1-square, just write it)}$$

$$6^2 + 2(6 \times 1) + 1^2$$

What's the value of 6 squared? (solve expression)

$$36 + 12 + 1 = 49 \quad \text{(write answer on paper too)}$$

How do we know 49 is right? Because 7 squared equals 49.

End

Epilogue

Some children will see the added bead as one or a 1-square. It is on the diagonal and this can be notated and written either way.

November 2, 2004

Exercise II

Material

One square for each of the numbers 1-10. Bead-bars 1-9. Felt mat, paper and pencil.

Prologue

Here we're building all of the squares – beginning with a square of 1 - we'll build into a square of 2.

Begin

What's the first thing to do? (superimpose) Yes, we need to superimpose the smaller square onto the larger one. We want to know what we need to add.

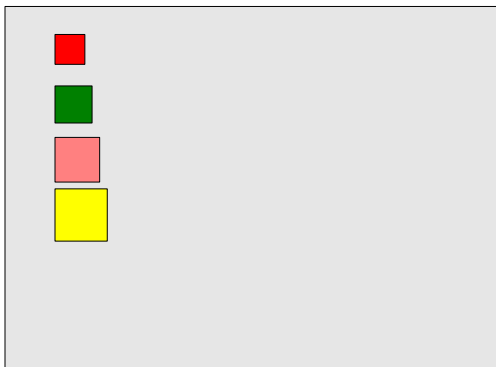
I need to add a 1 here (bottom), a 1 here (right) and a 1 here (corner).

Now, I'll do the square of 2 into the square of 3.

(as above – superimpose, add to bottom and right edges plus the corner as needed to complete each square)

Now, I'll do the square of 3 into the square of 4.

(as above, building column of new squares down left edge of table. Each time, larger square is used for the next one – moving down the table toward you).



See Exercise I for an illustration of how each square is transformed.

Now, I'll do the square of 4 into the square of 5. (repeat to square of 9 to square of 10)

(writes on board)

Now let's write down what we added on each time. We'll start at the top and we'll record the value of each part we added on each time.

(at top)

We added here, $1 + 1 + 1$ which equals 3

$$1+1+1=3 \text{ (written)}$$

We added here, $2 + 2 + 1$ which equals 5

$$2+2+1=5 \text{ (written)}$$

(continues to ten)

Written layout:

$$1+1+1=3$$

$$2+2+1=5$$

$$3+3+1=7$$

$$4+4+1=9$$

(what do we notice about the answers? Yes, they increase by 2!

$$5+5+1=11$$

$$6+6+1=13$$

$$7+7+1=15$$

$$8+8+1=17$$

$$9+9+1=19$$

So each time, the value of what we added on increases by 2. Just another interesting things about these numbers.

End

November 2, 2004

Exercise III - From One Square to a Non-Successive Square

Material

One square for each of the numbers 1-10. Bead-bars 1-9. Felt mat, paper and pencil.

Prologue

Now we can start with any square and build to any other size square.

Begin

Let's start with the square of 4 and let's build it to the square of 9.

So far, we've just built to the next square, but we don't have to stop there.

(lay square of 9 on mat)

We begin again by superimposing (align at upper left corner)

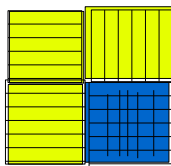
It looks like we have a lot to build this time.

Let's slide this one out of the way (put 9-square to the side at left)

I've got 4. So here's 5, 6, 7, 8 and 9 (added 5 4-bars vertically to right of the four square – which is laid horizontally)

I need to the same thing going down. I've got four here. (lay horizontally below square)

Here's 5, 6, 7, 8 and 9. Now I've got 9 coming down here (added at bottom left)



Now I need to add a 5-square at our bottom right corner. (horizontally laid)

Let's make sure I have 9 on each side now (count to 9 four times for each side)

Yes, I have 9 on each side. I have the square of 9.

Let's go right into writing. What did we start with (write $4^2 +$

Now, let's think about what we added. What did we add first? (write 4×5)

What's the second rectangle we added? Another 4×5 . (write 4×5)

Then what did we add? Yes, 5-squared (5^2)

Layout:

$$4^2 + (4 \times 5) + (4 \times 5) + 5^2 =$$

Do we have some like terms we can put together. Sure. So we can write it:

$$4^2 + 2(4 \times 5) + 5^2 =$$

What's the value of 4-square (fill in each time).

What's the value of 4 times 5 (20) - taken 2 times (40) ? (fill in)

What's the value of 5 square? (fill in)

$$16 + 40 + 25 = 81$$

Why is that the answer we're looking for. Yes, because 9 times 9 is 81.

End

November 2, 2004

Exercise IV

Material

One square for each number 1-10. Bead-bars 1-9. Felt mat, paper and pencil.

Prologue

Now we want to express what we've been doing each time. Let's just imagine the example we're thinking about is our previous one – square of 4 building into a square of 9. *These children are older than 6. They can follow calculations as we go now.*

Begin

The square we start with before we transform it, is our first square.

The square we build to is our second square. We have to have some way to refer to them. Now let's see if we can verbalize what we need to add to our first square, to get to the second square.

We know we start with the square of 4. What do we need to add to it?

How do express it in words, in a general way, that would apply to any example? We need to include the difference between the sides of the first and second squares.

So we need to build out the original square by – the length of the side of the first square - times - the difference between the lengths of the sides of the first and second squares.

And how many times did we do that? Twice.

So, we start with the first square, we add on 2 times the side of the first square times the difference between the sides of the first and second squares. $(2 \times (4 * (9-4)))$

What else do we still need?

The square of - the difference between the length of the sides of the first and second squares.

$$(9-4)^2$$

Suppose instead of all these words, we use algebraic notation. We call our first square a^2 and we build to b^2 . So 4 is 'a' and 9 is 'b.'

We start with a^2 . What did we add on to that first square?

Two times the side of the first square '2a' times the difference between the sides of the two squares (b-a)

Written layout:

$$a^2 + 2a(b-a)$$

But we're not quite done. Now we need the square of the difference between the sides of the first and second squares. So we have $(b-a)^2$

Layout:

$$a^2 + 2a(b-a) + (b-a)^2$$

So let's plug in our numbers and see if this really works. How do we write a^2 arithmetically?

$$4^2 + 2 \cdot 4 (9-4) + (9-4)^2 \quad \text{(Fill in numeric values for variables)}$$

Now let's plug in the answers

$$16 + 40 + 25 = 81$$

Is 81 the square of 9? Yes.

$$a^2 + 2a(b-a) + (b-a)^2 = b^2$$

add final answer b^2

$$42 + 2 \cdot 4 (9-4) + (9-4)^2 = b^2$$

$$16 + 40 + 25 = 81$$

Epilogue

This is a conclusion for exercise three. Not all children will get to this point but for those who can follow, it is good to bring them along to this point if possible.

November 2, 2004

Squaring a Sum

Exercise I - Binomial

Material

Bead bars 1-9, the squares of numbers from 1-9, white number cards, parentheses, blank tickets, felt mat paper and a pencil.

Prologue

Up to this point we began with a square already present, and we did something with it. Here we'll begin true squaring – which is a number taken by itself.

Begin

Suppose we take 4 plus 3. And we're going multiply that by 4 plus 3. (write it)

$$(4 + 3)^2$$

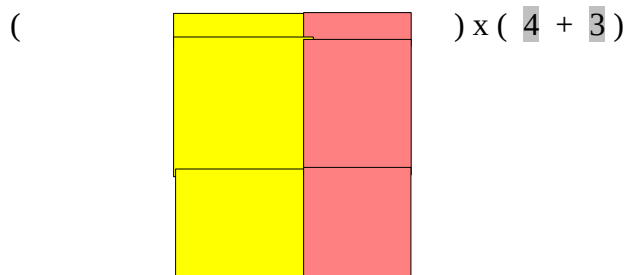
We've seen something like this in the distributive law of multiplication.

(invite the child to work this out. Some children will immediately set out square of 4 while others will set out 4-bars; either is okay)

(format multiplicand using beads and multiplier using cards - isolate multiplier as you go)

We need to take 4 four times. Then we'll take 3 4 times. I'm done multiplying by 4.

Now we need to multiply by 3. We need 4 3 times and 3 3 times.



So what do we have here. If we have a square, it should be equal sides. Let's see.

4+3, 4+3, 4+3 and 4+3. So on every side I can see 4+3. We have 4+3 squared.

Let's see what we have here. On our paper we can write:

$$(4+3)^2 = 4^2 + (3 \times 4) + (4 \times 3) + 3^2$$

What's the value of 4 squared? (write it)

What's the value of 3 times 4? (write it)

What's the value of 4 times 3? (write it)

What's the value of 3 squared? (write it)

So let's add them up

$$16+12+12+9 = 49$$

So what's our control of error? Yes, 7 squared is 49!

So let's do some more problems. What numbers would you like to square?

End

Epilogue

We're building up a pattern here.

November 2, 2004

Exercise II - Trinomial

Material

Bead bars 1-9, the squares of numbers from 1-9, number cards, parentheses, blank tickets, felt mat paper and a pencil.

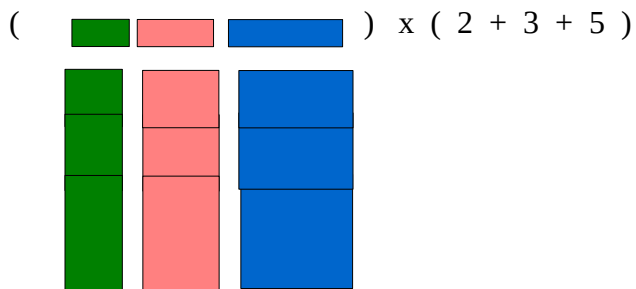
Begin

We're going to take $2 + 3 + 5$ and multiply that times $2 + 3 + 5$. This is a trinomial.

This is $(2+3+5)^2$

Invite the child to do it at this point.

Now let's check our sides: $2+3+5$, $2+3+5$, $2+3+5$ and $2+3+5$. We have a square of $2+3+5$.



Here we have 2 taken 3 times and here is 3 taken 2 times. Let's exchange (replace 3 2-bars with pink 2 3-bars.)

Here I have 2 taken 5 times and here I have 5 taken 2 times. Let's exchange.

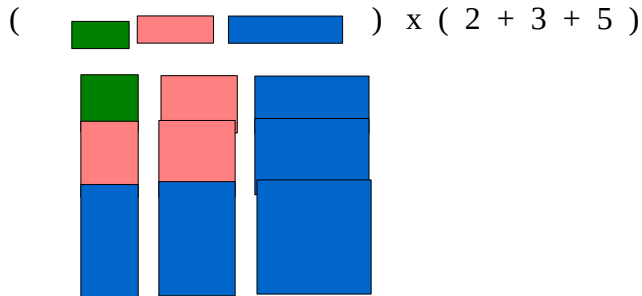
$$(2+3+5)^2 = 2^2 + \dots$$

How would we express those two pink rectangles? (3 times 2, 2 times 3)

Write:

$$(2+3+5)^2 = 2^2 + 2(3 \times 2)$$

What do we have next? A square of 3. What next? 2 times 5 – we'll make that 5 2 times. (two blue rectangles). (exchange) Then two times rectangles of 5 times 3. (exchange 5 3-bars for 3 5-bars at bottom)



So let's see what this all looks like on paper.

$$\begin{aligned}
 (2+3+5)^2 &= 2^2 + 2(3 \times 2) + 3^2 + 2(5 \times 2) + 2(5 \times 3) + 5^2 && \text{(read aloud)} \\
 &= 4 + 12 + 9 + 20 + 30 + 25 \\
 &= 100
 \end{aligned}$$

End

Epilogue

The children again see that the squares always appear on the diagonal (assuming we exchange correctly). Every time we do a binomial we get 4 pieces – two of which are squares. There are similar rules for trinomials and so on. If the children didn't get these impressions yet, they will hopefully realize these things here.

From here, we could go in two different directions: algebraic expressions or hierarchical value.

November 2, 2004

Exercise III - Algebraic Expressions

Material

Bead bars 1-9, the squares of numbers from 1-9, white number cards, parentheses, blank tickets, felt mat paper and a pencil.

Prologue

Begin with a binomial

Begin

Let's take 4 plus 5 and multiply that by 4 plus 5.

$$(\boxed{4} + \boxed{5}) \times (\boxed{4} + \boxed{5})$$

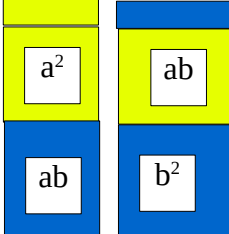
(invite the child to build the problem and to set the squares. If they use beads, we replace the squares where they exist afterward.)

Today, suppose we call this a (point to 4) and instead of 5, we're going to call this b.

I'll write some tickets to help me remember., We'll call 4 a and we'll call 5 b.

(write tickets and place above 4 and 5 in original beads and number cards – both)

(ticket the two squares in the bead results)

$$(\boxed{a} + \boxed{b}) \times (\boxed{a} + \boxed{b})$$


How will we call this rectangle? a times b. How about this rectangle, also a times b.

(either gather tickets on the mat or do the writing on paper – looks same either way)

We want to see what it looks like altogether. We took a plus b and multiplied by a plus b

$$(a+b) \times (a+b) = a^2 + 2ab + b^2$$

All of this all together equals a plus b squared. (write $(a+b)^2$)

End

Epilogue

Again, we can do a trinomial. We aren't limited.

Suppose we have $4+5+2$. Exactly the same as binomial lesson above except for the fact of using a trinomial. Delay all terms c in the algebraic expression to the statement end, for clarity. Lay out bead products with squares on the diagonal as usual – with commutative-pair rectangles surrounding each square. See Exercise II - Trinomial.

Some children like to do this on graph paper. Invite them to do quadrinomials and beyond. Note that our trinomial comes to 6 terms (instead of the canonical 9 terms) after we combine like terms. Ideas such as “like terms” is clarified through repetition – not through explanation.

Squaring with Hierarchical Values

November 3, 2004

Exercise IA

Age

Around 8 or 8 1/2 depending on extent of previous experience

Purpose

Direct purpose is squaring numbers with hierarchical value. Indirect purposes are preparation for square roots and development of the mathematical mind.

Material

Golden bead material, units, tens and hundreds (no thousands), decimal system number cards, parentheses, signs for operations, a large felt mat, paper and pencil.

Prologue

We begin with the golden bead material since it is most concrete

Begin

Today, let's make a square of 23.

That means we'll start with 20 plus 3

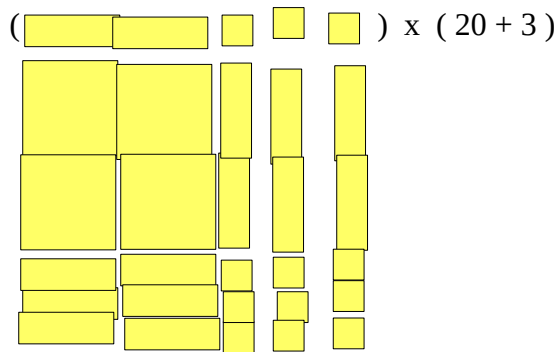
(Isolate the multiplier)

First we multiply each of the tens by 20. (place beads under)

Then we need to multiply each of the units by 20 (place beads under)

I'm done with the twenty (turn over the 20)

Now I need to multiply the 3. (lay product below multiplicand part being multiplied)



Here we have 23 squared. I can check and verify to make sure by checking each side.

If we look here across the bottom. (count sides, counting individual rows in the 100-squares as rows of 23 including the three to their right)

I wonder how much we have here altogether. Let's look and see.

(you can either add up in verbally in our heads or we can identify what we have here and **make a little addition problem**)

How many hundreds do we have? Four – let's write that down.

How many tens do we have? Six here but we have six more down here. 120. Let's write that down.

Then we need to look at how many units we have. How many? Yes, nine.

400

120

9

(see paper notes for layout.)

529

If we add these all up we see we have 529 beads.

What is our control of error? Yes 23 squared. If we multiply 23 times 23 we indeed see that we should get 529. So we have a square of 23.

Would you like to another problem like this? Sure. Now, if we have enough materials, we can do any number we want to.

End

November 3, 2004

Exercise IB

Material

Paper and pencil.

Prologue

We **work on paper** for this one. This can follow the previous exercise on the same day.

Begin

Now let's look at how we would do the writing for this problem only on paper.

Well 23^2 is the same thing as saying $(20+3) \times (20+3)$. That's what we had on our mat.

Remember we learned the distributive law - we're going to apply that here.

We start multiplying by what number? Yes, 20 – the first term.

So first we get 20 times 20.

$$23^2 = (20+3) \times (20+3)$$

$$= 20^2 + (3 \times 20)$$

$$(20 \times 3) + 3^2$$

(now we multiply by 3)

$$= 20^2 + 2(20 \times 3) + 3^2$$

Now we can fill in our values

$$= 400 + 120 + 9 = 529$$

So altogether, we get 529.

End

November 3, 2004

Exercise IIA - Squaring a Binomial Using the Pegboard

Material

Pegboard, box of pegs, small number cards, signs for operations, parentheses, paper and pencil.

Prologue

The first time you present this, show the child the **transition from a real square to a symbolic square (all green pegs in true square to using colored pegs)**.

Begin

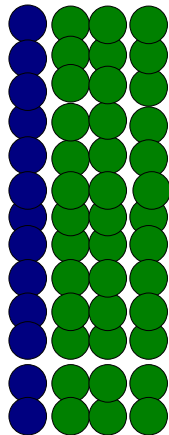
Let's first **build a true square of 13**.

(construct a square of 13 in center of pegboard using green pegs)

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 – (across top, then down 13 rows)

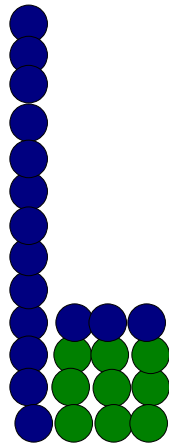
Now we'll **make our square symbolic using blue pegs for each group of ten green**.

(count along the top row, taking ten off in each, and putting them back into the box – replacing each group of ten with a blue bead at far right next to the remaining 3 in each row - for the ten per row we just removed)

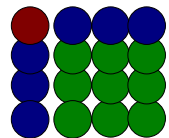


(Step 2, Replace green pegs vertically from first green column beginning top, with single

blue peg for each row)



(Step 3, Replace 10 of the blue pegs, working downward from top left, with a single red peg in upper left corner)



We arrive at a symbolic square of 13. We built a square of 13. We can still see 13 along the bottom and on the right side.

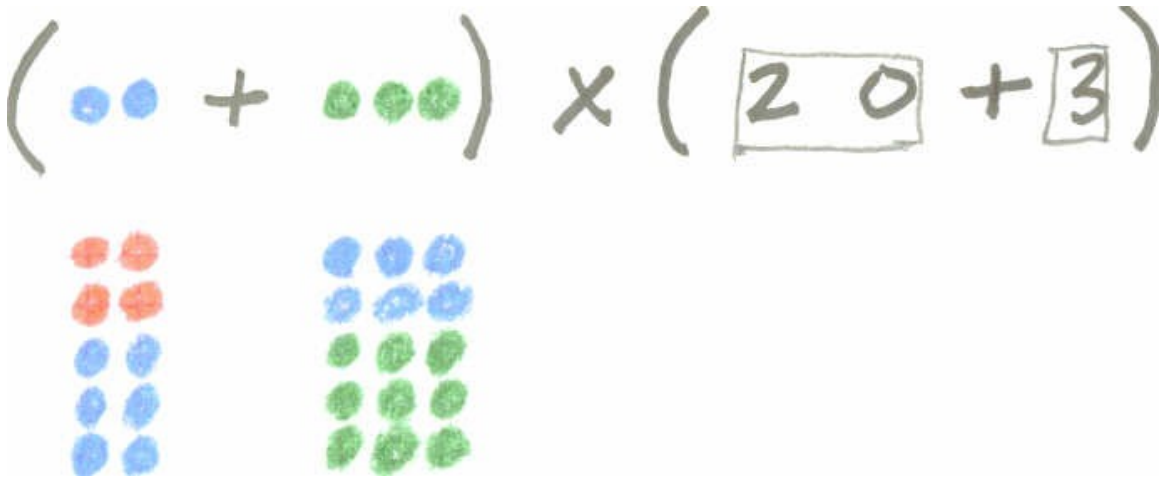
End

November 3, 2004

Exercise IIB

Prologue

We're building the expression on the pegboard here.



Choose a manageable size square to build. This **can follow the previous exercise on the same day**. We progress to symbolic squares at this point.

Begin

Now we're going to square a number on our pegboard

We'll choose 23. Let's isolate our multiplier.

We begin by multiplying 20. (20 times 20)

What do tens times tens give us. Yes, hundreds. 2 times 2 give us how many? Yes, 4. So 4 hundred. That's how I know I need to start with red pegs.

Next we multiply 3 times 20. Units times tens give us what category. 3 times 2 is 6 – so we need 6 tens.

We're done multiplying by 20. Now I'm ready to multiply by 3. So first we have 20 times 3.

Tens times units give us what category? Tens. And 2 times 3 is 6. So 6 tens

Now we have 3 times 3. Units times units gives us units. 3 times 3 gives us 9. And they go right here (see layout on paper notes)

We have built a square of 23. We can read that across the bottom and down the right side we can see 23.

What's our value of the square of 23. Let's add them up.

What's the value of our hundreds? 400 - let's write that.

How many tens do we have? We have 6 tens here and 6 tens there. So we have 120.

How many units do we have? Yes, we have 9 units.

If we add them all up, what do we have altogether?

$$400 + 120 + 9 = 529$$

Now you can do any 2-digit number you'd like, squaring on the pegboard.

End

Epilogue

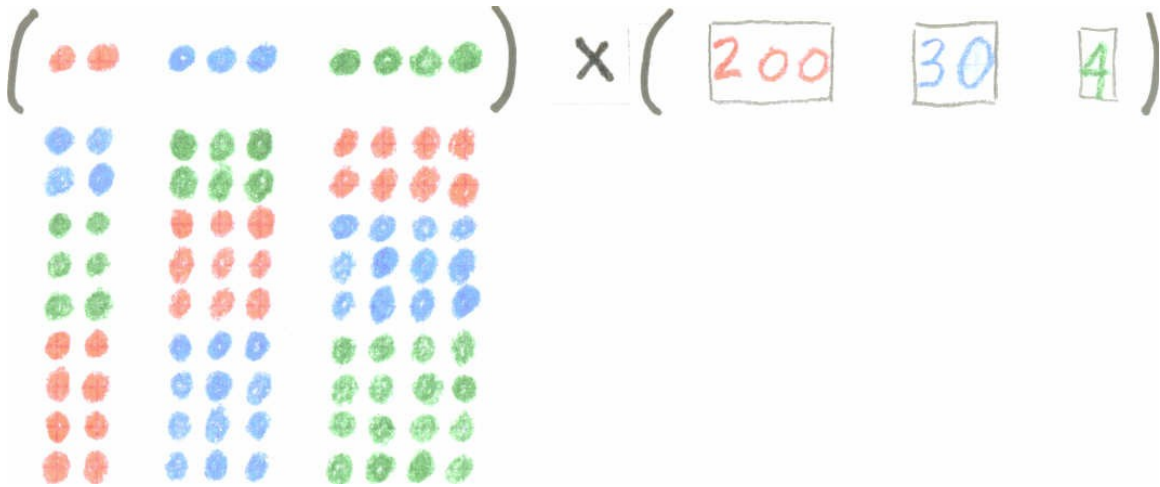
The pegboard is generally very popular with the children and this exercise is also a big draw.

November 3, 2004

Exercise IIIA - Squaring a Trinomial Using the Pegboard

Material

Same as previous except we use the **decimal cards** instead of the paper white cards.



Begin

Let's take 200 plus 30 plus 4 times 200 plus 30 plus 4. (set out the categories and respective values on the pegboard as depicted above)

First we need hundreds times hundreds, that gives us ten-thousands. 2 times 2 gives us 4, so I need 4 ten-thousands.

(help them count the zeros if they don't remember category multiplication)

Now we have tens times hundreds. That gives us thousands. 3 times 2 gives us – 6 thousands.

Hundreds times (continue)

:

(bottom row middle column)

Units times tens gives us tens and 4 times 3 is 12 – so we have 12 tens, and finally

Units times units (last) give us units and 4 times 4 is 16. They go right here (lower right corner now)

We have built the square of 2 hundred 34. We can read that across the bottom and again, down the right side. Now we have to calculate the value.

(begin upper left corner)

What's the value of this? 4 ten-thousands

A handwritten calculation showing the squaring of 234 using a grid method. The numbers are arranged in a triangular pattern, with each row representing a partial product. The rows are: 40,000; 6,000; 6,000; 800; 900; 800; 120; 120; 16. A horizontal line is drawn under the last two rows (120 and 16). To the right of the line, the sum 54,756 is written, followed by an equals sign and the expression 234².

$$\begin{array}{r} 40,000 \\ 6,000 \\ 6,000 \\ 800 \\ 900 \\ 800 \\ 120 \\ 120 \\ 16 \\ \hline 54,756 \end{array} = 234^2$$

Do we have thousands? Yes, here are 6 and here are 6.

How about hundreds. We have 8 here, 9 more in the center and 8 more at bottom left.

Do we have tens. Yes, we have 12 tens here and 12 tens there.

And do we have units? Yes, we have 16.

So let's add these up.

We can see that the value of 234^2 is 54 thousand 7 hundred 56. What could you do to check that answer? Yes, multiply 234 times 234.

(We could work this out in the checkerboard or bead frames if we have not mastered long multiplication. Help them to learn to check their own work)

End

Epilogue

This is the door to everything else in squaring. The children can do any 3 digit number they want to. For some children, they can be shown how to do this on paper. Remember that they get lots of practice along the way, so this is not as challenging to them as it first appears. They can use long-multiplication, the checkerboard, the large bead-frame or flat bead-frame, to check their work. Note: Remember to use the category multiplication language, followed by the simple multiplication. Looks like:

$$\begin{aligned}
 234^2 &= (200 + 30 + 4) \times (200 + 30 + 4) \\
 &= 200^2 + (30 \times 200) + (4 \times 200) \quad (\text{this was times } 200) \\
 &\quad (200 \times 30) + 30^2 + (4 \times 30) \quad (\text{this is times } 30) \\
 &\quad (200 \times 4) \quad (30 \times 4) + 4^2 \\
 &= 200^2 + 2(200 \times 30) + 2(200 \times 4) + 30^2 + 2(30 \times 4) + 4^2 \\
 &= 40,000 + 12,000 + 1,600 + 900 + 240 + 16 \\
 &= 54,756
 \end{aligned}$$

November 3, 2004

Exercise IV

Material

Graph paper, a ruler, a lead pencil, 3 colored pencils – green, blue and red.

$4000 \times 4000 =$ $4000^2 =$ $16,000,000$	$300 \times 4000 =$ $4000^2 =$ $1,200,000$	$20 \times 4000 =$ $80,000$	$5 \times 4000 =$ $20,000$	4000
$4000 \times 300 =$ $300^2 =$ $1,200,000$	$300 \times 300 =$ $300^2 =$ $90,000$	$20 \times 300 =$ $6,000$	$5 \times 300 =$ $1,500$	300
$4000 \times 20 =$ $80,000$	$300 \times 20 =$ $6,000$	$20 \times 20 =$ $20^2 =$ 400	$5 \times 20 =$ 100	20
$4000 \times 5 =$ $20,000$	$300 \times 5 =$ $1,500$	$20 \times 5 =$ 100	$5 \times 5 =$ $5^2 =$ 25	5
4000	300	20	5	

Prologue

As an extension, we can do this work on graph paper. Start with a binomial or a trinomial. For fun, we'll do a **quadrinomial**. This relies on the use of category multiplication.

This exercise reinforces category multiplication since we begin each step in the process from the top left square. This applies to when we begin to color and multiply as well as when we finally collect all the partial products into one addition problem.

Begin

Draw geometric square of quadrinomial

We'll square 4325.

(begin multiplying in upper left corner of our square)

First we count across and mark off our inner squares.

(use rule to mark 4 points on a horizontal line at top, continue marking points to right for 3, 2 and 5)

Now from the upper left corner, we count down in the same way

(leaving a pencil mark each time - count down 4 for the 4000, count down 3 for 3 hundreds, count 2 spaces for the 20 and 5 spaces for the 5)

Now wherever there's a dot we connect the lines. Let's put our numbers down so we know what we're working with.

Color Categories and Multiply

We start with thousands time thousands. What category do we get? Millions. So color this category green (upper left) and I'll do the writing in that square. 4×4 is 16 million.

Hundreds times thousands give us hundred-thousands. 300 times 4000. Let's color and write.

Tens times thousands give us ten-thousands. Let's color the square blue.

Next we have units times thousands, giving us thousands. So we color these green and also do the writing.

That's our first row done.

Thousands times hundreds gives us hundred-thousands. So we color the square red (second from top, far left). We have 4000 times 300, that makes 1,200,000

Next we have hundreds times hundreds giving us ten-thousands. Color blue.

Next we have tens times hundreds giving us thousands. Color in green and write.

Then we have units times hundreds. Color in red and write.

End of Second Row

Thousands times tens give us ten-thousands. Let's color in blue (3rd from top, far left)

Hundreds times tens gives us thousands. Color and write

Next is tens times tens, so color red since that gives us hundreds and write 2×2 is 4 hundreds.

Here we have units times ten giving tens

End of Third Row

Thousands time units gives us thousands. Color green and write.

Next we have hundreds times units giving us hundreds. Color red and write.

Next we have tens times units giving tens.

Units times units give us units. Color and write (last cell lower right)

That's our squaring on graph paper. Can you tell me our answer? Not yet! We have to add up all the answers in the cells. (begin upper left, proceed by category – all millions, all hundred thousands, etc.)

Collect Products

$$4325^2 =$$

$$\begin{array}{r}
 16,000,000 \\
 1,200,000 \\
 1,200,000 \\
 80,000 \\
 90,000 \\
 80,000 \\
 20,000 \\
 6,000 \\
 6,000 \\
 20,000 \\
 1,500 \\
 400 \\
 1,500 \\
 100 \\
 100 \\
 25 \\
 \hline
 18,705,625
 \end{array}$$

End

Epilogue

Determine category, color and then do the writing of the expression in the cell. This is the general procedure. We work from upper left and work to the right, then row by row downward.

November 3, 2004

Exercise V - Extracting Rules for Squaring

Material

Back to the pegboard (using hierarchical pegs), decimal system cards, signs, parentheses and blank labels.

Begin

Binomial

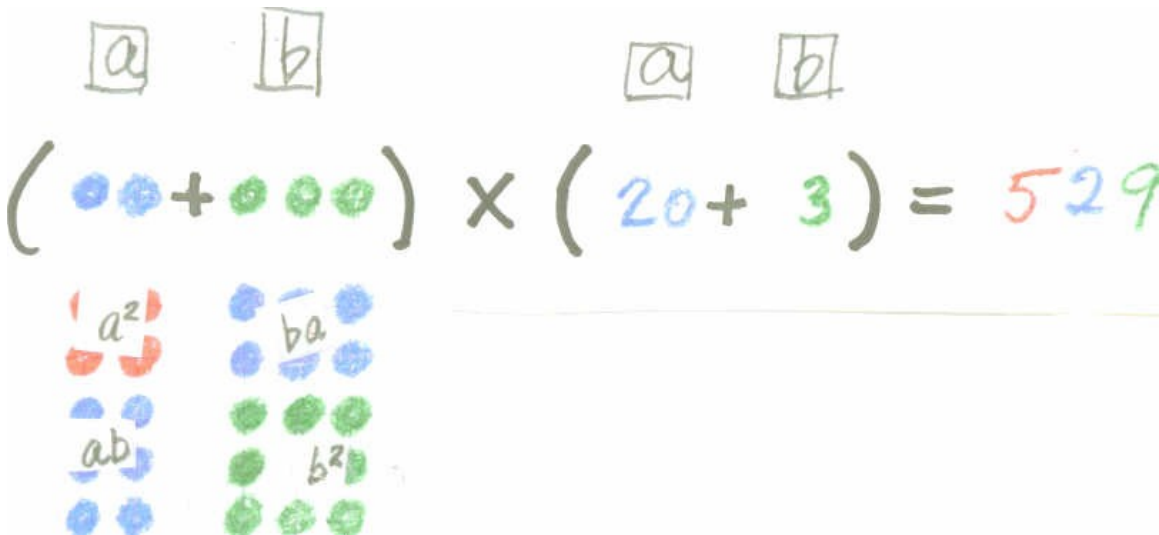
Let's start with a binomial. We'll use 23 again.

(invite the child to build the square of 23 hierarchically)

Let's call 20 'a' and let's call 3 'b.'

(write tickets above all terms, beads and cards – “a b a b”)

What does that mean the square would be. Yes, a^2 Let's write a ticket for that.



What would this rectangle be ab. How about this rectangle ab. Let's write tickets for those.

And the last would be b^2 . (place ticket on each sub-product group at lower left)
So let's see what we have altogether (pull signs down and make expression at bottom)

$$a^2 + 2ab + b^2$$

That's the same old thing we did before. Let's try another binomial and see what we get.
(choose number, set out square, ticket algebraically. They get the same expression again.)

Can we verbalize what happens when we square a binomial?
Let's see if we can express it in words. Let's start:

“When you square a binomial, first you take the square of the first term, then take 2 times the product of the first and second terms, and then we have the square of the second term at the end.”

That's quite a few words isn't it? That's why it's easier to express it as: $a^2 + 2ab + b^2$

That's what $a+b$ squared equals.



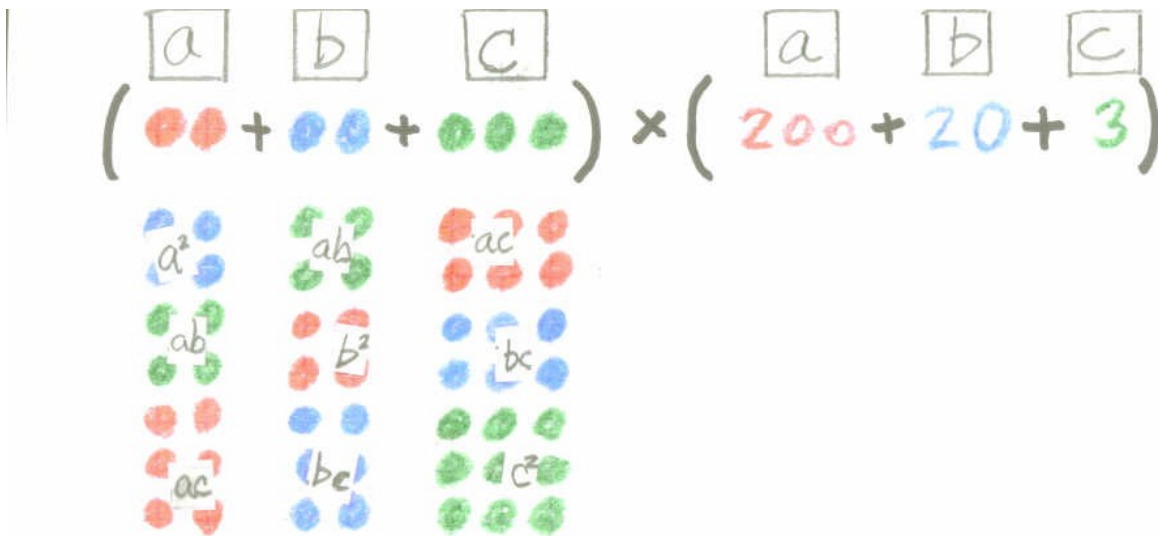
Trinomial

Let's take 223

(Invite the child to build the square. Based on past experience, she can do this.)

(Label a, b and c on top of both multiplier and multiplicand using blank tickets)

(Create products and ticket each sub-square)



$a^2 + 2ab + 2ac + b^2 + 2bc + c^2$
Pull tickets down to edge of table, combine terms and add for a final sum

Let's see if we can express in words what we do to square a trinomial:

“We take the square of the first term, plus 2 times the product of the first and second terms - plus 2 times product of first and third terms plus the square of the second term - plus 2 times the product of the second and third terms plus the square of the third term.”

That's a lot of words. That's why it's easier to express it as:

$a^2 + 2ab + 2ac + b^2 + 2bc + c^2$ (formulate tickets for the expression at left)

This is the same for any trinomial. Try it and see. It's the formula for squaring a trinomial.

End

Epilogue

This is putting the general form into language as was the previous. They get this all through repetition. Before we are done, there are generalizations about squaring we must cover. When bringing ticket down, begin upper left, complete that row and column, and work down diagonally from there, each time completing row and column.

November 2, 2004

Generalizations Regarding Squaring

These are things that you pick up one at a time when experience is built up and they've seen these things enough to state them directly. Not all children will arrive at that point at the same time. Some children may never be ready to come to these realizations. These children can do this work but will not be able to ahead to the next things. We need to have presented these realizations before going on to square roots.

- There is a strong relationship between numbers and algebra
- The squares always appear on the diagonal. Other pieces are rectangles.
- There are some product categories in the decimal system that, when squaring, always makes rectangles; they never make squares. One example is the category of tens – it never forms a square. Hundreds, as the product of two categories, on the other hand, always form squares and rectangles – both. Units always form a square.
- The square of the binomial will always have 4 pieces. Out of those 4, 2 are squares and 2 are rectangles. The square of the trinomial will always form 9 pieces. 3 are squares and the rest are rectangles. We can continue with quadrinomials and so on.

January 10, 2005

10. CUBING

Cubing a Binomial

Age

8 to 8 ½ years old.

Materials

Wooden cubing material, box of bead bars, large felt mat, bead squares, parentheses, number cards and signs, blank tickets and pencil. Thin strips of cardboard, approximately 2 inches long by 1/8 inch.

Prologue

Begin by building a square; the square will be deconstructed as the cube is built with correspondence between the part being deconstructed and the part of the cube being built. For example, if I am deconstructing the square's upper right quadrant, then the upper right quadrant of the cube's layer is the one being built.

We do the multiplication using the materials; the products are broken into squares (beads being replaced by the wooden cubing material) and used to fill in cube. Note that the current multiplier determines what kind of squares we make; if multiplier is 3, we want to reconfigure our beads into squares of 3.

Cubing a Bead Square - 1

Begin

If we take a square of 3 and multiply it by 3, what would we get? (cube of 3)

Suppose we take a square of 3 + 5 and multiply that by 3+5 (set out tickets on felt mat)

$$(3 + 5)^2 \times (3 + 5)$$

Let's build the square of 3+5. (child does so)



Let's ticket our squares.

We have a square of 3.

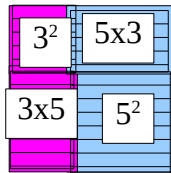
What do we have here (upper right of square)

Yes 5×3 – that's what my ticket will say

What do we have here (lower left of square)

Yes 3×5 – that's what my ticket will say.

And what do we have here? The square of 5 and that is what my ticket will say now that we have these tickets. (all quads are marked)



We can substitute these for 3×5 squared

(take all tickets on the square and put them into the first bracket – old tickets go aside)

$$(3^2 + (5 \times 3) + (3 \times 5) + 5^2) \times (3 + 5)$$

Now we can multiply.

We have to multiply each of these (parenthesized operand on left side) by 3

Then we need to multiply each of these (same operand on left) by 5

(pull down tickets for 3^2 at left side and the 3 at right side of expression)

$$(3^2 + (5 \times 3) + (3 \times 5) + 5^2) \times (3 + 5)$$

$\swarrow \quad \searrow$
 $3^2 \times 3$

So I'll build my 3 squares of bead bars down here beneath my ticket.

But **instead of these bead squares, we can use these wooden squares.**

(place wooden 3-squares above each of the 3 bead squares – remove bead squares)

If I do this (stack the wooden cubing material squares of 3 into a cube)

What do I get? A cube of 3. (stack the newly formed cube at bottom left mat)

I'm done with 3^2 , so I'll put that back into my brackets (of left operand)

Now we need to take (5×3) 3 times. (pull down tickets)

$$(3^2 + (5 \times 3) + (3 \times 5) + 5^2) + (3 + 5)$$

$$(5 \times 3) \times 3$$

Can I make a square of 3 from these 5-bars?



lay paper strips where the black lines appear forming squares of 3

(place paper strips between - **vertically** – this determines what kind of square to make)

There's oneI have 5 squares of 3

(replace with new wooden 3 squares - 5 of them)

Now I don't need my bead bars (take aside)

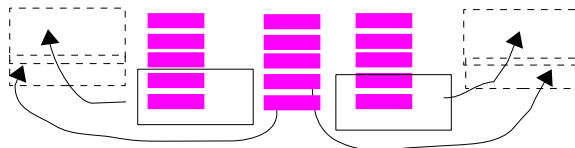
Now I'll put my new squares of 3 with the first cube of 3. (begin constructing 1st layer)

Now I bring down my 3×5 .

We need to take (3×5) 3 times. (bring down tickets – see ' (5×3) ' above)

I wonder if we can get any squares of 3 out of this?

Here's a square of 3 (move remaining excess from first column to new column)



form 5 squares of 3 from 3 rectangles of 5 3-bars – then place

wooden squares above the 5 groups of 3

I'm done with my bead bars (put them aside)

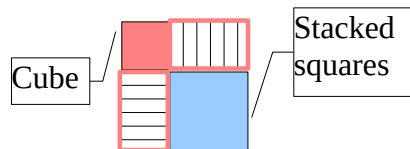
Let' gather my these squares and then build on to what I have down here (bottom left)

I need to take 5^2 taken 3 times (form the 3 5-squares placed horizontally as above)

Instead of these bead squares, let's use wooden squares.

(replace them - put bead squares aside)

I'll pile my wooden squares up (squares of 5) and we add them on to the cube we're building.



So now I'm done with the 3.

We, in fact, have the first layer of our cube

now.

Now we need to multiply our left operand by 5.

So we want to build our square of $3 + 5$ of beads again to work with.

Who would like to build this square for us?

So here's our square of $3 + 5$ and now we'll multiply by the 5 in our multiplier on the right.

So we start with 3^2 taken 5 times.

$$(3^2 + (5 \times 3) + (3 \times 5) + 5^2) + (3 + 5)$$

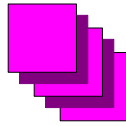
$\swarrow \quad \searrow$
 $3^2 \times 5$

I don't have enough bead square to take this 5 times so I'll start with wooden squares now.

(layout 3^2 5 times in same layout as for 'taken 3 times' - see above)



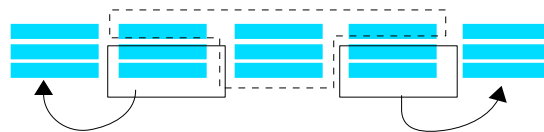
Now we'll pile these up and they'll go right here – just to the right of my cube's first layer.



Here's 5×3 taken one time, taken 2 times and a 5th time

(lay them out horizontally in beads as was done with 3 multiplier)

So like we did before – let's look for squares. What kind of squares? (of 5)



create 3 squares of 5 from 5 rectangles of 3 5-bars each

Why do we want squares of 5? (because our multiplier is 5)

So like we did with the first layer, I'll rearrange my bead bars looking for squares.

Since I took the **bead bars from the top right of my square**, I'll place my new wooden 5-squares onto my pile also at the upper right.

So now I need to take (3×5) 5 times.

(form 3-bars into (3×5) taken 5 times)

(form 3 squares of 5 out of these)



(replace bead bars with wooden squares)

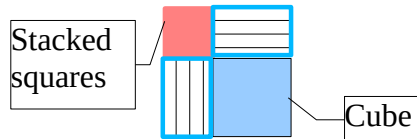
Now we need to take 5^2 5 times.

(take down tickets and form beads – or they might go directly to wooden squares or even to the cube)

(replace with wooden squares)

What do we have? (cube of 5)

And it goes here. (place lower right of my cube)



Now we need to put the second layer on top of the first layer.

The trick is to get the hands around the second layer and squeeze!

(lift and place atop first layer)

We now have the cube of $3 + 5$. We can check each side and see that.

Here we have $3 + 5$ (review all sides of the cube to show that all are $3 + 5$)

So we can say that all of these tickets are equal to this:

(form and verbalize final expression)

$$(3^2 + (5 \times 3) + (3 \times 5) + 5^2) \times (3+5) = (3+5)^3$$

(expression and cube are only things left on the mat)

Epilogue

Placement of wooden squares into the cube is based on the placement of the parts of the square; if we're working on 3^2 from the upper left corner, the squares we form to make the cube are placed into the upper left corner as well.

The second layer is the result of the second multiplier. Each part of a given layer will be stacked to be even with the initial cube-height of upper left corner – 3 cubed in this example for layer 1.

Note that when building a cube:

- Each layer has a height corresponding to the multiplier in use. So ab times b will yield a layer-rectangle of height b .
- If building from a bead square, the cube or rectangle is placed in the same location (e.g. upper right) as the bead square (e.g. Upper right) it was formed from.
- Cubes, rectangles and squares are built left to right and top to bottom for a given square or cube layer. This corresponds with the local standard practice of multiplying all terms on left times first term on right, followed by multiplying all terms on left with second term on right.
- Rectangles are built according to the multiplier. For example, a cube of $(3 + 5)$ will multiply both terms (upper right, 5×3) and (lower left, 3×5) times 3; therefore, both become stacks of squares of 3. Multiplying these two terms by 5 creates two stacks of squares of 5 for the second layer.

January 10, 2005

Building Cube of a Binomial Starting at Cube of First Term - 2

Exercises 2-4 are adapted from notes taken by Danielle Ruddock and Emily Burton

Materials

Wooden cubing material, felt mat, paper and pencil.

Prologue

The children would have been working for weeks or months with exercise one and would be very familiar with that work by now.

Begin

Suppose I were to ask you to **build a cube of $(6 + 3)$** . What do we need to start with?

Build First (Multiplier- 6) Layer

The cube of six (place the six cube on the mat), here it is.
Now, with what are we going to build on to this cube?

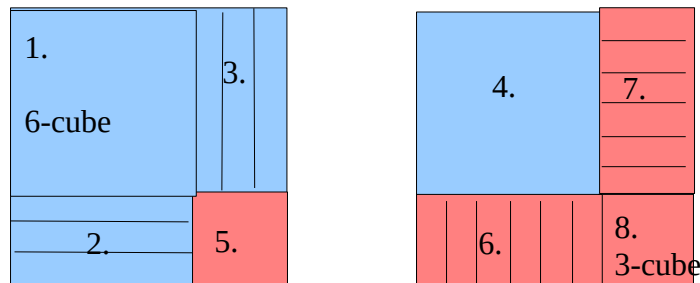
We'll build on **squares of 6**. Why? (**multiplier is 6**) In how many **directions?** (3)

How many squares of six build on each time?

Three, because we are building six plus three rectangles.

So we need three squares of six here (placed right of the cube), three squares of six here (below the cube), and three squares of six here (on top of the cube).

(we now have a cube of 6 and 3 identical rectangles (each a stack (3) of squares of 6) in 3 directions – extending from the 3 interior sides of our cube of 6. See steps 1-4 below)



Build 6-cube and its 3 legs followed by 1 leg of 3-cube (completing first layer) plus 3-cube at corner of layer two, followed by other two legs of 3-cube.

Build Second Layer (Multiplier- 3)

Now we have three more directions that we have to fill in. What do we need to use to fill them in? (squares of 3)

Each time it will be how many squares of three? (6)

We'll build rectangles of six squares of three. The first rectangle will complete our first layer. It will support the cube of 3 later (step 5 above). The next two will go along the edges of layer two, extending from the interior sides of the cube in layer two (see step 6 and 7 above).

I'm not done yet, what do we still need to place here? The cube of three in my second layer (see step 8 above)

We have built six plus three cubed (write this down on paper), that's what I have here on my mat.

Analyze and Verify

What is $(6 + 3)^3$ equal to? Let's take our cube apart so that we can see what we have here (take off the top layer and place like parts into corresponding rows).



What's the first thing that we have here? **Six cubed** (write this down on paper).

How would we describe one of these (hold up the six squared taken three times)? As six squared taken three times. That's a group. How many do we have here? Three of those groups. So we can say **we have three times, 6-squared times 3**.

Next we have these (hold up three squared times six). How many group do we have? Three. So **we have 3 times, 3-squared times 6**.

And here we have what? **Three cubed**.

Write all down on paper as describing each term:

$$(6 \times 3)^3 = 6^3 + 3(6^2 \times 3) + 3(3^2 \times 6) + 3^3$$

Now what's the value of six cubed?

$$(6 \times 3)^3 = 6^3 + 3(6^2 \times 3) + 3(3^2 \times 6) + 3^3 \\ = 216$$

Now let's look at the next group. What's the value?

$$(6 \times 3)^3 = 6^3 + 3(6^2 \times 3) + 3(3^2 \times 6) + 3^3 \\ = 216 + 324$$

Now let's look at the next group. What's the value?

$$(6 \times 3)^3 = 6^3 + 3(6^2 \times 3) + 3(3^2 \times 6) + 3^3 \\ = 216 + 324 + 162$$

What's the value of three cubed?

$$(6 \times 3)^3 = 6^3 + 3(6^2 \times 3) + 3(3^2 \times 6) + 3^3 \\ = 216 + 324 + 162 + 27$$

Now all we have to do is add these up.

$$= 729$$

How do I know 729 is the correct answer? It is the cube of 9.
(the children continue to work like this choosing what ever numbers they like)



Follow-up

Begin

After the children have done a lot of work with this we could ask how many pieces we have here on the mat? (8)

Out of these 8, 2 of them are what? (cubes)

The other 6 are what? (rectangular prisms).

Of these 6 prisms, let's see if we can identify more exactly what they are like (hold up 3).

These three have the square face of what? (square face of the first cube)

And if I set them on their face, how are they related to the second cube?

(they have the height of the second cube)

So they have the face of the first cube and the height of the second cube.

The other 3 (the squares of three) have the square face of the second cube.
If I set them on their face, how are they related to the first cube?
(they have the height of the first cube)

Epilogue

Through experience the children will see that every time you cube a binomial you end up with eight pieces; two are always cubes and six are always prisms.

January 10, 2005

Building the Formula - 3

Prologue

We might realize that we can introduce this to the children on a day when they are working on exercise two and they have built a cube of a binomial on their mat and disassembled it. Otherwise, invite the children to build a cube and disassemble it.

Material

Wooden cubing material, felt mat, paper and pencil. Blank tickets.

What cube have you built here today? You built the cube of six plus three.

Well today, instead of six, we are going to call it a , and instead of 3 we are going to call that b .

(hold up the cube of six)

What will this be then? (a cubed)

Let's write a ticket (place it right on top of the cube).

What will this be? (b cubed)

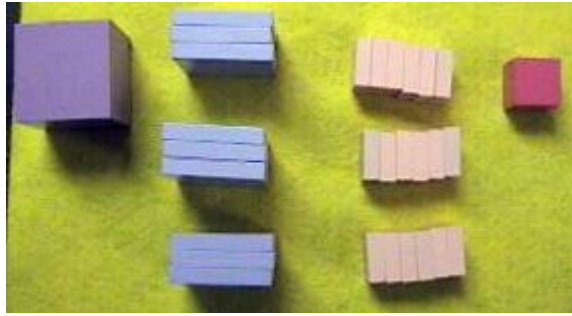
Let's write a ticket for that and place it on top of that cube.

Now, what will we call this piece (point to **six squared times three**)? This is going to be **a squared times b** . Do we need that times sign in there? No. Let's write a ticket for that (a^2b) and place it on top. We have how many of these groups? Three, so I need to write a ticket for each of them.

We have three more pieces (point to **three squared times six**) here, how will we identify those pieces? As b squared a . And I have how many of these pieces? Three, so I am going to write a ticket for each of them.

Now I want to know what we have here altogether, so let's bring the tickets down (**place the tickets in a row aligning like tickets under like tickets**).

This time **altogether we have:**



Label each cube and group of rectangles.

a^3	a^2b	b^2a	b^3
	a^2b	b^2a	
	a^2b	b^2a	

Write formula as you describe pieces.

$$(a + b)^3 = a^3 + 3a^2b + 3b^2a + b^3$$

What does that *a plus b cubed* equal altogether?

What's the first thing we have here? (*a cubed*)

And then we have *a squared b*. How many times do we have that? (3 times)

So it's *three a squared b*. Next we have *b squared a* taken 3 times, so it's *three b squared a*. The last thing we have is *b cubed*, so it's plus *b cubed*.

Why don't you try the same thing with different numbers and see what you get?

Epilogue

After some practice and experience in working with this the children will come to this realization.

January 10, 2005

The Formula on the Binomial Cube - 4

Prologue

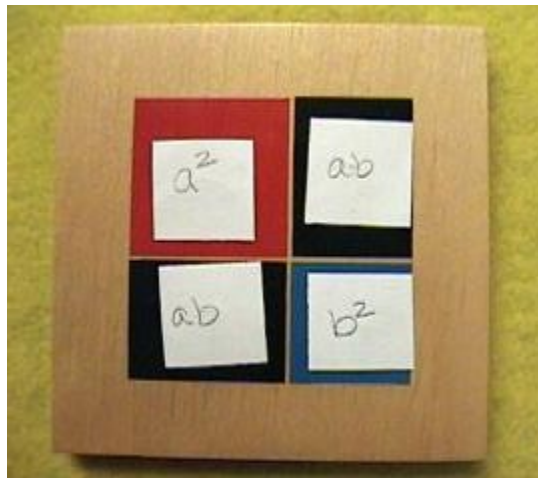
This exercise would come after sufficient experience with exercise three. Our assumption would be that the children have become somewhat familiar with this algebraic representations.

Materials

Binomial cube, a felt mat, blank tickets, and operation signs.

Let's see if we can build *a plus b cubed* in quite a different way today. We are going to **build it starting from the square of (a plus b).**

Let's **start off by looking at the lid of our binomial cube.**



Binomial cube lid illustrating the pattern of the square of the sum of a and b.

We are going to call this length a (point to the length of the red cube) and we are going to call this b (the length of the black prism).

That means **on our lid we have a plus b on all four sides** (point these out).

So we have a **square of (a plus b).**

Altogether we have what? The **square of a plus b**.

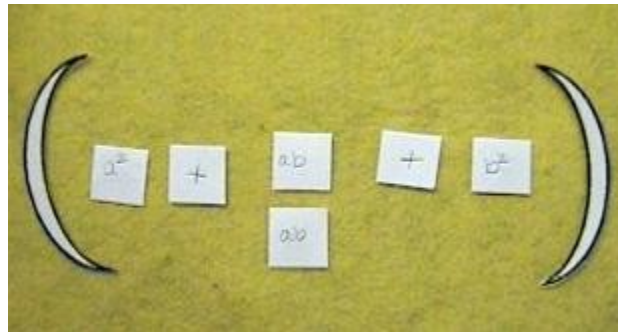
Let's write tickets for these pieces.

What would the first square be? (*a squared*) Let's write a ticket for it.

What would this piece be (the first prism)? (*ab*).

What about this one (second prism)? (*ab*). And what about this square? (*b square*)

Altogether this gives us the square of a plus b. We have a squared plus ab and another ab plus b squared (put all of that inside parentheses and add in the signs).



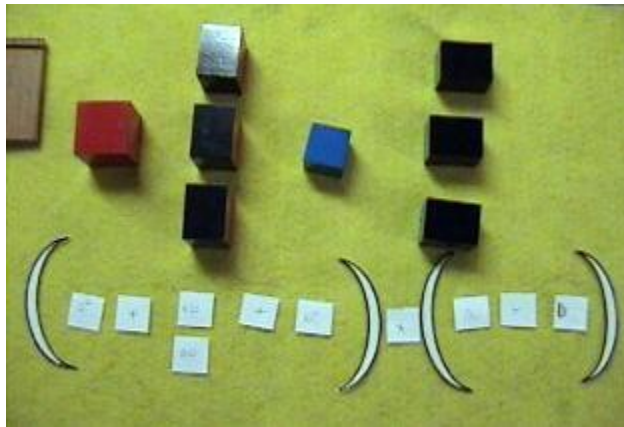
Now we are going to multiply all of that times a plus b (set this out in parentheses as well).

(On the mat set out all of the pieces from the binomial cube randomly)

Now we are ready to start multiplying.

We need to multiply all of our multiplicand by a and then by b. So we will start by bringing down a squared and a.

If we multiplied a squared times a, we get what? (a cubed) Do we have a piece that looks like a cubed? The red prism (place the cube on the lid).



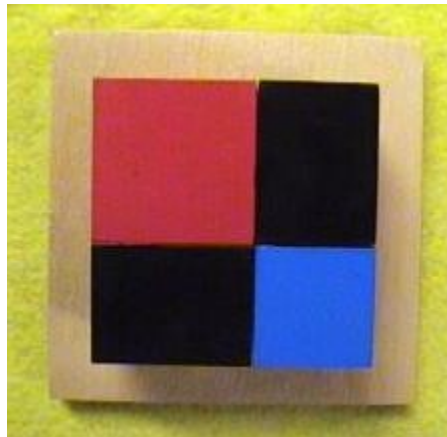
I am done with the a squared, we need to bring down ab .

So we have ab times a what does that equal? a squared b . Which piece looks like that? A piece where the face is a squared and the height is b (place it on the lid).

(bring down the next ticket of ab time a and repeat steps above: **pull down next tickets, identify the corresponding piece and place the piece onto the lid.** Complete the binomial cube pattern on the lid as you go.)

Our next ticket is b squared times a , what does that give us? (ab squared)

Which piece is that? (The rectangles with the blue face).



We are done multiplying by the a so we have the first layer of our cube (move it off the lid) now we are ready to multiply by b .

Let's bring down our first ticket of a squared b . What does that give us? We need to use the red and black cube, right?

Our next ticket is ab times b . What does that give us? It gives us ab squared. What does that look like? It is the black and blue prisms. Again we have another ticket of ab squared.

Our **last ticket is b squared**. Which piece looks like that? The blue cube.

Now we have the second layer of our cube so now we can move that off of the lid.

Now we want to put our whole cube together. We need to put the second layer on top of the first layer

(place the layers on top of each other; point out that we can see a plus b squared on each side. Open the cube from the front, back and top to show the children that we can see that pattern on the inside. **Altogether it gives us the cube of a plus b**)



Follow-Up

Begin

What do we have here altogether? **Let's take our cube apart** so that we can see what we have here (disassemble the cube and place the **corresponding pieces together**).

How do we identify this piece? (hold up a cubed) Let's **write a ticket** for that.

How do we identify this piece? (a squared b) In fact we have how many of those? (3)

How do we identify one of these pieces? (ab squared)

And how many of these do we have? (3)

So I have to write three tickets. How do we identify this piece? (*b cubed*)

I bet you would like to repeat this yourself. See if you can re-ticket the pieces

Epilogue

Encourage the children to keep practicing and identifying these pieces. The more readily they can do this the better it will be when they move on. The children can write tickets or we can have in the environment already prepared tickets.

Besides having the pieces lined up orderly and ticketed we can mix the pieces up and ticket them, this makes it a little more challenging. For variety we can have the pieces mixed up, show the child the piece and have him find the ticket that goes with it. Or do it the other way around, hand him a ticket and have him find the piece that goes with it. We have another opportunity here to make our observations. “How many pieces to do we have here altogether? Eight.

Out of those eight pieces, two are cubes and the others are rectangular prisms. Eventually after a lot of practice we can help them expression the verbalization that anytime we are cubing a binomial we start with the cube of the first term, take three times the square of the first term times the second term plus three times the first term times the square of the second term plus the cube of the second term. It is complicated with all those terms which is why it is easier to say that when we cube a binomial such as *a plus b cubed*, we get ***a-cubed plus three times a squared b, plus three times ab squared, plus b cubed***. That's the cube of the binomial. It is easier to say it algebraically than with all those words, isn't it?

January 11, 2005

Cubing a Trinomial

Materials

Trinomial cube, large felt mat, blank tickets, parentheses, paper and pencil, prepared tickets for trinomial in same box as binomial materials Tickets for each piece of a cube in algebraic notation.

Begin

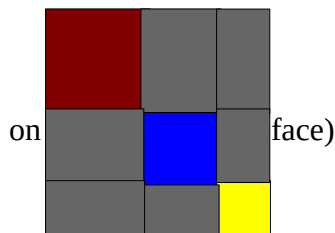
Ticket Trinomial

What happens if we multiply $a+b+c$ times $a+b+c$? (a square of $a+b+c$)

Let's look at this. This is a square of $a+b+c$. If we called this length a , this length b and this length c , we'd have $a+b+c$ on each side. (hold up cover of the box to see pattern)

What happens if we took the **square of $a+b+c$ and multiplied it by $a+b+c$** ? (**cube**)

(lay the square lid on the felt mat)



(fill in rest of square with black sides

(begin upper left – left to right, top to bottom)

Let's ticket each piece of our square. What's this piece? (a^2)

Let's write a ticket for it.

What's this piece? (ab)

Let's write a ticket for it.

What's this piece? (ac)

Let's write a ticket for it.

What's this piece? (another ab)
 Let's write a ticket for it.

What's this piece? (b^2)
 Let's write a ticket for it.

What's this piece? (bc)
 Let's write a ticket for it.

What's this piece? (ac)
 Let's write a ticket for it.

What's this piece? (bc)
 Let's write a ticket for it.

What's this piece? (c^2)
 Let's write a ticket for it.

(set out all pieces of the cube – use tickets for form expression below)

So from our tickets, we see we begin with:

$$(a^2 + ab + ac + b^2 + bc + c^2)$$

$$ab \quad ac \quad bc$$

and we want that whole term times...

$$\times (a + b + c)$$

(begin with first term in multiplier a – each subsequent time pull ticket down and back)

Multiply

$$a^2 \times a$$

That gives us a^3 (place a^3 piece – match pieces to outline on the trinomial cube)

ab times a (place a^2b piece)

ab times a (place a^2b piece)

(that ticket goes back, bring down the next)

ac times a (place a^2c piece)

ac times a (place a^2c piece)

b^2 times a (place ab^2 piece)

Which piece looks like that? (each time)

bc times a (place abc piece)

bc times a	(place abc piece)
c^2 times a	(place ac^2 piece)

So we have completed out first layer! Let's set it aside for the moment.

Let's continue multiplying our multiplicand times the second term of our multiplier. We bring down a^2 from our multiplicand and b from our multiplier. (bring down those tickets)

(Second term b – slide first layer off to the side)

$a^2 \times b$

That gives us a^2b	(place a^2b piece – match pieces to the outline on the trinomial cube)
----------------------	--

ab times b	(place ab^2 piece)
------------	----------------------

ab times b	(place ab^2 piece)
------------	----------------------

(that ticket goes back, bring down the next)

ac times b	(place abc piece)
------------	-------------------

ac times b	(place abc piece)
------------	-------------------

b^2 times b	(place b^3 piece)
---------------	---------------------

Which piece looks like that?	(each time)
------------------------------	-------------

bc times b	(place b^2c piece)
------------	----------------------

bc times b	(place b^2c piece)
------------	----------------------

c^2 times b	(place bc^2 piece)
---------------	----------------------

So we have completed out second layer! Let's set it aside for the moment.

Let's continue multiplying our multiplicand times the third term of our multiplier. We bring down a^2 from our multiplicand and c from our multiplier. (bring down those tickets)

(Third term c – slide second layer off to the side)

$a^2 \times c$

That gives us a^2c	(place a^2c piece – match pieces to the outline on the trinomial cube)
----------------------	--

ab times c	(place abc piece)
------------	-------------------

ab times c	(place abc piece)
------------	-------------------

(that ticket goes back, bring down the next)

ac times c (place ac^2 piece)

ac times c (place ac^2 piece)

b^2 times c (place b^2c piece)

Which piece looks like that? (each time)

bc times c (place bc^2 piece)

bc times c (place bc^2 piece)

c^2 times c (place c^3 piece)

What's left to do with our trinomial cube. Carefully we want to put the second layer on the first layer and then the third layer on the second.

When we look at this side, we can see the square of $a+b+c$. We can see it on all sides and even on top. But look here.

(open toward the center)

It's there too – still we see the same pattern of $a+b+c$ squared.

No matter how we open the cube, we see that pattern.



What remains to do is to ticket each part in the trinomial cube. We have to disassemble the cube for that. (take layers apart segregating pieces by a , b and a pile of c parts)



(label each cube)

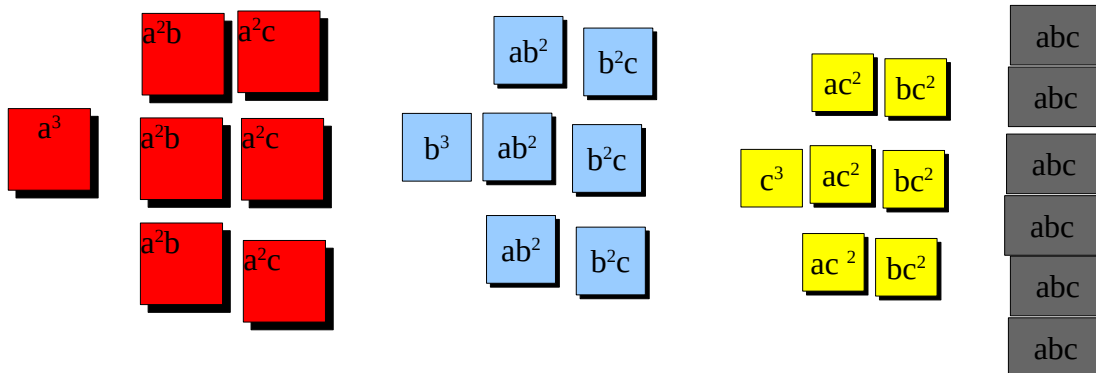
How do we identify this one? (a^3)

How many do we have? (1)

Let's ticket each of them

(repeat for each of the cubes – tickets are now on top of the cubes)

(continue to label all prisms)



How many pieces to we have altogether here? (27)

What's the cube of 3?

How many pieces did we have with the binomial cube? (8)

What's the cube of 2?

Is there a pattern there?

Epilogue

Suppose we mix all of our tickets and then relabel them? Or suppose we mix up the pieces and then identify and label them. These all help to readily identify the pieces without much thought.

We are concerned with sensorial experience of relating this algebraic notation to the trinomial cube from primary class – as well as being able to readily identify the pieces. Instead of setting out tickets for the entire expression as we did for Cubing a Binomial, we could begin writing these equations on paper.

$$(a+b+c)^3 = (a+b+c)^2 \times (a+b+c) =$$

$$a^3 + 3a^2b + 3a^2c + b^3 + 3ab^2 + 3b^2c + c^3 + 3ac^2 + 3bc^2 + 6abc$$

January 11, 2005

Cubing a Quadrinomial

Materials

Using graph paper and pencil and a ruler

Begin

Could we take a square of $a+b+c+d$ and multiply it by $a+b+c+d$? What would we get? (the cube of $a+b+c+d$) How many pieces would we get? (64) How did we know that? (the cube of 4)

When we did the binomial, we started from the square.

Let's draw the square of a quadrinomial on our paper.

(draw top horizontal line)

I'll take 5 for my first number, 3 for my next, 4 for my next number and 2 for my last.

Now I'll draw the same thing down the left side. (do so)

Now we want to outline our square. And we need to connect our dots.

Let's label our sides (a,b,c,d down left side and on top side)

(begin upper left corner and work horizontally)

	a	b	c	d
a	a^2	ab	ac	ad
b	ab	b^2	bc	bd
c	ac	bc	c^2	cd
d	ad	bd	cd	d^2

What is this one (a^2)

(continue labeling each rectangle on the square)

So we have the 16 terms for the square.

$$a^2 + 2ab + 2ac + 2ad + b^2 + 2bc + 2bd + c^2 + 2cd + d^2$$

And we want to **multiply it by** $a + b + c + d$ =

$$\begin{aligned} &a^3 + 3a^2b + 3a^2c + 3a^2d + 3ab^2 + 6abc + 6abd + 3ac^2 + 6acd + 3ad^2 + \\ &b^3 + 3b^2c + 3b^2d + 3bc^2 + 6bcd + 3bd^2 + \\ &c^3 + 3c^2d + 3cd^2 + \\ &d^3 \end{aligned}$$

Epilogue

This time we can use coefficient notation for repeating terms – i.e, $2ab$ instead of $ab+ab$. If we can work this out on our own, this is sufficient for the level we must reach to help the children.

The children's work could also go this direction at some point: “Supposing we take the cube of $a+b$ and multiply it again by $a+b$? What would that give us? ($a+b$ to the fourth power). This must follow the work with powers of numbers beyond cubing.

January 11, 2005

Cubing a Trinomial Having Numeric Value

Materials

Trinomial cube, large felt mat, paper and pencil.



Prologue

This work could come after the trinomial but before work with the quadrinomial.

Begin

(place trinomial cube pieces randomly on the mat)

We've been cubing our trinomial algebraically.

Today, instead of a – let's call that 5. Instead of b , let's call that 4. Instead of c , we'll call that 3. Instead of cubing $a+b+c$, we'll cube $5+4+3$.

(write)

$$(5+4+3)^3$$

Our trinomial cube will serve as our guide. We first start with this (hold up red cube) – so this is going to be 5^3 . Let's write that.

$$(5+4+3)^3 = 5^3$$

What's next? $5^2 \times 4$ – hold up the 3 a^2b .

We can build them onto our cube now.

What's next? $5^2 \times 3$ – hold up the 3 a^2c .

We can build them onto our cube now.

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3)$$

How about 5×4^2

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2)$$

Next piece is 4^3

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2) + 4^3$$

Now we have abc pieces since the cube needs them to continue. We have 6 of them.

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2) + 4^3 + 6(5 \times 4 \times 3)$$

(add pieces onto the cube each time)

Now we have b^2c pieces.

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2) + 4^3 + 6(5 \times 4 \times 3) + 3(4^2 \times 3)$$

Now we have ac^2 pieces.

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2) + 4^3 + 6(5 \times 4 \times 3) + 3(4^2 \times 3) + 3(5 \times 3^2)$$

Now we have bc^2 pieces.

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2) + 4^3 + 6(5 \times 4 \times 3) + 3(4^2 \times 3) + 3(5 \times 3^2) + 3(4 \times 3^2)$$

Now we have c^3 pieces.

(write)

$$(5+4+3)^3 = 5^3 + 3(5^2 \times 4) + 3(5^2 \times 3) + 3(5 \times 4^2) + 4^3 + 6(5 \times 4 \times 3) + 3(4^2 \times 3) + 3(5 \times 3^2) + 3(4 \times 3^2) + 3^3$$

Now we only need to find the values!

$$= 125 + 300 + 225 + 240 + 64 + 360 + 144 + 135 + 108 + 27 \quad (\text{add by digit and carry each time})$$
$$= 1728$$

How do we know this is correct? (12 cubed) If you add 5+4+3 you get 12!

(this is our control of error)

Epilogue

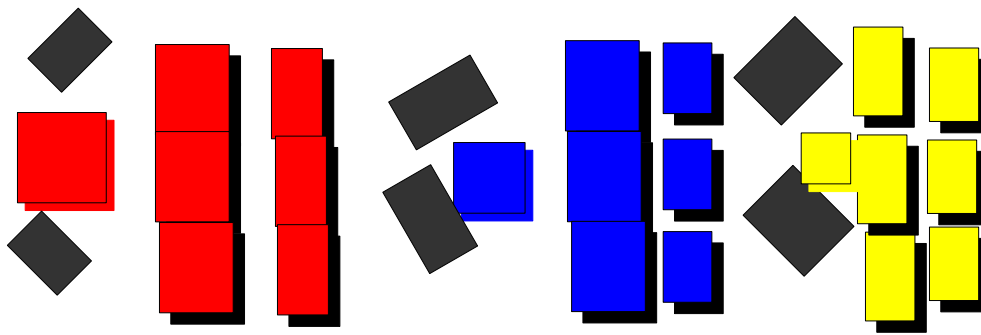
The trinomial cube is acting as a guide. The children can pick any numbers they want to at this point and continue practicing. First example, should use numbers that correspond to the actual size of the trinomial cube – then mix it up afterward as it becomes abstracted for the children.

January 12, 2005

Story of the Three Kings

Materials

Trinomial cube, hierarchical cube, large felt mat, blank tickets and pencil.
Two sets of prepared tickets (box labeled 3 kings): htu-notation on one side and abc-notation on the other side, htu-notation on one side and numeric value on other side.



Initial procession – build this layout during the opening of the story – before the revolt. Note that the height of the black prisms (above) surrounding a given king is the same as that king's height – in other words the top view features the two other relationships.

Purpose

Introduce decimal values.

Prologue

The words of the story are going to guide the usage of materials. Note that body-guards are always set at the height of their king. Also note that in this lesson, the trinomial cube is built in reverse since that is the most efficient transition to the hierarchical cubes. Be sure to order prepared tickets in the order of use.

This work is presented toward the end of the 6-9 or beginning of the 9-12 class generally. The work usually overlaps the two classes; present when child is ready.

Begin

I have a story today. (place trinomial-cube pieces randomly on table)

Once upon a time there were 3 kings.

(place the three cubes on table horizontally across top)

Each of the kings was allied to the others and together they formed an empire.

The importance of each king was shown by his size. And each had a **color of his own which he wore all over**.

Whenever there was an important event, all the kings marched in a procession with all their attendants behind them.

The **red king**, being the largest went first.

(place 3 a^2b and 3 a^2c prisms to right of red king as illustrated above)

He was followed by his 6 attendants. **Each of these attendants wore the color of their king – back to front**. Three of them were related to the blue king.

Three of them were related to the yellow king.

Next came the blue king followed by his 6 attendants

(place 3 b^2a and 3 b^2c prisms to right of blue king)

They also wore the color of their king – back to front.

Three were related to the red king and three were related to the yellow king.

And then came the smallest yellow king. He was also followed by his 6 attendants.

(place 3 c^2a and 3 c^2b prisms to right of yellow king)

They also wore the color of their king – back to front.

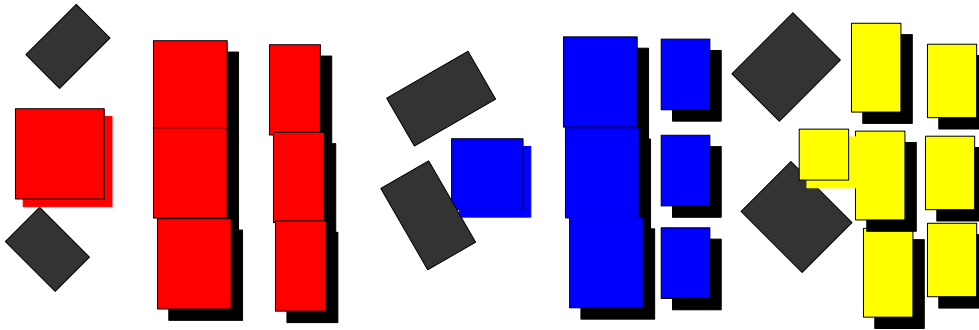
Three of them were related to the red king and three of them were related to the blue king.

Then came the 6 body guards dress in all black. They were each related to all the kings in the same way. Two of them went with the red king, two with the blue king and two with the yellow king.

(layout 6 abc cubes)

(we have now laid out all trinomial pieces in the army formation as illustrated above)

And this is how they marched. The harmony and order was beautiful to see.



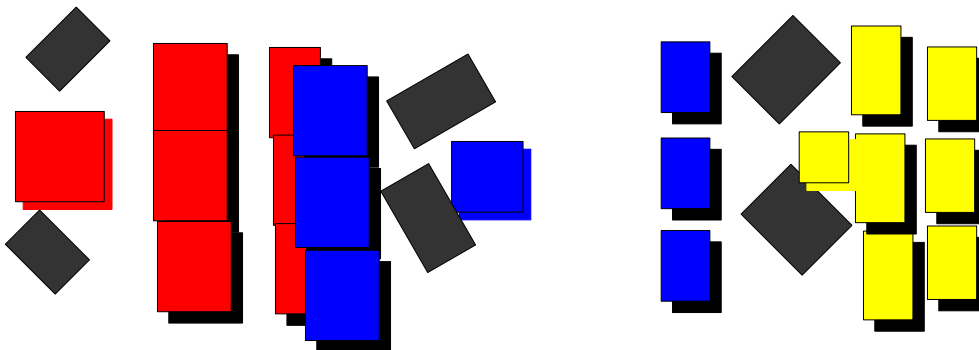
Revolt

But then something happened!

The blue king, who always had to go in the middle, got tired of being there. After all, he was the first to exert power – and he felt he deserved better.

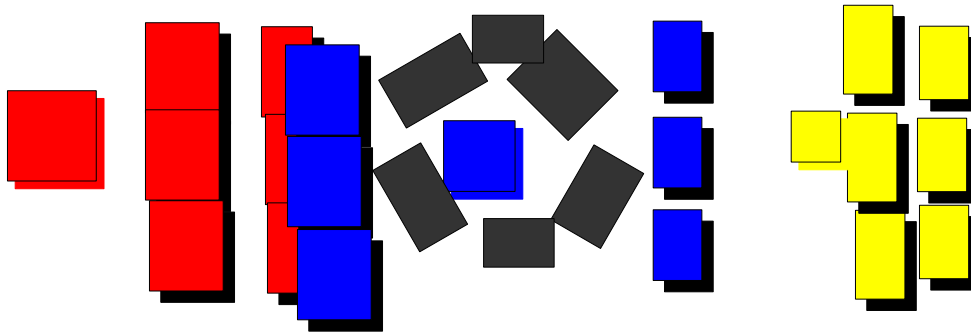
He wasn't like the yellow king who didn't want to make a bother, even though he was the initiator of them all. So what did the blue king do?

The **blue king sent his first 3 attendants (point to first column) up to capture the 3 closest** attendants of the red king.

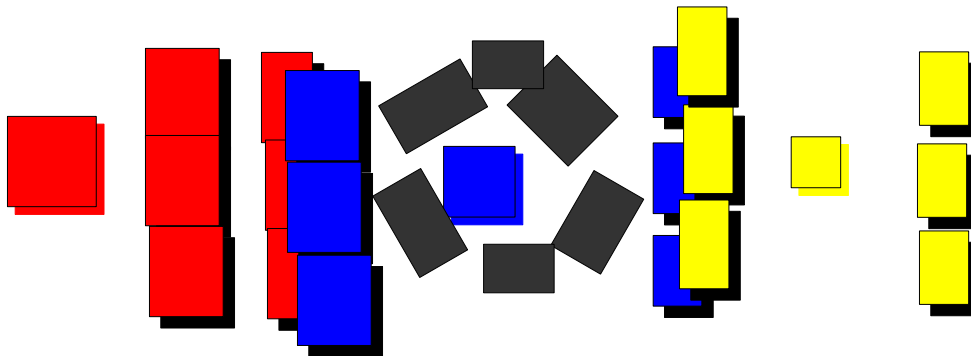


(place 3 ab^2 from left side of two blue columns- onto next closest column of 3 red, at left)

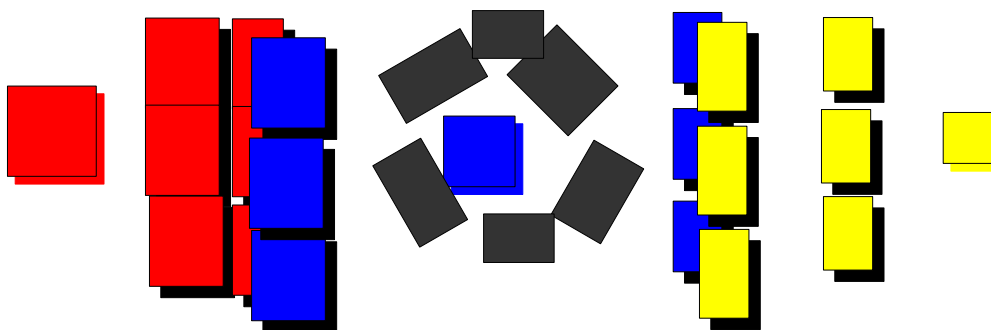
As soon as this happened **all the body-guards rushed to the blue king and surrounded him.** (place the 6 abc prisms around the blue cube)



The **yellow king** sent his first 3 attendants to capture the closest 3 attendants of the **blue king**. (place 3 c^2a cubes in left column of two yellow, onto 3 blue cubes at their left)



The other 3 attendants of the yellow king moved up to give support
And the yellow king moved all the way to the back to watch.



The red king was so large, he had no idea what had gone on behind him. And this is how

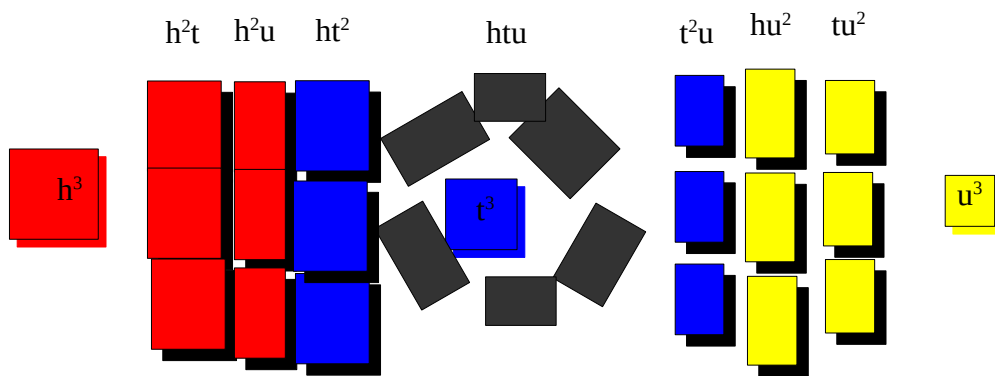
they marched on.

(note: captured prisms are face down and the capturing prisms are standing tall)

The red king continued in the lead (plus 3 plus stack of 3), the blue king still in the middle – but now surrounded by the 6 body-guards – and the yellow king came at the end preceded by blue-yellow stacked column, and then by a yellow column).

Kingdom of the Decimal System

Do you know why this happened? This happened because **they had passed into the kingdom of the decimal system.**



A cubed has become the cube of hundreds – h cubed.

B cubed has become the cube of tens – t cubed.

C cubed has become the cube of units – u-cubed.

The red cube is going to be *h cubed* (ticket it using prepared tickets ... continue for all)

The blue cube is the cube of tens so it's going to be *t cubed* (ticket it)

The yellow cube is the cube of units, we we'll call *u cubed* (continue to ticket each)

If that's the case, what will we call this piece? (h^2t)

And we have how many of these? (3) Let's ticket them.

How will we call this piece? (h^2u)

How many do we have of these? (3)

So let's ticket each of them.

(pieces that were stacked on top

are moved directly to the right)

How will call this piece (ht^2)

How many do we have? (3)

Let's ticket each of them.

How will we call these? (htu)

How many do we have? (6)

Let's ticket each of them.

How will we call these one? (t^2u)

How many do we have? (3)

Let's ticket each of them.

How will we call these one? (hu^2)

How many do we have? (3)

Let's ticket each of them.

How will we call these one? (tu^2)

How many do we have? (3)

Let's ticket each of them.

Now would you each like a chance to set out the tickets?

Epilogue

The 3 cubes were identified and ticketed first. Then you can ask the children for responses to make them think. If then need to check themselves, on the other side of these tickets is the familiar abc notation.



Ticket Values

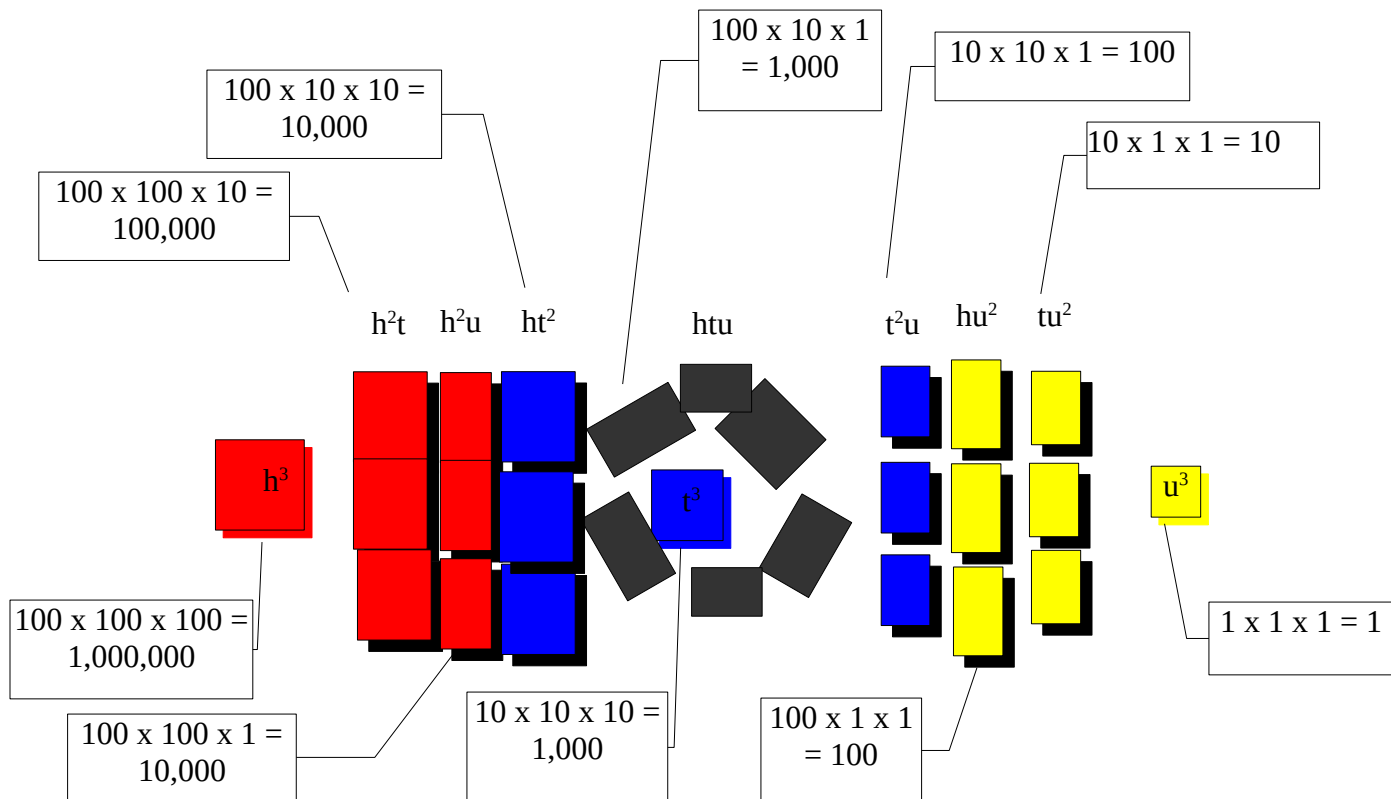
What would h^3 be worth? (use blank ticket and pencil to write ticket – place **atop** others)

h cubed means 100 times 100 times 100.

(write $100 \times 100 \times 100 = 1,000,000$)

And what does that equal – 6 zeros!

That equals 1 million.



Next we have h^2t – that's $100 \times 100 \times 10$.

That equals 100,000

(write ticket and place on the each cube)

I have 3 of them – so I'll do 2 more tickets.

Now we have h^2u – that's $100 \times 100 \times 1$.

That equals 10,000

(write ticket and place on the each cube)

I have 3 of them – so I'll do 2 more tickets.

Now we have ht^2 – that's $100 \times 10 \times 10$.

That equals 10,000

(write ticket and place on the each cube)

I have 3 of them – so I'll do 2 more tickets.

Now we have htu – that's $100 \times 10 \times 1$.

That equals 1,000 (write ticket and place on the each cube)

I have 6 of them – so I'll do 5 more tickets.

Now we have t^3 – that's $10 \times 10 \times 10$.

That equals 1,000 (write ticket and place on the each cube)

Now we have t^2u – that's $10 \times 10 \times 1$.

That equals 100 (write ticket and place on the each cube)

I have 3 of them – so I'll do 2 more tickets.

Now we have hu^2 – that's $100 \times 1 \times 1$.

That equals 100 (write ticket and place on the each cube)

I have 3 of them – so I'll do 2 more tickets.

Now we have tu^2 – that's $10 \times 1 \times 1$.

That equals 10 (write ticket and place on the each cube)

I have 3 of them – so I'll do 2 more tickets.

Now we have u^3 – that's $1 \times 1 \times 1$.

That equals 1 (write ticket and place on the each cube)

Now you see why the yellow cube moved all the way to the back.

He's one – the unit.

Here are the tens (point)

Here are the hundreds (point)

All of these are thousands (point, moving right to left)

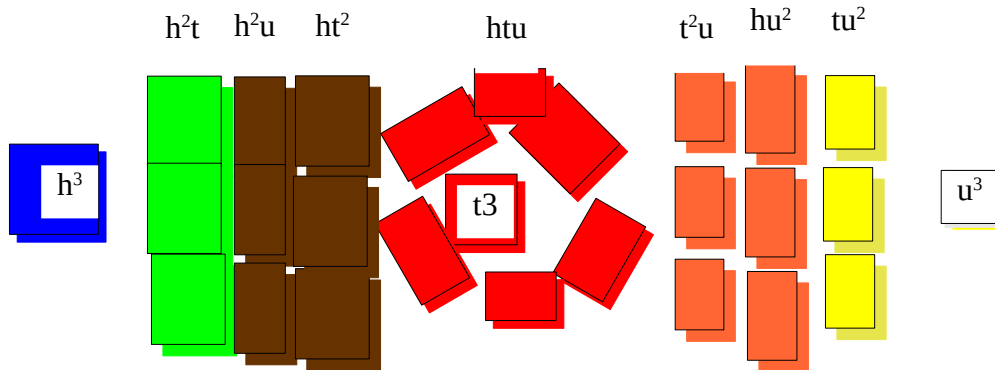
These are the hundred thousands

And here's the millions (pointing all the way left now)

Color By Value

Now suppose we color all the pieces of the same value in the same color.

(take new trinomial-cube box for new pieces – color-coded for value)



Here's our unit. (remove tickets and replace yellow cube for white)

Here are the tens. (remove ticket – 3 tu^2 yellow)

Here are the hundreds (3 hu^2 - orange)

Here are more hundreds (3 t^2u - orange)

These are thousands (continue working leftward – red htu)

This is also a thousand (replace t cubed – blue becomes red)

These are ten-thousands (brown cubes -3 ht^2)

These are also ten-thousands (brown cubes, different size same color -3 h^2u)

These are hundred-thousands (green prisms -3 h^2t)

And this is the millions (blue cube – h^3)

(put trinomial cube away)

(stack cubes upward that are of like value – like color)

Here we have our units (far right)

These are the tens. (point each time)

Here are the hundreds

All of these are thousands (moving left)

These are the ten-thousands.

Here are the hundred-thousands

And these are the millions. (far left)



I have another set of tickets – let's see if we can place these tickets.

What do we call this (h^3)

What do we call these (h^2t) How many do we have? (3) Let's ticket them.
(moving left to right)

What do we call this (h^2u)

How many do we have? (3) Let's ticket them.

What do we call this (ht^2)

How many do we have? (3) Let's ticket them.

What do we call this (htu)

How many do we have? (6) Let's ticket them.

What do we call this (t^3)

How many do we have? (1) Let's ticket them.

What do we call this (t^2u)

How many do we have? (3) Let's ticket them.

What do we call this (hu^2)

How many do we have? (3) Let's ticket them.

What do we call this (tu^2)

How many do we have? (3) Let's ticket them.

What do we call this (u^3) Let's ticket it.



Now maybe you want a chance to lay these tickets down too.

Epilogue

On the other side of these tickets is the numerical value (e.g., h^3 and 1 million). Use the same sorts of variations as in prior lessons – mix up tickets for the children to set them out again. Begin with the variable going toward the value or vice versa. We can mix up the pieces – mix the procession – for further repetition.

Putting the hierarchical cube away is done beginning with millions, progressing to next smaller category – units are last.



January 12, 2005

Cubing Decimal (Hierarchical) Values

Age

Upper elementary – 9 to 9 and a half.

Materials

Hierarchical cube, felt mat, paper and pencil.

Prologue

Most work is on paper. The cube serves as our guide.

Begin

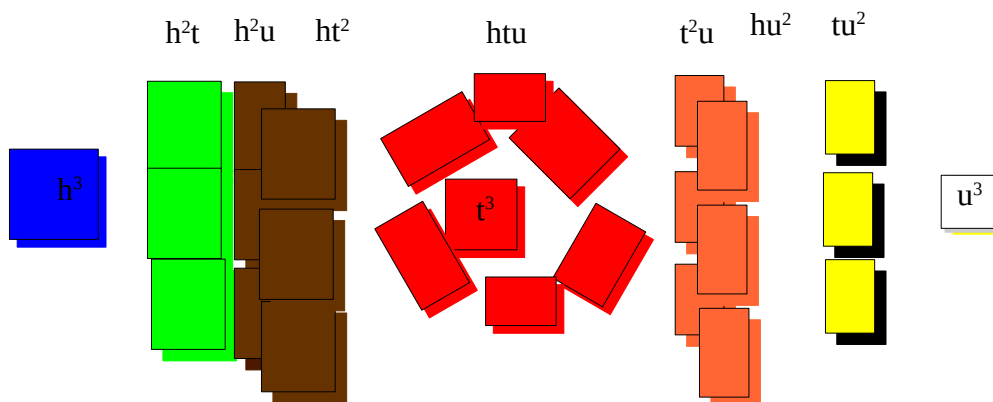
Now that we passed into the kingdom of decimal we can cube a number having a decimal value with hierarchical categories.

I wonder what the value of the cube of 234 is?

This means $200 + 30 + 4$ and we'll cube that number.

(take out hierarchical cube **pieces in procession after the revolt position** (3-kings))

(orange and brown are stacked, bodyguards surround middle king)



The first thing we have is h^3

(hold up h cubed)

Write on paper:

$$234^3 = h^3 = 200^3 =$$

(**ignore zeros** – use category after product calculation instead)

What is 2 cubed? (8) What category is this? (millions) So this is 8 million.

(add pieces onto cube at bottom left of table surface - each time we add an addend)

Write on paper:

$$234^3 = h^3 = 200^3 = 8,000,000$$

Next we have h^2t (hold up)

So h^2 is 2×2 – that's 4, times our t which is 3 – that's 12.

How many do we have? (3)

So that makes 12 times 3 which is 36.

What category is this? (hundred-thousands) So we have 36 hundred-thousands.

Write on paper:

$$\begin{aligned} 234^3 &= h^3 = 200^3 = 8,000,000 \\ 3h^2t &= 3(200^2 \times 30) = 3,600,000 \end{aligned}$$

Next we have h^2u (hold up)

How many do we have? (3)

3 times 200 squared times 4. So, 4 times 4 is 16, times 3 is 48.

What category are these? (ten-thousands) So we have 48 ten-thousands. (write)

Next we have ht^2 (hold up)

How many do we have? (3)

3 times 200 times 30 squared. So that's 2 times 3 squared is 18, times 3 is 54.

What category is this? (ten-thousands) So we have 54 ten-thousands. (write)

Next we have htu . (hold up)

How many do we have? (6)

6 times 200 times 30 times 4. So, 2 times 3 is 6, times 4 is 24, times 6 is 144. What category are these? (thousands) So we have 144 thousands.

(add pieces onto cube each time)

Next we have t^3 . (hold up)

How many do we have? (1)

That's 30 cubed. What is 3 cubed (27). What category are these? (thousands)

So we have 27 thousands.

(add pieces onto cube each time)

Next we have hu^2 . (hold up)

How many do we have? (3)

3 times 200 times 4 squared. So, 2 times 4 squared is 32, times 3 is 96.

What category are these? (hundreds) So we have 96 hundreds.

(write new row of expression and add pieces onto cube - each time)

Next we have t^2u . (hold up)

How many do we have? (3)

3 times 30 squared times 4. So, 3 squared is 9 times 4 is 36, times 3 is 108. What category are these? (hundreds) So we have 108 hundreds.

(write new row of expression and add pieces onto cube - each time)

Next we have tu^2 . (hold up)

How many do we have? (3)

3 times 30 times 4 squared. So, 3 times 4 squared equals 48 times, times 3 equals 144. What category are these? (tens) So we have 144 tens.

(write new row of expression and add pieces onto cube - each time)

Lastly we have u^3

In this problem, that means it's 4 cubed. So that's 64.

(write new row of expression and add pieces onto cube - each time)

Now we have to add these up.

(draw 'equals' line at bottom of vertical expression and add)

Write on paper:

$$\begin{array}{rcl}
 234^3 = h^3 = 200^3 = & & 8,000,000 \\
 3h^2t = 3(200^2 \times 30) = & & 3,600,000 \\
 3ht^2 = 3(200 \times 30^2) = & & 540,000 \\
 3h^2u = 3(200^2 \times 4) = & & 480,000 \\
 6htu = 6(200 \times 30 \times 4) = & & 144,000 \\
 t^3 = 30^3 = & & 27,000 \\
 3hu^2 = 3(200 \times 4^2) = & & 9,600 \\
 3t^2u = 3(30^2 \times 4) = & & 10,800 \\
 3tu^2 = 3(30 \times 4^2) = & & 1,440 \\
 u^3 = 4^3 = & & 64
 \end{array}$$

12,812,904

We get 12, 812,904

(control of error is the cube of 234)

Now you can cube any 3-digit number you'd like.

Epilogue

From this work, the children will move to paper and stop using the cube. We might introduce a calculator at this point to help them do control of error checking; it facilitates the work at this point instead of diverting their work.

January 18, 2005

11. SQUARE AND CUBE ROOTS

Square Root

Concept

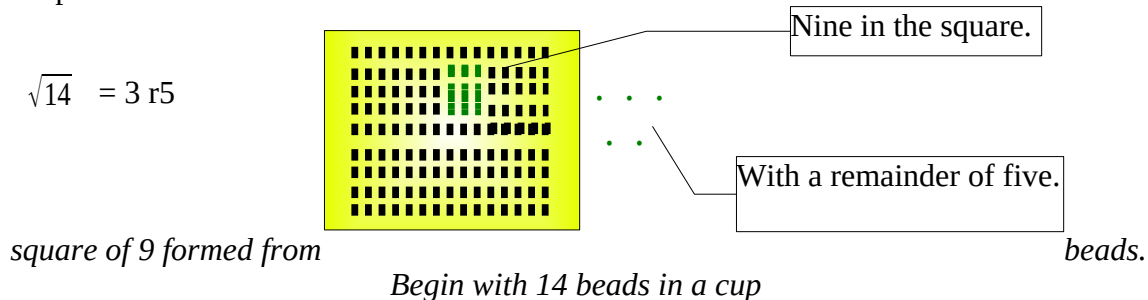
Age

8 to 9 years old.

Materials

A few bead squares from the bead chain cabinet, felt mat, the square roots board (a flat board with indentations for beads), green unit beads, an empty cup. Paper and pencil.

$$\sqrt{14} = 3 \text{ r}5$$



Prologue

The realizations from squaring prepare the children for square root work. The children have become familiar with what a square is and had practice building them. With numbers more than one hierarchical categories – certain categories, but only certain categories form squares – while others fill in rectangles and never form squares. Some make both squares and rectangles. Tens never make a square (refer to the square guide), hundreds form squares and rectangles – and so on. Thirdly, any time one builds a geometrical representation of a square, one will find the squares on the diagonal.

With these realizations, we can show the children how to extract the square root of numbers. First we present the concept and then three exercises.

Begin

(hold up a square)

What is this? (a square of 4)

How many beads are there altogether here? (16)

How many beads are there along here (point to bottom)? (4)

We can say that 4 is the square root of 16. The rest of the square grows from this root.

4 taken 4 times gives us 16. So 4 is the square root.

(hold up a square)

What is this? (a square of 8)

How many beads are there altogether here? (64)

How many beads are there along here (point to bottom)? (8)

We can say that 8 is the square root of 64. The rest of the square grows from this root.

8 taken 8 times gives us 64. So 8 is the square root.

(repeat once more, a third time with a third and last square)

How about the number 9? What's the square root of 9? (3)

How about the number 49? What's the square root of 49? (7)

(repeat until they've caught on)

Would you like to know how we write these down?

They used to write it this way



Radix is the Latin word for roots

Over the years the way it is written evolved into this.

(show current manner of writing)

The philosopher Descartes is credited for coming up with the symbol “square root”

So we would write $\sqrt{49} = 7$

What about the square root of 14?

(children can write it down if they like)

How about 32? (repeat for practice)

Any number contains a square root. Let's see how we find it.

===

Materials

Square root board, cup and green beads.

(count 14 beads into the empty cup)

With these 14 beads, let's see the largest square we can build.

(place 1 bead)

This is a square but I think I can keep going.

(build square of 2)

Here's another but I can keep going still.

(build square of 3 – 9 total)

Here's another but I think I can continue.

Ah, but not enough to finish another square! These extras then I'll move to the bottom.

We can see the square root of 14 is 3, with a remainder of 5.

$$\sqrt{14} = 3 \text{ r}5$$

Epilogue

They know how the **concept of square root, how to write them** and the fact that **any number contains a square root**. Note remainder can be larger than root.

January 18, 2005

Square Roots with More than One Digit - 1

Materials

Golden beads (units, tens and hundreds), felt mat, paper and pencil.

Prologue

Choose an exact square

Begin

(form 576 out of golden beads on the mat: Begin with four squares in upper left, then bars along the bottom and right edges – filling in the last little square at bottom-right with individual golden beads. show root on bottom and right edges)

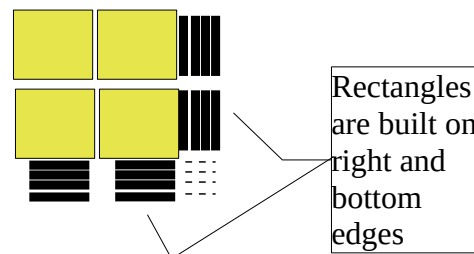
Now we're going to see what the largest square is that we can build out of 576.

(distribute the 4 hundreds)

I have one left. What do I need to do to keep going? (exchange for 10 tens)

Remember our 10s build on the rectangles

So I place them here and here.



I have one 10 left. What do I need to do to keep going? (exchange for 10 units)

Now with my units, I need to fill in this square. (do so)

I have no beads left over so I had an exact square.

I can read it along the edge (only count outermost bead layer). So the square root of 576 is 24. Let's write it.

$$\sqrt{576} = 24$$

What about 854, 961 or 676? You can try these on your own.

Epilogue

We are limited to numbers that go up to hundreds due to the materials. We don't let children choose their numbers yet.

The children can check answers by squaring. Show them how to check their work.

January 18, 2005

Deriving Number of Digits for a Square Root - 2

Materials

Paper and pencil. Guide squares (after introduction), peg board, box of hierarchically colored pegs, N/N^2 chart.

Prologue

We use the peg board to allow us to work with larger numbers and to work symbolically. Be sure to **bring down the cup(s) of the categories we're working with** to isolate what we're working with and organize that which awaits to be done. Explain the idea of the N^2 chart and why we break squares into periods when we find the root.

Begin

Introduction

What's the square of 4? (2)

What's the square of 3? (9)

One digit in our square.

What's the square of 5? (25)

What's the square of 9? (81)

Two digits in our square.

So for squares of 1-9, the squares can have 1 or 2 digits in them.

(show $N-N^2$ chart if needed, listing numbers 1-9 in column with their squares in next column)

Suppose these numbers from 1-9 were tens instead of units. If we square a ten, we get hundreds. So now it would be **1 or 2 digits in the hundreds** that would give us the square of these numbers.

The reason this is important to understand is because this tells us how many digits there will be in the square root of any number. Because the values of the squares have either 1 or 2 digits in them, when we look at any number, we'll **allow 2 digits in that number for**

each digit of the square root.

So 576 – taking that number – we start on the right and count over 2 places and make a mark like this (5'76). We know there will be 2 digits in the square root.

What about 1483? How many digits will there be in the square root here? (2)

Start on the right and count 2 places and make a mark (14'83). We have to allow 2 digits for the second pair of digits of the square.

So 57'86'54 will have 3 digits in the square root. For any number we can find how many digits are in the root using this method.

This tells us what kind of square – binomial for 2 digit root – trinomial square for a 3 digit root – or a quadrinomial for a 4 digit root.

Build Square

Let's take the example 625 (write down on paper)

First thing to do is to mark off our number to find the number of digits in the square root.

Count over 2 digits and make a mark (6'25)

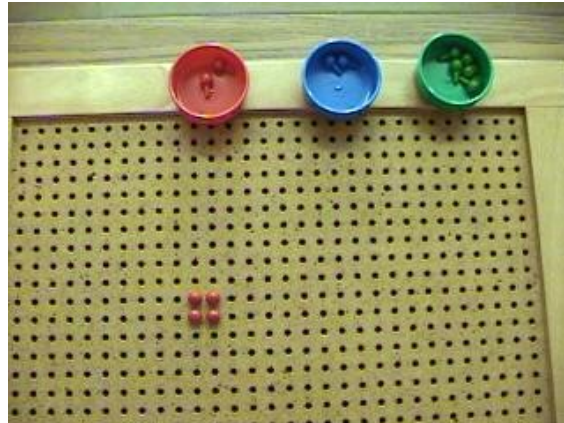
This number will have 2 digits in the square root. That tells me I'm going to build this square (hold up squaring guide chart for binomial version)

We form our number out of our pegs – separating our cups slightly by result digit.



Our highest category is hundreds and **we know that hundreds form squares.**

So we need to build the largest square than we can out of these 6 hundreds.



Next category is our tens – we build them onto our rectangles.

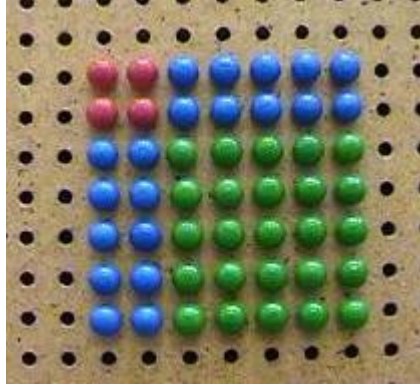
We also have two hundreds left that can be exchanged for tens as we go.

(build on 1 layer at a time)



With our **units we fill in the squares** – and I have **tens left over to exchange as needed**.

So we build on 1 layer at a time for our units.



We **find our answer across the bottom or along the right side.**

The square root of 625 is 25.

$$\sqrt{625} = 25$$

How can we check our answer? (multiply 25 times 25)

(encourage children to do several more examples – choose their number though)

Let's find the square root of 576. How many digits will I have? (mark off to find 5'76)

Our square root will have 2 digits. It will be a binomial (hold up square guide)

(form square out of pegs in the cups)

First we start with out hundreds. What is the largest square we can build?

(proceed layer by layer)

It's the square of 2.

(here's the new part for writing)

How many did I use up? I used 400 – so let's subtract that.

What do I have next? (10s)

Let's bring that down on our paper too.

How many tens did we use up? (16)

So I should have 1 left over and I do)

$$\begin{array}{r}
 576 \\
 - 4 \\
 \hline
 1 \\
 \\
 576 \\
 - 4 \\
 \hline
 17 \\
 16 \\
 \hline
 1
 \end{array}$$

What do we do now? We use our units and bring down our units on paper as well.

(fill in square)

$$\begin{array}{r}
 576 \\
 - 4 \\
 \hline
 17 \\
 16 \\
 \hline
 16 \\
 16 \\
 \hline
 0
 \end{array}$$

How many units did I use? (16) This leaves us nothing left over.

What is the square of 576? We can look right here(bottom edge) and see that it's 24.

===

Let's take the number 1156. How many digits will the square root have? Start on right, count over 2 and make a mark. Count over again and we see our root will have 2 digits.

(form 1156 on the cups – add new cup – separate into 2 groups as we do the number with the mark)

What do I do now? Why can't I build a square with these thousands? Here on my square guide – **we see thousands never form a square.**

So first I have to exchange this thousand for ten hundreds. (do so)

Now with all these hundreds, I'll build the largest square than I can.

Now how many did I use? (9) So, I'll subtract on paper and I see I should have 2 left.

Now I want to bring down my tens and I'll do that on my peg board and on paper.

$$\begin{array}{r}
 \sqrt{1156} \\
 - 9 \\
 \hline
 25 \\
 24 \\
 \hline
 1
 \end{array}$$

How many tens did I just use? I have a rectangle of 12 here and another rectangle of 12 here. When I subtract I should have 1 left over – and that's what I have here in my cup.

My square is already delineated so now I just need to fill it in with my units. We'll bring units down on the board and on paper.

$$\begin{array}{r}
 \sqrt{1156} \\
 - 9 \\
 \hline
 25 \\
 24 \\
 \hline
 16 \\
 16 \\
 \hline
 0
 \end{array}$$

How many units did we use up? (16) So we have none left over and what is the square root of 1156? (34) We can see across the bottom or down the right side our answer is 34.

Epilogue

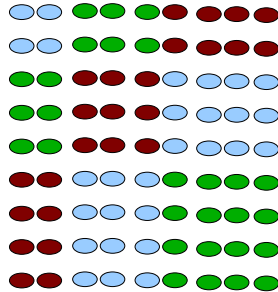
Get into the habit of exchanging when it is needed – not before. We can begin to introduce more writing as we go. We could go on to larger numbers: 3-digit and 4-digit roots.

January 18, 2005

Recording Analysis of Square Roots - 3

Materials

Paper and pencil. Guide squares (after introduction), peg board, box of hierarchically colored pegs, N/N^2 chart.



completed square of 54,756. note that the root is along bottom and right edges.

Prologue

We add in recording the analysis of what we used up. **Use the guide square for help in building the square as you go.** Problem is written vertically as you go. Interleave bottom and right edges while building outward from upper left corner.

Begin

We take square root of 54756.

Beginning on the right I mark off between the 7 and 5 and then between the 5 and 4. So this time, we'll have a square root with 3 digits (5'47'56)

So this won't work (hold up binomial guide square) (not enough digits)

So I switch to this one (hold up trinomial guide square)

So we need another cup too. Let's form our 54,756 in our cups.

We start with our first category – **ten-thousands**.

And looking at the trinomial square guide, we can see this is our first square (upper left).

The largest square we can build is 2. (because we have 5 ten-thousands)

The next thing is to bring down the next category – **thousands**. (on paper and pegboard)

Our thousands are used for what? (to build these two rectangles – see trinomial-square guide – begin building the square)

I'll exchange the ten-thousand for 10 thousands ... (build)

I can't go on – let's see what we built on to our square? How many rectangles? (2)

Each of them can be expressed as $2 \times 3 =$ so we built on two rectangles – each of which is (2×3)

So 2×3 equals how many? (6) times 2 is 12 – so I used up 12 thousands.

(write product into vertical expression)

When I subtract, I have 2 left over – and I do have 2 in my cup.

$$\begin{array}{r} \sqrt{54,756} \\ 2^2 \quad = 4 \\ \hline 2(2 \times 3) \quad = 12 \\ \hline 2 \end{array}$$

And we next bring down **hundreds** – on the peg board and on paper.

What about our hundreds – we have 3 areas to build (2 rectangles and a square)

If we look on the peg board, the spot for the square is already delineated so we'll fill that in first. (do so)

What did I build – a square of 3 – which uses up how many? (9)

So we subtract 9 and I should have 18 hundreds left.

Now with our 18 hundreds, we can see how many times we can build out the rectangles.

We built how many rectangles? (2) Each can be express by 2×4 .

2 times 4 is 8 – times 2 – is 16. Let's subtract that on paper. We should have 2 left.

$$\begin{array}{r}
 \sqrt{54,756} \\
 2^2 \qquad = 4 \\
 \hline
 2(2 \times 3) \quad = 12 \\
 \hline
 3^2 \qquad = 9 \\
 \hline
 2(2 \times 4) \quad = 16 \\
 \hline
 2
 \end{array}$$

Now bring down the **tens** – on the board and on paper.

We're filling in rectangles – they've already been delineated – we just fill them in. (do so)

What did we build? Two rectangles of 3 by 4.

3 x 4 – times 2 is 24. (write it under the problem) We should have 1 left over and we do.

$$\begin{array}{r}
 25 \\
 2(3 \times 4) \quad = 24 \\
 \hline
 1
 \end{array}$$

Let's bring down the **units** – on the board and on paper.

So now we bring down units and we're just filling in our delineated square.

What did we build? (square of 4) Let's record that. ($4^2 = 16$ – aligned in the problem)

$$\begin{array}{r}
 16 \\
 4^2 \qquad = 16 \\
 \hline
 0
 \end{array}$$

What is the square root of 54756? 234 is what we see here at the bottom.

We can check this by multiplying 234 x 234.

Epilogue

Try these numbers in practicals and for numbers to suggest to the children:
55,696; 273529; 1,764; 5,236; 12,564; 27,394,759.

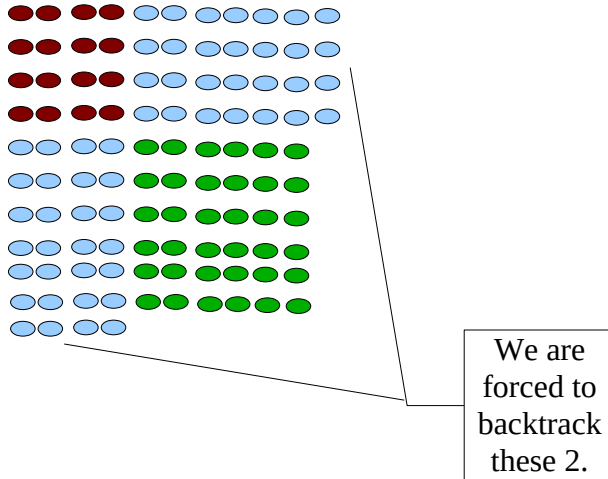
One thing that comes up sometimes – say we're building up squares of thousands. What's the most times you can build out? (9) Why – because you go to the next category. So sometimes, you will have enough pegs to keep building out further. Remember, we never build out (layers) more than 9 times. If we encounter it, exchange and the remaining beads on that category will go into filling on the remainder of the square.

January 19, 2005

Square Roots and Back-Tracking - 4

Materials

Paper and pencil. Guide squares (after introduction), peg board, box of hierarchically colored pegs, N/N^2 chart.



Prologue

This follows a lot of practice extracting 2-digit roots, recording analysis and extracting 3-digit roots. We continue to suggest numbers for the children – though they can have remainders. At some point they will choose their own numbers; when this begins we need to be available to them for unforeseen circumstances: what if you don't have enough pegs to fill in the square. In this case we use back-tracking.

Begin

Let's take 2174. We want its square root.

How many digits? **(21'74)** There are **two digits**, right?

Let's set up our pegboard, cups and build our squared number – 2174.

Since we have two digits we need the binomial guide square.

Now we can **start building our square out of hundreds – because thousands never make a square.**

I have to exchange my thousands for hundreds.

(share out hundreds)

Now I need to exchange my other thousand for hundreds. (continue sharing hundreds)

The square of 4 is the largest square I can make out of my thousands and hundreds.

We write our result ($4^2 = 16$) and subtract and we have 5 hundreds left.

Now we bring down our tens, both on paper and on the pegboards.

With our tens we build rectangles – the largest we can.

I can't go any further. We used up 2 rectangles, each of (4 x 7)

So 2 times 4 times 7 – is 56. So I used 56 tens.

Let's write that and subtract – the paper tells us we have 1 left in the cup.

$$\begin{array}{r} \sqrt{2174} \\ 4^2 \qquad = 16 \\ \hline 2(4 \times 7) \quad = 56 \\ \hline 1 \end{array}$$

Now let's bring down our units – on paper and on the pegboard.

We use out units to fill in the square – but look!

I don't have enough. I have run out of units.

Now we have to back-track.

I cannot build out rectangles this far. So **we pick up one row from each rectangle.**

We have to change our writing also – we built out only 4 x 6 – not 4 x 7

So 4 times 6 times 2 is 48. So now when we subtract we have 9 tens and 4 units.

$$\begin{array}{r} 57 \\ 2(4 \times 6) \quad = 48 \\ \hline 9 \end{array} \quad \begin{array}{l} \text{Correction due to} \\ \text{backtracking} \end{array}$$

So let's resume filling in our rectangle .

So I have built a square of 6. That's how many (36)

So that's how many units we used up.

Let's subtract on paper and we should have 58 left over. That's our remainder.

So what's our square root? (46) We read right across here (bottom and right side)

Our answer is 46 R58

$$\begin{array}{r} \sqrt{2174} \quad = 46 \text{ r}58 \\ 4^2 = \quad 16 \\ \hline \quad 57 \\ 2(4 \times 6) = \quad 48 \\ \hline \quad 94 \\ 6^2 = \quad 36 \\ \hline \quad 58 \end{array}$$

Epilogue

Sometimes we run across problems like this where we have to back-track, correct the paper problem and re-step through the following step. This was how that is handled. We may have to help the children realize that **one can never build a rectangle out more than 9 layers**. Also, they need to realize **we can only take away from rectangles** – never from a square that's already built on the pegboard.

We can also encounter a situation where we cannot complete a square; in this case, we need to back-track the surrounding rectangles.

January 19, 2005

Passages to Abstraction

Completing the Square While Going Along

Materials

Paper and pencil. Guide squares (after introduction), peg board, box of hierarchically colored pegs, N/N^2 chart.

Prologue

Back-tracking is fun initially – though it takes some of the fun out of building squares and we may run into lots of back-tracking. We hate erasing our problems after a time. How can we avoid this back-tracking? This presentation handles this issue.

Leave off the analysis for each partial square/rectangle.

Begin

We want to know how to extract square roots without the pegboard.

First, let's take 54756. We want the square root.

First let's see how many digits we'll have. (5'47'56) There are **3 digits**.

So we'll use our trinomial guide square.

I need my 10,000s cup this time. I'll build my number in the cups to begin. (do so)
(cups are filled and placed at top of pegboard as usual – separated slightly per result-digit)



Each time **we're going to complete each square – as we go**. So this is different now.

What do we do first? What can we build squares with? Yes, we'll use 10,000s.
 Let's build the largest square we can with our ten-thousands – that's a square of 2.
 We used 4 ten-thousands – let's write that.

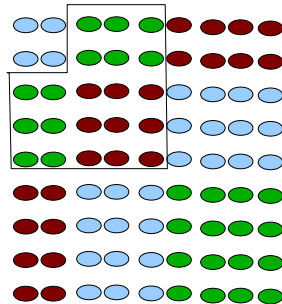
**Once we build a square on the pegboard, we know the first digit of the square root.
 (2)**

Let's record that part of our answer as well.

$$\begin{array}{r} \sqrt{54756} = 2 \\ 4 \\ \hline \end{array}$$

Now **we said we want to complete a square each time.** We did this with 10,000s.

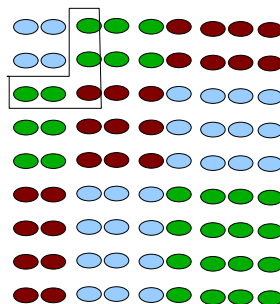
Looking at our guide square – this means we **need to build the first two rectangles as well as the next square.**



So we'll need what categories? (thousands and hundreds)

So, I'll bring down two categories at a time now – since we need both to complete the square. Let's bring them down on the board as well as on paper.

Now I'll build the 2nd, 3rd and 4th complete areas – peg layer by peg layer.



Build layer by
 layer rightward and
 downward, **across**
categories.

I don't have enough to continue. So what did we use up. We look at the square – which is the lowest category in the square so far.

How many have I used here (last build red square)? (900)

And what about my rectangles? (2 times 2 x 3) These are thousands.

So I used 12900 (write 129 aligned under like categories in original numbers)

Now we can subtract. **My paper tells me I should have 800 and 1 thousand left in my cups. Let's check the cups to be sure we're on track.**

We also know the next digit must be a 3. So we can write it in our answer.

$$\begin{array}{r} \sqrt{54756} = 23 \\ 4 \\ \hline 147 \\ 129 \\ \hline 18 \end{array}$$

To complete the next square I need to bring down the last two categories. On our guide square we can see the the bottom and right side of our trinomial square uses units and tens and hundreds. We already have our hundreds, let's bring down the tens and units now.

(note as we build layer by layer, **always increasing bottom and right side by one row of beads – we'll use hundreds, tens and units – per the guide square**)

How many did we use up in this square – that's 16 units.

Now we want to look at our tens. How many did we use up in our 2 rectangles. (24)

We used 24 tens – 240 – plus 16 units. That's 256.

We have 256 and now we look at our hundreds. How many did we use? (16)

We used 16 hundreds – plus our 256 – that makes 1856. Let's write that.

Subtracting we find we should have no remainder and there is no more pegs in my cups.

Looking at our last square, we see 16 units, so the last digit is 4. So let's add that.

Our answer is 234. We see that down the right side and at the bottom of our square. To check this, we can square 234.

$$\begin{array}{r}
 \sqrt{54756} = 234 \\
 4 \\
 \hline
 147 \\
 129 \\
 \hline
 1856 \\
 1856 \\
 \hline
 0
 \end{array}$$

Epilogue

We know the red rectangle of the trinomial is our last – so there is no other choice – use red pegs to complete the square. We build down and right until we run out of materials – layer by layer. At times, we may not be able to complete the square for a given layer – in which case, we remove the last partial layer and that becomes our remainder.

January 19, 2005

Calculating Next Root Digit

Prologue

We'll use the same number but we'll see if we can figure out what the next digit of the root is going to be before we build it on the pegboard.

Begin

Let's find the square root of 54756.

Looking at the category – 10,000 – can you tell me what the first digit will be?

It is the square of 2 since our first digit is 5 – the largest square possible is 4 – the square of 2. We can write down our first digit now.

We'll still complete a square each time plus surrounding rectangles; so which categories do we need to bring down?

(thousands and hundreds – since we're building two categories at a time)

Can we figure out how many times we'll be able to build out from our first square?

What category will we use to build out our rectangles? (thousands)

Thousands build rectangles. Hundreds make a square so I'll set them aside.

(set to right side of pegboard – other units and tens cups are at left)

I have 14 thousand pegs to build with.

It is those 14 pegs we can use for my rectangles. How many do we use up each time we build up 1 layer? (4 – 2 at each end of our square of 2)

How many times will 4 go into 14? (3) I think 3 is the next digit of our square root. Let's prove it on the pegboard.

Did we verify 3 was our next digit? Yes

And now we need to write what we used up.

We used 9 hundreds and we used 12 thousands – 12900.

$$\sqrt{54756} = 23$$

$$\begin{array}{r}
 4 \\
 \hline
 14 \div 3 = 147 \\
 129 \\
 \hline
 18
 \end{array}$$

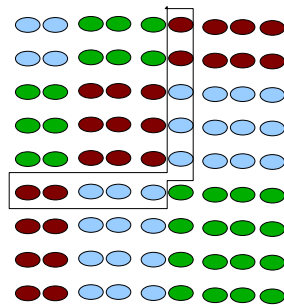
I should have 18 hundred left in my cups and I do.

Which categories build the rectangles now (see guide)? Hundreds and tens.

How many times we can build these out determines our next square of units.

So we set units aside for a moment.

How many will we use up each time we build one layer on? (46)



Build layer by
layer rightward and
downward, **across**
categories –
excepting units.

4 hundreds (because of these 2 red rectangles) and 6 tens (for these two blue rectangles)

How many times does 46 go into 185 – as our trial digit. (4 times)

So 4 is what we think the next digit of our square root is.

$$\sqrt{54756} = 234$$

$$\begin{array}{r}
 2^2 = (2) \quad 4 \\
 \hline
 14 \div 4 = (3) \quad 147 \\
 129 \\
 \hline
 1856 \\
 185 \div 46 = (4) \quad 1856 \\
 \text{plus } 4^2 \quad \hline
 0
 \end{array}$$

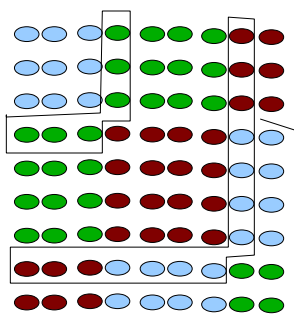
Trial Divisors 4 and 46 are
derived from the pegs
required to complete a new
layer around the bottom and
right edges – excepting the
corner that falls into the next
square to be built. See
graphic above for divisor 46.

What is the square root of 116965? (capture written only – nothing new)

$$\sqrt{116964} = 342$$

	9	Largest square possible.
	-----	Bring down next 2 categories.
	269	Which categories for rectangles?
26 ÷ 6 = (4)	256	Divide by 2 times 3 thousands...
	-----	(disregard hundreds for a moment)
	1364	Bring down next 2 categories.
136 ÷ 68 = (2)	1364	Divide by 2 times 34 tens...
	-----	(disregard units for a moment)
	0	

Divide by 2
times 3
thousands



Divide by 2
times 34
tens.

Epilogue

As we build on, the rectangles always determine the size of the surrounded square. We always look to our remainder on paper to see what rectangles we can build – which then determines what square will be surrounded by them. On paper, we leave off (or combine) the last square as we calculate against the size of the next two rectangles.

January 19, 2005

Calculating Next Root Digit Using the Rule

Prologue

Isn't there a way we can figure what has been used up without all that adding? Note that the **trial divisor** is derived from the number of pegs needed to complete a layer of pegs around all current squares and rectangles built so far; we omit the bottom right corner that falls into the next square to be built.

Begin

Let's find the square root of 55696.

(begin as in preceding lessons, determine number of digits - 3)

(what's the largest square we can make out of 5 ? - thus 2 is our first digit)

(we used up 4 and subtract - then bring down the next group – **called periods**)

(our sides will use 2 so we rectangles will use up 4. 15 divided by 4 is 3 – our next digit)

(fill in rectangles and a square with thousands and hundreds – verify 3 is the second digit)

I'm going to give you a bit of the rule for finding square roots. The rule says: you annex (put next to) to the **trial divisor (4 (of 15 divided by 4)) the new root (3)** that has been found and multiply it by the new root.

43 times 3 = 129

Is that how many I used up on my pegboard? Yes!

When we subtract, we should have 7 hundreds left and 2 thousands.

(verify 2700 is left in cups)

(bring down the next period on paper and pegboard)

(calculate new – 279 divided by 46 (sides are 23 plus 23))

How many times will 46 go into 279? (6)

(new part – 46 (trial divisor) plus annexed 6 (new root) times 6 (trial digit or new root))

(466 times 6)

(verify that it works by building the square)

(verify the answer of 236 by squaring it – is it 55696?)

$$\sqrt{55696} = 236$$

	4	

15 ÷ 4 = 3	156	Divisor 4 is derived from number of pegs needed for a single layer at bottom and right sides of the previous square. Divisor 46 is derived similarly but includes traversing a rectangle as well.
43 x 3 = 129	129	

279 ÷ 46 = 6	2796	
466 x 6 = 2796	2796	

	0	

Epilogue

We apply line from the rule: append (annex) trial divisor the new root to form a new number – then multiply that newly formed number by the new root. In this examples we had:

4 trial divisor and 3 as new root ==> 43 times 3
 46 trial divisor and 6 as new root ==> 466 times 6

General Pattern

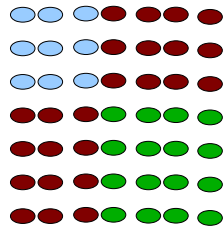
1. Look at first period
2. Largest square possible – build square
3. Subtract – bring down next period on board and pegboard
4. Predict next root (next period divided by (value allotted for each side of current square times two))
5. Fill in 2 board rectangles and next square on pegboard
6. Implement the rule (annex the previous trial divisor with the new root from that equation that has been found, to form a new dividend - and multiply it by the new root) and verify. We build rectangles: note trial divisor plus new root (answer) of current division equation - forms the multiplicand in the subsequent multiplication problem. The multiplier is the new root (also the answer from previous division problem). The dividend of previous equation is determined by the period brought down whose categories are used to build rectangles – we don't use categories for this that build squares – though we build the square as well when we actually build the rectangles.
7. Repeat steps 3-6 until all periods are processed

January 20, 2005

Special Cases

Materials

Pegboard, hierarchically colored pegs, guide squares, 2 strips of thin (to fit between rows of pegs) cardboard paper about 6 inches long, paper and pencil.



Note how categories meet that are not normally adjacent; an entire cross formation in the middle is left out.

Prologue

Here we see the periods are the same but we don't include all digits if a digit category is used to build square and we're building rectangles, for example.

Begin

Let's take the number 92416. I want its square root.

What do we do (find the periods 9'24'16). We will have a trinomial square, right?

Now we know which guide square we'll use – and I need units, tens, hundreds, thousands and ten-thousands cups and we'll form our number 92416.

(number is formed with pegs in cups at top of pegboard)

I want to build the largest square I can out of ten-thousands; we can build squares from ten-thousands, right?

Since our number begins with 9, we can see our first digit will be 9. Let's build that square now. (do so)

Subtracting, we bring down the next period (24 hundreds). What can we do with them here on the paper? (**ignore hundreds for now because they form a square and we're looking for rectangles**)

We only have 2 thousands and we can't take away from a square – I can't backtrack. If we don't build the first two rectangles, we cannot have a square either – so this means the second digit of our answer is **zero**.

We'll use these paper strips as a placeholder for our zero.

(place paper strips between pegs as able to show a zero value digit in root)

Our first square is the square of 3. Using the rule our problem is 2 divided by 6 which is zero times.

Bring down our next period. But we'll use 241 as our next dividend since units create a square. 241 divided by 60 is 4 so that is likely to be our next answer digit.

We take the trial divisor or 4, append (annex) it to 60 (604) and multiply by our new root. So our problem is 604×4 – and that gives us 2416.

Epilogue

The number on the right of each period is always used to build a square and is set aside when we are trying to build rectangles. Try 164,836, 42849 to find and handle zeros in the middle of the square root. Note that the paper strips are place-holders for a zero digit in the root.

A period always contains 2 rectangles and 1 square – except for the very first period from which we create the largest square possible. Anytime there is a zero in the square root, we interpret paper strip as a division operation against the root we have found.

===

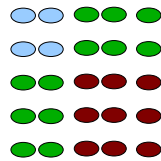
(special case of zero in units of root)

Let's take 52918 and find its square root. This is a trinomial example.

(5'29'18 means 3-digit root)

(build largest square possible for 10,000s – square of 2 – which uses 4)

(first digit of root is 2)



$$\sqrt{52918} = 230 \text{ r}18$$

$$2^2 = 4$$

$$12 \div 4 = (3) \quad 129$$

$$43 \times 3 = 129$$

$$1 \div 46 = (0) \quad 18$$

$$0$$

$$18$$

(build square up to this point – 3 does work as 2nd digit of root)

(place paper strips since next equation (1 divided by 46) is zero.

===

Let's find the square root of 900.

(complete a square of 9 in hundreds)

I used up my pegs.

I see 300 as my root, but is that correct? (no)

No – it's 30, isn't it?

When the root has a zero in the simple units digit, we drop the last zero of the result when building the square on the board.

===

By the rule:

$$\sqrt{46542} = 215 \text{ r} 317$$

$$2^2 = 4$$

$$6 \div 4 = (1) \quad 65$$

$$41 \times 1 = 41$$

$$244 \div 42 = (5) \quad 2442$$

$$425 \times 5 = 2125$$

$$317$$

Rule for Finding Square Root

We started with the fact that any number can contain a square root. Then we did exercise 1 which used the golden bead material, where we extracted 2 digit square roots – still sensorially.

At exercise 2 we went on to extracting square roots with the pegboard. There we worked category by category – initially just writing down the problem and answer. Then we introduced a bit more writing – subtracting the amount used up each time. And the analysis of how we used up what we subtracted was also added to the written page.

Then in exercise 3, Passages to Abstraction, we extracted roots using more than 1 category, completing the square as we went along (layer by layer using more than 1 category at a time). There were steps to the writing here also: first we noted amount used, subtracted the amount used, calculating that by adding up what was on the pegboard, then we added in the step of how to calculate the next digit of the root. Then we added the step of how to calculate how much we used up.

Finally, we saw some special cases. This is presented around age 10 or 11 generally.

1. Point off as many periods of two digits each as possible, starting on the right.
2. Find the greatest number whose square is equal to or less than the left hand period. Write this number as the first digit of the root.
3. Square the first digit of the root, subtract its square from the first period and annex (attach) the second period to the remainder in the vertical expression.
4. Double the part of the root already found (which is the side of the square we have built so far) for a trial divisor. Divide it into the remainder omitting from the latter the right hand digit. Write the quotient as the next digit of the root.
5. Annex the root digit just found to the trial divisor (which becomes subsequent multiplicand and which is the complete divisor of the entire remainder at hand) to make the complete divisor. Multiply the complete divisor by this root digit. Subtract the result from the dividend and annex to the remainder the next period for a new dividend.
6. Double the part of the root already found for a new trial divisor and proceed as before until the desired number of digits of the root have been found. Square the root found to check your answer.

January 31, 2005

Cube Roots

Introduction

Age

Upper elementary. Not all children will get this.

Purpose

Development of mathematical mind. Second purpose is to find the cubed root of any number. Note that periods are now 3 digits long.

Prologue

Cubed roots follows the same pattern as lessons in square roots. We introduce the concept, then extract cube roots of smaller numbers, then larger numbers and then we work our way into abstraction – arriving at the rule.

Since generally, we've done square roots, we explain less at the beginning that we did with square roots.

Prerequisites

If a child is attracted to cube roots, he need not complete square roots first; incorporate some of the reasoning from square root lessons, here. Cubing is required.

Materials

A few bead cubes from bead chain cabinet, felt mat, paper and pencil, the N^3 chart, box of 1-centimeter cubes.

Begin

(3 cubes are laid on center of felt mat – green, yellow, etc.)

Concept

What do I have here? (cube of 4)

How many beads are in this cube? (64)

How many are there along this edge (point) (4)

(point to 2 adjacent sides at bottom and then 3rd side going up)

How many are there along this other edge) (4)

And how many here (point to 3rd dimension) (4)

So we say that 4 is the cube-root of 64.

How many beads are in this cube? (216)

What's the cube root of this one? (6)

This is how we write this. $\sqrt[3]{216}$

If we use a square root, we don't write this little number. For cubed roots, we need to write that little 3 to tell us we're finding the cubed root.

How many beads are there here? (8)

What is the cube root of 8? (2)

What's the cube root of 27 (3)

===

Any number contains a cube root.

(place box of 1-cm. cubes on the felt mat)

What about 33? Let's find the cube root of 33.

Let's count out 33 little cubes. (count out 33 small white 1-cm. Cubes)

Now I want to know what the cube root is.

Here's the cube of 1. But I can still go on (form cube of 1 – single 1-cm. cube)

Here's a cube of 2, but I can still keep going. (form cube of 2 – 4 cubes together)

Here's the next cube – the cube of 3.

I have a few more cubes left but not enough to build to the next one.

So the cube root of 33 is 3 with a remainder of 6.

$$\sqrt[3]{33} = 3 \text{ r}6$$

Maybe you want to try one. What's the cube root of 72? Or 68?
(encourage them to try one and to write on paper what we've done)

===

Chart

To get even more practice, we can find cube roots using our $N-N^3$ chart.

Suppose we want to know the cube root of 100?

I could count out my cubes but that takes a while. Let's look at our chart here.

(hold up chart – left column=1-9, 2nd column=numbers squared, 3rd column=numbers cubed, 4th column is for later work – do not mention)

N	N^2	N^3	$3N^2$
1	1	1	3
2	4	8	12
3	9	27	27
4	16	64	48
5	25	125	75
6	36	216	108
7	49	343	147
8	64	512	192
9	81	729	243

Let's look in the 1st and 3rd column. I want to know the cube root of 100.

Let's look – do we have that number in this column? (no) We have 125 but that's too much.

So the cube root of 100 has to be 4 with a remainder of 36 (looking at columns 1 and 3)

What about 200? It's 5 (see 3rd column of row 5). The cube root of 200 is 5 (125 is listed in the column) with a remainder of 75.

So we can use this chart to find our cube roots. Which number should we find the cube root of – up to 999. Look in the 3rd column and find greatest amount that does not exceed the number we're after.

Epilogue

The N^3 -chart works for cube roots up to 1 digit. In subsequent lessons, the $3N^3$ column is used to calculate the quantity used up by building a given cube-root digit value in three directions.

January 31, 2005

Exercise - 1

Part A

Prologue

We'll find 2-digit cube roots.

Materials

Cubing (wooden) material, felt mat, paper and pencil, the N-cubed chart.

Begin

Let's start with 79507. We want its cube root.

Remember when we did square roots – the first thing we did after writing down our number, was to mark off periods – 2 digits per period.

We'll mark off periods here too, but we'll **mark off groups of 3** per root digit this time.

The value of the cubes have how many digits? (1-3 digits) This means we have to allow 3 digits in our number for each digit of the root.

(mark periods – 79'507)

Now we look at our first group on the left – our first period. What is the largest cube we can make out of 79? (cube of 4)

because the cube of 4 uses up how many (64). Yes, that close – let's write that and subtract.

$$\begin{array}{r} \sqrt[3]{79507} = 4 \\ - 64 \\ \hline 15 \end{array}$$

I need to build out from this cube. In how many directions? (3)

So from my box I'll pick up 3 squares of 4 and place 1 here, 1 here and 1 here.

3 times 16 is 48 so let's subtract 48. I still have 107

So pick up from the box 3 squares of 4 and we know where they go this time.

We have 1 here, 1 here and the last here (same 3 places as last time)

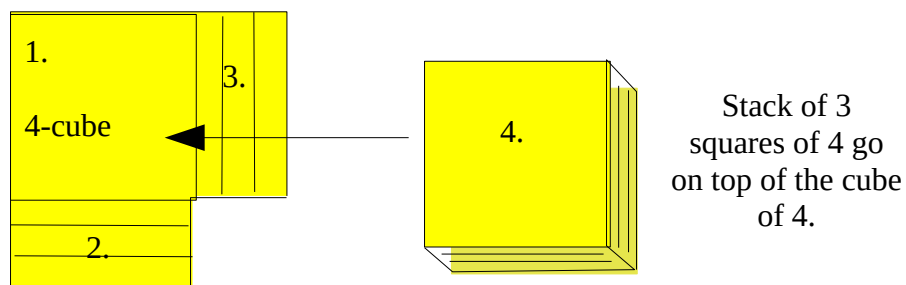
Three squares of 4 is how many (48) So let's subtract from our 107.

I still have 59 left over. Do I have enough to build on again? (yes)

(take another 3 squares from the box) So, again, I'll take 3 squares and place 1 here, 1 here and 1 here. On paper, again, I'll subtract 48.

$$\sqrt[3]{79507} = 43$$

$$\begin{array}{r} - 64 \\ \hline 155 \\ - 48 \\ \hline 107 \\ - 48 \\ \hline 59 \\ - 48 \\ \hline 110 \end{array}$$



Build 4-cube and its 3 legs

I have 11 left over. Does that give me enough to build on again? (no)

On our paper now, bring down the next category, tens, and let's look at our materials.

The spaces are determined. What kind of square do I need to use to fill in each of these

arms?

I need squares of 3. And in each arm, it will take how many squares of 3? (4)

It will take 4 squares of 3 to fill in each arm.

(One stack goes atop part 2 above, building perpendicularly to part 2. Second stack goes atop part 3 above, building perpendicularly to part 3. Third stack of 3-squares, four of them, are stacked upward, placed adjacent to part 2 and 3 in the open corner.)

In each arm we have 9 taken 4 times which is 36. We have 36 taken 3 times for the 3 arms – so we have 108.

Let' write down the 108 and subtract.

And we can bring down our units now. (bring down the 7 from the units category)

We need 3 cubed - which uses how many (27)

(place last cube into final corner of top layer)

$$\begin{array}{r} 110 \\ 108 \\ \hline 27 \\ 27 \\ \hline 0 \end{array}$$

We have no remainder. So we can see our answer along any of these edges. (43)

==

Let's find the cube root of 389017.

(mark off periods beginning with a group of 3 – 389'017)

What is the largest cube we can build out of 389? (7)

So we begin with the cube of 7. It uses up 343. Let's write that and subtract. (do so)

We have a difference of 46.

Now, we build on with squares of 7. And we build out in 3 directions.

So, take 3 squares of 7 – 1 goes heres, 1 goes here and 1 goes here.

(top, right side and bottom side)

How many did we use? (49 taken 3 times – 147)

Let's subtract our 147 – we have 313. Is that enough to build on again?

(build on another 3 squares of 7)

How many did we use up? 147 again – let's subtract. I have 166 left.

Is that enough to go on?

(add on another 3 squares, subtract)

We have 19 and let's bring down our next category.

We have to look at the material since it tells us what we need next. I need squares of 3.

Each arm is going to need how many squares of 3 . (7)

Let's add them. Now we need to add our final cube. (do so)

Written image:

$$\begin{array}{r}
 \sqrt[3]{389017} = 73 \\
 - 343 \\
 \hline
 460 \\
 - 147 \\
 \hline
 313 \\
 - 147 \\
 \hline
 166 \\
 - 147 \\
 \hline
 191 \\
 - 189 \\
 \hline
 27 \\
 - 27 \\
 \hline
 0
 \end{array}$$

First set of rectangles, layer 1.

Second set of rectangles.

Final cube.

Epilogue

Cube 73 to find the control of error. Bring down next category once the

physical cube corners are filled out, indicating we are moving to the next kind of rectangles to fill out the cube.

Part B

Prologue

Gradually, we want the children to come to a conscious accounting of the steps and the pieces and how they add up. Passing from part A to part B is determined when you see they are ready for more. We build the cube in the center of the mat.

Begin

Let's find the cubed root of 157464.

How many digits will our cube root have (2) (157'464)

Now we look at left hand period and ask what is the largest cube we can get from it?

It's the cube of 5. (place 5-cube on mat)

So let's write that and subtract. **(notate what was used as we go now)**

On our paper, we need to bring down the next category. We have 324 to build with.

We're building out from a cube of 5.

Do you think we can figure out how many times we can build this out – before we do it?

(we built one layer at a time last lesson – this is new)

Each time I build on one layer I add how many? (75 – squares of 5 times 3 directions)

So if we ask how many times 75 goes into 324 what do we get? (write margin 324 / 75)

We think it will be 4 times – let's see if it is. That would mean **we build 5 squared out 4 times** on this direction, also in this direction and also in this direction.

Now how many did we just add? We built on 3 arms. Each arm used 100.

Each arm is made of what kind of square? (5)

In each arm I have how many squares (4) **(circle to show its trial)**

So this gives us the multiplication we need to do. ($3(5^2 \times 4)$)
 (see paper notes for entire expression as it is built during problem solving)

So I've use up 300.

Now we need to bring down the next category (since we've filled in our first 3 arms)

Now we have 246. What do we do? We need to fill in the next 3 arms?

The material tells us – look at the material.

What kind of squares (4) and how many of each (5)

(add in 3 directions – 5 4-squares)

So how many did we use up? (3 arms, each 5 times 4-square)

So we have 240 to subtract. We have 6 left over.

Now we bring down our next category (the last 4). What do we need to do?
 (fill in last spot)

This happens to fill in the cube of 4. So we'll bring in our last cube of 64.

And subtracting, we find zero. Our cube root of 157464 is 54.

$$\begin{array}{r}
 \sqrt[3]{157464} = 54 \\
 \begin{array}{r}
 - 125 \\
 \hline
 324 \\
 - 300 \\
 \hline
 246 \\
 - 240 \\
 \hline
 64 \\
 - 64 \\
 \hline
 0
 \end{array}
 \end{array}$$

$5^3 = 125$
 $324 \div 75 = (4)$
 $3(5^2 \times 4) = 300$

$3(4^2 \times 5) = 240$
 $4^3 = 64$

Epilogue

If we encourage the children to choose their own numbers, they will have remainder often and that's okay. They will also run into situation where they cannot verify the trial digit.

===

Suppose we have 378783. We want its cube root.

(mark off periods 378'783)

We can see our cube root will have 2 digits.

(find largest cube of first period – cube of 7 – place cube of 7)

The cube of 7 is 343. We used 343.

(write it plus analysis, bring down next category)

When I build on here what kind of squares do I need (7)

Figure Trial Digit

Square of 7 is 49 – in 3 directions is 147. Three is my trial digit now. (circle it)

Now let's build those squares on – 3 squares in this direction, in this direction and in this direction.

(calculate what was used and notate what was used. Subtract)

$$\begin{array}{r} \sqrt[3]{378783} = 73 \\ - 343 \\ \hline 447 \\ - 441 \\ \hline 6 \end{array} \quad \begin{array}{l} 7^3 = 343 \\ 447 \div 147 = (3) \\ 3(7^2 \times 3) = 441 \end{array}$$

We have 6 left and we bring down our 8. We need to fill in the 3 arms.

Our material tells us we need to use squares of 3. It will take how many in 1 arm? (7)

Backtrack

But I only have enough to build 1 arm. So 3 won't work. We have to backtrack.

Take one layer off of each arm and put them away.

Now we change our 3s to 2s to match our arms.

We get a different used amount – 294. So subtracting we have 153 – and we bring down out 8.

Our material tells us we need squares of 2. How many squares do we need in each arm

(7)

So we've used 84. (write analysis using trial of 2 now)

Now we bring down our last category.

What do we need now – based on the materials? (cube of 2)

(add analysis on paper for the last cube)

Our cube root is 72 remainder 14,535

Written Image:

$$\begin{array}{r} \sqrt[3]{387783} = 72 \text{ r}14535 \\ - 343 \qquad \qquad \qquad 7^3 = 343 \\ \hline \quad 447 \qquad \qquad \qquad 447 \div 147 = (2) \text{ backtracked from} \\ 3 \\ - 294 \qquad \qquad \qquad 3(7^2 \times 2) = 294 \\ \hline \quad 1538 \\ - \quad 84 \qquad \qquad \qquad 3(2^2 \times 7) = 84 \\ \hline \quad 14543 \\ - \quad \quad 8 \qquad \qquad \qquad 2^3 = 8 \\ \hline \quad 14535 \end{array}$$

Epilogue

Try 15,625 for backtracking practice.

February 1, 2005

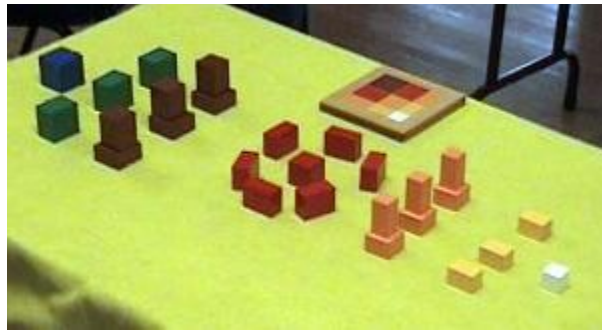
Three-Digit Cube Roots By Category - 2

Prologue

We'll find 3-digit cube roots. We work category by category to extract a 3-digit root. We begin with hierarchical trinomial cube in formation after the revolt. The cube is only the guide; layout is important because this way we automatically take the correct pieces as we go – layout guides. Most of the work is on paper.

Materials

Hierarchical cube, felt mat, N^3 chart, paper and pencil.



Begin

Let's take 143,055,667

(mark if off begin at right)

So our cube root will have 3 digits – a unit, a ten and a hundred.

So, I'll write h,t, u on top of where our answer should go later.

Look at first period on left – what is the largest cube we can get from it? (5)

Hundred cubed gives us millions so we'll have h^3 .

So I need to subtract 125 (5 cubed) – leaving 18.

So, the first digit of our root is 5. So I'll write it under my little h above my answer space.

If we look at our material, the next pieces are these green prisms. What category are these? (hundred-thousands)

So, that's my next category – I need to bring down hundred thousands.

How do we express this in htu notation (h^2t) How many do we have?

We know h but we don't know t . Can we calculate it?

We've got 180 (difference plus brought down)

Each time we build on a side, we would build on squares of 5 in 3 directions.

So we have 75 in each arm. How many times will 75 go into 180? (2)

Let's add that digit in the answer and circle it as a trial digit – for the t -digit.

Now can we calculate the value we've used? (3 times [$5^2 \times 2$])

That's 25 times 2 – that's 50 times 3 – is 150. Let's build those pieces on now.

I have 30 left over, looking at our material, the **brown prisms** are the next ones.

What category is this? (10,000s) So we need to bring down the 10000s on our paper.

In htu notation how do we express these pieces? (ht^2)

Calculating that – let's write the analysis on our paper. (3 times [5×2^2])

That gives us 60. Let's build these prisms on, and subtract in our problem.

(build on brown prisms)

What category is this prism? (also 10000) So I don't need to bring down the next category. How do we express these pieces (h^2u)

We have 3 of them, right? But we don't know the value of our units!

Can we calculate it? (we use the small brown prisms now – moving rightward)

These are h^2 and we build in 3 directions. So we need 245 divided by 75 ($5^2 \times 3$)

Let's add that to our paper problem.

I think 75 goes into 245 3 times – so 3 is our next trial digit. Let's circle it.

Now we have those $3h^2u$ – that's $5^2 \times 3$ – that's 75.

So we have 225 – let's subtract that. Let's build those pieces on (small brown)

I have 20 left over. The next category is 1000s so we need to bring down our next category. So we'll bring down thousands on paper – another 5.

How do we express these? (htu) How many do we have? (6)

So 6 htu (write at left of problem) – we have the values to plug in.

6 times 5 times 2 times 3 – gives us 180. Let's add that to paper. (to right side)

Let's build these pieces on (red htu pieces)

Let's subtract what we just used up. Is it time to bring down another category? Look at the material – this piece is also thousands – t^3 .

In this problem that means 2 cubed, which is 8.

Looking at the material, we have hundreds – expressed as hu^2 and we have 3 of them.

Let's bring down the hundreds on our paper – so we have 176.

So 5 times 3^2 - times 3 - give us – 135. So let's subtract and I have 41 left over.

Now I need to bring down the next 3 pieces since these are also hundreds. In our htu notation, how do we express these pieces? (t^2u) How many do we have? (3)

t^2 in this problem is 2^2 times 3. 2^2 is 4 times 3 – is 12 – times 3 – is 36.

Let's build on these pieces (last orange prisms) and let's subtract what we just used up.

I have 5 left over. My next category are these (yellow) tens category. So on our paper we need to bring down our tens. Wit material and on paper – bring down the next category.

How do we express these (tu^2) And we have how many of them? (3)

So, we have 3 tu^2 - in this problem, that 3 times [2 times 3^2]

3 squared is 9 – times 2 is 18 – times 3 – is 54.

Let's build the pieces on.

We have only this white piece left. What category is this piece? (units)

So let's bring down units on paper. In htu notation, how is this expressed? (u^3)

In this problem – its worth 3 cubed which is 27.

We can build on the cube and subtract what we've used up.

In fact, we have perfect cube.

(read the whole final expression)

Final written image:

$$\begin{array}{r} \text{h t u} \\ \sqrt[3]{143055667} = 523 \end{array}$$

Three-Digit Cube Roots By Category - 2

h3	- 125
----	-------

465

$5^3=125$

11. SQUARE AND CUBE ROOTS

Millions

3h ² t	- 150
-------------------	-------

180

$180 \div 75=(2)$
 $3(5^2 \times 2)=150$

Hundred-Thousands

Epilogue

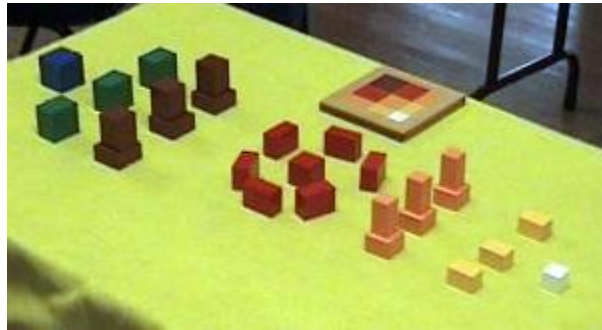
The children would spend lots of time on this. They always circle the trial digit to remind everyone that sometimes there will be backtracking. Recall the two rules for square roots which apply here also – you cannot backtrack from a square and you cannot build on a rectangle past 9. We have enough to build out 9 times. Trial digit ≤ 9 . Children are ready for next level of abstraction when they stop using the materials.

February 1, 2005

Calculate and Complete Each Cube - 3

Prologue

Begin with formation after the revolt.



Begin

We'll take the number 80,621,568.

(mark off periods beginning from the right - 80'621'568)

So our root will have 3 digits.

Instead of working category by category, we'll calculate the next digit of the root and then complete the cube up to that point.

With square roots, we calculated the next digit and then completed the square – so we'll follow the same pattern.

Let's add small h, t and u for the digits of our root. (add to top of page over answer)

We look at our millions (hold up blue cube). What is the largest cube we can make from 80? (4) So 4 cubed is 64 – let's write that at the right. We're not writing on the left this time.

We can fill in the 4 as the first digit of our root.

We have 16 left over. We need to bring down the next category. Why? We want to calculate the next digit of the root – and then **complete the cube up to that point**. We need to bring down the next category – but this because we need to calculate the next digit of our root. It is these pieces – hundred-thousands.

We want to know how many times they can be built out – that determines the value of t . So, I need to set up my division problem. What is it? Each time we build on a green prism – they are squares of 4. So h^2 is 16. We build these out in 3 directions – so that's 48.

How many times does 48 go into 166? (3 times) So we think 3 will be the second digit of our root – let's circle it.

Now looking at the material – and we know the value of t – I want to complete my next cube. That means I need not only the green prisms but I also need the brown (larger) pieces – plus I need this red t^3 – all the pieces that make my binomial cube.

Now I can build these on. (add green to blue, brown to green, red at corner – we now have a mini-cube)

Now we have to subtract the value of what we've used up here – now what we've added on – but what we've used up. Given this problem, we now have 43 cubed.

The value of 43 cubed is 79,507. So we can subtract that.

(hold up red cube)

What is the lowest category I've used? (1000s) The lowest category is this cube – 1000.

I'll put a squiggly line because I have to write this number (79507), aligning my categories of the original number. This line means we're going to ignore our 166 – and subtract the 79507 from our starting number. We'll subtract that instead of the 166 as we usually do.

Now we need to calculate the value of our next cube. To do that, we'll bring down the next category on our paper. That's our dividend and we need the divisor – which is the value of the prism we put on each side times 3.

(hold up 2 red prisms, 1 orange and 1 brown)

This is going to be the square I'll build onto each side. Given the numbers I build on – this is the square of 43. forty-red – three -orange – you can see it on each side.

So we need to divide this by (3 times (43 squared)). That's equal to 5547. How many times does that go into 11145? (2)

We, we think 2 is the third digit of our cubed root. (add to paper and circle it)

Now let's build on the rest of the pieces – up to the next cube (build on all the rest of the pieces) – then we have to do another squiggly line.

We now have to calculate the value of what we've added.

432 squared is 80,621,568 – so we'll put that after the squiggly line – aligned with our original number. And subtracting we find that we have a perfect cube.

Final written image:

	h t u
$\sqrt[3]{80621568} = 432$	
- 64	$4^3=64$

166	$1666 \div 48=(3)$
~~~~~	
- 79507	$43^3=79,507$
-----	
11145	$11145 \div 3(43^2)=$
~~~~~	$11145 \div 5547=(2)$
- 80621568	$432^3=80,621,568$

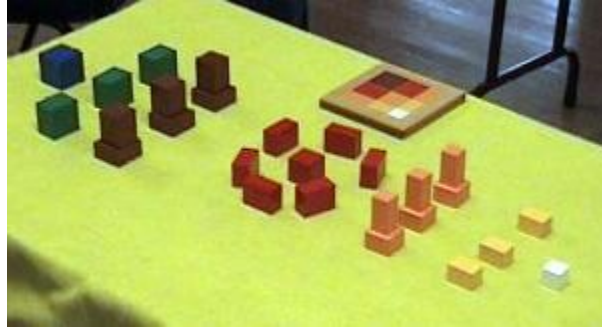
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February 1, 2005

More Completing the Cube - 4

Prologue

Begin with formation after the revolt again. Children do multiplication on the side of the page.



Begin

We'll take 143055667 again. We'll find the cube root by completing intermediate cubes again.

(write problem on paper and mark periods – beginning on right – 143'055'667)

So we know we have 3 digits – an h, a t and a u.

Largest cube we can make from 143 is 5 cubed.

(write it on paper and subtract) Let's write the first digit of our root and subtract - leaving us 18.

We are completing the cube as we go – so we'll build from this cube to the binomial cube. This means we need to know the value of t.

(hold up green prisms)

We can calculate the value of t by using this next category. Let's bring down the 100000s.

I need to set up a division problem. (180 divided by 75)

Divided by 75 because we have a cube of 5 – therefore when we build on the green prisms we're building on squares of 5 – in 3 directions.

75 goes into 180 how many times? (2) So we think 2 is our next digit.

(write it and circle it)

Now that we calculate that, let's look at the material and bring down the pieces for the next cube – the binomial cube.

We need these pieces (top brown pieces) and to finish it we need the red t^3 . Now I can finish off my binomial cube. (do so, blue, green, brown and red)

Now that we've built this, we need to calculate the value of what we built.

What have we built? It's 52 cubed – given this problem. That's 140608.

Let's add squiggly line.

What is the lowest category we've used? (thousands)

So that tells us we'll write this value lined up with the thousands category.

Now we'll ignore the difference we found earlier – we'll subtract our 140608 from our original number. And we get 2447.

Going on, we again want to complete the next cube – so we need to know the value of u.

We need to bring down the next category to give us our dividend. (hundreds)

(build the square as a separate figure)

We brought down hundreds. When I build this next square, notice that the lowest category is hundreds. We're using that category as well as what we have left over.

So our dividend is 24476. Square of 52 times 3. Square of the face is 52 – and in 3 directions. So that 3 times 52 squared. That's 8112. So 24476 divided by 8112 goes 3 times.

So I think 3 will be the next digit of our root. Let's write it and circle it.

Now let's build on the rest of the pieces completing that next cube.

Add a squiggly line and now we need to calculate the value of what we've built.

This is 523 cubed and if we cube that, we get 143,055,667.

	h t u
$\sqrt[3]{143055667} = 523$	
- 125	$5^3=125$

180	$180 \div 75=(2)$
- 140608	$52^3=140608$

24476	$24476 \div 3(52^2)=$
- 143055667	$24476 \div 8112=(3)$

0	$523^3=143,055,667$

February 1, 2005

Rule for Finding Cube Root

1. Divide the cubed number into periods of 3 digits beginning with the units and proceeding left.
2. Extract the cube of the first period. This is the first digit of the root.
3. Cube this root just found and subtract it from the first period.
4. Repeated step
 - a) To this remainder annex the first digit of the next period
 - b) Divide this number by (3 times the square of the root so-far-discovered).
 - c) This quotient is the trial next digit of the root.
 - d) Cube the entire root so-far-discovered, including this new trial digit and subtract this from the periods involved.
 - e) If the number is too large, reduce the trial digit by 1 and try again (backtrack).
7. Repeat step four, until all digits are found.

January 26, 2005

12. MEASUREMENT

Measurement is an interesting subject for children. It is the process of assigning some unit of measurement to an object or thing. Time is a measurement. Temperature is a measurement. Meters, liters and feet are all measurements.

The basic factor that one first needs to **convey to the children is that in order to measure and express that measurement, one has to have established a unit of measurement.** That unit is not a constant; we determine what it is. We could take a paper-clip and decide to use it as a unit of measurement. We could then proceed to measure the width of my table in lengths of paper-clips. It would be non-standard but still a valid measurement. When we measure we assign some unit of measurement to whatever object we are measuring. So this is the first key idea.

What do the children need to know before one introduces the concept of measurement. **One needs to know how to count** and that is about all one needs to begin **to understand the idea of measurement. This is possible in the primary classroom.**

In the elementary class one must consider how to convey ideas of measurement to the children. It should be done through little stories that **bring in the history of the development of measurement.** There are lots of books available for this today; history of mathematics, history of geometry, history of Egypt and so forth – all contain references to the history of measurement.

As we **look for the history of measurement and developing little stories,** we should make the child aware of how **people have used parts of their body to measure** things. Another idea is that, as communication amongst people increased, the need for a standard became evident. There are still places where, if one goes to the market to buy cloth, it is sold by arm's length. Naturally, we want a long arm when measuring in this manner – so it was not standard in a strict sense. This need has been felt in military and industrial development and that history can be brought out as well.

Another point of discussion with the children is about the different major measurement systems in-place. There is the British – or English standard system, also called the Customary system. There is also the metric system that is based on the decimal system. These **stories we develop about these systems should be 2-3 minute stories – very brief tidbits.** It is not a story in the sense of a Great Lesson. **This sort of story should permeate the classroom** – around all subjects.

When these stories are told to the children, it is **nice if activities are included that relate to the story.** If we tell a story about using parts of the body to measure with, an activity for the children might be to take a shoe off, trace around their foot on paper and cut out two of them from the group; write their name on their paper feet – and now they can be used to measure things in the environment. We can measure the length of the table in my feet and your feet. We discover that the measurements we took

differ – illustrating the basis for standardization. People had to figure out a way for those measurements to have a predictable meaning. Egyptian's resolved this problem by using the “Royal Foot” - which belonged to the Pharaoh. This was a small step toward standardization.

When the children are measuring, we can also talk to them about the element of error in measurement; it is considered to be inherent in measurement. We don't want to make a mistake but error is a function of the tools we use to measure. What did Egyptian's do if they were measuring in Royal Feet but it did not come out to an even foot? They used a span – the distance between the tip of the thumb to the tip of the small finger. Suppose it was less than a span? Then, they used a palm. If less than a palm, they used a finger.

Through our stories, we can bring in some of the **history of the two systems of measurement** that are mainly used. When was the yard or meter first used? What is considered the standard for measuring length today? Which system is older – the Customary or the metric system? What are the historical origins for each? What changes have been made along the way of development? Why do Americans still use the customary system. How does one convert a society from one to the other system? Which is easier to use – the Customary or the metric system?

Another thing to convey is that **every measurement includes two things: number and units**. If we just express number or just express the unit, the measurement is meaningless. Measurement means both by definition.

What do we measure? Just about everything it seems. We measure time and land. We measure space – both surface area and volume. We measure liquids and temperature. We measure money and angles. The mind is conditioned to measure – even inappropriately at times. There are 3 systems for measuring temperature; is there something which has only one way of measuring? Probably not.

There are some basic lessons on measurement that we want to present. Though we use measurement everyday, we don't realize how crucial it is to our lives – it's use and misuse. We might ask, “Do you think measurement is important to your life”?

January 26, 2005

Measuring with a Meter Stick

Age

About 5 years old.

Material

Meter stick, preferably one without subdivided markings and one with the marking.

Begin

We often measure length with something like this called a meter stick. Today, this will be the unit of our measurement. A meter is used for measuring length.

So we can measure the length of the table, for example, in meters. If I lay it along the edge (do so), we can see the table is just a little less than two meters long. Let's measure the height of the table (do so). The table is a little less than one meter high.

So we can use this meter to measure how long anything is we want to measure. We could measure our chalkboard. We could measure the classroom – and find out how many meters long the classroom is.

(encourage children to go and **measure the environment – until they find they need a finer source of measurement**)

One might want to measure small things so the meter stick may be too long. So, we have a solution.

Have you noticed that we have **numbers going down the length of this meter stick?**

We can read those numbers – beginning 1 and ending with 99 – meaning there are 100 places between these numbers. So our meter has been divided by these longer black lines in 100 equal parts. These are called **centimeters**. That prefix 'cent' means 100.

So we can measure things smaller than a meter. Let's measure the width of the table. We can see it measures 59 centimeters. Before we had a table length of less than 2 meters but now we can be more exact – our table length is 189 centimeters.

Epilogue

Find a meter stick with no divisions if possible – to prompt the idea in the child for a need for further granularity.

If in geometry, they have learned what perimeter is, could they then measure the perimeter of the classroom.

When introducing the yard stick, we can use the same lesson. The 1-foot ruler, done in the same manner.

January 26, 2005

Measuring Liquid Volume

Materials

Plastic containers 3-5 of around the 1-6 cup range.

Begin

Today let's talk about measuring liquid. We can use the liter as one of our units of measurement. One liter is filled right up to the top line (hold up container and point).

That's curious; the meter stick went all the way to the end. So why is that not the case here? (so we don't spill the liquid, the line is below the top)

If you fill this container up to this top line, you will have one liter. I wonder how many containers it would take to fill up a gallon milk carton? Maybe you'd like to experiment with this to find out!

When we have less than a liter, the container is broken into parts call millimeter. What does that prefix 'milli-' tell us? It means thousands! So this container is broken into 1000 milliliters. On this large container, the least liquid we could measure is 200 milliliters. So if we want to measure less than that, we can use a smaller container.

This small container starts at 100 millimeters and goes to 500 milliliters. And even smaller containers let use measure even smaller amounts of liquid.

So liquid measurement can be expressed in liters – or parts of a liter.

Epilogue

They can explore with the liter just as they did with the meter – measuring the environment.

January 26, 2005

Measuring Mass or Weight

Materials

Balance scale and an additional set of weights (in small box on-site) of 1 gram up to a kilogram.

Begin

Measuring mass or weight, we use the unit called **gram**. We've seen the balance scale before. We used it before to compare two amounts of substance – like sand and sugar – to make sure we had equal amounts.

But we can also weigh something in grams. How much does a half-cup of sand weigh in grams? Well we can find out.

(demonstrate how to find a weight that achieves balance)

Epilogue

We place whatever we want to measure on one side of the scale; then place a gram (or other value) weight into the other side to find a balance - in order to find a measurement of weight.

January 26, 2005

Metric Prefixes

Materials

Balance scale and an additional set of weights (in small box on-site) of 1 gram up to a kilogram.

Prologue

Once we've introduced basic lessons on measurement, we can present this lesson which helps them reflect on units of measurement. Much of the language is new to them and this help crystalize the content of our earlier lessons.

Begin

Remember we measured with meters?

(place ticket with “meter” in the units column on the decimal board)

A meter is the unit we use for a unit.

(each time place a ticket for each unit type – prefixes are in red and 'meter' is in black)

Sometimes we have 10 times a meter. And that's called a decameter.

But sometimes we have ten times less than the meter. That's called a decimeter.

Sometimes we have 100 times a meter. And that's called a hectometer.

But sometimes we have 100 times less than the meter. That's called a centimeter.

Sometimes we have 1000 times a meter. And that's called a kilometer.

But sometimes we have 1000 times less than the meter. That's called a millimeter.

When people set up these measurements, people used two sets of language. Did you notice prefixes were very different?

The multiples come from the Greek and the divisions come from the Latin.

When we write measurements and want to use these terms, we don't generally write the whole word out. We use abbreviations.

To help you learn the abbreviations, you will find it on the other side of the tickets we've laid down.

Epilogue

We didn't fill every column on our decimal board. Invite the children to find the rest of the terms for all the columns. We can make tickets for the other two forms of measurement: volume and mass (or weight).

We should have a thermometer for measuring temperature; one indoor and one outdoor that can be read through the window.

Living in a northern climate like this, one could graph daily temperatures for the whole school year – using red for temperatures above freezing and blue for those below. One can find thermometers that have both systems of measuring heat; a third unit for this is kelvins – used in science.

Measuring angles - we've done in geometry. Measuring time has been spoken about in history; Babylonians used a dial to track sunlight into twelve equal parts – eventually called hours. We can also bring in the number base of the Babylonians of 60. Time measurements of less than a second and longer than a year have their own vocabulary – decades, centuries and so on; one can do the same going into smaller gradations.

The measurement standard for time is the atomic clock. At one time, there was one single official atomic clock whereas now there are a number of them. Local atomic clocks actually pick up a signal of transmission versus operate as an atomic clock.

Lastly, money is something that is measured and the children need to learn how money is organized – coins and bills. Try to use real money when this is introduced.

January 27, 2005

13. SIGNED NUMBERS

Introduction

Materials

Felt mat, colored bead-bars in a box (1-9), gray bead-bars in a box (1-9), golden 10-bars in a box, black and white bead stair placeholders in a box. We have four boxes in this lesson. Second example adds gray 10-bars and negative red/white bead stair placeholders.

Age

7 or 7 1/2 years old – assuming they know math facts.

Prerequisites

Facility with addition and subtraction facts.

Prologue

This is important since it's something they need to understand and work with in the public curriculum and to make full use of our number system. The beginning is very sensorial. Introduce the concept of negative quantities before giving this lesson; use temperature or some other example that use negative numbers - “below zero – negative temperatures.” Maybe we have kept a graph of temperatures this year. It could be land that is above and below sea level that is used to illustrate the idea of a negative number. Football fans may relate to negative yardage. There may be a checking account for the class for going-out; this balance can be “in the red.”

We set up our 4 boxes. Then we build a snake from the colored (positive) and gray beads (negative) in the 2 boxes at far left. Then, until all original bars have been processed, we repeat these steps:

1. bring down the next 2 bars and set them next to each vertically at the bottom of the mat, aligned at top of bars to sensorially show the absolute difference.
2. determine the difference between these 2 bars
3. choose that difference between the two bead bars, from either the positive or negative stair beads and place the new stair beads into the snake to replace the 2 we've just taken. The stair beads are used represent our current working difference.
4. Place **original** snake bars we've just processed to the **left**: negative gray bars go in one pile and the colored positive bars go into a second. Placeholders – gold or gray 10 bars

as well as black and red stair bars – go back into their boxes always – once they are processed in the snake.

5. In subsequent lessons we introduce a check; sum the pile of original negative bars and sum the pile of original positive bead bars at our left. Put the larger of those 2 numbers at the top of a vertical expression on paper. Subtract to find the check answer – assigning to the answer the sign of the larger number at the top of the expression. Compare with any last beads left from our snake representing the same answer.

Begin

Today, we want to talk about positive and negative quantities.

Presentation of the Snake

I have some things on this tray we will use to work with positive and negative numbers.

(The four boxes are set out in order (see material list) at the top of the mat in a horizontal row)



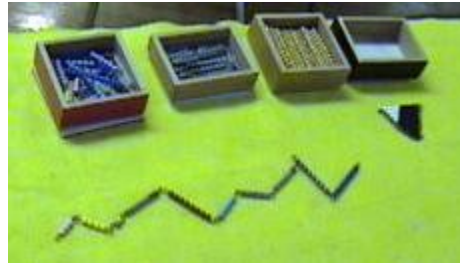
Who would like to help me? Hopefully we remember doing the snake game in primary. Do you remember how to set out that black bead stair? I can help you.



(sets out black and white bead stair as a pyramid)

If I say a positive number, take the beads from the colored box. If I say a negative number, take the bead from the gray beads. Let's build our snake.

$+3 +4 -1 -7 +8 +5 -5 -6 -8 +9$



Bring down $+3 +4$



What is the answer? ($+7$)

Let's take our answer from the bead stair and place it into the snake on its left end.

And our 2 bead bars, we done with them, so we place each them in one of the piles at left.



Our snake now begins at the left end with $+7$ and -1

Bring down +7 -1

What is the answer? (+6)



Let's take our answer from the bead stair and place it into the snake on its left end.

We are done with our 2 bead bars, so we place -1 at our left pile, and the stair-bar back.



Our snake now begins at the left end with +6 and -7

Bring down +6 -7



What is the answer? (-1)

We'll use this bead stair (introduce red and white) to keep track of negative answers.



We needed something different since she got a negative number answer of -1 (+6 -7)

Let's take our answer from the negative number bead stair and place it into the snake on its left end.



We are done with our 2 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair.

Our snake now begins at the left end with -1 and +8



Our snake is getting smaller now!

Bring down -1 and +8

What is the answer? (+7)

Let's take our answer from the correct bead stair and place it into the snake on its left end.



We are done with our 2 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair.

Our snake now begins at the left end with +7 and +5

Bring down +7 and +5

What is the answer? (+12)



Let's take our answer from the correct bead stair and place it into the snake on its left end.
This time we need to use our golden bead 10-bar too, since the **answer is more than 9**.



We are done with our 2 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair, the 10-bar back into its box.

Our snake now begins at the left end with +10 and +2 followed by -5

Our snake just keeps disappearing!

Bring down +10 plus +2 and our -5 (line 10 and 2 in vertical line, set 5 to side at top)



What is the answer? (+7)

Let's take our answer from the correct bead stair and place it into the snake on its left end.

We are done with our 3 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair, the 10-bar back into its box.

Our snake now begins at the left end with +7 and -6

Bring down +7 our -6

What is the answer? (+1)



Let's take our answer from the correct bead stair and place it into the snake on its left end.

We are done with our 2 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair, the 10-bar back into its box.



Our snake now begins at the left end with +1 and -8.

Bring down +1 and our -8

What is the answer? (-7)



Let's take our answer from the correct bead stair and place it into the snake on its left end.
We are done with our 2 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair, the 10-bar back into its box.



Our snake now begins at the left end with -7 and $+9$

Bring down -7 and our $+9$

What is the answer? $(+2)$



Let's take our answer from the correct bead stair and place it into the snake on its left end.



We are done with our 2 bead bars, so we put original bars in our piles at left, and the bead stair bars back into their stair, the 10-bar back into its box.



Now our snake is +2 long and we are done.

New Part

Problem is: -7 -6 +3 -6 -4 +5 -2 +3 -1 +8

(beads are laid out like a snake in center of felt mat)



I wonder how much our snake is worth altogether.

(repeat process above)

: :

(pull down first 2 bars set vertically and aligned for sensorial perception)



What bars do you have? (-7 and -6)

That's a negative 13!

Here's one more **box of gray 10-bars**

(this is a 6th box – gray 10-bars)

You need this to show -13.

(note that place holder beads include positive and negative ten bars and positive and negative bead stairs – they are always put back into their respective stores)

(If you accumulate 10 bars, the snake will turn golden – so optionally, you can leave the 10-bar in the snake and go on to the bars on the right)

(note that all bead bars accumulated on the left were from original snake – again, all others – placeholders - go back into the respective boxes when unused)

Check

So we have -7 left. Let's add up our pile on the left and see if our answer is correct.

I wonder what **all the positive bars on our left equal (add them up)**

Positive 19!

Let's add up all those negative bead bars on our left.

Negative 26. Let's write that above our 19.

The 26 goes on top since it's a bigger number

We need to subtract 19 from 26. And the answer gets the sign of the top number.

Epilogue

Encourage the children to check each problem; if they got the wrong answer, do the problem again with the same snake. The answer is important for this lesson.

If they have not seen a negative sign before, show it to them while you are introducing the introductory concept of negative numbers.

Allow children to make up their own numbers. Introduce the second case when it comes up with the children – this is a singular case. All bars are already introduced.

They may work with this material for a couple years. After they have had experience with this lesson, they can derive some realizations.

January 27, 2005

Operations

Prologue

The children have been adding and subtracting positive and negative numbers in the prior lessons.

Depending on the snake they have built, they will experience that when you add two positive number, the answer gets larger and stays positive.

When adding two negative numbers together, the answer also gets larger but the value or quantity is actually smaller.

When one is positive and one is negative, the smaller number is subtracted from the larger number and the sign of the larger number is used for the answer.

Addition

Begin

(we are using the colored and gray bars 1-9 to form each problem and answer)

Suppose we have a $+9 + +4$? (write it on paper and set it out in bead bars on the mat)

We always use our signs since we are dealing with signed numbers.

$$(+9) + (+4) = 13$$

We get a positive 13. That's one example of addition with signed numbers.

Suppose we have $(-9) + (-4)$? (set out on the mat and write expression)

(set problem (horizontal) and answer (vertical) out on the mat)

$$(-9) + (-4) = 13 \quad \text{(fill in answer)}$$

Suppose we have $(+9) + (-4)$? (set out on the mat and write expression)

(set problem (horizontal) and answer (vertical) out on the mat)

$$(+9) + (-4) = +5 \quad \text{(fill in answer)}$$

Suppose we have $(-9) + (+4)$? (set out on the mat and write expression)

(set problem (horizontal) and answer (vertical) out on the mat)

$$(-9) + (+4) = -5 \quad \text{(fill in answer)}$$

(we can move the problem beads vertically to compare and see the answer sensorially if needs be)

So there are the four cases we can encounter adding signed numbers.

When **adding numbers that have the same signs**, which example am I referring to?
(the first two)

How do I get my answer?

(add them together and the answer takes the sign of those two numbers)

When **adding numbers of difference signs** (last two examples), how does one find the answer?

(subtract the smaller from the larger number and the answer takes sign of larger number)

What if we have an answer of zero? (zero does not use a sign)

Follow-up to Paper

(besides generalizations with addition, show the children how to do a problem like the snake – on paper.

(write out problem $-4 + 9 - 7 + 3 - 6 + 8$ horizontally)

I wonder what the value of our snake is? Can we find the value on paper?

When we add the first two we get a positive 5.

Now we add in a negative 7, giving us a negative 2

Now we have a positive 3, giving us a positive 1

(continue with this verbal pattern)

... gives us a positive 3

Remember if we have a negative 4 and a positive 4 in our problem, we can just cross those numbers out before we begin, since they equal zero together.

Subtraction

Begin

(indicate problem to the children and have them set it out in bead bars)

$+7 +3 +4 +3 -4$ (form snake with these numbers)

Let's write that down $(+7 +3 +4 +3 -4)$

Who can tell me the answer? (positive 13)

(place answer beads horizontally under the problem)

Now I'll write the problem again but I'll change something a bit.

$+7 +3 +4 +3 -4$

This time I'll take this away (hold up $+4$ bar). I'll show this on my paper.

$(+7 +3 +4 +3 -4) - (+4)$

(literally, take the $+4$ bead-bar off the mat)

Now what's my answer? (positive 9)

Isn't that the answer on the mat? (add what remains to find $+9$)

Now I'll put this back (put $+4$ bead bar back on snake)

Write it down again.

This time I'll take this away (hold up -4 bar). I'll show this on my paper.

$(+7 +3 +4 +3 -4) - (-4)$

(literally, take the -4 bead-bar off the mat, held in the hand)

Now what's my answer? (positive 17)

Let's do another.

Let's take a $+7 +8 -3 +4 +3$

Let's write it down on paper:

$+7 +8 -3 +4 +3$

What's the answer this time? (positive 19) (set out below the snake)

Now what if I take this one away $(+3)$

Let's write it down again.

$$(+7 +8 -3 +4 +3) - (+3) =$$

What's our answer? (+16)

We can set out that answer (set +16 below the snake)

Let's take our original problem again.

$$(+7 +8 -3 +4 +3) - (-3) =$$

This time we'll take away the -3. I've done that with bead bars so I need to do it on paper.

What's our answer? (+22)

Is that what we have left on the mat? (check to be sure)

(do the following two sets for additional repetition)

$$\begin{aligned} +7 +3 +4 +3 -4 &= 13 \\ (+7 +3 +4 +3 -4) - (-3) &= 16 \\ (+7 +3 +4 +3 -4) - (+3) &= 22 \end{aligned}$$

$$\begin{aligned} +9 +6 -7 -4 +4 &= 8 \\ (+9 +6 -7 -4 +4) - (+6) &= 2 \\ (+9 +6 -7 -4 +4) - (-6) &= 14 \end{aligned}$$

Let's look at our first problem.

What happened to our answer when we took away a positive number? (answer is smaller)

When we **took away a negative** number, what happened to the answer compared to the first example (got larger instead)

Did the same thing happen in our other 2 examples. (yes)

When we took away a positive number - yes, our answer got smaller.

When we took away a negative number, our answer got larger.

Positive Minus Positive

Now suppose we have a +9 and we take away a +4.

$$(+9) - (+4) =$$

(write this and hold up +9 bead bar)

I can take it away like this

(cover 4 beads with hand)

What do I have? (+5)

Negative Minus Negative

Now suppose we have a -9 and we take away a -4.

$(-9) - (-4) =$ (write this and take -9 bead bar)

I can take it away -4 like this (cover 4 beads with hand)

What do I have? (-5)

Positive Minus Negative

Suppose we have +9 minus a -4?

$(+9) - (-4) =$ (hold up +9 bead bar)

Do I have a negative 4 to cover up? (no)

Let's set this aside for a minute then. (set 9 bar to left on mat – take +4 and -4)

Here is a positive 4 and negative 4 I'm setting together on my mat.

What the value of this? (zero)

So now if I move my 9 bar back over next to them (all vertical) what's the value on my mat? (+9) That's what we started with.

Our problem says to take away a negative 4 and I now have one of those (hold up -4 bar)

And there on the mat is our answer. So what is it? (+13)

Negative Minus Positive

Now what about -9 minus a +4?

$(-9) - (+4) =$

Here is our negative 9 – our minuend – what we started with.

Our problem says to take away a positive 4. I don't have a positive 4 to just cover up!

So let's set that aside for a minute and again set +4 and -4 on our mat together.

(-9 is at left)

What's the value of this (point to the 2 fours) (zero)

So now I can move this back over (move -9 next to the 2 fours)

What's the value on my mat? (-9) Now we can work out our problem.

We want a -9 minus a +4. Now I have a +4 I can take away. (take the +4 bead bar away)

There on our mat is the answer. What is it? (-13)

These are all the ways we can subtract signed numbers.

If you take away a negative number, the answer will be larger than what we began with.

If you take away a positive number, the answer is smaller than what we began with.

If you subtract a positive from a negative, we add them, taking the sign of larger number.

If you subtract a negative from a positive, we add them, taking the sign of larger number.

Epilogue

If working in an elementary class with children that did not practice enough with the snake, they will respond to do more examples on paper than with the bead bars. Introduce the beads as we've done above, but once they've caught on, let them do this on paper and they may like this better.

Multiplication

Begin

Let's start with +5 taken +3 times

write $(+5) \times (+3) =$

Positive Times Positive

Here's a positive 5 1 time (set down 5 bead bar), 2 times, 3 times

(they are laid in center of mat horizontally)

What is this altogether (+15)

(add answer to written problem)

Negative Times Positive

Suppose we have this problem.

(write it)

$(-5) \times (+3) =$

Here's -5 taken 1 time (put gray 5-bar on table), 2 times and 3 times.

(there are now 3 gray 5-bars laid horizontally in center of table – others were put away)

What does this equal? (-15)

Positive Times Negative

How about this problem.

(write the problem)

$(+5) \times (-3) =$

(set out on the mat 3 +5 bars and 3 -5 bars)

What's the value of what I have here? (zero)

But now I have something to work with in solving my problem.

We know that in multiplication, the multiplier tells us how many times to take the multiplicand. In this case, the multiplier tells us how many times to take the multiplicand away.

(take away +5 bar 3 times)

Give us what is left on the mat – negative 15.

Negative Times Negative

If we have (-5) times (-3), how would we handle this?

We have that negative multiplier telling us we need to take this away that many times.

I need some 5 bars. (place 3 negative and 3 positive 5 bars on the mat)

What's the value of what I have here? (zero)

And, now I have something to work with to work out my problem.

It is telling us to take a negative 5 away 1 time, 2 times ... 3 times. (each time take a bar)

A negative 5 taken away 3 times give us how many? (+15)

(point to negative sign of the 3 when we say 'taken away')

So just remember what our multiplier is telling us to do. It is telling us how many times to take a certain quantity away. When we have a negative multiplier, it tells us to take that quantity (multiplicand) away the number of times in the multiplier.

So how could we state rules for multiplying signed numbers?

With same sign, the answer will be positive.

If they have different signs the answer is negative.

Written Image at End:

$$(+5) \times (+3) = +15$$

$$(-5) \times (+3) = -15$$

$$(+5) \times (-3) = -15$$

$$(-5) \times (-3) = +15$$

Epilogue

Explain “we need something to work with” when we add the multiplier – negative and positive versions. Then do the take away of negative multipliers and tell them we are taking the multiplicand away – that number of times. Then we look at what results in bead bars for an answer.

Division

Materials

Add two green skittles borrowed from Racks and Tubes, two cups that have black lines on them. All signed number material is on a tray; the cups are saved and introduced only here.

Begin

Let's start with (+28) and divide that by (+2) $(+28) \div (+2)$

(take 4 7-bars)

What else do I need? (2 skittles)

Share out the 7-bars and what is our answer? (+14)

(there are 2 skittles and 2 7-bars vertically below each one on the mat now)

(boxes of bars are still at the top of the mat though and have been constantly there)

$$(+28) \div (+2) = (+14)$$

For the second case, we'll take a negative 28, represented by 4 -7 bars, and divide that by positive 2.

$$(-28) \div (+2)$$

(set out 2 skittles and share them out)

What the answer? (-14)

$$(\text{write } (-28) \div (+2) = -14)$$

$$(-28) \div (+2) = -14$$

What if I have this problem. This time I'll ask a group division question.

How many groups of a -14 can we make out of this amount? (make 2 groups)

It's positive 2 since there are 2 groups that we've made.

$$(-28) \div (-14) = +2$$

This time I'll start with positive 28 and I have a little story for you.

Two men are debtors – people in debt – they owe something. What they owe is a positive 28 dollars.

These two cups (hold up) represent the 2 debtors. (black lines represent negative sign)

The two of them are in debt for this amount (place 2 7-bars in each cup)

So that means they are each in debt a negative 14 (-14)

$$(+28) \div (-2) = (-14)$$

When we have a positive divided by a negative, our answer is negative – just like when we have a negative number divided by a positive number – the answer is negative.

So if the two signs are different when we divide, the answer is negative.

When our two signs are the same, the answer is positive.

Epilogue

We don't have enough material to make 2 groups of 14, instead of 14 groups of 2 – this is the one exception of that form. So we make it work out using the cups.

February 2 , 2005

14. POWER OF NUMBERS AND NUMBER BASES

Non-Decimal Bases

Base Two

Prologue

The children are prepared for this lesson by working with squares and cubes. When we have a number squared, it's a number to the second power. When we have a number cubed, it's a number to the third power. The "powers of two" materials allow us to look at numbers beyond powers of 2 and 3. From smallest to largest, it is comprised of:

- yellow cube – power of zero (unless small 1 centimeter cubes is used as unit instead as is done in this first lessons; then all that follow shift upward to the next category)
- white prism – power of 1
- green prism – power of 2
- larger yellow cube – power of 3
- larger white prism – power of 4
- larger green prism – power of 5

Prerequisites

Work with bead chain cabinet and wooden hierarchical material.

Materials

Powers of Two box, box of 1 centimeter cubes, felt mat, blank tickets and strips of paper on a tray about 1x3 inches, pencil.

Begin

(spread pieces from powers of two box on mat center)

Here's our unit (set at left on mat)

Being alone, it has no power. (ticket 2^0)

But now I'm going to do something. I'm going to form a group. (place 2 white 1-cm.)

It's our first group and it's a group of 2. So it has some power.

We can write that like this.

(ticket it - 2^1)

We read this as two to the first power.

(we now have 3 cubes, 1 ticket)

But I want to keep going.

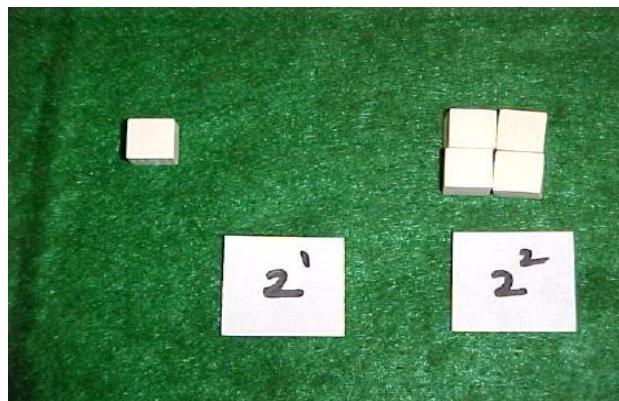
(each time, double previous group and combine the two for a new group)

Here is 2 to the first power again (put 2 separate groups of 2^1 side by side, then combine)

When I put them together they make a square. We write that like this.

It's two to the second power.

(ticket - 2^2)



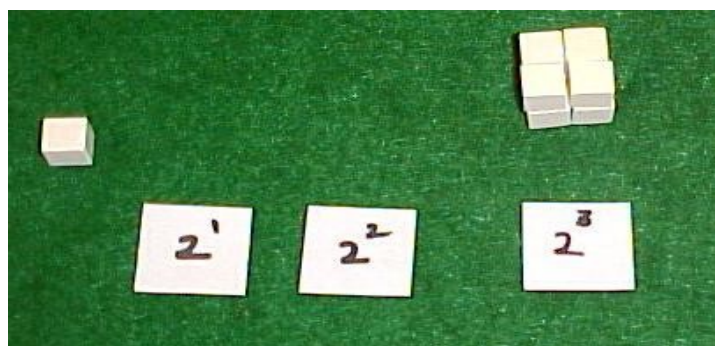
Here is two to the second power again.

(place 4 more cubes)

When we put them together – it's a cube.

It's two to the third power and this is how we write it.

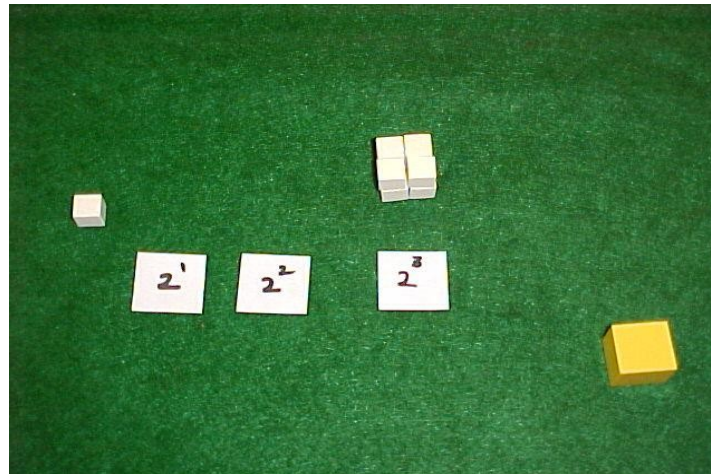
(place ticket 2^3)



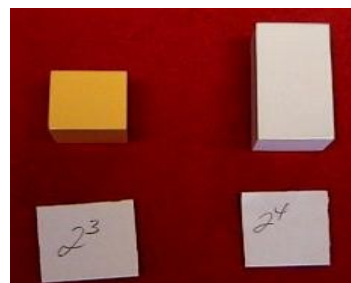
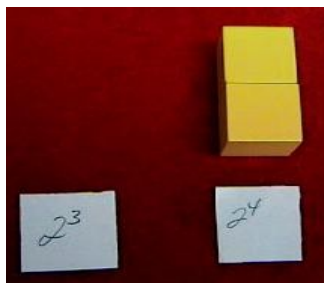
each time, use previous group, create second like it, and combine them for next group

But I want to keep going. This is the same as these. (hold up yellow cube)

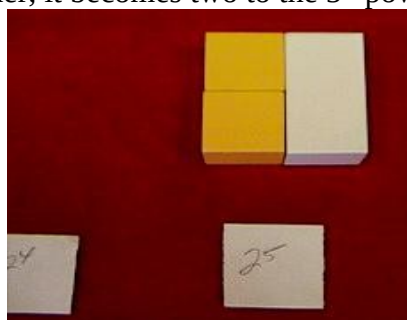
I'll use this instead now. (place a second yellow cube)



Here is two to the third power again. When we put them together, we have 2 to the 4th power. I can use this white prism instead since it is the same as two of these yellow pieces.



But I want more power. Here is two to the fourth power again (place white prism)
When we put them together, it becomes two to the 5th power.



2 yellow cubes and 1 white prism - 2^5

Here is 2 to the 5th power again (green prism)

When we put them together, we have two to the 6th power. (green prism on top)



But I want to keep going. Here is two to the 6th power again (yellow cube)



(place it next to the other group we have formed)

Now we have 2 to the 7th power.

Here is 2 to the 7th power again (place white prism next to the group)



Now we have two to the 8th power. But I want more power.

Here is two to the 8th power again., (place green prism next to, then on top)



Now we have two to the 9th power.

Could I keep going? I've run out of material. But we could go on otherwise.

Let's see again. **(take pieces off, going from 9th power to 1st power, dismantling each)**

Here is two to the 8th power **(take top off and pass down to the left, next lower power)**

Here is two to the 7th power (take top off and pass down)



Here is two to the 6th power (take top off and pass down)

Here is two to the 5th power (take top off and pass down)

Here is two to the 4th power (take top off and pass down)

Here is two to the 3th power (take top off and pass down)

Here is two to the 2th power (take top off and pass down)

Here is two to the 1th power (take top off and pass down)

And here is my unit where I started from. Remember we said it is a unit and therefore it has no power. There is a way we can write that down.



Two to the zero power. That means 1. Any number to the zero power is 1.



Ticket Exponent

Let's see what this means. Here's our unit. We said it has no power so it is just 1.

Next we have 2 to the 1st power – and we took it 2 times. (ticket $2^1 \times 2$)

Here's 2 to the 2nd power. Remember we took 2 to 2nd power taken 2 times. (ticket $2^2 \times 2$)

Here's 2 to the 3rd power. Remember we took 2 to 3rd power taken 2 times. (ticket $2^3 \times 2$)
(continue pattern)

Here's 2 to 8th power. Remember we took 2 to 7th power taken 2 times.

Ticket Value

Now let's look at the numerical value of each of these.

First value is 1.

Two the first power has how many? (2) So it's value is again 2.

(refer to ticket in second row of tickets)

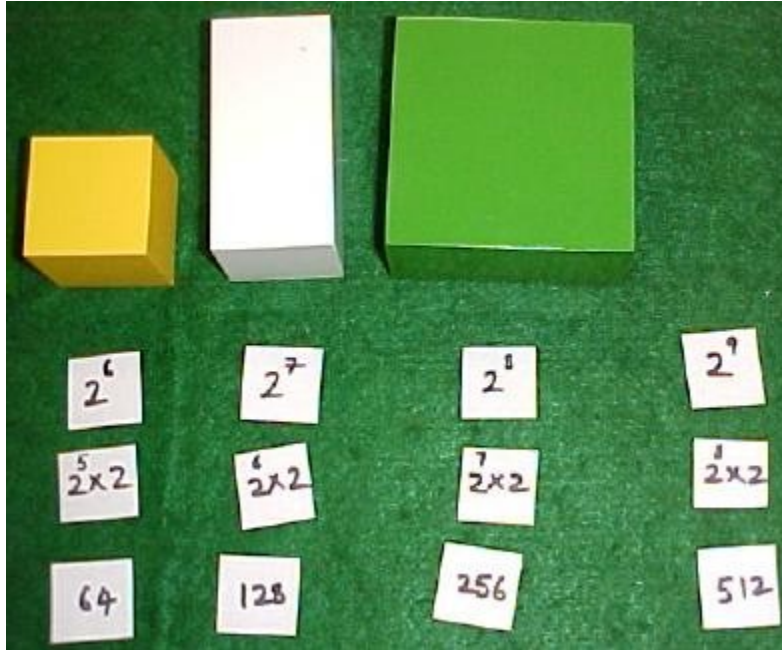
Next we took 2^1 taken 2 times. That means 2 times 2 which equals 4.

Next we took 2^2 taken 2 times. That means 4 times 2 which equals 8.

(continue pattern – 16, 32, 64, 128)

Next we took 2^7 taken 2 times. That means 128 times 2 which equals 256.

Next we took 2^8 taken 2 times. That means 256 times 2 which equals 512.



This means if we build up all the way to 2 to the 9th power with our 1 centimeter units, we would need 512 of them to build 2 to the 9th power.

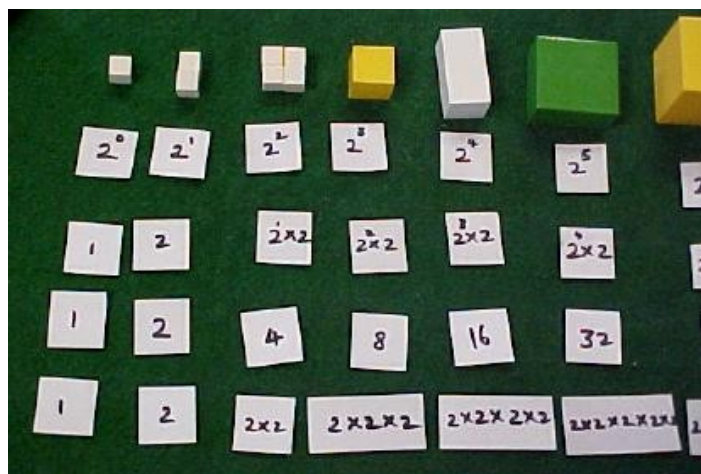
Ticket Factors

We have 1 more row of tickets!

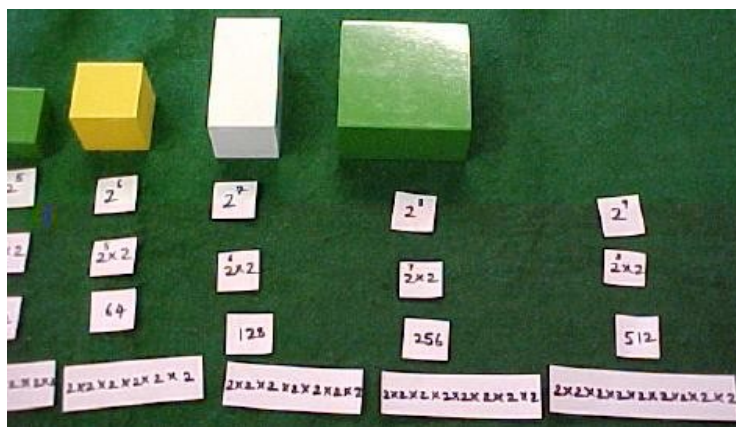
Next we have 2 to the 3rd power. That means $2 \times 2 \times 2$ – 3 factors of 2.

Next we have 2 to the 4th power. That means $2 \times 2 \times 2 \times 2$ – 4 factors of 2.

Next we have 2 to the 5th power. That means $2 \times 2 \times 2 \times 2 \times 2$ – 5 factors of 2.



(continue pattern)



Next we have 2 to the 9th power. That means $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$ – 9 factors of 2.

So what do we notice about the number of factors and the value of the exponent?
(exponent tells how many factors we'll have)

Epilogue

The number in the first group we make, in this case 2, is called the base and the little number to its right and superscript is called the exponent – from Latin exponere and it means 'to tell something about.' The exponent is telling something about the base. This terminology can be given to the child at some point along the way.

In the classroom, any line of ticketing could be a logical stopping point. If you stop after presenting a row of tickets, when you pick it up another day, build up to

that point again as review – before going on.

Substitute materials when you reach the cube.

February 2, 2005

Base Three

Materials

Add wooden cubing material to powers of two box, box of 1 centimeter cubes, felt mat, blank tickets and strips of paper on a tray about 1x3 inches, pencil.

Begin

(set white unit cube alone on the felt mat and build up as done with Base Two lesson)

Today, I'll write this ticket – 3 to the zero power.

That tells us how many we're going to put in our first group. (ticket 3^1)(make group of 3)

To go from 3 to the 1st power – to the next power, I need to take this 3 times (point to 3^1)

Here's 1 time (point original group), here's two times and 3 times. (place 2nd, 3rd group)

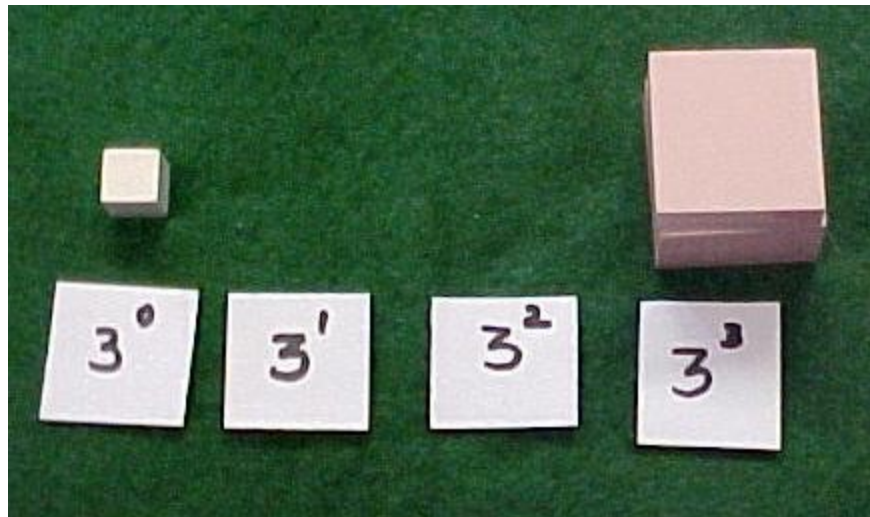
If I put them all together, they make 3 to the 2nd power. (ticket 3^2)

This is the same as these squares, so I'll use them instead (substitute square for 9 cubes)

To go from 3 to the 2nd power – to the next power, I need to take this 3 times (point to 3^2)

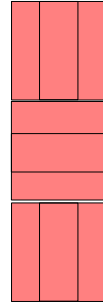
Here's 1 time (point original square), here's two times and 3 times. (place 2nd, 3rd squares)

If I put them all together, they make 3 to the 3rd power. (ticket 3^3)



To go from 3 to the 3rd power – to the next power, I need to take this 3 times (point to 3³)
 Here's 1 time (point original cube), here's two times and 3 times. (place 2nd, 3rd cubes)
 If I put them all together, they make 3 to the 4th power. (ticket 3⁴)

(build to contrast the “cubes”)



(continue pattern until materials run out)

I've run out! I can only go to 3⁶ before I run out of materials.

Now it's your turn – what number would you like to pick between 2-9 (since that's what we have in wooden cubing material) to use as base.

Epilogue

Purpose is to show that powers of numbers is not limited to base 2.
 Substitute materials at the square and cube.

February 2, 2005

Changing the Size of the Unit

Materials

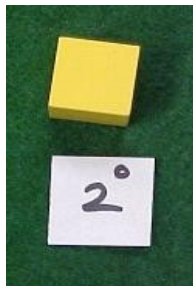
Power of two material, felt mat. Wooden cubing material. One centimeter cubes, blank tickets and a pencil.

Begin

(pieces on felt mat – small at left graded to largest at right)

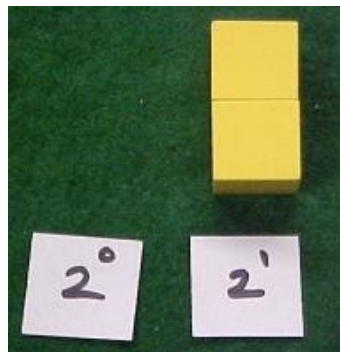
(place yellow cube)

This is my unit. So I'll write 2^0 power on my ticket.



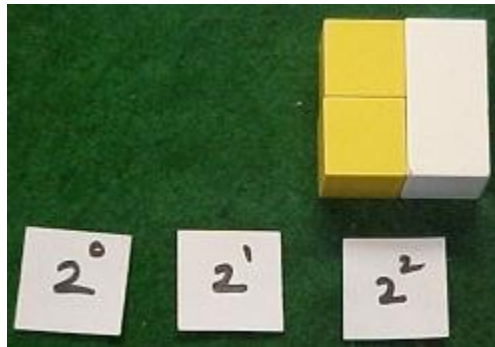
This is how many we need in 1^{st} group. I need two. So here's another.

When we put them together (do so vertically), we have 2 to the 1^{st} power.



Here is 2 to the 1^{st} power again. When I put them together, we have 2 to the second

power. (ticket each time)



(continue building up each power of 2 – ticketing as we go – until we run out of material)
So when we use that yellow cube as our unit – we can only build up to 2^6 .

Let's look at these again. (deconstruct each power of 2 stack as we go)

Remember this is 2^5 . (move larger green prism from top of stack of 2^6 group)

This is 2^4 (move larger white prism)

Here is 2^3 (move larger yellow cube)

This is 2^2 (move small green prism)

This is 2 to the 1st power (move small white prism)

And here's our unit – 2 to the zero power. (point to small yellow cube)



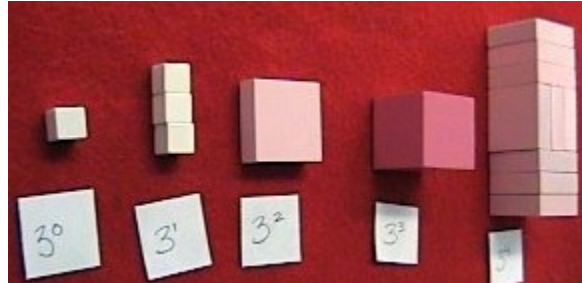
To go from 3 to the zero power – to the next power, I need to take this 3 times (point 3^0)

Here's 1 time (**add group this time using white cube**), here's two times and 3 times.
(place 2nd, 3rd group)

If I put them all together, they make 3 to the 1st power. (ticket 3^1)

(continue pattern)

To go from 3 to the 3^{rd} power – to the next power, I need to take this 3 times (point to 3^3)
Here's 1 time (add group), here's two times and 3 times. (place 2^{nd} , 3^{rd} group)
If I put them all together, they make 3 to the 4^{th} power. (ticket 3^4)



I've run out of materials. Does it remind of of something? Just like the Wooden hierarchical materials, I have a point, a line, a square – and then another point, line and if I had more materials... well, could we go on?

Epilogue

Purpose is we want children to know a unit can be a different piece – not just the little white cube.

February 2, 2005

Powers of Ten

Materials

Wooden hierarchical materials – laid out in linear formation – with the gaps between categories. Two floor mats, blank tickets and pencil, plus prepared tickets from wooden hierarchical material.

Begin

You remember this. It's our wooden hierarchical material. We have unit, tens and hundreds in our simple numbers. Here is the unit, tens and hundreds in the thousands. And finally, the unit of millions.

We're going to talk about the decimal system today. These categories are categories of the decimal system. That word 'decimal' starts with 'deci' and that means 10 in Latin.

We've worked with powers of numbers. I'm going to write a ticket for unit. The ticket I've written is 10 to the zero power. (place ticket 10^0 below small unit at far right)

That tells me how many I'll put in my first group. We'll put 10. It's 10 to the 1st power. (place ticket 10^1 below the simple ten piece)

We've said for a while, things like 10 units gives us 1 ten – or 10 tens give us 1 hundred. Well, this is why – our decimal system is base ten!

Well here's tickets for the rest of my decimal system.

Here is 10 to the 2nd power. Here is 10 to the 3rd power. (ticket each 10^x)

Here is 10 to the 4th power. Here is 10 to the 5th power.

Here is 10 to the 6th power (ticket million cube)

10^6	10^5	10^4	10^3	10^2	10^1	10^0
--------	--------	--------	--------	--------	--------	--------

first row of tickets, each aligned underneath the appropriate piece

(ask for 2 volunteers to place **prepared value tickets**)

Now, I would like you to place these prepared tickets where they belong.

(for example, under million cube is ticket saying 10^6 followed by one under that one which says 1,000,000 – the prepared ticket)

(all prepared tickets are placed under respective pieces – values and exponential notation)

10^6	10^5	10^4	10^3	10^2	10^1	10^0
1,000,000	100,000	10,000	1,000	100	10	1

second row of prepared tickets provides the value for each category

Do you notice, the number has the same number of zeros as the exponent has each time? So 10 to the zero power has a 1 and no zeros. And 10 to the 4th power has 1 and 4 zeros.

So – we started with a unit – it's just a dot – that little tiny cube. (point to unit at right)
Ten to the first power is a line – it has length - one new dimension added to our dot.
Ten to the second power is a square – it has length and width - two dimensions.
Ten to the third power is a cube – we have length, width and height - three dimensions.
We'll call it a dot again – though a larger dot.

When we get 10 to the 4th power, it's made of ten of these thousand cubes. It is a line again but larger this time – with length added to our dot.

Ten to the 5th power – we've again added width – giving us length and width again.

For 10 to the 6th power – we've again added height – and we have another dot that is again larger than the previous one.

Epilogue

“Powers of Ten” (by author Morrison) is a fun book to have in the classroom; bring it out after this presentation. ISBN is 0716760037.

February 3, 2005

Operations Using Exponential Notation - 2

Notes compliments of Carson Wright

Material

Paper and pencil, wooden cubing material and a felt mat.

Prologue

We have already done some addition and subtraction and multiplication and division of numbers with exponents.

Addition and Subtraction

Begin

Today lets do addition and subtraction using some bigger powers, not limiting ourselves to squares and cubes, we will write numbers with exponents.

Write: $1^0 + 2^1 + 3^2 + 4^3 + 5^4$

First we need to find (convert) values of each number $1 + 2 + 9 + 64 + 625 = 701$

(do other addition combinations before moving onto combining addition and subtraction done the same way)

We could also mix up the signs like this (write) $2^2 + 4^3 - 7^0 + 5^2 - 3^2 + 8^2 =$
Lets first find their value $4 + 64 - 1 + 25 - 9 + 64 = 147$

* At some point we can also introduce parenthesis in to the problems. *Anytime we have a problem with parenthesis we have to do what is inside the parenthesis first.* Make child aware of this

What about multiplication and division, they are presented similarly as before.

Suppose we have a problem like (write) $2^4 \times 3^6 =$
We find the values $16 \times 729 = 11,664$

Multiplying Exponential Notation with Common Base

Begin

This is what we do for multiplication in exponential numbers when they have the same bases and different exponents.

Find the value and work out the multiplication.

(take out wooden cubing materials, felt mat)

I have something interesting to show you, (write) $3^3 \times 3^2 =$

Here is our multiplicand is 3^3 .



3^2 is our multiplier and is telling us how many times to take our multiplicand. So let's set out 3^3 and we want to take it 3^2 times. Here it is 1 time, 2 time, 3 times = 9 times.

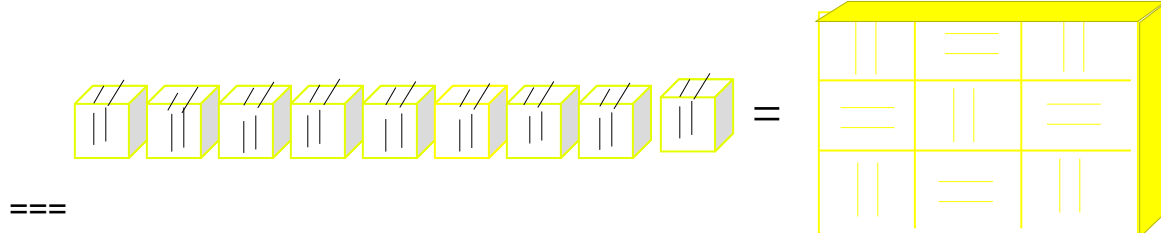


Now if I want to express my answer as a power of three. Let's see if our materials will help us, to go from 3^3 to the next power (3^4).

I need I need to take 3^3 three times. (place the pieces together, to make a cube of three) This is 3^4 .

To go from 3^4 power to the next power, I need to take it how many times?

Three. Form a square that equals 3^5 .



(now we turn the terms around, build each in the same fashion as above and show the answer remains the same)

(write) $3^2 \times 3^3 =$ We need to take 3^2 twenty seven times (we get that from 3^3)

(lay out 27 squares on the felt mat at side and build in the center)

But I want to express my answer as a power of three.

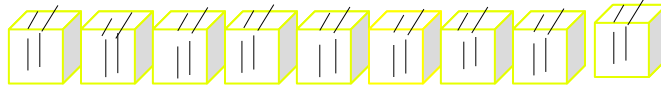
To go from 3^2 to the next power, what do I have to multiply it by? 3 (place three squares together.)



To go to the next power I have to take this (the cube now formed) three times, put them together to form a 3 power of four.



Put the rest together to equal 3^5 . The answer is the same



===

(all on paper)

Lets look at $4^2 \times 4^4 =$ lets do this on paper today. What is this problem asking us?

It means we will start with 4^2 & take it $4 \times 4 \times 4 \times 4$ times when we break it down into its factors. We can write it as $4^2 \times (4 \times 4 \times 4 \times 4) =$

If we multiply $4 \times$ one of those factors we get (write) $4^3 \times (4 \times 4 \times 4) =$

We get rid of one of those factor then multiply it again using our exponent, $4^4 \times (4 \times 4) =$

If we do this one more time we would write $4^5 \times (4) = 4^6$

Lets do some more problems, but first, this is an easier way (**a rule**) when we are multiplying numbers with exponent bases. **All we have to do is add the exponent.**

What about a problem like $6^3 \times 6 = 6^4$ or $7^4 \times 7^0 = 7^4$

This is because even though there is not always an exponent on the number, say 6, it is understood to mean 6^1 . And 6^0 is a non-existent place value of 1.

Division with Exponential Notation

Suppose we have $3^5 \div 3^3$ and we ask how many 3 cubed fit into 3^5 power?

We will build (set out) 3^5 power on the mat, how many 3^3 are contained in here or we want to divide it by 3^3 ?

How many 3^3 do we have (separate into groups of 3 cubed) 9 We want to express our answer of nine as a power of three.
write the answer $= 3^2$

===

$$3^5 \div 3^2 =$$

Let's form 3^5 on our mat; this time our problem is asking how many squares of 3 are there in 3^5 ? Separate them into 27.

Again - if I want to write my answer as a power of 3

(once the children have had that opportunity to explore the material, try doing this sort of problem $4^6 \div 4^4 =$)

Lets look at this on paper: $4^6 \div 4^4 =$

We know that 4^6 power is how many factors of 4? (6) ... and 4th power is how many factors of 4? (4).

Let's rewrite that listing factors.

$$\frac{4 \times 4 \times 4 \times 4 \times 4 \times 4}{4 \times 4 \times 4 \times 4} =$$

The idea is we are subtracting exponents of the divisor from the exponent of the dividend.

$$\frac{4 \times 4 \times 4 \times 4 \times 4 \times 4}{4 \times 4 \times 4 \times 4} = \frac{4 \times 4}{1} = 4^2$$

===

Zero and Negative Exponents

Suppose we have $3^2 \div 3^2 = 3^0$ This is because how many squares of 3 are contained in a square of 3? One.

How about this division problem $3^3 \div 3^5 = 3^{-2} = 1/3^2 = 1/9$

The dividend's exponent is smaller than the divisor's. So we get a fraction then.

Epilogue

A negative number means we have a reciprocal (numerator and denominator are swapped creating the multiplication identity-value for a given number).

Bring over the decimal board and describe the decimal categories in exponential notation. Place both labels, showing common fraction and exponential notation that lays out the following information by category.

$$\begin{array}{ll} 1,000,000 & = 10^6 \\ 100,000 & = 10^5 \\ 10,000 & = 10^4 \\ 1,000 & = 10^3 \\ 100 & = 10^2 \\ 10 & = 10^1 \end{array}$$

$$1 = 10^0$$

$$1/10 = 10^{-1}$$

$$1/100 = 10^{-2}$$

$$1/1,000 = 10^{-3}$$

$$1/10,000 = 10^{-4}$$

$$1/100,000 = 10^{-5}$$

$$1/1,000,000 = 10^{-6}$$

Final Verbiage

When a number has a negative exponent, we can say it is on the wrong line.

For example:

$$3^{-2} = 1/3^2 = 1/9 \quad \text{or} \quad 1/3^{-2} = 3^2/1 = 9$$

February 3, 2005

Expanded Power Notation - 3

Notes compliments of Carson Wright

Material

Paper and pencil.

Begin

I will show you something new today. Suppose I have the number 200 and I want to think about that number in terms of power notation.

We write it in power notation $200 = 2 \times 100 = 2 \times 10^2$

With our number system we can take a base and raise it to a power.

So we can take a number like: $8000 = 8 \times 1000 = 8 \times 10^3$

Lets do another number like this $9,000,000 = 9 \times 10^6$

With really big numbers it much easier to write them like this.

Suppose we have a number like $230 = (200 + 30) = 2 \times 10^2 + 3 \times 10^1$
(do a lot of these with the children)

Or, how about this:

$$4373 = 4 \times 10^3 + 3 \times 10^2 + 7 \times 10^1 + 3 \times 10^0$$

Once the children have had a lot of experiences do others with decimal fractions

$$23.64 = 2 \times 10 + 3 \times 10^0 + 6 \times 10^{-1} + 4 \times 10^{-2}$$

Epilogue

They **need lots of practice**; keep giving them numbers until they can write the numbers in expanded number notation and the reverse of that. Don't go on otherwise.

February 3, 2005

Operations Using Expanded Power Notation - 3

Notes compliments of Carson Wright

Addition

Let's take a problem like $4235 + 1509$. We will work this out in expanded number notation.

It is important that we keep the same categories lined up according to the appropriate power of ten. Note that we can add any category because there is no carrying until the end, when we add the actual values.

$$\begin{array}{rcll} 4235 = & 4 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 5 \times 10^0 \\ +1509 = & 1 \times 10^3 + 5 \times 10^2 + & & + 9 \times 10^0 \\ & 5 \times 10^3 + 7 \times 10^2 + 3 \times 10^1 + 14 \times 10^0 \\ & 5000 & 700 & 30 & 14 \end{array}$$

We used **three steps**: we rewrote the numbers in expanded notation, we add the numbers that fell into each power category, and we calculated the value for each sum including the power category – adding all values together at the end.

Subtraction

$$\begin{array}{rcll} 1. & 216 & 2 \times 10^2 + 1 \times 10^1 + 6 \times 10^0 \\ - & \underline{127} & 1 \times 10^2 + 2 \times 10^1 + 7 \times 10^0 \\ & & (1 \times 10^2) + (-1 \times 10^1) + (-1 \times 10^0) \\ & & = 100 + (-10) + (-1) \\ & & = 89 \end{array}$$

When we are breaking numbers apart like this, we call the 0-9 integers coefficients – to distinguish them from the power categories. **Rule: when adding or subtracting numbers in expanded power notation, you simply add or subtract the coefficient.** The coefficient is the value-digit in the power category.

Multiplication

$$\begin{array}{r} 42 \\ \times 3 \\ \hline \end{array}$$

42	=	4 x 10 ¹	+	2 x 10 ⁰	
<u>x 3</u>	=			<u>3 x 10⁰</u>	
		12 x 10 ¹	+	6 x 10 ⁰	
		120	+	6	= 126

The first thing we do is break it up into notation.

And we know that when we multiply, we multiply each digit in the multiplicand by the multiplier. We start with 2 x 10 and 3 x 10.

===

How about this problem?

135		1 x 10 ²	+	3 x 10 ¹	+	5 x 10 ⁰
<u>x 42</u>				<u>4 x 10¹</u>	+	<u>2 x 10⁰</u>
		2 x 10 ²	+	6 x 10 ¹	+	10 x 10 ⁰
	4 x 10 ³	+	12 x 10 ²	+	20 x 10 ¹	

We start with 2 times 5 times 10 to the zero power. (... complete first partial product)

We have taken care of multiplying by the units we need to multiply by the tens

20 x 10¹ power (... complete second partial product)

Now we only need to add to find our product.

===

What about this problem?

$$\begin{array}{r} 2.34 \\ \times 1.6 \\ \hline \end{array}$$

We are multiplying the coefficients and you multiply the powers of 10. But when you multiply numbers in the same base, you add the two exponents.

2 x 10 ⁰	+	3 x 10 ⁻¹	+	4 x 10 ⁻²		
1 x 10 ⁰	+	6 x 10 ⁻¹				
		12 x 10 ⁻¹	+	18 x 10 ⁻²	+	24 x 10 ⁻³
2 x 10 ⁰	+	3 x 10 ⁻¹	+	4 x 10 ⁻²		

So beginning with my first digits of multiplier and multiplicand ... 4 x 6 is 24. And I'll add my exponents (-2 + -1) to get 10⁻³. I'll make a new column to hold this result (24 x 10⁻³). (continue multiplication process as usual, multiplying coefficients and adding exponents)

Division

Lets start out by dividing 10,000 by 100 or $\frac{10000}{100} = \frac{10^4}{10^2} = 10^2 = 100$

Here, we had no coefficient. We only had to deal with the exponent.

Let's do 6000/20.

We will start out by writing the number in expanded number notation again.

$$\frac{6000}{20} = \frac{6 \times 10^3}{2 \times 10^1} = 3 \times 10^2 = 300$$

How about this problem:

$$\frac{30000}{500} = \frac{3 \times 10^4}{5 \times 10^2} = \frac{3}{5} \times 10^2 = \frac{3}{5} \times \frac{100}{1} = 60$$

Looking back at our original problem is that the answer we would get? Check long division also yields 60 as the answer.

$3/5 \times 10^2$ – this was our answer, three-fifths of 100 or 60. This is the same as 30,000 divided by 500 – also 60.

(verbalize that we need to divide both coefficients and subtract the exponents)

So we have divided the coefficient and also subtracted exponents to find the category for the quotient's coefficient. We also call this **scientific notation**.

February 9, 2005

Number Bases General Application

Numeration Across Powers of a Given Base

Age

Upper elementary

Material

Teacher-made board called the Number-base Board. Along with this board, we'll use golden or green unit beads, bead-bars 1-9, squares and cubes from the bead-chain cabinet. Blank tickets, pencil and a roll of adding machine tape about 2 inches wide.

On the right of the board, we have units; second column is bars, third column is squares and fourth column is cubes.

Prologue

Get into this after working with powers of numbers. The decimal system is base 10. Any other number except 1 can be used as a number base. The children often become interested in this through history studies – Babylonians for example, who used a base system of 60. Mayans used a base 20 system. First we'll work on counting in other bases.

Begin

We're going to count in another base system. We'll use 5 – so I'll write 5 here at the top of my board.

(place ticket with 5 in the “Base” field at top of board)

I want you to do the writing. I will say the numbers – I want you to write what I said.

(student writes, teacher says number and places beads)

1 (place bead in units column), 2 (place bead), etc. in units column up to 4.

(beads are added to the units column – now we move on to bars)

10 (place bead), 11 (place bead), etc. ... up to 14.

(5-bar is now in the bars column (at 10) and beads are being added to the units column – exchange)

20, 21, 22, etc.

(there are now 2 5-bars (at 20) in the bars column, beads are added to units up to 24)

(continue in this pattern – adding 5-bars into the bars column when units digit turns over – then adding individual beads into the units column)

: : :

100, 101, etc.

(now we have a square in the square column (at 100), no bars in the bars column – and add beads to the units column up to 104)

: : :

110, 111, etc.

(now we have a square in the square column, 1 5-bar in the bars column – and add beads to the units column) up to 114.

: : :

210, 211, etc.

(now we have two squares in the square column, 1 5-bar in the bars column – and add beads to the units column) up to 214.

: : :

(count up to 1000)

(count in another base for repetition)

Epilogue

Notice we never use the number 5 while we count when the base is 5. This is a good group activity since one can count and another can do the writing. Point out with experience when turning over of digits occurs.

The children may want to combine the information they generate. “Numeration in Various Bases” (features bases 10, 2, 4, 5 and 7), a work board, allows children to compare the same value representation for different bases.

February 9, 2005

Operations

Prologue

We have a finger chart for addition similar to the finger charts from primary. The numbers go over to 9 and down to 9 – i.e., up to the number just under “base number” - whatever it is.

Material

Number Base Board, paper and ruler.

Addition

Begin

We'll work with base 5 and construct a finger chart in base 5.

<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
1	2	3	4	10
2	3	4	10	11
3	4	10	11	12
4	10	11	12	13

Finger chart for addition in base 5 system

Suppose we want $3 + 3$. But that's base 10. We'll use our number board.

(place 3 and 3 on the units – count up to 4 and add the 5-bar into the bars column by exchanging 5 of the units for the bar)

So we get 11 for our answer.

How about $4+4$.

(same process – place beads in units column – count out each, exchanging each 5 for 5-bar and placing it into the bars column – then taking the column result after counting all)

That's equal to 13 in base 5.

Now that we have our finger chart, you can do any addition problem you like in base 5.

Epilogue

The children can create their own problems with their finger chart.

===

Suppose we have the problem $232 + 143$ and we want to add this in base 5.

$2+3$ – looking at our finger chart, we see that's 10. So we put down our zero and carry the 1.

$3+4$ gives us in the finger chart 12. So we add the one we carried to give us 13. We can write out 3 in the answer and carry our 1 to the next digit. (continue)

$$\begin{array}{r} 232 \\ +143 \\ \hline 430 \end{array}$$

The other way to write a number is to write showing its base is 31_{Five}
(continue for repetition)



Subtraction

Prologue

Note that the top row, then moving onto a diagonal, is the guide for minuends. It begins top row second cell from left and proceeds rightward – from “2 x (base-1)” to 1. It moves onto the diagonal band when the values enters units – single digit

values. We typically color this band of cells red.

A diagonal is the guide for subtrahends. It begins second row from top first cell at left and proceeds down diagonally from “base-1” to 1. We typically color this band of cells blue.

Begin

We'll stay with base 5 and look at subtraction now.

(see paper notes for table)

First thing we do is to make a finger chart.

(red squares on the chart are the initial number – decreasing down to 1 less than “base number.” It goes across top, then proceeds down the diagonal moving rightward)

We work in base 5 so on the left we'll start our chart with 13 – **twice the number of the number that is less than “base number.”**

(blue numbers are the subtrahends – decreasing down to 1 as it moves down diagonal rightward)

So we start on second cell from left at top row – with 13. Then we will in moving rightward – 12, 11, 10 – then moving down the diagonal – 4, 3, 2, 1.

These are our minuends – numbers in red.

The numbers in blue are our subtrahends. In base 5 those numbers start at 4.

	13	12	11	10	4			
4						3		
	3						2	
		2						1
			1					

Now we need to fill it in. How about $12 - 3$. I'll place 12_{five} on my number-base board and I want to take 3 away. What do I need to do? (exchange)

So now I've exchanged my bar for units – and I can take away 3. What's my answer? (4)
(fill in chart in the appropriate cell for $12_{\text{five}} - 3_{\text{five}}$)

How about $10 - 2$. I'll place 10 on my number-base board and I want to take 2 away.
What do I need to do? (exchange)

So now I've exchanged my bar for units – and I can take away 2. What's my answer? (3)
(fill in chart in appropriate cell)

Now I want $11 - 3$. So I set 11 on my board (1 bar and 1 bead). First we exchange the bar for units.

Now we can take away 3. (do so) How many are left on the board? (3)

So that's my answer. (fill in chart).

(we can fill in cells that are less than 5 in both operands – e.g., $4 - 2$ is 2 in both base systems – so just fill those in on the base 5 chart)

(continue to fill in cells to completion)

	13	12	11	10	4			
4	4	3	2	1	0	3		
	3	4	3	2	1	0	2	
		2	4	3	2	1	0	1
			1	4	3	2	1	0

This color difference was filled in using rules for base 10.

This color difference was filled in using the materials and deriving the difference.

Now we can do subtraction $24 - 13$. This problem is easy. $4 - 3$ is 1. $2 - 1$ is 1. So our answer is 11. But there is no carrying here so it's the same as we use with base 10.

How about $21 - 3$. We need to borrow a bar and exchange it for units – if we do that, we'll have $11 - 3$ and that gives us 3 (look at the chart)

I borrowed a bar, so how many are left? (1)

So 13 is my answer.

$$\begin{array}{r} 21 \\ - 3 \\ \hline 13 \end{array}$$

Epilogue

Use simple examples. They will make up complicated examples.

Multiplication

Prologue

Top blue band is the multiplicand. Left red band is the multiplier.

Ranges for both go from 1 to “base.”

Begin

This **chart starts with 1 and goes to “base”**. We" construct our base 5 chart in the same way as we did with addition – in a regular rectangle. We'll with 1 and go to base, which is 5.

How many rows and how many columns do we need? (5)

We'll start at 1 and go to 10 – rightward.

We'll start at 1 and go down to 10 – downward.

Why do you think the multiplication chart goes to “base”?

(Note that finger charts for multiplication is preparation for squaring)

1	2	3	4	10
2				
3				
4				
10				

So we'll fill in our number board now. Let's take 3 taken 3 times

On our board, we set 3, once twice and three times. Now we need to count and exchange. 1, 2, 3, 4 and this is a bar. So 3 taken 3 times in base 5 gives us 14. (fill in on chart)

How about 4 taken 2 times? On our board, I'll set out 4 two times. Now I can count and exchange – 1, 2, 3, 4 and this makes a square. I'll add a 5-bar. And we have left, 1, 2, 3. So 4 taken 2 times is 13.

And since the commutative law still applies, we also know that 2 taken 4 times is also 13. So we can fill that in also. (do so)

(continue pattern as above to fill in table)

1	2	3	4	10
2	4	11	13	20
3	11	14	22	30
4	13	22	31	40
10	20	30	40	100

Now we can do base 5 multiplication.

Suppose we have 123 times 3 in base 5.

3 x 3 is 14. So what do I write? 4 and we carry the 1.

Then 2 x 3 is 11 – plus the 1 we carry – gives us 12. I write my 2 and we carry the 1.

Then we have 1 x 3 plus our carry – that makes 4.

So 123 times 3 in base 5 gives us 424.

$$\begin{array}{r}
 123 \\
 \times 3 \\
 \hline
 424
 \end{array}$$

(continue several problems for repetition - 203 x 42)

(be sure to show how to add the partial products using the addition finger chart)

Epilogue

If coding problems in mixed bases, use the other notation – with the base written in letters as subscript to the number. You don't need to include those subscripts on partial products – since they are derived and the base is a given then.

Division

Prologue

The division chart in base 10 is another odd one like the subtraction chart – but it can be done.

Begin

Suppose we have 20342 divided by 3. We want to work this out in base 5.

3 won't go into 2 so we start with 20. How many times will 3 go into 20? (3 times)

We know 3 times since we looked in multiplication chart for 3 on multiplicand, then go down the column to find a number that is 20 or less – so it has to be 3, looking at our multiplier at the left.

That will use up what? (14) So we need to subtract 14.

Now we have to use our base 5 subtraction. We borrow a bar to take 4 away.

How many are left? (1) Then we bring down the 3.

Now we ask how many times will 3 go into 13? (2 times) We can see that looking under 3 in our multiplication chart.

We use 11 so we subtract 11 and we have 2 left over.

What do we bring down on paper? (4)

Now we have 24 and we're asking how many times will 3 go into 24? (4)

So we write our 4 in the answer – it goes 4 times using up 22.

We subtract 22 from our 24 and we get 2. Now we bring down our 2.

3 goes into 22 how many times? (4)

So our answer is 3244!

$$\begin{array}{r}
 3 \overline{) 20342} \\
 \underline{- 14} \\
 13 \\
 \underline{- 11} \\
 24 \\
 \underline{- 22} \\
 22 \\
 \underline{- 22} \\
 0
 \end{array}$$

Epilogue

If doing problems with mixed bases on same piece of paper, identify dividend, divisor and quotient using subscript notation. All else is assumed.

February 9, 2005

Conversion

Prologue

When they encounter expression systems of greater than base 10 – expressing 11 in base 12 for example, we use symbols. The children can use any symbol they like for this as long as they make a key e.g., “e = eleven”).

Material

Number base board, unit beads and bead bars, squares and cubes from the bead chain cabinet, blank tickets and pencil, an empty cup.

Begin

Suppose we have the number 1432 in base 5 and we want to know what this number is in base 10.

(write $1432_{\text{five}} = ?_{\text{ten}}$)

Let's start by placing this on our number board.

1 is 1 cube, 4 is 4 squares, 3 is 3 bars and the 2 is 2 units.

(place all on the board)

So I've set out 1432 in base 5 on my board.

Using our knowledge from Powers of Numbers, let's write down the values for these amounts.

(build problem paper image as we add lines)

I'll start on the left with my cube. I have how many of these? (1)

So it's 1 x what? How would we express this in power notation? (5^3)

So it's 1 times 5^3 and that equals 125.

(write $1 \times 5^3 = 125$)

Next I take my 4 and that's 4 squares of 5. So that $4 \times 5^2 = 100$

Next we have 3 and that's $3 \times 5^1 = 15$

Next we have our 2 in units – so that's 2 times $5^0 = 2$.

So adding up that makes $125 + 100 + 15 + 2$ – that equals 242.

So 1432 base 5 is equal 242 in base 10.

Written image:

$$\begin{array}{r} 1 \times 5^3 = 125 \\ 4 \times 5^2 = 100 \\ 3 \times 5^1 = 15 \\ 2 \times 5^0 = 2 \\ \hline 242 \end{array}$$

Let's do another.

(repeat $1223_{\text{four}} = ?_{\text{ten}}$ and others for practice (107))

===

What about going from base 10 to another base?

Let's take 54_{ten} and I want to know what it's worth in base four.

I have counted out 54 beads in my cup (from golden materials)

We can't have this many in the units – so we have to count and exchange)

(count 1, 2, 3 and a bar – 1, 2, 3 and a bar, etc.)

(add bars and units to replace the 54 units – 13 bars – then exchange for cubes)

(count 1, 2, 3 and a square, etc. - we now have 3 squares, 1 bar, 2 units)

We can see that 54 in base ten is worth 312 in base 4.

Now you can try your own example.

(continue to do examples for repetition)

===

But I wonder if there's a way to do this on paper instead of counting these beads each time?

Let's think about what we did. We started with 54. (write 54 at top left of paper)

I began making groups of 4. I divided the 54 by 4. (write $54 \div 4 = 13$)

When we did that we got 13 bars. And we had a remainder of 2.

Then we began to make groups of 4 from bars to exchange them into squares.

If we divide 13 by 4, we get 3 times – 3 squares, with a remainder of 1.

If we then take that 3 and ask how many groups of 4 we can make out of 3, the answer is zero – but we have a remainder of 3.

There is your answer reading the remainder from the bottom up. You just keep going until you get a quotient of zero – the remainder gives the answer.

$$\begin{array}{r} 54 / 4 = 13 \text{ r } 2 \\ 13 / 4 = 3 \text{ r } 1 \\ 3 / 4 = 0 \text{ r } 3 \end{array}$$

===

Suppose we have 6821_{ten} and I want to know what it equals in base 3.

(write $6821_{\text{ten}} = ?_{\text{three}}$ at top of paper)

We start with 6821 and divide it by 3. (write $6821 / 3 = 2273$ r 2)

Now we take the 2273 and divided it by 3 (write $2273 / 3 = 757$ r 2)

We take 757 divide by 3 (write $757 / 3 = 252$ r 1)

Now we have 252 divided by 3 (write $252 / 3 = 84$ r 0)

Note that we have to write down 0 remainders since they are place holders.

Now 84 divided by 3 (write $84 / 3 = 28$ r 0)

And 28 divided by 3 (write $28 / 3 = 9$ r 1)

Then 9 divided by 3 (write $9 / 3 = 3$ r 0)

Then 3 divided by 3 (write $3 / 3 = 1$ r 0)

And 1 divided by 3 (write $1 / 3 = 0$ r 1)

So our answer is 100100122 base 3.

(start at bottom and read remainders going up)

Suppose we start with the same number in base 10 – 6821 - but this I want to convert it to base 12. Can we do that?

(write $6821_{\text{ten}} = ?_{\text{twelve}}$)

$$6821 / 12 = 568 \quad \text{r } 5$$

$$568 / 12 = 47 \quad \text{r } 4$$

$$47 / 12 = 3 \quad \text{r } 11$$

Whoa! I can't write a 2-digit number since it would throw off our columns.

What do we do? We need another way to represent 11 – another symbol for it.

I'm going to use the letter e since it's the first letter of eleven.

So we'll have a remainder of 'e' representing a remainder of 11 in base ten.

$$6821 / 12 = 568 \quad \text{r } 5$$

$$568 / 12 = 47 \quad \text{r } 4$$

$$47 / 12 = 3 \quad \text{r } \mathbf{e}$$

$$3 / 12 = 0 \quad \text{r } 3$$

So 6821 base ten equals 3e45 in base 12.

Epilogue

The children can do lots of these. We limit ourselves to 4-digit numbers due to the board and materials. The children can use larger numbers – moving into 4th power and so on. They can practice converting from whatever base to whatever base they like.

February 10, 2005

15. RATIO AND PROPORTION

Concept

Prologue

Ratio is a Latin word meaning computation. The origin of the word is from the Latin word 'reri', meaning to consider or reckon. **Ratio is the relative size of two quantities expressed as one divided by the other.** So a ratio is always a comparison and must have two things to compare.

We should also **remember it is a comparison by division.** One thing the children would likely have had already involving ratio is pi. Pi is the comparison of a circle's circumference to its diameter. We say the ratio of the circumference to the diameter is pi to 1. Or we can say the ratio of the diameter to the circumference is 1 to pi. So the **order is important.**

We begin by introducing concept, language and notation.

Material

Peg board, box of pegs, number cards, felt mat, blank tickets, paper and pencil and various objects we have gathered from the environment.

Begin

I'll put some beads on my mat – 2 green beads and 3 red beads. The **comparison I want to make is this: the ratio of green beads to red beads is 2 to 3.** That's 2 green beads to 3 red beads.

(beads were placed in center of mat – put them away and now put out skittles)

I'll put out skittles now – 4 blue and 3 red skittles. I want to express this comparison now. I can say the ratio of blue skittles to red skittles is 4 to 3.

We can compare all kinds of things like this. We can compare lead pencils to colored pencils. (lay out pencils for an example)

I have 3 lead pencils to 8 colored pencils – so my ratio is 3 to 8.

We could look around the classroom and compare the ratio of people wearing glasses to those not wearing glasses. I see 12 with glasses to 13 not wearing glasses. So we have 12

with glasses to 13 without glasses. We don't always have to compare a thing to a like thing. We could compare the number of tables versus chairs in the room, for example.

Let me show you how to write ratios. I have 3 blue felt squares.

(lay them on the mat at left)

And I have 5 red felt squares. (lay them down)

So the ratio of blue felt squares to red is 3 to 5. **We write that like this: 3:5**

When we read the dots, we read it as 'to' so we say 3 to 5 when we read this.

Epilogue

The economic geography studies is a good area in which to get some further experience with ratios.

February 10, 2005

Using Ratios - 1

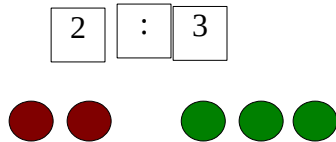
Material

We are using the peg board here, along with blank tickets.

Begin

Suppose you have a ratio of 2 red pegs to 3 green pegs.

(place peg board on table, place 2 red pegs at top left, 3 green pegs next to red. Write the ratio on a blank ticket 2:3)



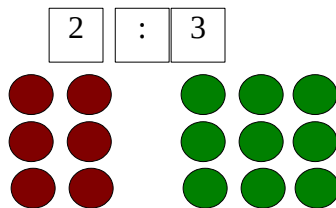
this ration tells me for every two red pegs here, I'll have 3 green pegs here. So, suppose I do this.

(add 2 more red and 3 more green pegs into a second row)

I can say I still have a ratio of 2 red pegs to 3 green pegs. I still have a ratio of 2 to 3.

Why? I still have what is more or less a fraction; for every two here, I still have 3 there. The ratio has not changed.

Suppose I add another row. (do so) Now I have 6 red pegs here and 9 green pegs over here. But I'm going to say that I still have a ratio of 2 to 3.



We can say this because for every 2 red on our left, we have 3 green beads on our right.

(continue with different numbers and ratios for **repetition**)

The **first number in a ratio is called the antecedent. The second number is called the consequent.** So here, 2 is my antecedent and 3 is my consequent.

Epilogue

Generally it takes some time for the child if you multiply one side, like we did when we added red beads, you have to do the same thing to the other side. Or in fractional terms, what one does to the numerator has to be done to the denominator also. Fraction work has prepared them for this – reduction of fractions for example.

February 10, 2005

Problem Solving with Ratios - 2

Begin

The ratio of 2 numbers is 2:4. The smaller of the two numbers is 8. What is the larger number?

I want to show you how to work this out on the peg board.

I have my ratio on the peg board now. What other information does our problem give?

Yes, the smaller number is 8 – and I can represent that.

(place 6 more green pegs to form 8 total)

So here is the smaller number – 8 – of green pegs. We **set it in groups of two because our antecedent is 2.**

Since our ratio is 2:4, for every 2 green pegs on our left, we need 4 red ones on the right.

(format the red beads on the right to conform to the ration)

So what is the larger number? (16) Yes, our consequent is 16.

===

There are two numbers given in a ratio of 2:3. This time I know the larger number is 9.

(place pegs – 2 green at left and 3 red at right – separated by a hole or two on the board, plus tickets for 2:3 a placed at top margin)

So I form my 9 on the right side. But then there have to be 2 on the left for each 3 on the right – so we can see the smaller number is 6. And we **always set out pegs in groups that reflect the numbers in our ratio.**

===

Let's go back to this ration of 2:4 and we're given that the smaller number was 8. So our our paper we can write the ratio of 2:4 at the top of our page. (do so)

We know the smaller number is 8. What can we do to 2 in order to arrive at 8? (multiply by 4)

$$2:4$$

$$2 \times 4 : 4 \times 4$$

$$8 \quad 16$$

What do we have to do to the right side? Multiply by 4, right? So we can see our larger number is 16 using our paper too.

===

Suppose we have the ratio of 3:7. We know the larger number is 21.

What do we need to do to the 7 to get to 21? (multiply by 3)

$$3:7$$

$$3 \times 3 : 7 \times 3$$

$$9 \quad 21$$

When we multiply our left side by 3 also, we see our antecedent is 9.

Epilogue

This is a simple problem but allows the child to see how ratios can be used. Establish the ratio, create a problem using either the antecedent or consequent, solving for the other. After they work out problems on the peg board, talk to them about working a problem like this on paper – no peg board; how it could be done.

February 10, 2005

Relating Ratio to Prior Work - 3

Begin

I have a ratio of 2:3 (set out at top of peg board)

I'm going to set out a number of pegs in this ratio.

(we have 3 rows of red (6) and 3 rows of blue (9) plus tickets 2:3 at top border)

We said a ratio is a comparison of two numbers. Look carefully at what this is comparing – what it means.

If I look just across the top row and I want to compare red pegs to blue pegs, **I can say my red pegs are $\frac{2}{3}$ of the blue pegs.** Let's write that on paper.

Write: $\frac{2}{3}$

We've been writing ratios as 2:3 – watch.

(add line to create)

$$2 \div 3$$

We just added a little line but the ratio really means 2 divided by 3. That's what our fraction really means too. It is the antecedent divided by the consequent. That's what this ratio really is.

Remember when we did ratios before, solving those word problems, we know we can set these up as fractions now.

Say we have the ratio of 2:3 but I'll write it this way $\frac{2}{3}$

And we know the larger number is 12. Do we know what to do to our fraction on the left to get the correct fraction on the right? What do we do? (multiply denominator by 4 to get to 12)

So we also have to multiply our numerator too!

$$\frac{(2 \times 4)}{(3 \times 4)} = \frac{8}{12}$$

So everything we know about equivalent fractions comes in here too. Multiples and fractions apply to working with ratios!

Ratios can also be larger numbers to smaller numbers. So for example we might have a ratio of 3:2.

(switch numbers on the peg board and set up new ratio of 3:2 using the same 9 red beads and 6 blue beads – grouped according to the original ration of 3:2)

We're comparing red pegs to blue pegs and they are in the ration of 3 to 2.

Look just across the top row – **we can say the red pegs are 3 halves of the blue pegs.**

(point to top row of pegs)

So we have our ratio of 3 over 2. Suppose we know the larger number is 27. We want to know the value of the smaller number. What can we do? (multiply by 9)

Right, and if we multiply the numerator by 9 we also have to multiply the denominator by 9 too. So we can see the smaller number is 18.

$$\frac{(3 \times 9)}{(2 \times 9)} = \frac{27}{18}$$

===

Suppose we have this problem:

A man left some money when he died. It was to be distributed to his wife and only son in the ratio of 2 to 3. The wife received 2400 dollars. How much did the son receive?

We know at least two ways to work this problem out. We could use the peg board.
(each peg can represent a hundred or whatever we determine)

On Peg Board

We have a ratio of 2:3 so I'll set out the tickets at the top of our board. We will set our pegs in groups of 2 on the left side – up to 24 in 12 rows – since each of our pegs are each worth 100 dollars.

Now, what will we do with pegs on the right? They must lie in groups of 3 since our consequent was 3 in this problem. And we also need how many rows? (12)

For every two pegs on the left, there has to be 3 pegs on the right – 12 rows of each.
So how much did the son receive? (3600 dollars)

The son received 3600 dollars since we have 12 rows of 3 pegs – each 100 dollars.

On Paper

Can we work this out on paper? We begin by writing our ratio as a fraction. To get 2400 from 2 we need to multiply our antecedent by 1200. So we also need to multiply our consequent by 1200 as well. So we get 3600 dollars on paper too.

$$\frac{(2 \times 1200)}{(3 \times 1200)} = \frac{2400}{3600}$$

===

We aren't always just looking for one thing in these kinds of problems. Sometime we're looking for two things.

On Peg Board

Two numbers are in a ratio of 4:7. The sum of the two numbers is 55. What are the two numbers?

Let's set up our peg board. We set our tickets at the top for our ratio. ("4:7")

And my problem also gives me that the sum of the two numbers is 55. How much do we have on the peg board with our ratio? (11)

(set ticket at end of row "11")

(pegs are placed already in 4:7: configuration)

Well that's not 55, so let's add another row to each. (add 4 on left and 7 on right side)

(set ticket "22" at end of second row)

How many do I have now? (22)

(continue for 5th row – 55 total – place ticket at end of each row 33, 44 and the 55)

So what is our final answer? We can see the smaller number is 20 and the larger number is 35. Is 20 and 35 equal 55? (yes)

On Paper

Is there a way we can use the information given to us to work it out on paper?

We have the ratio of 4:7 and we know the sum of the two numbers is 55. We can add the two numbers together to get 11. If we divide the sum of 55 by 11, we get 5.

This tells us we need to multiply each side of the ratio by 5.

4:7

4 x 5 : 7 x 5

20 35

===

Two number are in a ratio of 3:5. The difference of the two numbers is 10. What are the

two numbers?

On Peg Board

Let's set out our ratio and our pegs for this ration of 3:5.

This time what do we need to ask. We want to find a difference. What is the difference between 3 and 5? (2)

Let's write that on a ticket and place it at the end of our first row of pegs.

Well the problem says the difference should be 10 so let's add another row.

So I have 6 on the left and 10 on the right and the difference is 4.

That's not 10 so let's add another row.

(continue to add rows and tickets for 5 rows total until we have difference of 10)

What's our two numbers? (15 and 25) Is there a difference of 10 between those numbers? (yes)

On Paper

(find difference between 5 and 3 and divide 10 by that number – i.e., $10 \div 2$)

(multiply antecedent and consequent by that quotient)

Written image:

$$\begin{array}{ccc} & & 3:5 \\ & & 3 \times 5 : 5 \times 5 \\ & & 15 \quad 25 \end{array}$$

February 10, 2005

Proportion

Concept

Prologue

The children have been working with proportions without knowing it. If we have an equivalent fraction such as $\frac{2}{3} = \frac{6}{9}$, that is a proportion. We have **a statement that two ratios are equal – that's what a proportion is.**

A **continual proportion** would be $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$ and so on.

Bring the children's attention to what a proportion is, give them the name and then we extend their understanding.

Material

Felt mat, some objects from the environment.

Begin

(place skittles – 2 green up top of felt mat and 1 blue one below those – all far left)

I'm going to set out a ratio of green skittles to blue skittles. I have set a ratio of 2:1, green to blue.

Now I'll set out another ratio – this time 4:2 – 4 green skittles to 2 blue skittles.

(again green are at top of mat – 4 placed horizontally with row of blue below)

If I were to write these as fractions, the first would be $\frac{2}{1}$

The second would be $\frac{4}{2}$

Are these ratios equal? Yes – so we can put an equal sign between them since 2 over 1 is equal to 4 over 2.

$$\frac{2}{1} = \frac{4}{2}$$

This relationship has a special name. **If two ratios are equal, we say they are in proportion.**

It says the values are equal. When the ratios are expressed in this order and have the same value, we say the ratios are in proportion.

Let's look at some beads now. I'll set on my mat.

Here I set out 1 to 5 (1 above, 5 below – both at left)

On the right side 5 to 25 (5 above, 25 below – both at right)

So we have the ratio of 1 to 5 on the left and the ratio of 5 to 25 on the right.

If we write them as fractions we have $\frac{1}{5}$ and $\frac{5}{25}$.

Are my fractions equal? $\frac{1}{5} = \frac{5}{25}$

We see they are. For every bead up here (point left) there are 5 beads down here.

For every bead up here (point right) there are 5 beads down here (lower right 25)

So we can say these two ratios are in proportion.

New Ratio in Proportion

Now, I'll switch the right side. Suppose we make the right side a ratio of 25:125.

So now we have $\frac{1}{5}$ on the left and $\frac{25}{125}$ on the right? Are my two ratios equal?

How do we know? Again, for every one bead here (top left) there are 5 beads here (bottom left) And for every bead here (top right) there is 5 beads down here (bottom right).

Can we check this?

If we multiply on the left side – both numerator and denominator – by 25, we get to the ratio on the right.

How else could we check it?

We could divide numerator and denominator of the ratio at right by 25 – divide on the right instead of multiplying the left side – and we arrive at the same answer.

Concept $\frac{(1 \times 25)}{(5 \times 25)} = \frac{25}{125}$

Epilogue

Note that verbally antecedent and consequent are expressed positionally.
Make sure the children know this.

February 10, 2005

Proportion and Similarity

Material

2 rectangles and 3 squares from circle quadrilateral fraction material. Power of 2 material.

Begin

(set out fraction pieces – various rectangles and squares)

What do we know about these figures? (they are polygons and have 4 sides – quadrilaterals)

Some of them are similar. Which ones? (the two rectangles) What else? (the 3 squares)
Similar means what? (they have the same shape but have different sizes)

There is something special about similarity we want to talk about today. **For two figures to be similar, they have to be in proportion.**

Not just any two rectangles are similar, they have to be in proportion to be similar.

These figures have how many dimensions? (2) Yes, a base and width – or length and height.

So for these two rectangles to be similar, it means that both their lengths and widths have to be in proportion. They have to form equal ratios.

Let's check it out. I'll say my smaller rectangle has a length of 1 (point to rectangle length)

Let's compare it to the larger rectangle (set side by side lengthwise)

We can see two shorter ones would make up the longer one – so we have a ratio of 1:2

But we also have to look at width. We'll say my smaller rectangle has a width of 1 again. Let's compare it to the width of the larger rectangle. (set side by side)

For every one on the smaller, how many are there on the larger? (2) So again, if this is one (smaller width) then this is two (larger rectangle width)

So are these equal ratios? Sure. We can really say **these rectangles are similar now because we know that length and width are both in proportion.**

We can also look at the squares in the same way. But since they are squares, we only have to compare one side.

Let's start with the middle size and largest one. Should we take smaller to larger or larger to smaller – we can express our ratio either way. Let's choose larger to smaller.

(set squares side by side)

For every one on the smaller square, there are how many on the larger square? (2)

So expressing it larger to smaller, we'll say the ratio is 2 to 1 (2:1)

Since we know the sides are the same, we can say these squares are in proportion.

(repeat for others if needed to make it more concrete for the child)

(introduce powers of 2 pieces)

How many dimensions do we have here? Can you organize them by like shape?

(child does so – tall prisms, wide prisms and cubes)

I wonder if these pieces are in proportion? We have 3 dimensions to compare now.

Let's start with the two white prisms. We'll compare the 3 dimensions of each to see if they are similar or not.

First let's compare lengths. Let's do it smaller to larger.

(put smaller and larger side by side)

If the smaller is 1, we would say the larger length is 2. So our ratio is 1:2.

Now let's compare the width. If the smaller width is called 1 and we line up the widths of both pieces, we can see the width on the larger is 2. So, again we have a ratio of 1:2.

Finally, let's compare height – our 3rd dimension. If the height of the smaller prism is 1 what could we say about the larger? (set both pieces height-wise, side by side)

It's also 2. Again, we have a ratio of 1:2.

So we've compared all 3 dimensions and saw they each have the same ratio between the 2 pieces. So we verified they are truly similar.

===

(further repetition for those children who've not got it yet)

On my peg board, I'll set out 2 ratios: 2:3 on my left and 4:6 on the right

(we have same layout as previous lessons but have created two sets of ratios – one left and one right)

I would like to know whether these two ratios are in proportion. Watch.

(make group of 4 antecedent on right side into 2 rows of 2)

(make group of 6 consequent on right side into 2 rows of 3)

This helps us see these are in proportion. We have 1 group of 2 on the left antecedent and 2 groups of 2 on the right antecedent.

===

There is something about proportions you don't yet know that is really neat.

Suppose I write down these two proportions $\frac{1}{2}$ and $\frac{3}{6}$

Are they of equal value? Yes – we know since they could be rewritten as a decimal fraction to see their equality that way – $\frac{1}{2}$ is equal 0.5 and $\frac{3}{6}$ is also equal 0.5.

We could multiply both numbers on the left (1:2) by 3 and we would find the ratio on the right $\frac{3}{6}$. So we can see it this way as well.

Well there is another way I'll show you today. We have an equation $\frac{1}{2} = \frac{3}{6}$.

We know that we can do the same thing to both sides of an equation and keep it balanced.

Suppose I multiply both sides of this equation times $\frac{2}{1}$. Can I do this without changing the value? (yes)

Now let's simplify the left side.

$$\begin{aligned}\frac{1}{2} \times \frac{2}{1} &= \frac{3}{6} \times \frac{2}{1} \\ \frac{2}{2} &= \frac{3}{6} \times \frac{2}{1} \\ 1 &= \frac{3}{6} \times \frac{2}{1}\end{aligned}$$

Let's look at our numerators. Where are they in our original equation?

(3 and 2)

Let's look at our denominators. Where are they in our original equation?

(6 and 1)

This means when you have an equation you can do **cross-multiplying** and you don't change the value.

So when you have an equation which is a proportion, we can use cross-multiplication to solve problems.

===

Suppose we have $6:9 = x:12$

$$\frac{6}{9} = \frac{x}{12}$$

To work this out cross-multiplying, we have $9x = 6 \times 12$. What does that give us? (72)

To find just x , we need to divide by 9 – and we see that $x = 8$.

So let's try this word problem using cross-multiplication.

A painting 4 feet wide by 6 feet high appears in a photograph to be 2 inches wide by 3 inches high. The eye in the photograph is 5 millimeters wide. How wide in

millimeters is the eye in the painting?

What is our first ratio to set up? We want to compare the painting to the photograph.
So 4 ft to 2 inches is a good beginning point.

We can't have feet and inches in the same ratio – so we need to make them the same unit of measurement. So before we go further, let's change those feet into inches.

So we have 48:2 inches. That's our ratio of width of the painting to the width of the photograph.

Now what other ratio do we need? We don't need height since we know the painting is similar to its image in the photograph. We're **interested in the width of the eye in the painting.**

We don't know the width in the painting but it is 5 millimeters in the photograph. So we can set up a proportion and solve using cross-multiplication.

$$\text{So we have } \frac{48}{2} = \frac{x}{5}$$

Then we have $2x = 240$ and solving we find $x = 120$ millimeters.

===

Let's try this one. First let's recognize this fact.

If you travel at a constant speed, we know the distance traveled in is proportion to the time spent.

Now, if you traveled 8 miles in 10 minutes, how long will it take you to go twelve miles traveling at a constant speed.

What's the first ratio we need? (8 miles to 12 miles)

What's the second ratio we need? (10 to x)

So our proportion looks like this:

$$\frac{8}{12} = \frac{10}{x}$$

So we cross-multiply and we get $8x = 120$ and solving for x we find $x = 15$ minutes.

===

If an item is on sale for 20 percent off and the original price was 120 dollars, what is the new price, the sale price?

What is our first ratio? (20:100) This represents 20 percent off.

What is the second ratio? ($x:120$)

So we have this proportion:

$$\frac{20}{100} = \frac{x}{120}$$

So we have $100x = 2400$ and solving we find $x=24$. Since that's our discount, we need to subtract 24 from our 120 – so our sale price is $120 - 24$ which is 96 dollars.

Or – we could choose a ration of 80:100 to begin with instead of the 20:100 and we don't have to subtract from 120 dollars at the end.

February 7, 2005

17. ALGEBRA

Material

Felt mat, bead bars 1-10, box of number cards and signs, parentheses, blank tickets and pencil, several long cardboard division bars.

Age

Upper elementary.

Prologue

We're going to give the children some ideas about how to approach algebra – what an equation is and how you can manipulate an equation. If we can help the child understand those ideas, hopefully, if they study algebra later, they will understand it. There is a reason one can do what one can do.

We use simple equations for illustration; the children can then take up more complex problems on their own. Note that except for the first example, **cards are the workers and are used to restablish equality on both sides of the equal sign.**

Begin

Adding to Balance an Equation

I'm going to set something on this mat. I have set out $5+3=8$



I want you to notice the equal sign – we have what is called an equation. Because $5+3$ is equal to 8.

This equal sign tells us that 2 things have the same value. Do you agree these are the same value?

So – this is an equation.

But now I'll do something; on the left side I will add a 2-bar – so now I have $2+5+3 = 8$.

$$- \boxed{+} - \boxed{+} - \boxed{=} -$$

I unbalanced it didn't I – they aren't equal anymore.

But suppose on the right side, we also add 2.

$$- \boxed{+} - \boxed{+} - \boxed{=} - \boxed{+} -$$

So they are now equal again – both sides are equal 10.

So when you have an equation, if for any reason you have to add something to one side that unbalances the equation, you can make it balanced again by adding the same amount to the other side.

This preserves that statement of equality by the equal sign. Now you can make up your problems like this.

(do as many problem as needed to make the key points clear – idea of an equation and how an equation can be manipulated using addition)



Subtracting to Balance an Equation

Here I've set out on my mat $5+3=8$.

$$- \boxed{+} - \boxed{=} -$$

Is this an equation? (yes) Why?

Yes, it's an **equation** because it has the same value on each side of the equal sign.

But suppose we do this. (take the 3 and the plus sign away)

$$- \boxed{=} -$$

This is not equal, right? But I can do something to fix it. Watch.
 (add to right side “-3” using cards)

$$\text{---} = \text{---} - 3$$

Is it an equation now? (yes) Why? (because both sides of equal sign are equal value)
 8 minus 3 equals how many? (5) So we can add that “- 3” on the right side to equal 5.

Here we see that when we have an equation, if we need to subtract a value from one side, in order to keep it balanced, we need to subtract the same value from the other side.



Multiplying to Balance an Equation

I've set out $5+3=8$. Is this an equation? Why? (same value both sides)

$$\text{---} + \text{---} = \text{---}$$

Suppose I do this. Now I have multiplied the left side by 2. Is this an equation? (no)

$$(\text{---} + \text{---}) \times 2 = \text{---}$$

But I can fix it. Suppose on the right side I also multiply by 2. (do so)

$$(\text{---} + \text{---}) \times 2 = \text{---} \times 2$$

Do we have an equation now? Why? (same value each side)

So when you have an equation, if you need to multiply one side by a value, in order to keep the equation balanced, multiply the other side by the same value.

(carry out the multiplication on both sides if the children need it)



Dividing to Balance an Equation

I've set out $5+3=8$. Is this an equation? (yes) Why? (there is the same value on both sides)

But now suppose I do this. (divide left side by 2)

$$\frac{\text{blue bar} + \text{red bar}}{2} = \text{yellow bar}$$

Now it's unbalanced. But I can fix it.

Suppose we add a division bar and divide the right side by 2 also.

$$\frac{\text{blue bar} + \text{red bar}}{2} = \frac{\text{yellow bar}}{2}$$

Now is it an equation? (yes)

So when you have an equation, if you need to divide one side by a value, in order to keep the equation balanced, divide the other side by the same value.

Epilogue

There are two ideas you bring to the children through these examples. First is in order to have an equation, what is the left of the equal sign has to have the same value as what is to the right of the equal sign. Secondly, if for any reason, you need to perform an operation on one side of an equation – either side – you need to do the same thing on the other side to keep the equation balanced.

February 7, 2005

Solving for an Unknown in an Equation

Begin

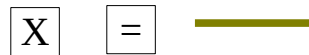
I'm going to set something out on my mat. $5 + x = 8$ (use beads for 5 and 8)



Why am I using x? The convention is we **use the letters at the end of the alphabet for unknowns** and the **letters at the beginning of the alphabet as constants**.

I bet you know the answer to this. But sometimes you won't know the answer just by looking. This is why I'll show you how you can solve for the answer.

We've got "x=" here. But we also have this 5-bar and this addition sign – we want to get rid of that. I can do that by taking them away. But now I have $x=8$.



I unbalanced it since I only subtracted from the left side.

To keep it balanced what do I need to do? (subtract the same value from the other side)

So on the right side, we need to add "-5".

If we solve this we see that $x=3$.



To check and see if we are right, we write out original equation down on paper and plug in the value for x that we found. We can see that $5+3$ is equal to 8 so it checks out.

So here I used what operation to solve for an unknown? (subtraction)

Let's fill in x on our paper to see if the answer completes the equation.



Let's do one more here. Let's take $9+x=16$

$$\underline{\hspace{2cm}} + \boxed{X} = \underline{\hspace{2cm}}$$

What do I do to solve. (take away the 9)

But if I take away 9 from the left side what has happened? (it's unbalanced)

What do I need to do then? (subtract 9 from the other side too)

So now I have $x = 16 - 9$ and we find $x = 7$.

$$\boxed{X} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Let's plug that in to check our answer. (do so)

Our original equation then with our value for x is “ $9+7=16$ ”

Does that check? (yes)

So 7 is indeed our correct answer.



Suppose I have $x-6=4$

$$\boxed{X} - \boxed{6} = \underline{\hspace{2cm}}$$

I want to solve – but this is a minus 6 so I can't just take it away.

Suppose on the left side I add “+6” - what does “ $-6+6$ ” equal? (zero)

$$\boxed{X} - \boxed{6} + \boxed{6} = \underline{\hspace{2cm}}$$

But now it's unbalanced. What can I do? (add +6 to other side too)
 (add in beads since it is a quantity)

$$\boxed{X} \boxed{-} \boxed{6} \boxed{+} \boxed{6} \boxed{=} \text{yellow line} \boxed{+} \text{purple line}$$

Our 6s on the left cancel out and we are left with $x = 10$.

To check we plug in the value of x into our original equation. (write on paper)

So we have " $10 - 6 = 4$ " – does it check? Yes.

Here we use which operation to solve for the unknown? (addition)



I'm going to set something on my mat. $2x = 6$.

$$\boxed{2} \boxed{X} \boxed{=} \text{purple line}$$

We want to know the value of x .

I can get rid of the 2 – I need to get rid of that. So I can divide by 2 to do that.

$$\begin{array}{r} \boxed{2} \boxed{X} \boxed{=} \text{purple line} \\ \hline \boxed{2} \end{array}$$

We know that 2 over 2 is 1 right? But now I've unbalanced my equation.

What do I need to do? (divide other side by 2 also)

$$\begin{array}{r} \boxed{2} \boxed{X} \boxed{=} \text{purple line} \\ \hline \boxed{2} \qquad \boxed{2} \end{array}$$

2 over 2 equals 1 so I can take all of this away leaving only x on the left.

6 divided by 2 equals 3.

Now let's check our equation. Plugging in our x value we have $2 \times 3 = 6$
Is that correct? (yes) So, it checks out.



Suppose I have this on my mat: $\frac{x}{2} = 4$

We want to know what x equals. That means I need to get rid of the 2 and the division.
I can get rid of that by multiplying by 2.

But now my equation is unbalanced so I need to multiply the other side by 2 also.

2 over 2 equals 1 so we can get rid of that on the left side. We have $x = 4 \times 2 = 8$.

(check as in previous – plugging in value of x into original equation)

Does it check? Yes, so our answer is 8.

Epilogue

Check is always on paper each time. Introduce the dot for multiplication at this time so as not to confuse the x as a multiplication sign.

February 7, 2005

Word Problems

Age

Upper elementary

Begin

Let's find the number $\frac{1}{5}$ of which is equal to 15.

How can I write this equation?

$$\frac{x}{5} = 15$$

There's our equation. We want to find the value of x . What do we do?

We have to multiply by 5 to end up with just x .

If we multiply the left by 5, we have to multiply the right side by 5 also.

$$5 \times \frac{x}{5} = 15 \times 5$$

The 5s cancel out so we have 75 as our answer.



What is the width of the rectangle that has an area of 1200 square feet and a length of 40 feet.

The first thing we need is the formula for the area of a rectangle. We learned that as base times height.

The formula for a rectangle is also represented as length times width – so we should be aware of both of these. We'll use length times width since that's the information given.

So we have “ $A = lw$ ” - then we need to plug in the information given in the problem.

What are we given? (area of 1200) (write “ $1200 =$ ”)

We are given the length which is 40 also. So we get this equation:

$$1200 = 40w$$

What do we need to do to solve for the width? (divide by 40)

Why? This is how we get rid of 40 on the right side, right?

But if I divide the right by 40 I have to divide the left by 40 also.

$$\frac{1200}{40} = \frac{40x}{40}$$

We can cancel our zeros and we end up with

$$\frac{120}{4} = w$$

So we have $w=30$. We can check that by plugging in our value of 30 for x in the original equation. (do so)



The area of a rectangle is 200 square feet. The length is twice as much as the width.

Find the two dimensions.

We again need our rectangle formula – so we start with $A=lw$.

We can plug in the area so we have “ $200 = lw$ ”

What other information do we have? (length is twice the width)

Suppose we represent that as $2w$ for the length times w for the width. So we have:

$$200 = 2w \times w$$

If we multiply on the right side we get this.

$$200 = 2w^2$$

We can divide the right side by 2 to get rid of that. But we also have to divide the left side by two also – this leaves us.

$$\frac{200}{2} = \frac{2w^2}{2}$$

We cancel on each side and we get $100 = w^2$ and we can find the square root of 100 – which is 10.

(cancel the 2s on the left and the 2s on the right)

So we have 10 feet = w and 20 feet is our length.

So **let's check** by plugging our numbers in. $200 = 20 \cdot 10$

You can make up endless problems like this using formulas in our work with area or volume and so forth.



Material

Add material – 4 x 12 inch laminated cardboard piece with 4 inch lines about 1 inch apart across the whole page – along with little paper shovels.

Sometimes we have two unknowns. What about this problem:

This represents a garden. (hold up board)

The garden needs to be dug up in the spring to get ready for planting.

The garden is divided into 12 equal sections. The people who dig this garden are

represented by these little tools.

The reason the garden is divided into 12 equal sections is because the boy who normally digs the garden takes 12 days to do.

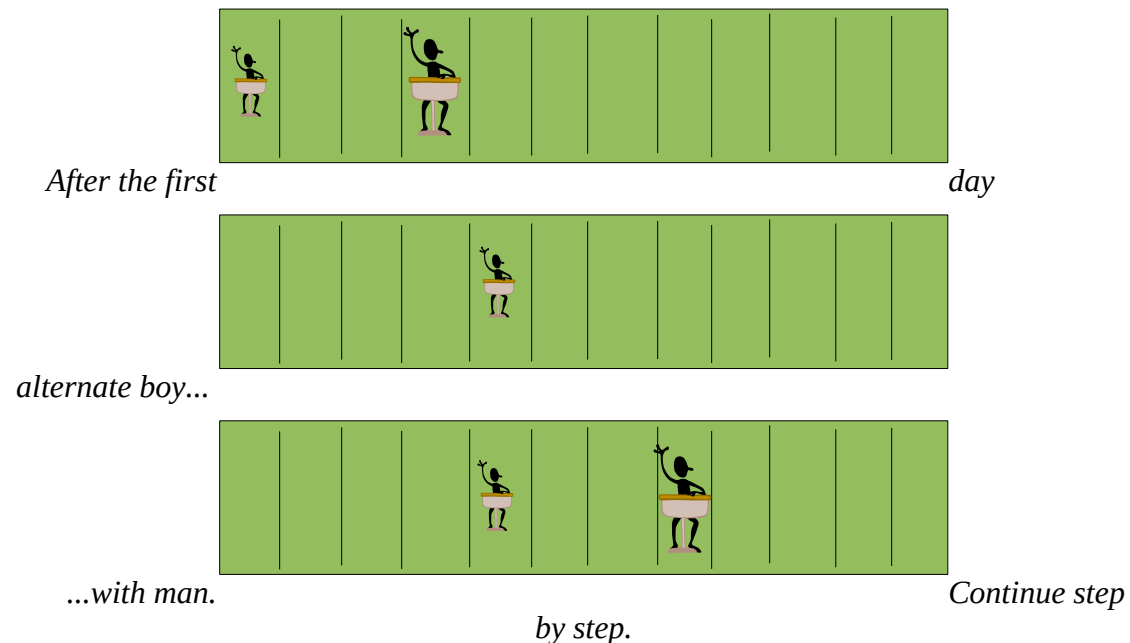
One week the boy was busy. So a man was hired to come and dig the garden. **The man works 3 times as fast as the boy. How many days does it take the man to dig the garden?**



The man and boy are working together for two days. But then the man has another job to go to. The boy is left to finish the garden. How much is left for the boy to finish?

This time, they are working together.

(move shovels for first day, second day and so forth as before to find answer)



First day, the boy did one (move shovel) and the man dug 3 sections (move second shovel)

On the second day ... etc.

So on the third day, the boy will have 4 sections remaining.



A garden is divided into 12 parts. A boy and a man who works twice as fast as the boy dig the garden. If each works for 2 days, the work is finished. Find the number of sections that each digs in one day.

When we approach a problem like this we first want to know what is given.

We know the garden is in 12 parts. The man works twice as fast as the boy. They both work for two days.

What is it we want to find? (number of parts each digs in a day)

Now we need to know what equation we need. We are solving for two unknowns, right?

There is the number the boy digs and the number the man digs. **When there is two unknowns you generally have to write two equations.**

First we decide what letters we'll use to represent the unknowns: the part dug by the boy will be b for boy. We'll use m for the part the man digs. This helps us keep track of what the variables mean.

So b equals number of parts dug by the boy and m is the number of parts dug by the man.

(write this on paper)

b = number of parts dug by the boy in a day.

m = number of parts dug by the man in a day.

Can we write an equation to represent the relationship between these two? ($2b=m$)

Yes, $2b = m$ since the man works twice as fast as the boy.

(write $2b=m$ on paper)

Now we need to write an equation that will enable us to solve the problem.

How can we do this? We know the garden was 12 so we start with " $12=2b + 2m$ "

since they work together for 2 days. This represents the given information.

Write on paper: $12 = 2b + 2m$ (2 is derived from two days)

Now we can use that first equation to help us solve the second equation. What can I do?
(substitute using the first equation)

We can write $12 = m + 2m$. (write this on paper)

Now we have an addition problem we can do: $12 = 3m$

We can divide each side by 3 – and we see that **m is equal 4 parts** of the garden.

4 parts is what the man digs in 2 days. How many does the boy dig? (2 parts)

How do we know? (because of the first equation – it says he works half as much)

So **2b equals 4** – if we divide both sides by 2 we find that b is 2.

So in one day, the boy digs 2 and the man digs 4 parts.

Final writing on paper:

$$\begin{array}{l} 2b = m \\ 12 = 2b + 2m \\ 12 = m + 2m \\ 12 = 3m \\ 12 \div 3 = 3m \div 3 \\ 4 = m \end{array} \quad \begin{array}{l} 2b = m \\ 2b \div 2 = 4 \div 2 \\ b = 2 \end{array}$$

Epilogue

Make a variety of word problems like these. Eventually move away from props and do those problems on paper too. Point out different people solve differently.

February 6, 2005

16. WORD PROBLEMS

Introduction

Prologue

The children should be prepared in primary for word problems – beginning with the golden bead materials. When the teacher poses word problems, she asks the children if they have to do addition, subtraction, multiplication or division. Then they use the golden bead material to work out the answer.

We mentioned periodically that we should give word problems with each area of math. However, we don't stop and give word problems each time. In class, we **should give word problems to the children each time a new skill is introduced** and they have become comfortable with it. Word problems should be part of every day in the class.

Give the children **strategies for solving word problems**. Have sample texts for word problems. Use books adopted by local public schools. If it is poor supplement it with other samples. **Give them practice materials or standard tests used**; cut them into cards for use like flash cards. They may use certain terms not known by the children so we need to explain this.

The word problems we'll do are ones we can use to teach the children effective problem solving skills for word problems. Keep in mind there is often more than one way to solve a word problem – depending on how one looks at the problem.

Example

Jim divided a small pie into 6 equal pieces and ate 5 of them. Write a fraction to show how much of the pie he ate. How do we begin?

Begin by noting the information given. Help the child find what is given and if there is any information they do not need.

Then, identify what is to be found.

Identify the process to be used. Solve and identify the unit to be used in the answer.

So in this case, we begin by using the family of sixths fraction pieces – take away the 5 pieces. This could also be drawn – taking care to keep the pieces equal. He ate $\frac{5}{6}$ of the pie.

We can make problems like this when the children are young. Any time they need to use sensorial material to solve a problem, this is fine. Drawing the material may help some children solve problems more easily. Help them to discover which strategies

help them.

Example

The girls at a birthday party have eaten $\frac{2}{3}$ of the cake. Later, the birthday girl's younger brother ate $\frac{1}{4}$ of the leftover cake. What fraction of the cake is left?

In this case, we can draw the cake and mark off the part eaten. Then, we can break up the remainder of the cake and break it into fourths. So the answer is $\frac{3}{12}$ or $\frac{1}{4}$ of the cake.

There is another way to solve this depending on their skill level. We have $\frac{1}{3}$ leftover. So we can multiply $\frac{1}{3}$ times the $\frac{3}{4}$. This also gives us the fraction of the cake left.

Another way is $\frac{1}{3}$ of the cake was available and the boy ate $\frac{1}{4}$ of that. We can multiply those to find what he ate. So he ate $\frac{1}{12}$ th. Then we have to convert the $\frac{1}{3}$ into twelfths and subtract: $\frac{4}{12} - \frac{1}{12} = \frac{3}{12}$ or again $\frac{1}{4}$ cake.

Example

$\frac{5}{6}$ of a pole is 10 meters long. How long is the pole? Again, we can make an illustration – draw the pole and mark off the unknown portion as well as the 10 meters we know of.

Let's divide the part we know into 5 parts of a total of 10. We find each section we know of is 2. And we can find that the $\frac{1}{6}$ th piece also must be 2 – since that would be a $\frac{5}{6}$ th equal piece.

We can solve it arithmetically. We can divide 10 by $\frac{5}{6}$. This is group division. We invert and multiply to get this answer – 10 times $\frac{6}{5}$ gives us the 12 meters answer.

February 6, 2005

Distance, Velocity and Time

Level I

Material

Envelop labeled “Distance, Velocity, Time.” Felt mat, golden beads – units through hundreds. Green skittles from racks and tubes. Blank tickets and pencil. Pre-printed cards whose general initial setup is as follows:

What is known?	Time	t	=
	Velocity	V	÷
What is wanted?	Distance	d	x
		?	

Prologue

In the classroom, we present specific word problems which involve ticketing and labeling what is know and what we must find. These give the children an opportunity to explore strategies for word problems in an organized manner using the golden bead materials as a sensorial basis. We can prepare different examples. These prepared examples are ones the children often run into in textbooks and tests. Give them preliminary experience.

Before exploring these word problems, give the children some experience of what the terms mean. The **children may run a foot race; help them to measure off a distance**, establishing a starting and finish line. Help the children understand they must run in as straight a line as possible because that is called the **distance**.

Use a stopwatch to time the race. Time how long it takes from start to finish line. Tell the children this is called **time**. How fast they run is called the **velocity**. Run several races. Help the children make a **graph of their speeds** – since graphing is required in most public schools and older children can help the younger ones do the timing and graphing. Do this preparation with them around a week before beginning these word problems.

Begin

Remember the race we ran. I'll show you something else about those terms. How far you ran was the distance. How fast you ran was the velocity. How long you took was the time.

I have an envelope labeled velocity. Inside are tickets. They say “what is known”, “what is wanted”, “time”, “velocity”, “distance”, “v”, “d”, “t” and a question mark “?”.

(place labels vertical column at right while speaking as term is introduced)

Distance

We have a plane that travels 500 miles per hour. How far can it travel in 3 hours?

So what is known? I know the velocity. I also know time.

What do I want to know. I want to know distance. (place labels on mat)

I'll write 500 on my ticket since we're traveling 500 miles per hour.

I'll write 3 on my ticket since we're traveling 3 hours.

What is known	Velocity	v	500
	Time	t	3
What is wanted	Distance	d	?

Let' see if we can use our beads to find the answer.

We can see that in one hour we go how far. (500 miles)

(lay out 5 squares stacked with 1 skittle on top of each)

This skittle represents 1 hour.

But how many hours did I go? So that was the first hour. Here's the second and here's the third

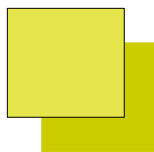
(place two more stacks of 500 – 5 squares stacked – each with one skittle on top – stacks placed horizontally across the middle of the mat)

What is known	Velocity	v	500
	Time	t	3
What is wanted	Distance	d	?



So what can we do with our materials to find the answer? (put them together)
(move all skittles aside and stack all squares together in center of mat)

What is known	Velocity	v	500
	Time	t	3
What is wanted	Distance	d	1500



stack of 15 squares stacked

So I count 15 squares – that's 1500 miles. So I add my ticket that says 1500 in place of my question mark. We found the plane went 1500 miles in 3 hours.



Velocity

A plane goes 1500 miles in 3 hours. At what speed is it traveling?

Start out with what is know. What is 1500 miles? (distance)

It's done in 3 hours? What is that? (time)

I want to know how fast the plane goes. What's that? (velocity)

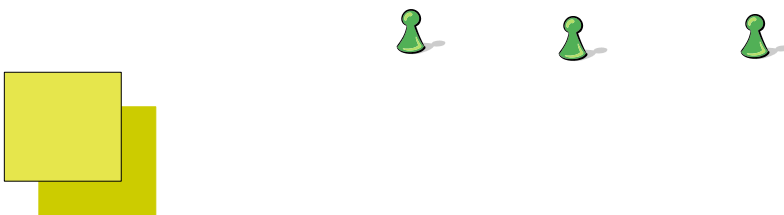
What is known	Distance	d	1500
	Time	t	3
What is wanted	Velocity	V	?

Let's use golden beads to find our answer. Here's my 1500 hundred miles.

(stack up 15 squares – we're going to share out each square one by one - to each skittle)

I took 3 hours to go 1500 miles so here are my 3 hours.

(place 3 skittle across mat)

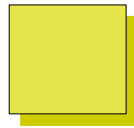
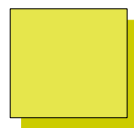
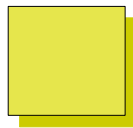


So now let's share out. I'll share out the distance to the time. (share out all squares laying each under each of the skittles – completing **one-by-one** per each skittle)

So how far did the plane go in 1 hour? (500 miles)

I can write 500 miles per hour on a blank ticket and I'll replace my question mark.

What is known	Distance	d	1500
	Time	t	3
What is wanted	Velocity	V	500



Time

A plane traveled 1500 miles.

If the velocity was 500 miles per hour, how many hours did the plane travel?

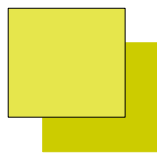
What is known? (distance) (lay out label)

What is known? (velocity) (lay out label)

What is wanted? (time) (lay out label)

What is known	Distance	d	1500
	Velocity	V	500
What is wanted	Time	t	?

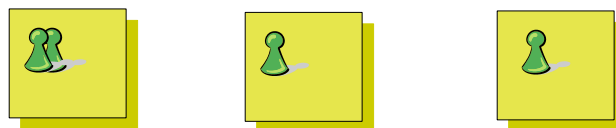
Let's set out our 1500 miles. (set out stack of 1500)



The velocity is 500 miles per hour. What question do we ask ourselves to find the hours? How many 500s are there in that 1500 hundreds.

(use group division – make groups of 5 until all are exhausted)

So how many hours would the plane travel (3) (place 3 skittles atop stacks)



So, my plane travels 1500 miles at the speed of 500 miles per hour and traveled for 3 hours.

(fill in '?' with '3')

Epilogue

To facilitate make problems, create problem cards. In this case, we have green solving for distance, blue for time and yellow for velocity. First distance, we took from store **500 3 times** putting skittle on each. For velocity, we took stack of 1500 put

out 3 skittles and then did **distributed division** one by one. For time, we began with stack of 1500 and did **group division** – taking 500 it and placing under first skittle, taking 500 placing it under second skittle, take 500 for 3rd skittle)

February 6, 2005

Level II - Arithmetical Solution

Prologue

When the children's skills are ready, introduce problems like those we just did using an arithmetical solution. Look at the what is known and what is wanted as before. Then work it out using an arithmetic solution – through building an expression that represents the material level I impressions we have built up.

Distance

(put out labels as in level 1)

A plane travels 500 miles per hour. How far can it travel in 3 hours?

What is known? (velocity and time)

(place cards)

What is wanted? (distance)

Place ticket for velocity and time (500 and 3)

(Same as level 1 – place 3 stacks of 500, one for each hour, each with a skittle on top)

What did we do to find our answer? What process did we use? (multiplied)

We took velocity and we multiplied it by times 3.

(place 'v', 'x', and '3' in expression on mat at bottom of workspace)

Let's plug in the velocity of 500 mph.

And that equals what? (1500)

1500 what? (miles)

Final expression: $500 \times 3 = 1500 \text{ miles}$



Velocity

A plane goes 1500 miles in 3 hours. At what speed does it travel?

What is known? What is wanted? (place tickets as in level 1 solving for velocity)

So what is my velocity? (500 miles per hour)

(same up to here – now add the expression at bottom of mat)

$$1500 \div 3 = 500 \text{ mph}$$

What process did we use to find our answer? (division)

So we divided our distance by time. And what did that equal? (500 miles per hour)

And we can introduce that as “mph”



Time

A plane traveled 1500 miles. If velocity was 500 mph, how many miles would the plane travel?

So what is known? (velocity and distance) (set out upper labels as you speak)

What is wanted? (time)

(same as level 1 – **use group division** with materials)

So how many hours did my plane travel? (3 hours)

So, what process did I use? (group division)

What did I start with? (distance and velocity)

(form new expressions at bottom) $1500 \div 500 = 3 \text{ hrs.}$

February 6, 2005

Level III - Using Formula

Age

8 or 9

Prologue

Here in level 3 we express the formula. This is the first time they will see number variables.

Distance

A plane travels 500 mph. How far will it go in 3 hours?

(lay out tickets as you speak, for what is known and what is wanted as in level 1 and 2 - fill in written tickets for variables – all like was done in levels 1 and 2)

(bead materials are optional for level 3 – use if the children are unsure)

What is the velocity in this problem? (500 mph)

What is the time? (3)

What do I need to do with those numbers to get distance? (multiply)

How can we express what we did here to get our answer?

We'll use these letters to help us? We'll make a formula.

Let's start with what is wanted – our distance. (move d ticket)

(move tickets to bottom of mat while talking)

(move equal sign in front of d)

We then took that 500. (move v ticket down)

What process did we use? (move x sign)

We multiplied the 500 by what? (3) (move t sign)

This is a way to use abbreviations to express what we do.

$$d = V \times t$$

We **always start with what is wanted first.**

We can think of this as a short-hand way to find the distance. We call it a formula.

But people made a rule that in a formula, when you have 2 letters together, it means to multiply them. **So I don't need this multiplication sign.**

(take multiplication sign away - leaving “d=Vt” at bottom of mat)

If we plug in our numbers, we have the expression $d = 500 \times 3$. That gives us 1500 miles.



Velocity

A plane goes 1500 in 3 hours. How fast is it traveling?

What is known? What is wanted? (place tickets at top of mat as always)

What process do we use? (division)

What do we begin with? We begin with what is wanted. In this case – velocity.

(move label 'v' at bottom of mat plus an equal sign)

We take distance and divide by time.

(place tickets at bottom of mat for each new expression)

$$V = d \div t$$

We never use this division sign ('÷') in a formula.

But remember we can write a division problem as a fraction. So I can take this division sign out and write it like this.

$$V = \frac{d}{t}$$

So **plugging in our numbers** we have: 1500 divided by 3 equals 500 miles per hour.

(substitute the tickets for written numbers to complete the expression)



Time

A plane traveled 1500 miles. If velocity was 500 mph, how many hours would the plane travel?

(format labels for what is known, what is wanted, fill in known values with written tickets)

What are we solving for? (time)

(format “t=” at bottom of mat)

What process did we use?(division)

So we took distance and divided by velocity (place expression while speaking)

$$t = d \div V$$

But we have to use our division bar – so let's do that – write the division as a fraction

$$t = \frac{d}{V}$$

Let's fill in our numbers now. (substitute distance and velocity numbers for variables)

So what's our answer? (3 hours)

Epilogue

Children can do more problems like these; we supply the problems until they are comfortable with the process. Then they can make up their own problems.

February 7, 2005

Principal, Interest, Rate and Time

Level I

Prologue

Begin with an oral introduction. Discuss the bank, loans, savings accounts – things they may already know of. We are doing **simple interest – not compound interest**. The class may also have a bank account for outings.

In some places, people use capital instead of principal. We begin using the terms since the children have had them already. This work begins in lower elementary but the children will only begin to use formulas abstracting in upper elementary class.

Material

Felt mat, color-coded prepared problems, golden bead material (using green unit beads instead of gold unit beads), green skittles (from racks and tubes), blank tickets and pencil, envelope marked “interest” (containing “what is known?”, “what is wanted”?, “time”, “interest”, “rate”, principal”, “years”, “\$” (2), “100”, ‘?’ , ‘/’ , ‘x’ (2). ‘=’, ‘(’, ‘)’ , ‘t’, ‘I’, ‘r’, and ‘P’ – plus two division bars).

Begin

When we put money in the bank, it is called principal. The bank pays us money for the use of our money. The money they pay us is called interest. They pay interest according to a rate. The rate is a percent which means, they pay a certain amount of money per 100 parts. The time is how long the money is in the bank. The bank thinks in terms of years. The rate then, is always expressed as per year.

Interest

I leave 1300 dollars in the bank for 3 years. The bank paid me 10% interest per year. How much money did the bank pay me for 3 years?

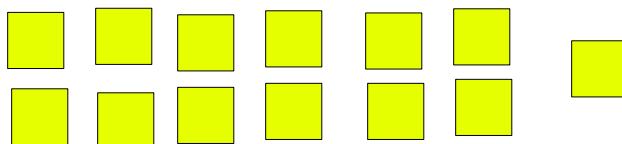
We know the principal, the years and the rate.

We said the rate is always expressed per 100. Let's fill in our numbers.

What is known?	Principal	p	\$	1300	
	time	t		3	years
	rate	r		10	100
What is wanted?	Interest	I	\$	?	100

So let's start with principal. I'll get that out with gold bead material.

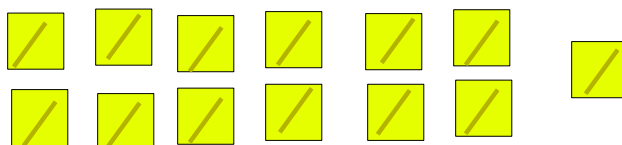
(create stack of 13 squares – then spread them over the mat into rows of 6, 6 with 1 in 3rd row)



The rate paid was 10 percent which is 10 per 100. That means for each one of these hundreds, the bank paid me 10 dollars.

For this hundred I got 10 dollars. And for this 100, and for this 100 ...

(place a 10-bead on top of each square)

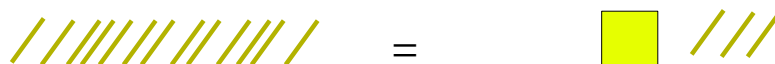


So here's the ten the bank paid me for each 100. That's for 1 years. I'm going to collect all the 10s. (take all 10 bars aside in pile at far left)

I don't need these hundreds now so I'll put them away. (do so)

So how much did the bank pay in 1 year? (count 10-bars – exchange at 100 for a square)

(we now have a square and 3 10-bars placed at left of mat – labels at top)



They paid me 130 for 1 year. But I left the money in the bank for 3 years.

So I'll set out 130 a second time and a third time. (do so)
 (we now have 3 groups – each with 1 square and 3 10-bars – across bottom horizontally)

We put them altogether to find interest for 3 years. 390 dollars is my answer.
 (**fill in 390** for the question mark on the labels)

Epilogue

In beginning, choose problems. Allow children to make problems after some practice.



Rate

I left 700 dollars in the bank for 3 years. The total interest was 168 dollars. What was the rate?

What is known?	Time	t	3	years
	Interest	I	\$	168
	Principal	P	\$	700
What is wanted	Rate	r	?	100

Ticket layout

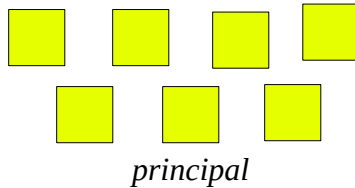
I'll use green beads for my 168 dollars (1 square, 6 10-bars and 8 green units)
 That's for 3 years but I need 1 year's worth.
 (place 3 green skittle horizontally across mat and share out 10-bars)
 (exchange the square and share out to skittles)

So what did I get for 1 year? (56)

56 dollars. I only need one of these. (leave 1 group of 56 – put other two away)

I'm going to spread out my principal on my mat.

(place 7 squares in two rows 4 then 3)



I want to find out how much interest I got per hundred.

So I'll share out my 56 to each of my hundreds.

(share out beads on top of the squares if possible)

So I put 8 on each 100 – each square has 8 units. (these are green units)

So I got 8 parts per hundred from the bank. My rate is 8; I'll replace my question mark.



Principal

I received 165 dollars for deposit left in the bank for 3 years at 5 percent interest. What was the amount of the original deposit?

What is known and what is wanted? (format labels)

What is known?	Time	t	3	years
Level I	Interest	I	\$ 593165	
	Rate	r	5	100
What is wanted	Principal	P	\$	?

label layout

We got interest for 3 years but we need to know what we got for 1 years.

Let's get out our 165 (using square, 6 10-bars and 5 green units)

So I need to share this out for 3 years.

(spread 3 skittles across mat – share out the beads to each skittle)

Now I can tell the interest I got for 1 year – it's 55 dollars.

We don't need these other 2 years worth – so I'll put these beads away.

(one group of 55 remains)

We want principal. We know the rate was 5 parts per 100.

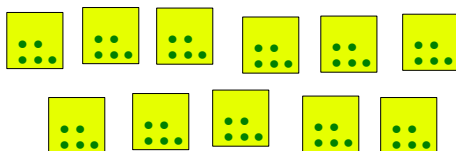
That means for every 100 dollars, I got 5 dollars.

So here's 5 dollars. (place 1 square on mat with 5 green beads on top)

I'll take another 100 and it also gets 5.

(using group division)

(continue placing hundreds with 5 greens on top until exhausted)



I got 5 dollars for each of these hundreds.

So what do I do to find the total principal? (add them up)

I don't need these units now.

(put away green units)

My principal was 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 – 11 hundred dollars.
 (stack the hundreds at bottom left of mat)

I'll write a ticket for that and replace my question mark.



Time

I deposited 1500 dollars in the bank. At the rate of 8 percent, the bank paid me 360 dollars. How many years was the money in the bank?

What is known and what is wanted?

What is known?	Principal	P	\$	1500
	Interest	I	\$	360
	Rate	r	8	100
What is wanted	Time	t	?	years

Labels layout

Our problem tells us 1500 dollars in principal so I'll get that out. The first thing I have to find is how much interest each 100 got.

(spread out the 1500 dollars across the mat – 15 squares)



I need to share out the interest to each 100. I got 360 in interest. I'll get that out.

(**share out one-by-one** each 10-bar onto each hundred, then exchange and share out each green bead)



So how much did each hundred get? (24)

I need only one of these 24s so I'll put the rest away. (do so)

For each hundred I got 24 dollars. For each hundred – this was for some number of years.

We know the rate we got paid was 8 percent - or 8 per hundred.

So I need to make groups of 8 out of these 24 – since 8 percent is my rate.

I'm looking for the time. (use **group division** to create groups of 8)

How many groups did I make? So what was the total time I left my money in the bank?

(3 years) So I'll replace my question mark with 3.



February 7, 2005

Level II and Level III - Arithmetic and Formula

Prologue

We should work through all of level II for all components (interest, etc.) and then all of level III; for convenience here, we combine them.

Begin

Interest – Level II

I leave 1300 hundred dollars in the bank for 3 years. The bank paid 10 percent interest per year. How much money did the bank pay me for 3 years.

What is known and what is wanted?

(as in level I, put out tickets of what is known and what is wanted)

Given principal is 1300 dollars and I was paid 10 percent interest over 3 years.

What did we do to solve this problem?

We got our 1300 hundred dollars and placed interest onto each square. I really multiplied my 1300 by my rate which is 10 over 100

(pull down tickets while speaking)

$$\begin{array}{r} 10 \\ 1300 \times \frac{\text{-----}}{100} \end{array}$$

Then I gathered my ten bars – that was just one year. My money was in the bank 3 years.

So I multiplied that 3 times.

$$\begin{array}{r} 10 \\ 1300 \times \frac{\text{-----}}{100} \times 3 = \end{array}$$

So let's do the math. I can cancel – this is 1 over 10 (cancel 10 over 100)

I can also cancel here (1300 and the other zero from the 100)

$$130 \times \frac{1}{1} \times 3 = 390$$

So my answer is \$390.

Interest – Level III

(we have our tickets setup as always – using same problem as before)

You've been working on this for along time now. Can we develop a formula for this?

What is the first thing I pull down? (what I want to know)

That's the interest – so I'll pull that down.

I started with my 1300 hundred dollars – that's my principal.

I multiplied that by my 10 percent – that's my rate.

And I multiplied that by my time – my 3 years.

$$I = P \times r \times t$$

Remember if letter are multiplied together, we can take them out so we have:

$$I = Prt$$

If you looked in math books, this is how you would see it. Let's try some numbers and see how it works. (children try out some new problems, plugging in numbers)

Rate – Level II

I left 700 dollars in the bank for 3 years. The total interest was 168 dollars. What was the rate?

What is known and what is wanted? (lay out tickets as in level I)

In this problem 700 dollars is my principal, 3 is my time in years and my interest is 168 dollars. We don't know the rate.

We started with total interest of 168 dollars. (pull down ticket '168' to bottom of mat)

I got our my 3 skittles to divide it by the number of years. (pull down ticket '3')

I got 56 when I did that.

Then I divided that answer by my 700 dollars. (place tickets)

$$168 \div 3 \div 700 = \text{interest rate}$$

I divided the 56 into 3 skittles - that gave me 56 divided by 700.

$$56 \div 700 = \text{interest rate}$$

Let's think of that as a fraction. All we can do now is reduce. We can divide numerator and denominator by 7 in this case.

$$\frac{56}{700} = \frac{8}{100}$$

Rate – Level III

(we have our tickets setup as always – using same problem as before)

Let's express the formula.

We begin with what we want to find – rate. (format “r =”)

I started with interest, right? I took it and divided it by my years – which is my time.

$$r = I \div t$$

Instead of using this division sign, let's use our division bar.

$$r = \frac{I}{t}$$

And I divided that by what? (principal)

$$r = \frac{I}{t} \div P$$

We want to get rid of this division sign. We could think of this principal as P over 1.

$$r = \frac{I}{t} \div \frac{P}{1}$$

So what do we do. We invert and multiply.

$$r = \frac{I}{t} \times \frac{1}{P}$$

I don't need the multiplication sign so I can do this. Notice that in formulas we always write capitol letters first. This is my final formula.

$$r = \frac{I}{Pt}$$

Principal – Level II

I received 165 dollars for deposit left in the bank for 3 years at 5 percent interest. What was the amount of the original deposit?

What is known and what is wanted?

(we have our tickets setup as always – using same problem as in level I)

So my interest is 165 dollars, my time is 3 years, my interest is 5 percent. I don't know my principal.

The first thing I did was set out my 165 dollars. Then I divided it by 3. (format tickets)

$$\frac{165}{3}$$

That was interest for 3 years. Then we shared that out – 5 per 100.

$$\frac{165}{3} \div \frac{5}{100}$$

That equals

$$55 \div \frac{5}{100} = \text{original deposit}$$

So we invert and multiply – and we can cancel out our 55 and 5 for 11 and 1 so we have:

$$11 \times 100 = 1100 \text{ dollars}$$

Principal – Level III

I received 165 dollars for deposit left in the bank for 3 years at 5 percent interest. What was the amount of the original deposit?

(we have our tickets setup as always – using same problem as in level I)

Let's see if we can develop the formula. What is wanted? (principal)

(lay out (P ="))

What did we do first? We divided interest by time.

$$P = \frac{I}{t}$$

Then we divided that by the rate – 5 per 100.

$$P = \frac{I}{r} \div t$$

But I don't want that division sign so let's make it into a fraction.

$$P = \frac{I}{t} \times \frac{1}{r}$$

So our final answer is:

$$P = \frac{I}{rt}$$

Time – Level II

I deposited 1500 dollars in the bank. At the rate of 8 percent, the bank paid me 360 dollars. How many years was the money in the bank?

We know principal, rate and interest and we want time. (layout tickets)

So in this problem my principal is 1500 dollars, my rate is 8 per 100, my interest is 360 dollars.

What did we do to solve? We divided our interest and divided it by the principal.

We start with out 360 and divide it my 1500.

$$\frac{360}{1500}$$

What did I do to that? I divided that by my 8 per 100 – my 8 percent.

$$\frac{360}{1500} \div \frac{8}{100}$$

We can invert and cancel. This leaves us:

$$\frac{360}{1500} \times \frac{100}{8} = \frac{36}{15} \times \frac{10}{8}$$

Multiplying across I get 360 over 120 – I can cancel again and I get 3.

That's 3 years for my time.

Time – Level III

I deposited 1500 dollars in the bank. At the rate of 8 percent, the bank paid me 360 dollars. How many years was the money in the bank?

We know principal, rate and interest and we want time. (layout tickets)

So in this problem my principal is 1500 dollars, my rate is 8 per 100, my interest is 360 dollars.

We divided out interest 360 by our principal.

$$t = \frac{I}{P}$$

Then we divided that by the rate.

$$t = \frac{I}{P} \div r$$

But I can't use that division sign so I'll express it as a fraction.

$$t = \frac{I}{P} \div \frac{r}{1}$$

When I divide by a fraction – can invert it and multiply instead.

$$t = \frac{I}{P} \times \frac{1}{r}$$

So multiplying this through I get this formula.

$$t = \frac{I}{Pr}$$

Epilogue

If the children need to, use the materials. They may be confident enough without them so don't use them unnecessarily. Perhaps do the first one to see how they do – with materials – then without for subsequent problems. Review verbally what we did in all cases. Stay with the child for a while after introducing something new.

Often times these lessons begin by seeing children already working.

February 11, 2005

18. REMEDIAL MATH

Quantity

Prologue

We are trying to fill in gaps. The immediate goal is to get the children reading and writing – so they can take part in cosmic education. But at the same time, we need to begin remedial math work if it is needed. The aim is to, as quickly as possible, help the child understand each of the four basic math operations, to memorize math facts and to know which operation needs to be used to solve a given problem.

In London, in 1958, on the elementary course he gave there, Mario Montessori spoke about one of Dr. Montessori's books on math “Psycho-Arithmetica.” He said, “The problem often presented itself of children who came to school when already in the second age group. The tendency of the teachers was to make them go through all the successive steps and apparatus used with the first group. As they lacked the incentive given by the urge of the sensitive periods, the children were bored.”

Mario went on to say that Dr. Montessori advised giving only what was essential to acquiring the necessary knowledge – so they understood the operations and how to do them – and that this needed to be done according to the second plane characteristics. Therefore, we will make **use of a few selected pieces of primary material** but only the ones that are absolutely essential. We'll use them in a different way than they are typically used in the primary classroom.

Some of the materials are already in the elementary environment, that are needed from the primary class for remedial work. Considering what is necessary for the children to know, to go on to elementary work, they children first of all **need a knowledge numbers from 1-10**. They need a **knowledge of the categories and hierarchies** – how zero is used as a placeholder. Then, they need knowledge of the **four basic operations**, as well as which one is used in a given circumstance. Finally, they need the **math facts** at their finger-tips.

We need to help them acquire these skills through **second plane characteristics: reasoning mind, large scale, working in groups, imagination, and a need for variety and expansion of environment**. We want to make it fun – to engage the children.

Short frequent lessons are more effective than a long lesson that drags until you wear them out and leave them as a failure. Short lessons build up their confidence. “Look at how much you know already!” They need us to be their cheerleader – to **help them build internal confidence and self-esteem**.

Hopefully, records are available about what they've had, what they need and so on. Regardless of these records, the **teacher will need to do some assessing to figure out exactly where each child is at**. The records are a good starting point for this.

Through observing and the records, through talking to other teachers, hopefully one is also able to develop some idea of how many children need remedial work – and the type of remedial work required. This allows planning for the right work and allows the use of groups for this work.

The **numbers from 1-10** are generally known – but not always. Use the fingers to count, making sure they can count. See if they can write the numbers and if they can recognize the numbers. One way to do this in a non-threatening way is to write a number in the air – then say, “can you write 2”? There is no pencil, there is no paper – write it big in the air for them and ask if they can do this too. If one finds it necessary to test quantities, bring in number rods just for this test; mix them up on a mat and ask them if they can find 2, if they can find 5 and so on.

For **decimal categories**, the first thing we introduce is quantity. It could be to a small group of children or it could be with a single student.

Material

Golden bead material, unit, ten, hundred and thousand from glass or plastic material, felt mat and store of golden material – hundreds, thousands and so on.

Begin

(tray with some bead material is on a felt mat)

Remember those numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 10?

We'll do more work with numbers today.

Here, we have a unit (solitary gold bead on the mat) This is one unit.

(place 10-bar next to it, at left of the unit, - point to each bead on the bar)

1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Ten units make ten.

1 ten, 2 tens, 3 tens, 4 tens, etc. (put hundred square on mat to left and count)

Ten tens – that's 1 hundred. (count and trace each ten)

(place golden cube at left of all so far)

Here is 1 hundred, 2 hundred, (...up to 10)

Ten hundreds – that's 1 thousand.

(point to each as you speak)

One unit, one ten, one hundred, one thousand.
(they are arrayed horizontally – thousand cube at left)

(do second period work)

Can you please show my 1 hundred? Can you show my 1 unit?

Is this 1 hundred? (hold up ten) (keep them on their toes)

Epilogue

As you feel they are catching on, introduce the wooden material for the hundreds and thousands. “This is also 1 hundred” and show the wooden square. “We also have more units and tens”!

Take the child to the shelf and show them the material on the shelf, pointing out the units, tens, hundreds and thousands again. Show them where the trays are, since they now can continue this work on quantity. “Sally, bring me 6 thousands please.” When they come with their tray and beads ask, “How many did you bring, let's count them.” Then ask for 2 categories at a time – and build up from there. You can put a quantity on a tray and ask, “Who knows how much I have here”? These **lessons can last 2-3 minutes**. Build in the repetition, high frequency of short duration.

February 11, 2005

Symbol

Material

Felt mat, set of large decimal system cards 1-9000 (borrowed from a primary classroom). Do the work on a table as opposed to the floor as in primary.

Prologue

Sitting at a chair at a table helps the child focus and manage their growing body. We emphasize that zeros determine category using language like “6 tens.” We are focused on the symbols, the zeros and the category names.

Begin

(cards are upside down, separated by category, thousands at the top, hundreds below, etc.)

(select bottom set of cards – units, and turn up at chest level – begin drawing 1-by-1)

What do we have here? (draw card) One!

(continue to draw) Two, Three, ... etc.

How many zeros do these numbers have? (none) Oh, interesting! No zeros.

(select tens stack and do same)

(cover the zero of the 10)

What do I have here? (1) But its 1 ten **(remove hand from zero)**

What do I have here? (2) (cover zero again) But it's 2 tens!

(continue as above to 90)

: : :

What do we have here? (9) but how many zeros does it have?

So how many zeros does the tens category have? (1) Yes, 1 zero.

(select hundreds stack and do same)

(cover the zeros of the 100)

What do I have here? (1) But it's 1 hundred **(remove hand from zeros)**

How many zeros are there here? (2)

What do I have here? (2) (cover zeros again) But it's 2 hundreds!

(continue as above to 900)

: : :

What do we have here? (9) but how many zeros does it have? (2)

So how many zeros does the hundreds category have? (2) Yes, 2 zeros.

(repeat same process for thousands)

What do we have here? But it's 1 thousand (remove hand)

How many zeros are here? (3)

: : :

So how many zeros do the thousands have? (3)

Epilogue

We don't set the cards out as in primary – they remain in their stacks; thousands at top, hundreds below, tens and then units at bottom. **Check for understanding** after this lesson. “Who thinks they can find 500 hundred in these cards? Can you find 4 tens”? Get them looking through the stacks to find the right card.

“I wonder who can tell me how much this is”? (set 300 and 60, two cards, next to each other – tens at right as usual). “How do you know this is 300? How many zeros do we have for hundreds”? **Reinforce that the number of zeros determine the category.**

The **language is secondary** to their understanding categories and instances of categories. We use language “6 tens” at this level versus sixty.

February 11, 2005

Symbol and Quantity

Material

Large decimal system cards, felt mat, supply of golden bead material, an empty tray and a cup for each child in the group to retrieve materials.

Prologue

We are again working at a table. Number cards are set out face up – thousands at top of a column at right, hundreds below, tens, then units at bottom. Show the child how to put the material under the highest category; we are beginning to sort the material and cards by category now.

Begin

Please choose a card and put it on your tray. Go to the golden bead material and bring back this much. (she chooses 300)

How much did you bring? Let's count it. One hundred, two hundreds, three hundred.

Let's pick 2 cards now. (200 and 70)

2 hundreds and 7 tens. Can you get that amount from the bead materials? (does so)

How many did you bring? (I brought 2 hundreds and 7 tens)

(she has placed each bead piece under its respective card, all inside the tray)

So you brought 2 hundreds and 7 tens. We can put these together like this.

(do magic slide to show the number is 270 altogether)

So this is 2 hundred and 7 tens. (continue to use this language probably at this point)

Epilogue

You can choose the cards and have the child get the material. You could put the quantity on the tray and have the child get the card. This step is important in their understanding of the decimal system – the ability to put symbol and quantity together. Make sure to do enough repetition to be secure.

February 11, 2005

Counting and Changing

Material

Large decimal system cards, felt mat, supply of golden bead material, an empty tray and a cup for each child in the group to retrieve materials.

Prologue

This work is required before going into operations. If working in a small group, give the children turns: counting tens, then hundreds, etc.

Begin

(set out a quantity of units, tens and hundreds)

(hundred squares are at far left, then tens rightward, with units at far right)

I wonder how many are here. Who would like to help me count it?

(stop child after he has taken his 10th bead)

Stop! We can only go to 9 before we have to exchange them for a 10-bar.

Let's set that new 10 down here at the bottom of the mat under our other 10-bars.

Let's continue counting now.

(stop child before reaching 20 – add another 10-bar as above – and go on counting)

Stop! We can only go to 9 before we have to exchange them for a 10-bar.

Let's set that new 10 down here at the bottom of the mat under our other 10-bars.

(when units are counted, point to the 10-bars at bottom we've counted and add a card for the units we've counted -below the units that were left over in the units pile.)

So we have a lot of units – 2 10-bars and 7 more.

I'll place a card to say we had 7 units and put these 2 10-bars with our other original 10-bars now.

.

So we had 2 tens and 7 units from our pile of units.

(repeat the process for tens and hundreds. Count tens. Place tens into hundred squares when passing from 9 to 10 tens; placing that material below the original group of hundreds to keep it isolated until at tens have been counted. Then put these hundreds with the other original hundreds and put a large decimal card below the number of tens left over)

Epilogue

This understanding that ten of one category makes one of the category to the left, is fundamental and has to be mastered before going on. Make it a game for them and repeat it until they've caught on. Risk a bit of their boredom to make sure it is very solid.

February 11, 2005

Operations

Addition

Material

Golden bead material, large decimal number cards, felt mat, 3 trays and cups for 3 children and 3 sets of small decimal system cards 1-3000, plus paper and pencil.

Prologue

Each of the four operations will follow the same process – the same 4 steps. **First step** is to introduce the process with the materials. **Second step** is after the materials, for each category, after we do that category with the material, do the same on paper and ask if our “heads agree with our hands.” **Third step** is that for each category, we’ll do the work on paper first and then verify it with the material. The **fourth step** is to do the work abstractly on paper. Most of the time is spent with step two.

Step I

Begin

The process follows:

1. Each child chooses and gathers a number from the bead material, placing each into rows of materials on the table – as if combining 3 of the previous lesson's materials – one underneath the last each extending from thousands at left to units on right.
2. Write each of the numbers in a column on paper
3. combine all materials on the table by moving all upward; separate carry-over material at bottom aligned by category until the current category is counted. Place card below each category before counting the next category
4. at end, add the plus sign and equals bar under the 3 numbers in the column
5. explain what the plus sign and equal bar do for us and describe how we added by combining all materials together
6. Add the final answer on paper by referencing the cards at the bottom at each category

Step II

What is done with the material is the same as in step one. On paper is where the difference comes up.

Begin

(unit beads are already counted, number card placed under each category)

Eric found that there were 9 units. Let's write that under the units.

(add 9 on the vertical written expression on paper)

Let's see what we have (look at numbers in the category)

$1+1+7$. Does your brain agree that this is the right answer?

(next child counts up tens)

Jasmine found we have 3 tens. (write 3 in answer under tens category)

We have 3 tens left. What happened to the other tens? (they were exchanged into hundreds)

Let's look at the numbers on paper and see if our brains agree.

$1+8$ give us how many? (9) and $9+3$ give us what? (13) So yes, our brains agree since we have 3 for our tens and we know the other 10 tens were exchanged for hundreds.

(next child counts hundreds)

Barb found 1 hundred on her card. Let's look on paper and see if we agree.

$1+7$ is 8. $8+2$ is how many? (10) Shouldn't I have a zero here? (no, we had a carry from the tens so we have 11)

Oh, so we had eleven with the carry from the tens category. So our brains do agree with what we found using the beads.

Step III

Begin

(we've combined materials for 3 children but have not yet begun to count)

Let's see if we can add up our numbers on paper first this time. (do so)

Now can we verify our answer using the materials? (do so)

Step IV

Entire process is on paper.

Epilogue

In steps 2 or 3, when addition is on paper, perhaps many addition facts are known but not quite all of them. If this happens, have the finger addition chart close at-hand – so this becomes memorization work as well. If they don't know many facts, this lesson is not really possible.

We don't typically use carry-over marks on paper. For those children who need it, use it. But normally we try to handle carry-over in our heads.

February 11, 2005

Subtraction

Material

Golden bead material, large decimal number cards, felt mat, 3 trays and cups for 3 children and 3 sets of small decimal system cards 1-3000, plus paper and pencil and a set of small decimal system cards going up to 9000 used in primary.

Prologue

Begin with one child for the first example. Then subsequent exercises can be done in small groups,

Step 1

Begin

(hand tray with small decimal card set to the child – this lesson begins for one child only)

Here's a tray. You can make up any number you want to make up using these small decimal cards.

I'm going to make another number using the large decimal cards and I'll form my quantity from the bead materials too. (do so)

Okay, I brought 7899. That's a lot. (move materials from the tray to the felt mat)

What number did you bring? (I brought 3768)

(place large decimal cards for 7899 to left of minuend materials)

(place small decimal cards for 3768 to below minuend cards to form vertical expression)

But you don't have beads for your number. That's because I'm going to give you the amount in your number from my beads.

How many units do you need? (8 units) (count out 8 from the minuend materials)

How many tens do you want? (6 tens)

What about hundreds? (I'd like 7 hundreds) (count out 7 from the minuend materials)

And what about thousands? (3 thousand)

So you have enough to make your number now.

So let's count how many are left in each category.

(set small decimal cards below each category as we determine how many are left in minuend remaining)

(write the problem down)

I started with 7 thousands 8 hundreds 9 tens and 9 units.

But I gave you 3 thousands 7 hundreds 6 tens and 8 units.

So I took them away from what we had in the beginning. We did subtraction.

How many are left? (4131)

That's are difference or remainder – what is left.

(write in the answer in the vertical expression on paper and fill in cards at bottom of decimal card expression – to complete that expression too – minuend, subtrahend and different at bottom of the vertical expression)

Epilogue

As you go on, introduce borrowing and exchanging and give them lots of practice on that. Begin with one child and then let groups begin to work together – one getting the minuend, the other getting the subtrahend.

Make sure you are still around to help after this, though you gradually allow the children to take over.

===

(get two volunteers; one forms number out of large decimal cards and gets material quantity for that number, second child forms a number from small decimal cards on a tray)

Emily what number did you bring? (6653)

Carson what number did you bring? (2448)

(write the numbers on a vertical expression on paper)

Carson needs 8 from Emily's materials but she only has 3. What can we do? (exchange)

So now we can give Carson the 8 units he would like (do so)

How about tens - how many does Carson need? (4 tens) (take them from minuend)

How about hundreds - how many does Carson need? (4 hundreds) (take them from minuend)

Carson needs two thousand. (take them from minuend onto Carson's tray)

So now how many are left Emily? How many units do you have? (place decimal cards under each category as we go from right to left – units to thousands)

How many tens? How many hundreds? Thousands?

We have 4 thousands 2 hundreds and 5 units.

Emily started with 6653 but along came Carson. He took 2448. And adding my subtraction sign and equals bar, I'll add in my answer of 4205 at the bottom of our paper.

Step II

(materials are handle the same way and difference lies on the paper – we merely stop at each category to handle the paper updates)

Emily gave Carson those 8 units and we have to stop. **How many units are left? (count)**

There are 5 left so let's write it down in our units place on our paper.

Does that make sense? Do our brains agree with that answer?

How many tens are left?

Does your brain agree? Does it really make sense to you? (repeat for each category)

Step III

(look on paper first this time and predict what the answer will be for each category – then verify using the materials as we did for step 1.

Step IV

(we work on paper only, forming numbers, subtracting by category, finding difference)

February 11, 2005

Multiplication

Material

Golden bead material, large decimal number cards, felt mat, 3 trays and cups for 3 children and 3 sets of small decimal system cards 1-3000, plus paper and pencil and a set of small decimal system cards going up to 9000 used in primary.

Prologue

This is much like addition except all 3 children will take the same amount as the multiplicand.

Step I

Begin

(we delay writing on paper – in contrast with all other prior lessons)

(give each child a tray, there is already 3 sets of small decimal card stores available)

Today I would like each of you to get 1486 in small decimal cards.

Can you set out bead material for the same value on the mat now?

(each child forms 1486 in bead materials. There are 3 rows, one for each child: cards at left, thousands, hundreds, tens and units at far right)

(cards go down left side of mat and bead material go down right side – as in all prior)

We want to know how many we have altogether. What can we do? (combine them)

(the children combine all 3 sets of materials now)

Now we need to start to count.

(child begins counting units, exchanging at tens, placing carry-over tens at bottom of mat underneath the combined tens category)

Now how many are left in the units cup? (8 units)

Now we need to take a large decimal card '8' to show we had 8 units left over after creating those new tens from our units.

(proceed counting tens, hundreds and thousands as was done in prior lessons – see addition for most complete write-up)

(each time we place a small decimal card below the counted category for the amount that did not carry-over to the next category on its left)

(the 3 card representations of our multiplicand are in a column at far left of materials still)

Now we just have to slide our cards together to find the answer. Let's look at what we have – 1486, 1486 and 1486 here (point to each at left).

We took the same number 3 times – so we really only need that number 1 time. We can write it down like this. We can write it like this: (write “**1486 x 3 =**”)

And what did we get? Let's add our answer too. (write “**1486 x 3 =4458**”)

This is called multiplication. When we take a number a certain number of times – it's called multiplication. We can say we are multiplying. We multiplied 1486 times 3.

Step II

(stop at every category and write down the answer, asking if it makes sense)

Step III

(stop before finding each category answer, predict answer on paper, check with materials)

Step IV

On paper only. If you write vertically, we worry about categories. Here, write vertically.

Epilogue

We reinforce the idea right away that they only need the multiplicand 1 time and that we can use a multiplier to show how many times we will take it. Apart from this, this lesson is identical to addition.

February 11, 2005

Division

Material

Golden bead material, large decimal number cards, felt mat, 3 trays and cups for 3 children and 3 sets of small decimal system cards 1-3000, plus paper and pencil and a set of small decimal system cards going up to 9000 used in primary.

Step I

Begin

(make large number from large decimal cards and then form that amount at top of mat)

Let's start with this number: 9789. I have a large number and a lot of beads.

I'm going to share out my beads to each of you. I'm going to give the same amount to each of you. So watch and make sure everyone gets the same amount.

(tease them by giving uneven amounts from time to time)

Let's start with thousands. Here's a thousand for you, a thousand for you, and a thousand for you. **(sharing out 1-by-1)**

Did you each get the same number of thousands? (yes) Okay, put your trays down and I want each of you to go get a small number card for the number of thousands you received.

How about hundreds? (proceed in the same manner)

I've shared out my hundreds and I have no more. So I can't just give this extra one to Priyanka? What could I do to make it fair? (exchange it for 10 tens)

Okay, I'll do that. In the meantime, please get the small decimal card that tells how many hundreds you received and put it on your tray under your hundreds category.

(share out tens and units in same manner, form vertical expression with large cards at top)

Okay, set your trays down and put your number cards together and set them here underneath the big number cards.

We began with 9789 and I shared it out to the 3 of you. You each received 3263. We call this process division. It is equal sharing out.

Let me show you how we write that on paper. $9789 \div 3 = 3263$

This little **sign means we're dividing** this big number. And **how many people did we share out to?** (3) So, that means I'm dividing my number by 3 so I'll fill that in. I'll add my equal sign and my answer here at the end of the expression.

Step II

Stop and write answers as we go, after each category, record the answer on paper. Ask, "Does that answer make sense"?

Step III

Predict answer by writing on paper, then verify using the materials.

Step IV

Derive answer on paper only.

Epilogue

Even when working with operations, one must realize this work should go parallel – one to the other. Once the children have understood the process of addition, and they are working productively on addition facts at perhaps step 3, one should introduce subtraction. There is overlap among these lessons – so timing is critical to getting them through this and moving into other work. The children will only learn if they have repetition and interest; one way we provide for those is through variety and this parallelism provides some of that variety.

Be sure with each of the operational procedures, as soon as it is introduced with golden bead material, the memorization work begins – also in parallel. As soon as the child understands at least 2 of the processes, do a third period lesson by throwing out a little word problem for them. Then ask, "How would you find the answer to that problem? Would you use addition or subtraction?" **Get them used to identifying which operation is used in application to actual problems. This is the 3rd period work.**

February 11, 2005

Memorization

Material

The four addition finger-charts, two subtraction finger-charts, three multiplication finger-charts and two division finger charts. For each process, there are also control charts. For each operation, there are also plastic tickets that have problems printed on them – all in a red box. There are tiles that can be used on the empty chart and can be used as needed. Paper and pencil for each child. Colored paper, hole punch, ribbon or yarn, colored pencils or inks - for making booklets.

Prologue

We focus on addition here but each operation / process is the same. Depending upon how many children need this work you may use many or few charts – one for each child. First thing to know is how finger movements work.

First and fourth charts work the same. Find first number at top (4) using right hand. Second number is found on the left side with left hand (7). Right finger comes down to row where left finger is and left finger goes rightward to meet right finger on a joining row–column junction of the two; the junction cell lists the answer (11).

Second and third charts are more involved.

For **second chart**, which looks like a half chart with number guides only down the left side. We have to find both numbers on the left side. If doing $4+7$ on this chart, place fingers on 4 (right hand) and 7 (left hand) as before but both at left side. Move right finger (top) finger to end of the row, then move downward to meet the row that the left finger is on – that cell contains the answer. What if adding the same number twice? Go to the end of the row and there is the answer.

The **third chart**, which looks like a diagonal strip of the chart, we have to use walking and jumping – go to end of each row for each addend. Right (top) finger goes rightward off the chart by one cell and left finger (bottom finger) goes up and off the chart by one cell. Then each finger jumps over a corner step of a cell, moving toward each other. Where the fingers meet, turn diagonally down and left and that cell contains the answer.

The child with the blank chart needs to use the control chart for answer.

Make the child aware of why the second chart has half as many numbers as the first and fourth charts. “If Becky has an apple and an orange and Carrie has an orange and an apple – don't they have the same thing? When we think about our numbers, if we have $2+5$, isn't that the same answer as $5+2$ ”? Let them use the charts to find out. Make them aware of the commutative law in this context.

Begin

(give each child a chart and his own paper and pencil)

We're going to write a math fact – $4+9$ – Eric, can you write that problem?

We need to find the answer to this problem. Don't say the answer out loud but **use your chart to find the answer and then write it on your paper.**

Make sure to write your name at the top of your paper. If you want to write the addition sign or the equal sign in different colors, you can do that if you like.

If that answer is not in your head already, that's where we want to put it. So when you write down your answer, be sure to put it in your mind and save it for later.

(repeat above for each child – do four problems)

Now let's do these same problems again, but this time let's do it in our heads only.

Turn your paper over now and here we go: $8+4$

Do you have that answer in your head? Okay, not quite memorized so you can use your chart one more time. Remember, we want to remember this answer for this problem.

Draw a little star by this problem so you remember this is a problem you need more work to memorized.

The next problem is $4+9$. Can you remember that answer without using the chart? Okay, write it down.

(continue, reinforcing the attempts to file the answers permanently in the mind)

Epilogue

Show the children how to use the finger-charts so they can work independently. The next time we get the work out, they should have recorded which problems need more effort to memorize. The children can make a booklet of other facts which have been memorized. “Look how many you've learned already”! Be positive and build up their confidence to go forward. Some children are more auditory and so they

need to say them as well as to write them; in these cases, as they continue to work and repeat work on the few they still need to memorize, suggest they say the answer softly to themselves.

These lessons focused on four facts, should be little 5 minute lessons. Next time they get this work out, they will remember some and not all. Keep the ones that need work and add new ones so we always work with four problems.

February 11, 2005

Teens - Quantity

Prologue

The other area of work the children might need help in is teens and tens. This work is going to run parallel to the other work in remedial math. So there can be a variety of lessons going on in remedial math.

Material

Teen board and beads that go with it, the tens board and beads that go with it, a felt mat, 3 small colored movable alphabets.

Begin

(begin with teens – quantity)

In our box we have some 10-bars and we'll put them on the upper right corner of our mat. All of our 10-bars will go together there.

Here's a unit (hold it up). Let's put that here to the left of the 10-bars.

(build up the colored bead-stair – 1 at top, 2 next, then 3, etc. into a triangle)

Now how many did we say this is? (10) (hold up 10-bar)

And how many is this? (1) (hold up unit)

(count and point)

When we put them together we have 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 – we have eleven.

(lay the 11 at bottom left laid vertically)

(hold up 10-bar) What's this? (10)

(hold up 2-bar) What's this? (2)

(count and point)

When we put them together, we get 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 – we get twelve.

(lay the 12 at bottom left laid vertically to right of 11)

(hold up 10-bar) What's this? (10)

(hold up 3-bar) What's this? (3)

(count and point)

When we put them together, we get 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 – we get thirteen.

(lay the 13 at bottom left laid vertically to right of 12)

Second Period

Can you please show me 11? Can you show me 12?

(repeat for plenty of repetition)

Third period

What is this? (point to 12)

What is this? (point to 13)

(repeat for plenty of repetition)



(Use the same process for numbers 14-19)

February 11, 2005

Teens - Symbols

Material

Seguin boards: two tall wooden planks with a column of “10”.

Begin

What number is this? (point to first 10 on the board) (10)

What number is this? (point to second 10 on the board) (10)

So these are all tens, right?

Just like with the beads, I'm going to put this small wooden number here on our first ten.

(slide the wooden “1” into the right side of the first “10” on the board)

This makes eleven.

I'll slide this 2 in the second one to cover the zero in the same way.

What did I make now? This is 12 now.

(repeat up to 19)

Second Period

Can you show me 14? Can you show me 17?

(take all number cards out)

Can you make 15 for me?

Third Period

What is this number? (point to a number)

(continue for repetition)

February 11, 2005

Teens - Quantity and Symbol

Material

Golden bead material. Seguin boards: two tall wooden planks with a column of “10”. Note that the beads are placed to the left of the board number's position; when we have bars and beads, the beads lie to the right of the bar – as done normally when we lay out mixed categories of beads.

Begin

(have bead stair set out and have the board material close at hand)

Eric can you please make 11 using the cards?

Now let's make 11 out of the beads too and set them to the left of the number 11 on the board.

(repeat for 12-19 in order, then vary the order among those numbers for practice)

February 11, 2005

Tens - Quantity and Symbol

Material

Golden bead material. Seguin boards: two tall wooden planks with a column of "10". We have 10s this time, 10-90. Note that the beads are placed to the left of the board number's position; when we have bars and beads, the beads lie to the right of the bar – as done normally when we lay out mixed categories of beads.

The beads that go with this Seguin board are 9 10-bars and 10 units beads.

Begin

(Seguin boards are laid on the mat. 10-50 on left and 60-90 at right)

What's the name of this number? (10) (lay 10-bar to its left)

This will make 20. (hold up a second 10-bars) If I add a second ten-bar, I have 20. There it is. (lay it to left of 20 and pull the 10-bar next to '10' down to '20')

This is 30 (point to 30 on the board – hold up another 10-bar)

If I add another 10-bar and pull these other 10-bars down, we have 30.

Do you hear the sound 'ty' on the end of my words? That means tens. So if I say forty – there are 4 tens. Forty.

(continue up to 90, 10-50 on left board and 60-90 on right board)

(start at beginning again – at first position at top of board on left – 10 position)

I can slide in this '1' – and we have eleven. (place beads, 1 10-bar and 1 unit, at left)

(slide out the '1' and slide in the '2')

If I slide this one out and slide this '2' in, now we have 12. (place 12th bead at its left)

Here's 19. If we slide this (number) out and add one more unit bead, how many do we have here? (20)

We have 10 unit beads don't we. Let's exchange these for a 10 and slide both 10-bars down to the second spot on our board – to 20.

(stack of number is ready to begin again – 21-29 – take care at transitions between 29 and

30, 39 and 40 and so on)

Epilogue

Be sure to be around to help them – **don't leave them – especially while they are mastering transitions.** They will pack up the boards and leave them on the shelf for good. Be playful when possible to get them involved in what otherwise might be boring for them.

The children can **use the small movable alphabet to write the number's name.** Eleven and twelve in one color and when you go on to other teens, they can make the 'thir' in one color and the 'teen' in a second color. Use the same idea with the tens; 'twen' in one color and 'ty' in a second color, 'thir' in one color and 'ty' in same second color. Sometimes the children will like to write the words for the numbers to the left of the beads which lie to the left of the board.

Remember, the **first thing to find out is exactly where each child is in their math mastery;** use personal observations, direct working with them and past records to begin this investigation. Make sure to start each child at the point they are currently at then. Be very targeted to avoid boring and losing the child on one hand, versus beginning at too advanced a place on the other.

Be flexible with your groups of children. Pull 3 together and begin with addition. Two may catch on but the third may need more repetition. Get that third one working with other children who've not yet had a first presentation.

This **work starts at the beginning of the school year** and needs to be done everyday throughout the day – short frequent lessons. We then expect to see the children progressing 6 weeks into the school year; if not, be concerned. Ask yourself whether the lessons have been regular because some children need consistent repetition.

Multiples and factors, commutative and distributive laws, measurement, divisibility, fractions – all these things can be worked on without having memorized their math facts. So don't hesitate to **allow parallel work** in this regard. The **children need variety.**

When it comes to remedial work, **if something works for a child – use it.** Try the flash cards, work sheets or whatever it is – if they work use it. The older they get, the harder it is to make up their math remediation. Make use of any residual aspects of the absorbent mind they may still have.

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